

Instance-dependent certification complexity in branch-and-bound

Abstract

We study the size of an optimality certificate for a fixed relaxation and specified branch-and-bound operations. For continuous problems with a quadratic relaxation gap that vanishes at box vertices, we compare minimum cover, partition, and tree certificates with a running supremum of near-optimal covering numbers, up to a logarithmic factor. For unconstrained objectives with Lipschitz gradients and matching vertex-vanishing upper error, integrals over all positive-dimensional root-box faces characterize these sizes within constant factors. Analytic sublevel asymptotics then give polynomial and logarithmic rates, including rates governed by boundary faces. We also analyze McCormick exact faces, relaxed constraints, objective-cutoff propagation, and branching decisions. A fixed path family separates spatial and decomposition certificate sizes exponentially in dimension while preserving the local relaxation; one-dimensional relaxation-minimizer branching is 4-competitive under an exact-gap oracle. For integer problems, a convex-class covering number gives a relaxation-dependent lower bound and an exact semantic certificate size. Gaussian sparse regression exhibits linear certificates below root exactness. Gaussian binary least squares has matching certificate-size laws at divergent sublinear signal-to-noise ratio and distinct thresholds for a sufficient linear-tree criterion and equality of box and orthant relaxations. A finite nonnegative least-squares coefficient bound proves the simultaneous criterion for square and tall systems. We distinguish certificate size, search nodes, relaxation evaluations, and propagation rounds throughout; archived computations illustrate the mechanisms and their practical limits.

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1 Introduction

Branch-and-bound produces a feasible solution and a proof of its quality. The proof excludes every region that might contain a substantially better solution, using relaxation bounds, disjunctions, and domain reductions. Its size can vary greatly between instances of the same dimension. A large search tree can reflect a weak relaxation, poorly chosen splits, expensive propagation, or difficulty finding a good incumbent. These causes require different explanations.

This paper studies the size of the proof for a fixed instance, relaxation, and class of permitted operations. We call this size the *certification complexity*. The central question is how the relaxation’s error interacts with the instance’s near-optimal points. We develop geometric characterizations for continuous box relaxations, combinatorial lower bounds for convex integer relaxations, and probabilistic laws in two Gaussian models. We then examine how propagation, branching, and decomposition change the certificates that can be constructed.

1.1 Certificate size and search work

Let f_* be the global minimum and let $\epsilon > 0$ be an additive optimality tolerance. A lower-bound certificate covers the points that must be considered by regions on which a specified test proves that the objective is at least $f_* - \epsilon$. A finished branch-and-bound tree supplies such a certificate when its incumbent is optimal. With any feasible incumbent of value $U \geq f_*$, pruning at $L(B) \geq U - \epsilon$ still implies $L(B) \geq f_* - \epsilon$, so the same lower bounds apply. Constructive upper bounds usually fix an optimal incumbent; the relevant exceptions state an explicit search convention.

For spatial certificates, three minima are useful: N_{cover} counts a smallest cover by valid boxes, N_{rect} a smallest rectangular partition, and N_{tree} the leaves of a smallest tree formed by coordinate cuts. They satisfy

$$N_{\text{cover}} \leq N_{\text{rect}} \leq N_{\text{tree}}.$$

These finite minima need not be equal. A lower bound for covers and an upper bound realized by a tree can nevertheless identify their common asymptotic order. A complete binary tree with L leaves has $2L - 1$ nodes.

Domain reduction requires additional care. A closed discarded slab can contain a retained feasible face, whose points the reduction did not exclude. We therefore pair each closed test box with the subset of points it *owns*; retained boundaries continue into the surviving region. These owned families need not be ordinary covers by boxes valid on their whole closures. The geometric lower bounds apply to them directly through per-test estimates. [Section 2](#) gives the ownership construction and charges removed pieces, probe tests used in stopping bounds, and relaxation evaluations ([lemma 2.3](#)). Lower bounds after reduction require the stated pointwise gap assumptions. A weaker bound-gap assumption suffices for raw certificates without reduction.

We distinguish four quantities:

- (i) the size of a smallest certificate in the chosen class;
- (ii) the number of nodes processed while finding one;
- (iii) the work needed to check its regions, including relaxation evaluations, tightening rounds, and propagation rounds;
- (iv) the work needed to find an incumbent of the required quality.

The branching results compare item (ii) with item (i). Propagation and probing can shift work from nodes to item (iii). A geometric lower bound on owned events becomes a node lower bound only when the relevant rounds per node are bounded and the model accounts for probe tests. Every result fixes its relaxation and operation class; a stronger relaxation or an additional operation must satisfy the hypotheses anew.

1.2 Main results

The continuous results concern $\epsilon \downarrow 0$ for a fixed instance. Write $D \subset \mathbb{R}^n$ for the compact root box, $F \subseteq D$ for its feasible set, $m = f - f_*$, and $E(\eta) = \{x \in F : m(x) \leq \eta\}$. Constants may depend on the instance, relaxation, and dimension. The random-model results instead let the problem size grow and state their probability and scaling assumptions separately.

Near-optimal geometry. For a box $B = \prod_i [l_i, u_i]$, put

$$q_B(x) = \sum_i (x_i - l_i)(u_i - x_i).$$

Suppose the bound satisfies $L(B) \leq f(x) - \alpha q_B(x)$ at every feasible $x \in B$, for a fixed $\alpha > 0$. A valid box then obeys $\alpha q_B(x) \leq m(x) + \epsilon$. Each summand of q_B is at least the squared distance to the nearer endpoint, so every point of $E(\eta)$ certified by that box lies within sup-norm distance $\sqrt{(\epsilon + \eta)/\alpha}$ of a box vertex. This gives the covering lower bound that drives the continuous theory.

Let $N_\infty(S, r)$ count the least number of axis-aligned cubes of side r covering S , and define

$$\Phi_\alpha(\epsilon) = \sup_{\eta \geq 0} N_\infty\left(E(\eta), 2\sqrt{(\epsilon + \eta)/\alpha}\right). \quad (1)$$

Under second-order upper error in both objective and constraint violation, a local violation-to-distance error bound near the full feasible set, and a Lipschitz objective, dyadic subdivision yields

$$c\Phi_\alpha(\epsilon) \leq N_{\text{cover}} \leq N_{\text{rect}} \leq N_{\text{tree}} \leq T_{\text{bis}} \leq C_0 + C \log(1/\epsilon) \Phi_\alpha(\epsilon) \quad (2)$$

for sufficiently small ϵ ([theorem 3.6](#)). The supremum over levels is essential: one near-optimal level can miss many separated wells. With quadratic growth away from the optimal set and the specified

density conditions, positive-dimensional optimal manifolds give the familiar $\epsilon^{-p/2}$ rate for dimension p . An isolated degenerate minimizer can also give a polynomial rate, so the dimension of the optimal set alone does not determine complexity.

Integral lower bounds sharpen this geometry on strata with controlled coordinate-projection multiplicity. Relaxed constraints can supply further witnesses: slightly infeasible points with objective below $f_* - \epsilon$ must be excluded too. Under an inner-tube hypothesis, a global KKT minimizer with a positive-dimensional active stratum and a nonzero multiplier for a relaxed constraint forces $\Omega(\log(1/\epsilon))$ all-owner events. This result excludes reductions using original feasibility. It allows nonlinear exactly kept constraints, with a C^2 objective and compatible original exact and relaxed constraint data, and LICQ and multipliers for that same description. Changing only the description of the feasible set does not preserve these hypotheses ([theorem 3.12](#)).

All root-box faces can matter. For unconstrained f with a Lipschitz gradient, a lower bound gap αq_B , and pointwise upper error $\alpha' q_B$, the three spatial minima and dyadic node count are within constant factors of

$$1 + \sum_{G: \dim G \geq 1} \int_G (m(x) + \epsilon)^{-\dim G/2} d\mathcal{H}^{\dim G}(x), \quad (3)$$

where G ranges over the faces of D , including D itself ([theorem 4.1](#) and [remark 4.2](#)). This characterization needs no quadratic-growth assumption. For analytic objectives, classical analytic sublevel asymptotics convert each face integral into a polynomial or logarithmic rate. On a nonzero face restriction with zeros, small sublevel volumes have order $t^{\lambda_G} \log^{\theta_G-1}(1/t)$; λ_G is the real log canonical threshold and θ_G its multiplicity. That face contributes $\epsilon^{\lambda_G - \dim G/2} \log^{\theta_G-1}(1/\epsilon)$ when $\lambda_G < \dim G/2$, and $\log^{\theta_G}(1/\epsilon)$ at equality; positive-minimum and identically-zero faces are treated separately. When all minimizers are interior, the full-dimensional integral suffices, and the rate is logarithmic exactly when all minimizers are nondegenerate. At the boundary a proper face can give a larger exponent than the full box ([corollary 4.4](#) and [example 4.9](#)).

McCormick relaxations have a different exactness geometry. For separately relaxed bilinear terms, the zero-gap faces correspond to vertex covers of the interaction graph. [Section 3](#) derives matching, transversality, and fractional-cover bounds from this structure. A two-dimensional convex example with an aligned optimal segment has a two-leaf exact certificate ([proposition 3.21](#)), while an aligned three-dimensional example still requires $\Theta(\epsilon^{-1/2})$ boxes ([example 3.22](#)). Alignment alone therefore does not supply a general characterization. These results explain why convergence order alone can miss the placement of exact regions that determines certificate size.

Changing the certificate representation. [Section 5](#) retains a fixed factorization and local relaxation, but allows local bag certificates connected by affine separator minorants. For a tree decomposition with M bags and width w , global quadratic growth, $C^{1,1}$ bag objectives, vertex-vanishing factor error, and fixed width, incidence, and conditioning parameters give a certificate of size $O(M \log(M/\epsilon))$, with explicit dependence on those parameters ([theorem 5.3](#)). This is an existence result using the minimizer and separator slopes. A nonconvex quartic path family has a unique nondegenerate minimizer and an exponential spatial-cover lower bound in dimension, yet admits an $O(n \log(n/\epsilon))$ decomposition certificate under the same separate unary and bilinear oracle ([theorem 5.4](#)). The comparison concerns proof size. Construction, local checking queries, and solution time carry separate costs.

Propagation and branching. [Section 6](#) models objective-only cutoff propagation by the greatest hull-consistent fixed point of a specified finite expression graph. Local exactness gives an ideal

certificate count independent of tolerance; one-sided graphs supply a sufficient condition. Conversely, specified coordinatewise derivative cancellation with a uniform no-dominance margin transfers geometric lower bounds through charged propagation phases. These are sufficient conditions in different classes. The round count remains schedule dependent, and a bounded ideal node count can coexist with a diverging number of rounds (theorems 6.5 and 6.7).

For the exact oracle $L(B) = \min_B(f - \alpha q_B)$ on an interval, splitting at any relaxation minimizer is 4-competitive against the smallest finished tree, with an optimal incumbent and continuous f (theorem 7.1). Convexity is unnecessary for this counting theorem. Smooth nonanalytic examples give information lower bounds for node-local rules, and fixed interior safeguards impose an unbounded worst-case loss on kink families. A recentring safeguard attains the shortest safe chain that hits the kink; its finite-tolerance bound is stated separately. In several dimensions, splitting every available coordinate can lose a logarithmic factor even on a separable quadratic. Repeated coordinate ambiguity forces an exponential dependence on dimension for node-local rules with coordinatewise minimizer selection, nonanalytic germ information, and no objective-dependent information from siblings (proposition 7.6). The rule selecting the largest coordinate contribution to q_B at the relaxation minimizer has a constant ratio for one arbitrary separable coordinate and a fixed number of sharp coordinates, with an exact coordinatewise oracle satisfying the sharpness assumptions (theorem 7.7). Its general fixed-dimensional competitiveness remains open.

Integer certificates. For a fixed convex projected relaxation ϕ , a required integer set P , and threshold τ , call a class admissible if $\phi \geq \tau$ on its convex hull. The minimum number of such classes covering P is the *class number* $\kappa_\tau(P)$. It bounds the leaves of every P -covering convex-piece certificate. An unrestricted semantic binary certificate attains $\kappa_\tau(P)$ leaves when its root may be any convex set containing P and its convex children cover the required integer points. This existence statement asserts no efficient construction or checking algorithm (theorem 8.3). Convex cuts preserving every required point in P preserve the lower bound. Certified incumbent removals require an augmented leaf-and-removal count.

The class number can be much smaller than the best variable-branching tree. For a Haar random unimodular lattice with a uniform torus target, vanishing tolerance relative to the Gaussian-heuristic squared radius gives class-number bounds between $2^{(\frac{1}{2} \log_2(3/2) - o(1))n}$ and $2^{(1/2 + o(1))n}$ with high probability. They are invariant under basis changes (theorem 8.9). A separate block-curvature family has exponentially many classes for a perspective relaxation and an exact root bound from small block hulls.

Gaussian statistical models. Condition C1 requires every single-variable fixing that disagrees with a specified optimal binary point to have relaxation bound at least OPT. When the relaxation is exact at fully fixed assignments, it gives a path certificate with at most $2p + 1$ nodes for p binary variables if the optimal incumbent is available. Strict bounds above OPT give the same search bound under exact best-bound selection when a fully fixed optimal node recovers its feasible extension (lemma 8.12). Failure of C1 alone gives no lower bound on tree size.

For sparse ridge regression, section 9 identifies separate planted-support thresholds for root and single-variable certification. They are governed by $2 \log p$ and $2 \log(p\lambda/n)$ for an explicit ridge-dependent statistic, under the stated Gaussian high-dimensional scaling and fixed margins. With $k = p^{\gamma+o(1)}$ and $n = ak \log p$, a suitable ridge yields a unique planted optimum, an inexact root, and a linear path certificate when $2 - \gamma < a < 2$ (corollary 9.5). Selected local second-moment lifts preserve these first-order thresholds under explicit sandwich and growth assumptions. In a separate pure-noise or low-total-signal regime, conflict packings force $\binom{p}{k}^{c+o(1)}$ leaves. The pairwise-lift hard

Certificate model	or search	Governing structure	Scope of the comparison
Spatial box certificates		Near-optimal covering profile; all root-face integrals	Specified lower gap and upper-error assumptions
McCormick certificates		Interaction graph and placement of near-optimal strata	Separate lower bounds and exact examples
Decomposition certificates	certificates	Bag width, incidence, and quadratic growth	Fixed local oracle; size and query costs distinguished
Cutoff propagation		Expression graph and hull-consistent fixed point	Objective-only phases; rounds counted separately
Branching search		Relaxation minimizers and available node information	Exact-gap oracle; incumbent and selection hypotheses
Convex integer certificates	certificates	Admissible classes of required integer points	Fixed projected relaxation and retained point set
Gaussian models		Correlation extremes, convex conflicts, and NNLS coefficients	Distinct asymptotic regimes and certificate events

Table 1: The certificate models compared in this paper. Each chapter states its hypotheses and cost measure; a bound in one row does not automatically transfer to another.

bound requires that zero-fixed helper columns remain in the lift.

For binary least squares, let $A = \sqrt{\rho/N}H$ with H an $M \times N$ standard Gaussian matrix, independent standard Gaussian noise, and $\beta_N = M/N \rightarrow \beta \geq 1$. For $\rho \rightarrow \infty$, $\rho = o(N)$, and the tolerances $0 \leq \epsilon_N \leq \rho_N$, the logarithm of the minimum convex-piece certificate size and of the minimum variable-tree leaf count is $\Theta((N/\rho) \log \rho)$; sufficiently large fixed ρ gives $\Theta(N)$ instead (theorem 10.8). A finite Gaussian nonnegative least-squares coefficient bound proves the simultaneous C1 threshold $\theta_c = 1/[4(2\beta - 1)]$ when $\rho = \theta N$ with fixed $\theta > 0$ separated from θ_c (theorem 10.6). It also yields an exact coefficient mixture and a random root threshold asymptotic to $\log N/(2\beta_N - 1)$ for equality of box and orthant values (theorem 10.7). This root event concerns upper-box inactivity. Planted recovery, rounding correctness, and an integral root optimum are distinct events.

1.3 Relation to prior work

The cluster problem in global optimization concerns the accumulation of boxes near minimizers. Earlier analyses relate this accumulation to the convergence order and constants of bounding schemes (Du and Kearfott, 1994; Wechsung et al., 2014). Constrained analyses also identify nearly feasible points with low objective values as a source of clustering (Kannan and Barton, 2017). Our profile comparison and root-face characterization address the size of adaptive certificates, with lower bounds applying to every valid cover and constructive upper bounds realized by subdivision. The nonlinear constraint-gap argument retains the original exact/relaxed partition and makes the operation restrictions explicit.

Instance-dependent certified Lipschitz optimization also uses near-optimal covering numbers (Bachoc et al., 2021). That work measures function evaluations under a Lipschitz oracle. Here the local relaxation specifies the regions that can be certified and the relevant resolution. Classical analytic singularity theory supplies the sublevel-volume asymptotics used in section 4 (Lin, 2017); our result connects these asymptotics on all box faces to spatial certificate sizes.

McCormick and multilinear envelope formulas are established tools (McCormick, 1976; Rikun,

1997). We use them to relate gap-zero faces and graph projections to certificate counts. Interval overestimation and validated exclusion regions are also established tools (Schichl et al., 2014). Our objective-cutoff propagation analysis fixes the expression graph and distinguishes its ideal bound from its round cost. Competitive comparison with an optimal certificate has analogues in other approximation problems, including analysis of the chord algorithm (Daskalakis et al., 2016). Our branching results compare tree counts under an exact quadratic-gap oracle, with explicit information and incumbent assumptions.

Tree decompositions and reuse of conditioned subproblems are established in discrete AND/OR branch-and-bound and weighted constraint optimization (Marinescu and Dechter, 2009; de Givry et al., 2006). Treewidth also controls approximate LP formulations for polynomial optimization (Bienstock and Muñoz, 2018). The focused decomposition comparison here fixes the continuous factor relaxation and proves a separation between global-box and affine-separator certificate sizes. Its existence and checking statements keep the costs of local optimization explicit.

Integer branch-and-bound lower bounds already use midpoint conflicts and bounds on the number of integer points that one leaf can contain (Dey et al., 2023). Hiding sets give a related obstruction to small formulations (Kaibel and Weltge, 2015). The class-number formulation records the full convex-class covering obstruction for a specified projected relaxation. It also isolates the additional restriction imposed by variable or split branching. The statistical applications use these distinctions to compare conflict bounds, convex-piece certificates, and path certificates.

The Boolean and perspective formulations of sparse regression are prior work (Pilanci et al., 2015; Xie and Deng, 2018; Dong et al., 2015). Gaussian support-recovery thresholds for other estimators, including the Lasso, concern different optimization and certificate events (Wainwright, 2009). Our sparse-regression results compare root certification, single-variable certification, and selected local lifts in explicit asymptotic regimes. We also prove a narrow fixed-dimensional limitation of the Gaussian exactness assertion in Pilanci et al. (2015, Theorem 2) under its printed per-entry noise and ridge normalization: the true support becomes optimal, while the exactness probability approaches a constant below one. The underlying reformulation and deterministic exactness conditions remain valid.

For binary least squares, maximum-likelihood recovery and box-decoder recovery have been studied previously (Hansen et al., 2009; Hu and Lu, 2020). A theorem about rounding one relaxation supplies no simultaneous conclusion for all wrong-fixing nodes without an appropriate probability bound. Gaussian cone face-dimension and chi-square projection mixtures are classical (Hug and Schneider, 2016; McCoy and Tropp, 2014). We give a direct sign-orbit proof and use a finite coefficient tail to control all single-wrong-fixing nodes, including square systems. The resulting C1 threshold, root inactivity threshold, and certificate-size law address distinct branch-and-bound questions.

1.4 Scope and organization

The continuous constants can grow exponentially with dimension, and some explicit statistical lower-bound constants make the bounds vacuous at the sizes of the archived computations. Section 11 presents those computations as finite-size evidence with their recorded solver settings, precision, and provenance. They illustrate mechanisms and limitations; they establish no general practical speedup.

Section 2 defines the certificate classes and operation accounting. Section 3 proves covering, stratified, constraint-gap, and McCormick results. Section 4 gives the root-face integral characterization and analytic consequences, and section 5 compares spatial and decomposition certificates under the same local oracle. Sections 6 and 7 study propagation and branching. Section 8 develops convex

integer certificates, followed by sparse regression and binary least squares in [sections 9](#) and [10](#). [Sections 11](#) and [12](#) give the computational evidence, limitations, and open questions. Long proofs are collected in the appendices.

2 Certificates and the work used to produce them

Fix a compact axis-aligned box $D \subset \mathbb{R}^n$, a nonempty closed feasible set $F \subset D$, and a continuous objective $f : D \rightarrow \mathbb{R}$. Write

$$f_* = \min_{y \in F} f(y), \quad m(y) = f(y) - f_*, \quad E(\eta) = \{y \in F : m(y) \leq \eta\}.$$

The function m is nonnegative on F ; it can be negative outside F . For each closed test box $B = \prod_i [l_i, u_i] \subset D$, the fixed relaxation scheme supplies a set $R_B \supset F \cap B$ and a projected objective f_B with $f_B(y) \leq f(y)$ on $F \cap B$. Its bound is

$$L(B) = \inf_{z \in R_B} f_B(z), \quad \inf \emptyset = +\infty.$$

For a lifted relaxation, $f_B(y)$ is the infimum over auxiliary variables at the fixed original point y . All statements concern exact certificates for this specified scheme. They do not include numerical errors or the cost of computing an individual bound.

Definition 2.1 (Three spatial certificate classes). A box is valid at additive tolerance $\epsilon > 0$ when $L(B) \geq f_* - \epsilon$. Let $N_{\text{cover}}(\epsilon)$ be the minimum size of a finite valid-box cover of D , and let $N_{\text{rect}}(\epsilon)$ be the minimum size of such a cover whose box interiors are pairwise disjoint. Let $N_{\text{tree}}(\epsilon)$ be the minimum number of leaves of a finite axis-parallel split tree rooted at D , all of whose leaf boxes are valid. These trees split a box into two closed children at one coordinate value. A missing finite certificate gives the corresponding minimum the value $+\infty$.

Thus

$$N_{\text{cover}} \leq N_{\text{rect}} \leq N_{\text{tree}}. \quad (4)$$

An arbitrary rectangular partition need not itself be a split tree. Refining it to a coordinate grid may increase its size. We will obtain lower bounds for covers and upper bounds by constructing trees, which can give the same asymptotic rate for these three different finite minima.

A complete binary tree with ℓ leaves has $2\ell - 1$ nodes. A complete b -ary tree has $1 + b(\ell - 1)/(b - 1)$ nodes. These identities count tree nodes only. They count neither repeated relaxation evaluations nor work performed by tightening or probing. A run with a feasible incumbent $U \geq f_*$ may prune at $L(B) \geq U - \epsilon$; every such pruning test is valid in [definition 2.1](#). Upper bounds below use the ideal incumbent $U = f_*$ and therefore measure certification after an optimal solution is available. They do not measure the work needed to find that solution.

2.1 Quadratic gaps and owned tests

For $y \in B$, put

$$a_i^B(y) = (y_i - l_i)(u_i - y_i), \quad q_B(y) = \sum_{i=1}^n a_i^B(y), \quad d_i^B(y) = \min\{y_i - l_i, u_i - y_i\}. \quad (5)$$

We will use $a_i^B \geq (d_i^B)^2$ and $a_i^B \leq (u_i - l_i)d_i^B$. If $C \subset B$, then $q_C \leq q_B$ on C . Zero-width factors are zero. Dimension-collapsing reductions are permitted: the surviving degenerate box remains a node, evaluated in its affine hull.

Assumption 2.2 (Objective lower gaps). For a fixed $\alpha > 0$, the bound gap is

$$L(B) \leq f(y) - \alpha q_B(y) \quad (y \in F \cap B). \quad (6)$$

The stronger pointwise gap is

$$f_B(y) \leq f(y) - \alpha q_B(y) \quad (y \in F \cap B). \quad (7)$$

Both are required uniformly over the boxes allowed by the model.

Every valid raw test satisfies

$$m(y) + \epsilon \geq \alpha q_C(y) \quad (y \in F \cap C). \quad (8)$$

After tightening, the same inequality may hold only on a subset of the closed test box. To handle this distinction, an *owned test* is a pair (C, A) , where C is a closed box and $A \subset C$ is a rectangle with possibly open endpoints. Its validity means [equation \(8\)](#) for $y \in F \cap A$. Owners are Borel and convex. An owned certificate is a finite such family whose owners cover F . It is a different certificate class from [definition 2.1](#).

Lemma 2.3 (Boundary-safe event accounting). *Consider a finite terminating run using axis-parallel splits and the following reductions. An original-feasibility reduction $B \mapsto B'$ retains $F \cap B$. A same-relaxation reduction retains*

$$\{z \in R_B : f_B(z) \leq U - \epsilon\}, \quad U \geq f_*.$$

An emptying reduction is a terminal event. Expand any stopping bound obtained as a minimum of probe-child bounds into the finite child partition on which those bounds rely, recursively until raw successful tests are reached. Let ℓ_{aug} count the terminal events in this augmented event tree, and let R_{rel} count nonempty same-relaxation reduction rounds. Under the pointwise gap there is an owned certificate for F with

$$K_F \leq \ell_{\text{aug}} + 2nR_{\text{rel}}. \quad (9)$$

The bound gap alone suffices when there are no same-relaxation reductions. Original-feasibility reductions add no owners meeting F .

Proof. Initially the root owns D . At a split in coordinate i at s , assign $A \cap \{y_i \leq s\}$ to the lower child and $A \cap \{y_i > s\}$ to the upper child. Suppose next that the current closed test is $B = \prod_i [l_i, u_i]$ and a nonempty reduction retains $B' = \prod_i [l'_i, u'_i]$. Carry $A \cap B'$ forward, including its boundary. For each i , assign the discarded lower and upper owners

$$A_i^- = A \cap \bigcap_{j < i} \{l'_j \leq y_j \leq u'_j\} \cap \{y_i < l'_i\}, \quad A_i^+ = A \cap \bigcap_{j < i} \{l'_j \leq y_j \leq u'_j\} \cap \{y_i > u'_i\}.$$

Use the closed slabs

$$C_i^- = \prod_{j < i} [l'_j, u'_j] \times [l_i, l'_i] \times \prod_{j > i} [l_j, u_j]$$

and their upper analogues as tests. Empty owners are omitted. The first coordinate outside B' gives a disjoint partition of $A \setminus B'$, with at most $2n$ convex rectangular owners. This remains true when B' has zero widths. An emptying reduction uses (B, A) as one terminal event. Induction shows that all terminal owners partition D .

A discarded feasibility owner contains no feasible point. If a feasible y is discarded by a same-relaxation reduction, then $y \in R_B$ and $f_B(y) > U - \epsilon \geq f_* - \epsilon$. For its slab $C \subset B$,

$$f(y) - \alpha q_C(y) \geq f(y) - \alpha q_B(y) \geq f_B(y) > f_* - \epsilon.$$

At a successful raw terminal bound from a containing source box B , the bound gap similarly gives $f(y) - \alpha q_C(y) \geq L(B) \geq f_* - \epsilon$. A relaxed-infeasibility test owns no feasible point. The expansion of probing bounds ensures that each terminal successful test has such a raw source. Filtering the partition by nonempty intersection with F proves the assertion and the count. $\square$

Closed discarded slabs need not be valid at their retained boundary. For example, let $D = [-1, 1] \times [0, 1]$, $F = \{0\} \times [0, 1]$, $f = 0$, $R_B = F \cap B$, and $f_B = f - \alpha q_B$ on R_B . Exact feasibility reduction retains $\{0\} \times [0, 1]$. Both closed frame slabs contain $(0, 1/2)$, where their q is $1/4$, although their discarded owners contain no feasible point. This example satisfies the pointwise gap. It shows why [lemma 2.3](#) gives an owned certificate and does not imply $N_{\text{cover}} \leq K_F$.

2.2 Which costs the accounting controls

Let T_{aug} count processed nodes in the augmented event tree. Then [equation \(9\)](#) gives

$$K_F \leq T_{\text{aug}} + 2nR_{\text{rel}}. \quad (10)$$

Consequently a geometric lower bound $K_F \geq H(\epsilon)$ bounds this combined event and reduction work. If at most r relevant reduction rounds occur per augmented node, it also gives $T_{\text{aug}} \geq H(\epsilon)/(1+2nr)$. Without that hypothesis it does not give a node lower bound. When no stopping test uses a probe partition, actual search nodes provide T_{aug} . Otherwise virtual probe tests must be included or separately charged: a root can certify itself using many child probes while committing no split.

There is a separate evaluation-cost conclusion. Suppose every relevant reduction round is charged to a distinct counted relaxation evaluation, and every raw source needed by a successful terminal bound, including probe sources, is counted among S_{eval} evaluations. Keep each successful source box once, and keep the discarded slab owners. This new family covers F by valid source boxes and valid owned subsets, possibly with overlap. Its count satisfies

$$K_{\text{eval}} \leq S_{\text{success}} + 2nR_{\text{rel}} \leq (2n+1)S_{\text{eval}}. \quad (11)$$

Every surviving feasible point is covered by a successful source; every discarded feasible point has a slab owner. A source box satisfies [equation \(8\)](#) everywhere on its feasible subset, so the per-test lower bounds below apply to this new family. The count K_{eval} is separate from K_F : deduplicating inherited evaluations need not decrease the number of terminal owners. Uncharged reductions or uncharged probe bounds are outside [equation \(11\)](#). Tightening rounds, removed-piece certificates, and relaxation evaluations remain distinct quantities.

3 Continuous geometry of certificates

The cluster problem in continuous global optimization concerns the boxes that remain near an optimum. Its classical analyses give upper estimates for specified subdivision methods ([Du and Kearfott, 1994](#); [Wechsung et al., 2014](#); [Kannan and Barton, 2017](#)). Here we first bound the contribution of one valid test, so the resulting lower bounds apply to every permitted cover and split tree. Near-optimal covering counts also appear in certified Lipschitz optimization ([Bachoc et al., 2021](#)). The scale here is determined by the relaxation gap. A quadratic gap gives a $\sqrt{\epsilon + \eta}$ scale; it is not an assumption that the objective is quadratic.

3.1 Covering and arcsine lower bounds

Let $N_\infty(Y, \delta)$ denote the minimum number of closed cubes of side δ covering Y , with $N_\infty(\emptyset, \delta) = 0$. A raw valid cover under [equation \(6\)](#), or an owned certificate satisfying [equation \(8\)](#), will be called an α -valid family in this subsection. Its size is denoted by K .

Theorem 3.1 (Vertex covering bound). *For every $\eta \geq 0$ and every α -valid family covering $E(\eta)$,*

$$K \geq 2^{-n} N_\infty \left(E(\eta), 2\sqrt{(\epsilon + \eta)/\alpha} \right). \quad (12)$$

No smoothness, convexity, or manifold hypothesis is needed.

Proof. If y belongs to an owner with test C , validity gives $(d_i^C(y))^2 \leq q_C(y) \leq (\epsilon + \eta)/\alpha$ for each coordinate. Choosing the nearer endpoint in every coordinate puts y in a cube of the stated side about a vertex of C . Its 2^n vertices cover the family's contribution. For a raw cover use C itself as its owner. $\square$

For integration over a feasible manifold, the orientation and number of coordinate sheets matter. Let S be an embedded C^1 manifold of dimension $1 \leq d \leq n$. At $y \in S$, choose an $n \times d$ orthonormal tangent matrix $U(y)$. For a d -coordinate subset I , put $J_I(y) = |\det U_I(y)|$. Cauchy–Binet gives $\sum_I J_I^2 = 1$. Assign y to the lexicographically first maximizing I , giving Borel sets S^I . Define

$$M^I(S; A) = \sup_{x \in \mathbb{R}^I} \# \{y \in S^I \cap A : y_I = x\}, \quad M(S) = \sum_{|I|=d} M^I(S; \mathbb{R}^n).$$

The definition is independent of the orthonormal tangent basis. It counts only the sheets on which that projection is chosen.

Lemma 3.2 (Per-test arcsine estimate). *For every closed box B and Borel set A ,*

$$\int_{S \cap B \cap A} q_B^{-d/2} d\mathcal{H}^d \leq C_{n,d} \sum_{|I|=d} M^I(S; A \cap B), \quad C_{n,d} = \left(\frac{\pi^2}{d} \right)^{d/2} \binom{n}{d}^{1/2}. \quad (13)$$

The inequality is understood for nonnegative extended integrals.

Proof. On S^I , $J_I \geq \binom{n}{d}^{-1/2}$. Arithmetic–geometric mean gives

$$q_B^{-d/2} \leq d^{-d/2} \prod_{i \in I} [(y_i - l_i)(u_i - y_i)]^{-1/2}.$$

Apply the coordinate area formula on $S^I \cap B \cap A$. It changes the integral with the factor J_I into the integral over y_I , counted with the number of points in each fibre. Bound that number by $M^I(S; A \cap B)$. Each nondegenerate interval contributes

$$\int_{l_i}^{u_i} [(t - l_i)(u_i - t)]^{-1/2} dt = \pi.$$

The factor $d^{-d/2} \pi^d \binom{n}{d}^{1/2}$ and summation over I give [equation \(13\)](#). If a projected coordinate is fixed in B , the portion of $S^I \cap B$ has zero d -measure: its Jacobian is bounded below and its projected image has zero Lebesgue measure. Thus positive-measure terms use only nondegenerate intervals, which also proves the claim for degenerate boxes. $\square$

Theorem 3.3 (Stratum lower bound). *Suppose $S \subset F$ and $M(S) < \infty$. Every α -valid family covering S satisfies*

$$K \geq \frac{\alpha^{d/2}}{C_{n,d}M(S)} \int_S (m + \epsilon)^{-d/2} d\mathcal{H}^d. \quad (14)$$

For $d = n$, this becomes $K \geq (\alpha n / \pi^2)^{n/2} \int_S (m + \epsilon)^{-n/2}$. These conclusions hold for owned subsets, so shared closed boundaries of positive stratum measure cause no overcounting.

Proof. Validity gives $(m + \epsilon)^{-d/2} \leq \alpha^{-d/2} q_C^{-d/2}$ on each owned contribution. Apply [lemma 3.2](#) and sum. A cover can overlap, since summing nonnegative contributions only increases their integral. For disjoint owners the contributions partition S . $\square$

Finite coordinate multiplicity follows locally from a C^2 graph, with uniform constants for graphs of uniformly bounded second derivatives. [Lemma A.1](#) proves this fact. Some such control is necessary: the planar union of k parallel segments inside a fixed box has zero curvature on every segment but an integral proportional to k . An assertion using curvature alone would omit this number of sheets.

3.2 A constrained covering comparison

For the upper bounds, let D be a cube of side s_0 . Specify the original constraint model by

$$F = \{z \in D \cap P_{\text{ex}} : g_j(z) \leq 0, h_k(z) = 0\}, \quad v(z) = \max\{0, \max_j g_j(z), \max_k |h_k(z)|\}.$$

Empty maxima are omitted. The closed set P_{ex} contains exactly kept restrictions, and $R_B \subset B \cap P_{\text{ex}}$. The functions in v are the original relaxed constraint functions.

Assumption 3.4 (Second-order upper error and feasibility error bound). There are constants $\tau > 0$, $\kappa \geq 0$, $v_0 > 0$, and $L_f \geq 0$ such that

$$f_B(z) \geq f(z) - \tau w(B)^2, \quad v(z) \leq \tau w(B)^2 \quad (z \in R_B), \quad (15)$$

$$\text{dist}_2(z, F) \leq \kappa v(z) \quad (z \in D \cap P_{\text{ex}}, v(z) \leq v_0), \quad (16)$$

and f is L_f -Lipschitz on D in Euclidean distance. Here $w(B)$ is the largest side length.

$$\Lambda = \tau(1 + L_f \kappa).$$

The distance bound is for the full feasible set, including the root and exact constraints. A constraint qualification asserted only at global minimizers does not supply [equation \(16\)](#). It must be established on the region used by relaxed witnesses or assumed directly.

Let $T_{\text{bis}}(\epsilon)$ be the number of processed nodes of uniform 2^n -ary dyadic subdivision, with $U = f_*$. Write $s_j = s_0 2^{-j}$ and let $N_j(Y)$ count closed level- j grid cubes meeting Y . A 2^n -way split can be realized by n successive coordinate splits; thus its leaf partition is an admissible binary tree partition.

Lemma 3.5 (Localization of unpruned boxes). *Under [assumption 3.4](#), there are j_0 and C_0 , independent of ϵ , such that*

$$T_{\text{bis}}(\epsilon) \leq C_0 + 2^n 5^n \sum_{\substack{j \geq j_0 \\ \Lambda s_j^2 > \epsilon}} N_j(E(\Lambda s_j^2 - \epsilon)). \quad (17)$$

In particular this tree terminates at every positive tolerance.

Proof. Choose j_0 so that $\tau s_j^2 \leq v_0$ and $\kappa \tau s_j^2 \leq s_j$ for $j \geq j_0$. An unpruned cube B has $L(B) < f_* - \epsilon$, so some $z \in R_B$ satisfies $f_B(z) < f_* - \epsilon$, even if the infimum is not attained. By the upper error and the distance bound a nearest feasible y obeys

$$|y - z|_2 \leq \kappa \tau s_j^2 \leq s_j, \quad 0 \leq m(y) < \Lambda s_j^2 - \epsilon.$$

Hence $\Lambda s_j^2 > \epsilon$. Assign B to a grid cube containing y . At most 5^n level- j cubes lie within sup distance s_j of a point in that cube. Coarse levels give a fixed finite count, and each unpruned cube generates 2^n children. This proves the displayed bound and finiteness. $\square$

Theorem 3.6 (Running-supremum covering comparison). *Assume the bound gap [equation \(6\)](#) and [assumption 3.4](#). Put*

$$\Phi_\alpha(\epsilon) = \sup_{\eta \geq 0} N_\infty \left(E(\eta), 2\sqrt{(\epsilon + \eta)/\alpha} \right). \quad (18)$$

For sufficiently small $\epsilon > 0$, with constants depending on the fixed instance and scheme,

$$2^{-n} \Phi_\alpha(\epsilon) \leq N_{\text{cover}} \leq N_{\text{rect}} \leq N_{\text{tree}} \leq T_{\text{bis}} \leq C_0 + C(1 + \log(1/\epsilon)) \Phi_\alpha(\epsilon). \quad (19)$$

If an owned α -valid family covers F and has size K_F , then also $T_{\text{bis}} \leq C_0 + C(1 + \log(1/\epsilon)) K_F$.

Proof. [Theorem 3.1](#) gives the lower bound at every η . For a term in [equation \(17\)](#), set $\eta = \Lambda s_j^2 - \epsilon$. A cube of side s_j meets at most 3^n closed grid cubes of that side. Refining a cube of side as_j to cubes of side s_j takes at most $\max\{1, \lceil a \rceil\}^n$ cubes. Apply this with $a = 2\sqrt{\Lambda/\alpha}$ to bound the level count by a fixed multiple of $\Phi_\alpha(\epsilon)$. There are $O(1 + \log(1/\epsilon))$ nonempty levels in the sum.

For the last assertion, validity puts every point of $E(\Lambda s_j^2 - \epsilon)$ within $\sqrt{\Lambda/\alpha} s_j$ of a test vertex. Each test has 2^n vertices; each such cube meets a bounded number of level- j cells. This bounds the same level sum by a fixed multiple of K_F . It does not identify the owned family with a raw cover. $\square$

Define $\psi_\alpha(\eta) = N_\infty(E(\eta), 2\sqrt{\eta/\alpha})$ for $\eta > 0$. Then

$$2^{-n} \sup_{\eta \geq \epsilon} \psi_\alpha(\eta) \leq \Phi_\alpha(\epsilon) \leq \sup_{\eta \geq \epsilon} \psi_\alpha(\eta). \quad (20)$$

Indeed the two scales differ by at most $\sqrt{2}$ when $\eta \geq \epsilon$, and splitting each coordinate interval in two accounts for the factor 2^n . When $\eta < \epsilon$, set inclusion and monotonicity bound the term by $\psi_\alpha(\epsilon)$.

The supremum cannot generally be replaced by its value at $\eta = \epsilon$ with constants uniform over an instance class. [Example A.2](#) gives a smooth one-dimensional family with uniform relaxation and derivative bounds, a bounded single-scale count, and certificate size at least a constant times the number of near-optimal wells.

Under the pointwise gap, [equation \(10\)](#) and [theorem 3.6](#) show that same-relaxation tightening can improve on uniform subdivision by at most a logarithmic factor when comparison is made to augmented event and reduction work. A statement about nodes alone needs a bound on rounds per counted node; a statement about evaluations uses [equation \(11\)](#). The logarithmic comparison does not assert that tightening takes a bounded number of rounds.

3.3 When optimal-set dimension gives the rate

Let $M = E(0)$. An optimal set does not determine the rate by itself: flat objective growth can make an isolated minimizer expensive. The next statement makes the needed growth condition explicit.

Proposition 3.7 (Quadratic growth and covering dimension). *Under the hypotheses of [theorem 3.6](#), assume*

$$m(y) \geq c_g \operatorname{dist}_\infty(y, M)^2 \quad (y \in F), \quad c_g > 0. \quad (21)$$

Then

$$2^{-n} N_\infty(M, 2\sqrt{\epsilon/\alpha}) \leq N_{\text{cover}}(\epsilon), \quad T_{\text{bis}}(\epsilon) \leq C_0 + C \sum_{\Lambda s_j^2 > \epsilon} N_\infty(M, s_j). \quad (22)$$

If M has box-counting dimension $p > 0$, each of the three certificate minima and T_{bis} has logarithmic growth exponent $p/2$: $\log N(\epsilon)/\log(1/\epsilon) \rightarrow p/2$. If M is finite, the upper bound is $O(1 + \log(1/\epsilon))$. The upper assertion requires only [assumption 3.4](#) and quadratic growth; it does not require an objective lower gap.

Proof. The lower bound takes $\eta = 0$ in [theorem 3.1](#). By quadratic growth, $E(\Lambda s_j^2) \subset \{\operatorname{dist}_\infty(\cdot, M) \leq \sqrt{\Lambda/c_g s_j}\}$. A cover of M by s_j -cubes therefore bounds the level count in [equation \(17\)](#) by a fixed multiple of $N_\infty(M, s_j)$. For every $\delta > 0$, the definition of box dimension bounds this count above by $s_j^{-p-\delta}$ and below by $s_j^{-p+\delta}$ at sufficiently small scales. The geometric upper sum stops at $s_j \asymp \sqrt{\epsilon}$. Taking logarithms and letting $\delta \downarrow 0$ proves the exponent statement. A finite set has a uniformly bounded covering count at every scale. $\square$

Corollary 3.8 (An optimal manifold). *Under the hypotheses of [proposition 3.7](#), suppose M is a compact embedded C^1 manifold of dimension $p \geq 1$, possibly with boundary, and with a finite regular coordinate-chart cover. Then*

$$N_{\text{cover}} \asymp N_{\text{rect}} \asymp N_{\text{tree}} \asymp T_{\text{bis}} \asymp \epsilon^{-p/2}.$$

Proof. Finite bounded-slope charts give $N_\infty(M, s) \leq C s^{-p}$. A single relatively open chart contains a p -dimensional coordinate cube; a sup-norm cube of side s projects to a cube of that side, so $N_\infty(M, s) \geq c s^{-p}$. Apply [equation \(22\)](#); the geometric sum is $O(\epsilon^{-p/2})$ for $p > 0$. $\square$

A stratum integral can give the complete rate under direct geometric regularity conditions. Suppose finitely many strata $S_k \subset F$, of dimensions $d_k \geq 1$, have finite $M(S_k)$. For fixed positive t_0, r_0, θ and finite constants K_R, K , assume that for $0 < t \leq t_0$: every $y \in E(t)$ is within $\sqrt{K_R t}$ in sup norm of a point $z \in S_k$ with $m(z) \leq K_R t$; on each stratum,

$$m(x) \leq K(m(z) + |x - z|_\infty^2), \quad \mathcal{H}^{d_k}(S_k \cap \{|x - z|_\infty \leq r\}) \geq \theta r^{d_k}$$

whenever $m(z) \leq K_R t_0$, $|x - z|_\infty \leq r_0$, and $0 < r \leq r_0$. These are proximity, quadratic doubling, and lower density hypotheses, respectively.

Theorem 3.9 (Constrained stratum integral comparison). *Under these hypotheses and those of [theorem 3.6](#), put $I_k(\epsilon) = \int_{S_k} (m + \epsilon)^{-d_k/2} d\mathcal{H}^{d_k}$. Then*

$$1 + \sum_k I_k(\epsilon) \asymp N_{\text{cover}} \asymp N_{\text{rect}} \asymp N_{\text{tree}} \asymp T_{\text{bis}}.$$

If some strata are finite sets, their contribution to the upper bound is instead $O(\#S_k(1 + \log(1/\epsilon)))$; the statement does not supply a matching logarithmic lower bound from those points alone.

Proof. [Theorem 3.3](#) bounds the count below by each I_k and hence by their sum up to a fixed factor. For the upper bound, at level j choose a maximal sup-norm s_j -separated family in $\{z \in S_k : m(z) \leq K_R \Lambda s_j^2\}$. Proximity and [lemma 3.5](#) assign every unpruned cube to one of these points within Cs_j , and each point receives at most a fixed number of grid cubes. Their relative balls of radius $s_j/3$ are disjoint. Density gives mass at least $\theta(s_j/3)^{d_k}$ in each; doubling puts all this mass in $\{m + \epsilon \leq C's_j^2\}$. Thus the level count is at most $C \sum_k s_j^{-d_k} \mathcal{H}^{d_k}(S_k \cap \{m + \epsilon \leq C's_j^2\})$. Tonelli's theorem changes the sum over levels to an integral of $\sum_{s_j \geq \sqrt{(m+\epsilon)/C'}} s_j^{-d_k}$. For $d_k \geq 1$ this geometric sum is bounded by $C(m + \epsilon)^{-d_k/2}$. Coarse levels add a constant. A finite zero-dimensional stratum gives at most its cardinality at each level, which explains its stated contribution. $\square$

3.4 Gaps caused by relaxed constraints

An objective can be kept exactly while relaxed constraints still force large certificates. Feasible-point objective gaps do not describe this case. Its witnesses can be infeasible points with objective below f_* , the mechanism already identified in constrained cluster analysis ([Kannan and Barton, 2017](#)). For a lower bound we need such points to be admitted by the specified relaxation.

Assumption 3.10 (Inner tube). For fixed $\alpha \geq 0$ and $\beta > 0$, every test box satisfies

$$z \in B \cap P_{\text{ex}}, \quad v(z) \leq \beta q_B(z) \implies z \in R_B, \quad f_B(z) \leq f(z) - \alpha q_B(z). \quad (23)$$

Exactly enforced constraints must be included in P_{ex} , since [equation \(23\)](#) cannot promise admission of their violators. The case $\alpha = 0$ permits an exact objective.

Lemma 3.11 (Owned tube dichotomy). *Every raw valid certificate under [assumption 3.10](#) satisfies*

$$v(z) > \beta q_C(z) \quad \text{or} \quad f(z) - \alpha q_C(z) \geq f_* - \epsilon \quad (z \in C \cap P_{\text{ex}}). \quad (24)$$

For runs using splits, same-relaxation reductions, and raw or expanded probe pruning, the ownership construction in [lemma 2.3](#) gives the same dichotomy for $z \in A \cap P_{\text{ex}}$. All terminal owners, including owners with no feasible point, have count

$$K_{\text{all}} \leq \ell_{\text{aug}} + 2nR_{\text{rel}}. \quad (25)$$

Original-feasibility reductions and other independent ways to remove infeasible tube witnesses are excluded in this run statement.

Proof. If $v(z) \leq \beta q_C(z)$, then for a containing source B , $q_C(z) \leq q_B(z)$ puts z in its inner tube. Hence $z \in R_B$ and $f_B(z) \leq f(z) - \alpha q_B(z) \leq f(z) - \alpha q_C(z)$. A successful source bound gives the second alternative. An infeasible source cannot contain such a tube point. For a discarded same-relaxation owner, the removal predicate gives $f_B(z) > U - \epsilon \geq f_* - \epsilon$ instead. Emptying reductions use the same argument on their parent. The owners partition all of D , and their count is the one in [lemma 2.3](#) before filtering by feasibility. Filtering is not allowed here because the witnesses can be infeasible. $\square$

For $\nu > 0$ define $W(\epsilon, \nu) = \{z \in D \cap P_{\text{ex}} : v(z) \leq \nu, f(z) < f_* - \epsilon\}$. Every family of K test-owner pairs covering these points and satisfying [equation \(24\)](#) obeys

$$K \geq 2^{-n} N_{\infty}(W(\epsilon, \nu), 2\sqrt{\nu/\beta}). \quad (26)$$

Indeed the objective alternative is impossible, so each coordinate's distance to the nearer test endpoint is less than $\sqrt{\nu/\beta}$.

Theorem 3.12 (Compatible nonlinear constraint transfer). *Let z_* be a global minimizer. On an open neighborhood of z_* , suppose f , a finite constraint description of the unchanged exact set P_{ex} , and every original relaxed function defining v are C^2 . Combine that description, the root inequalities, and these original relaxed functions. Assume LICQ for all equalities and active inequalities in this combined tuple and KKT multipliers for the same tuple. Suppose an active relaxed inequality has a positive multiplier, or a relaxed equality has a nonzero multiplier.*

There are a neighborhood U_ and constants $c, \epsilon_0 > 0$ such that for $Y \subset E(\eta) \cap U_*$ and $0 < \eta + \epsilon \leq \epsilon_0$, every dichotomy family covering $D \cap P_{\text{ex}}$ satisfies*

$$K \geq 2^{-n} N_\infty(Y, c\sqrt{\eta + \epsilon}). \quad (27)$$

If the manifold on which all active constraints equal their values at z_ has dimension $d \geq 1$, then on a fixed relatively open feasible patch S of that manifold,*

$$K \geq c_1 \int_S (m + \epsilon)^{-d/2} d\mathcal{H}^d \geq c_2 \log(1/\epsilon) \quad (28)$$

for sufficiently small $\epsilon > 0$. Neither strict complementarity of all active inequalities nor a second-order sufficient condition is needed for this lower bound.

The complete proof in [appendix A.2](#) constructs constraint-preserving curves by the implicit function theorem. It proves uniform injectivity, coordinate multiplicity, and area bounds for their tolerance-dependent shifted patches before applying the arcsine estimate. The compatibility requirement preserves the original exact/relaxed assignment and violation function. A LICQ description of F alone cannot replace it. For example, let P_{ex} be the unit ball, relax $1 - |z|^2 \leq 0$, and minimize $-z_1$ over their intersection. The feasible set is the unit sphere and its equality description is LICQ at e_1 , but the original exact and relaxed gradients are opposite. With $R_B = B \cap P_{\text{ex}}$ and the exact objective, the root is already exact and satisfies the tube condition with $\alpha = 0$. No point in P_{ex} has objective below -1 . The combined LICQ hypothesis correctly excludes this instance.

For a run, [equation \(28\)](#) applies to K_{all} , and hence to augmented event and reduction work by [equation \(25\)](#). Under the distinct-evaluation charging model, retain every successful source, including infeasibility sources, and the discarded owners. They cover $D \cap P_{\text{ex}}$ and satisfy the dichotomy, giving a fresh count $K_{\text{eval,all}} \leq (2n + 1)S_{\text{eval}}$. Thus the same lower bound applies to evaluations in that model. Exact propagation of original constraints can remove the infeasible witnesses and is outside these conclusions.

Corollary 3.13 (Finite regular KKT minimizers). *Assume [assumption 3.4](#) and that the set of global minimizers is finite. Near each minimizer, suppose f and a finite description of the full feasible set are C^2 , the description satisfies LICQ and KKT, every active inequality has a positive multiplier, and the Lagrangian Hessian is positive definite on the common nullspace of the active inequality and equality gradients. Then $T_{\text{bis}} = O(1 + \log(1/\epsilon))$.*

If at least one active stratum has dimension at least one and the objective bound gap holds, all three raw certificate minima and T_{bis} are $\Theta(\log(1/\epsilon))$. The same matching rate holds under the tube hypothesis when that minimizer satisfies the compatible original-tuple and relaxed-multiplier hypotheses of [theorem 3.12](#).

If instead there is one minimizer z_ and its active stratum has dimension zero, a finite exact split-tree certificate exists under the stronger vertex-vanishing upper error for some $\alpha' \geq 0$,*

$$f_B(z) \geq f(z) - \alpha' q_B(z), \quad v(z) \leq \alpha' q_B(z) \quad (z \in R_B). \quad (29)$$

Thus $N_{\text{tree}}(\epsilon) = O(1)$ even at $\epsilon = 0$.

Appendix A.4 proves quadratic growth from these KKT assumptions, the local stratum logarithm, and sharp growth at a zero-dimensional stratum. The error bound for the relaxation remains an independent assumption. For the feasible-objective branch, a local description of the full feasible set suffices for these geometric facts; the tube branch specifically requires compatibility with the original exact/relaxed tuple.

Width-squared upper convergence alone does not give the last conclusion. For $F = [0, 1]$, $f(x) = x$, and $f_B(x) = x - \tau w(B)^2$ on $R_B = B$, the minimum at zero is sharp, but every positive-width interval containing zero has a negative bound. A finite closed interval cover of $[0, 1]$ must contain such an interval; singleton tests at zero cannot cover all small positive points. Hence it has no finite exact certificate. Under a positive objective lower gap, uniform bisection still requires $\Omega(\log(1/\epsilon))$ when a minimizer coordinate stays in a fixed middle portion of its dyadic interval at every level. A root-coordinate position with an odd rational denominator greater than one has this property. Merely being nondyadic does not imply it.

Example 3.14 (Isotropic and directional constraint gaps). Let $D = [-6/5, 13/10]^2$, $f(z) = z_2$, and let F be given by $-z_2 \leq 0$, with $P_{\text{ex}} = \mathbb{R}^2$. Keep f exact. For $R_B = \{z \in B : -z_2 \leq q_B(z)\}$, the tube coefficient is $\beta = 1$. The strip $[-6/5, 13/10] \times [-2\epsilon, -\epsilon]$ lies in $W(\epsilon, 2\epsilon)$ for small ϵ . Its horizontal projection has length $5/2$, so equation (26) gives $N_{\text{cover}} \geq c\epsilon^{-1/2}$. The upper error, $\text{dist}(z, F) = v(z)$, and quadratic growth to the optimal segment give the upper part of proposition 3.7 without an objective lower gap. The segment has $N_{\infty}(M, s) = O(s^{-1})$, so the geometric level sum is $O(\epsilon^{-1/2})$. Combining the upper and infeasible-cover lower bounds gives all three raw minima $\Theta(\epsilon^{-1/2})$.

For the directional relaxation $R_B = \{z \in B : -z_2 \leq a_2^B(z_2)\}$, three horizontal slabs give an exact certificate: $[-6/5, -3/5]$, $[-3/5, 0]$, and $[0, 13/10]$ in the second coordinate. The first is infeasible since $-z_2 \geq 3/5 > 9/100 \geq a_2^B$; in the middle slab a negative feasible z_2 would require $1 \leq z_2 + 3/5$, which is impossible; the last has minimum zero. This scheme has no isotropic tube coefficient $\beta > 0$. Finally, exact propagation of the original inequality retains $z_2 \geq 0$ and makes even the isotropic relaxation exact in one node.

3.5 McCormick gaps and graph geometry

The objective-gap hypothesis equation (6) requires loss in every coordinate not fixed by a box vertex. Bilinear envelopes have a different zero set. In this subsection, fix the decomposition

$$f(x) = g(x) + \sum_{ij \in G} c_{ij} x_i x_j, \quad F = D \cap P, \quad (30)$$

where g is convex and continuous, P is a polyhedron in original variables, G has an edge ij for every nonzero coefficient, and each bilinear term is relaxed separately by its convex envelope on B . The polyhedron is kept exactly. Write $f_B = f - \Gamma_B$ for this relaxation, so

$$L(B) \geq f_* - \epsilon \iff \Gamma_B(x) \leq m(x) + \epsilon \quad (x \in F \cap B). \quad (31)$$

The fixed decomposition matters even when the full objective is convex. For an owned test (C, A) in this subsection, *validity* means $\Gamma_C(x) \leq m(x) + \epsilon$ for $x \in A \cap F$. This is the McCormick validity condition, rather than the isotropic q condition used earlier. Every owned-family statement below requires this condition. The formulas below are the classical bilinear-envelope formulas (McCormick, 1976); we give their derivation to identify the geometric properties used by the certificate bounds.

Lemma 3.15 (Bilinear gaps and exact faces). Put $\delta_i^- = x_i - l_i$ and $\delta_i^+ = u_i - x_i$ on B . The gap for the term $c_{ij} x_i x_j$ is

$$\gamma_{ij,B}(x) = |c_{ij}| \begin{cases} \min(\delta_i^- \delta_j^-, \delta_i^+ \delta_j^+), & c_{ij} > 0, \\ \min(\delta_i^- \delta_j^+, \delta_i^+ \delta_j^-), & c_{ij} < 0. \end{cases} \quad (32)$$

It satisfies

$$|c_{ij}|d_i^B d_j^B \leq \gamma_{ij,B} \leq |c_{ij}| \min(d_i^B w_j, d_j^B w_i), \quad (33)$$

$$\gamma_{ij,B} \leq \frac{1}{2}|c_{ij}|(a_i^B + a_j^B), \quad \max_B \gamma_{ij,B} = \frac{1}{4}|c_{ij}|w_i w_j. \quad (34)$$

Moreover $\Gamma_B = \sum_{ij \in G} \gamma_{ij,B}$ vanishes at x precisely when the coordinates of x at box endpoints form a vertex cover of G . Thus it vanishes on an entire coordinate face precisely when that face's free coordinates form an independent set.

Proof. Normalize the two nondegenerate intervals to $s, t \in [0, 1]$, removing affine terms. The convex envelope of st is $\max(0, s + t - 1)$. For $s + t \leq 1$, express (s, t) as a convex combination of $(0, 0)$, $(1, 0)$, $(0, 1)$, whose weighted product is zero. For $s + t \geq 1$, use $(1, 0)$, $(0, 1)$, $(1, 1)$, whose weighted product is $s + t - 1$. Every convex underestimator is bounded above by these combinations, and the two affine minorants are valid. Similarly the concave envelope is $\min(s, t)$, using, when $s \leq t$, weights $s, t - s, 1 - t$ on $(1, 1)$, $(0, 1)$, $(0, 0)$ and reversing coordinates otherwise. Scaling and the sign of the coefficient give [equation \(32\)](#).

Each endpoint distance is at least its nearest-endpoint distance; choosing the product involving the nearest endpoint in one coordinate gives [equation \(33\)](#). The product of the two candidate products in the minimum is $a_i^B a_j^B$, so the minimum is at most $\sqrt{a_i^B a_j^B} \leq (a_i^B + a_j^B)/2$. It is also at most $w_i w_j / 4$, with equality at the common interval centers. Degenerate intervals give zero directly. Each edge gap is nonnegative and vanishes exactly when one endpoint coordinate is fixed; their sum therefore has the claimed graph zero set. $\square$

The upper inequalities give

$$\Gamma_B \leq \frac{1}{2} \left(\max_i \sum_{j: ij \in G} |c_{ij}| \right) q_B, \quad \max_B \Gamma_B = \frac{1}{4} \sum_{ij \in G} |c_{ij}| w_i w_j. \quad (35)$$

All edge maxima occur at the same box center. The first inequality supplies a vertex-vanishing upper error, but supplies no positive isotropic objective lower gap.

A joint envelope of the bilinear sum is pointwise stronger than the sum of separate envelopes. It has the same exact-face criterion: any convex representation of a point on a box face uses points of that face, by a supporting affine functional. Its restriction is therefore the envelope of the restricted sum. That restricted bilinear sum is convex exactly when it is affine: a surviving free edge has the indefinite Hessian submatrix $\begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}$. Sums of termwise edge losses below are not claims about a joint envelope.

The ownership and cost construction also applies to this model. Indeed, restricting a convex underestimator from B to $C \subset B$ gives an admissible underestimator on C , so each child envelope is at least the restricted parent envelope. Hence $\Gamma_C \leq \Gamma_B$ on C . At a discarded same-relaxation feasible point, $f_B(x) > U - \epsilon$ gives $\Gamma_B(x) < m(x) + \epsilon$, and therefore the owned child test is McCormick-valid. Successful containing source bounds give the weak inequality in the same way. Original-feasibility owners contain no feasible point. Thus the rectangular ownership proof gives $K_F \leq \ell_{\text{aug}} + 2nR_{\text{rel}}$ for this validity condition as well. The augmented-node and separately charged evaluation conclusions in [equations \(10\)](#) and [\(11\)](#) follow with the same distinct costs. They do not require an isotropic objective gap.

Theorem 3.16 (Flat near-optimal sets). *Suppose a selected nonempty graph $H \subset G$ satisfies $|c_{ij}| \geq \alpha > 0$ on every edge. Let V have orthonormal columns spanning a p -flat, $p \geq 1$, and let v^i be*

its rows. For a full-dimensional bounded convex body $K \subset \mathbb{R}^p$ put $W_i(K) = \sup_{t \in K} v^i \cdot t - \inf_{t \in K} v^i \cdot t$ and define

$$\tau(V, H) = \inf_K \frac{(\max_{ij \in H} W_i(K) W_j(K))^{p/2}}{\text{vol}_p(K)}.$$

For every bounded convex $S \subset F \cap (z_0 + \text{range } V)$ with $m \leq \eta$ on S , every raw valid cover or rectangular-owner family covering S has size

$$K \geq \tau(V, H) \left(\frac{\alpha}{(p+1)^2(\eta + \epsilon)} \right)^{p/2} \mathcal{H}^p(S). \quad (36)$$

If every vertex-cover projection is injective on $\text{range } V$, then $\tau(V, H) > 0$. This sufficient condition is called graph transversality.

Proof. Intersect S with an owner $A \subset C$. For a positive-volume intersection its centroid belongs to that convex intersection even when some endpoints are open. The elementary supporting-plane estimate proved in lemma A.3 puts its nearest distance to each pair of box faces at least $W_i/(p+1)$. At that owned centroid, validity and equation (33) give $\max_{ij \in H} W_i W_j \leq (p+1)^2(\eta + \epsilon)/\alpha$. The definition of τ bounds the intersection volume; summing over the family proves equation (36).

For transversality define

$$\Delta = \min_{J_0 \text{ vertex cover of } H} \max_{\substack{J \subset J_0 \\ |J|=p}} |\det V_J|.$$

Injectivity makes $\Delta > 0$. For any body K , put $b = \max_{ij \in H} W_i W_j$. The coordinates with $W_i \leq \sqrt{b}$ form a vertex cover. Among them choose p rows with determinant at least Δ . Their coordinate ranges enclose K in a parallelepiped of volume at most $b^{p/2}/\Delta$. Thus $\tau(V, H) \geq \Delta$. $\square$

For a unit line direction v , the formula reduces to $\tau(v, H) = \max_{ij \in H} \sqrt{|v_i v_j|}$. In higher dimension transversality is stronger than positivity of τ : on $\mathbb{R}^2$ with one edge, $\tau = 1$ because area is at most the product of coordinate widths, yet projecting to one endpoint is not injective on $\mathbb{R}^2$.

Proposition 3.17 (Curved transversal strata). *Let $S \subset F$ be a compact subset of an embedded C^1 p -manifold, and suppose every tangent space is graph-transversal for the selected graph H above. If $m \leq \eta$ on S , then for small $\eta + \epsilon$ every valid family covering S satisfies*

$$K \geq c_S \mathcal{H}^p(S) (\alpha/(\eta + \epsilon))^{p/2}.$$

In particular an optimal stratum of positive measure forces $\Omega(\epsilon^{-p/2})$ tests.

Proof. For each vertex cover J_0 , injectivity on each tangent space gives locally a p -coordinate projection with coordinates in J_0 and an inverse chart of bounded derivative. A finite chart cover gives

$$\mathcal{H}^p(S \cap \{|x_{J_0} - \xi|_\infty \leq r\}) \leq C r^p$$

uniformly in ξ and small r . This follows by projecting each chart intersection into the p chosen coordinate intervals and bounding the inverse chart's area factor. At an owned point of a valid test, put $\rho = \sqrt{(\eta + \epsilon)/\alpha}$. The edge bounds imply $d_i d_j \leq \rho^2$ on H . Consequently $\{i : d_i \leq \rho\}$ is a vertex cover and contains a minimal cover J_0 . The point's J_0 coordinates lie within ρ of one of that test's $2^{|J_0|}$ endpoint combinations. Sum the slab measure estimate over the finitely many minimal covers and endpoint choices. Each test contributes at most $C' \rho^p$ measure, proving the assertion. $\square$

Proposition 3.18 (Fractional vertex-cover gap budget). *Suppose an n -dimensional feasible cube R of side W has $m \leq \eta$ throughout, and $H \subset G$ has coefficients at least $\alpha > 0$ in magnitude. Let*

$$\tau^*(H) = \min \left\{ \sum_i z_i : z_i \geq 0, z_i + z_j \geq 1 \ (ij \in H) \right\}.$$

If $\rho^2 = 4(\eta + \epsilon)/\alpha \leq W^2$, every raw valid cover of R has size at least

$$K \geq (W/\rho)^{2\tau^*(H)}. \quad (37)$$

Conversely, on a root cube D of side W , for the full relaxed graph G there is a grid tree with

$$K \leq 2^n (W/\rho')^{2\tau^*(G)}, \quad 0 < \rho' \leq W, \quad \frac{\rho'^2}{4} \sum_{ij \in G} |c_{ij}| \leq \epsilon. \quad (38)$$

This is a statement about the specified gap budget and near-optimal cube. It does not determine the complexity from the optimal set alone.

Proof. The center of $C \cap R$, when this intersection has positive volume, is at distance at least half each intersection width w_i from the corresponding faces of C . Validity gives $w_i w_j \leq \rho^2$ on H . Thus its volume is at most

$$\max \left\{ \prod_i w_i : 0 < w_i \leq W, w_i w_j \leq \rho^2 \ (ij \in H) \right\}.$$

For $W > \rho$, substitute $w_i = W(\rho/W)^{2z_i}$. The constraints become exactly those defining $\tau^*(H)$, and the maximum product is $W^n(\rho/W)^{2\tau^*(H)}$. The equality case $W = \rho$ is direct. Summing intersection volumes proves [equation \(37\)](#).

Choose an optimal fractional cover z of G with $z_i \in \{0, 1/2, 1\}$; [lemma A.4](#) proves its existence. Grid side lengths $b_i = W(\rho'/W)^{2z_i}$ satisfy $b_i b_j \leq \rho'^2$ on every relaxed edge. [Equation \(35\)](#) bounds the gap in every grid cell by ϵ , so $m \geq 0$ makes every cell valid. The grid is realizable by coordinate cuts, and its size is at most $\prod_i \lceil W/b_i \rceil \leq 2^n (W/\rho')^{2\tau^*(G)}$. The upper construction must use every relaxed edge; a subgraph used for the lower bound cannot omit edges from its gap budget. $\square$

Proposition 3.19 (Concave matching slices). *Let $\{i_r j_r : r = 1, \dots, k\}$ be a matching in G , with $k \geq 1$, and put $v_r = e_{i_r} - \text{sign}(c_{i_r j_r}) e_{j_r}$, $Vt = \sum_r t_r v_r$, and $T_z = \{t : z + Vt \in F\}$. Every valid raw cover, or owned valid family covering this slice, obeys*

$$K \geq \left(\frac{k}{\pi^2} \right)^{k/2} \prod_{r=1}^k |c_{i_r j_r}|^{1/2} \int_{T_z} (m(z + Vt) + \epsilon)^{-k/2} dt. \quad (39)$$

Proof. The preimage of a test box is a rectangle $\prod_r [\alpha_r, \beta_r]$, because the pairs are disjoint. On the chord in direction v_r , the selected term has concave quadratic coefficient $-\kappa_r$, where $\kappa_r = |c_{i_r j_r}|$. Its envelope is no greater than the endpoint chord, so its gap at t_r is at least $\kappa_r(t_r - \alpha_r)(\beta_r - t_r)$. Nonnegative termwise gaps can be summed, giving on the owned feasible slice

$$m + \epsilon \geq \sum_r \kappa_r(t_r - \alpha_r)(\beta_r - t_r).$$

Arithmetic-geometric mean and the one-dimensional arcsine integral bound each test's contribution to the integral by $k^{-k/2} \pi^k \prod_r \kappa_r^{-1/2}$. Integrate over its owned subset and bound by the full preimage rectangle. Degenerate rectangles have zero k -measure. Summation proves the claim. $\square$

For one such feasible chord through a minimizer, $z_* + tv$ with $|t| \leq t_0$ and $t_0 > 0$, an upper growth bound $m(z_* + tv) \leq Mt^2$ with $M > 0$ yields a logarithmic lower bound by [equation \(39\)](#): the integral of $(Mt^2 + \epsilon)^{-1/2}$ is $2M^{-1/2} \operatorname{arsinh}(t_0 \sqrt{M/\epsilon})$. An unconstrained interior $C^{1,1}$ minimizer supplies this upper growth. With constraints the feasible chord must be established separately.

Proposition 3.20 (One-sided concave ray). *Suppose $z_* + tv$ is feasible for $0 \leq t \leq t_0$, with $t_0 > 0$, a selected edge satisfies $c_{ij}v_iv_j < 0$, and $\liminf_{t \downarrow 0} m(z_* + tv)/t = 0$. Then $N_{\text{cover}}(\epsilon) \rightarrow \infty$ as $\epsilon \downarrow 0$, including for constrained problems.*

Proof. If covers with at most N boxes existed along $\epsilon_\ell \downarrow 0$, their intersections with the ray would give at most N closed intervals covering $[0, t_0]$. Pad empty intersections by singleton intervals and pass to a subsequence along which all endpoints converge. The limit intervals still cover $[0, t_0]$, since the distance of any point to their union is the limit of its distances to the finite unions. For a limiting interval $[a, b]$ and $a < t < b$, endpoint interpolation of the concave selected edge gives

$$m(z_* + tv) + \epsilon_\ell \geq |c_{ij}v_iv_j|(t - a_\ell)(b_\ell - t).$$

For large ℓ , t lies in that interval. Passing to the limit gives the same inequality at tolerance zero. Among finitely many limit intervals covering points arbitrarily close to zero, one has left endpoint zero and right endpoint $b > 0$. Consequently $m(z_* + tv)/t \geq |c_{ij}v_iv_j|(b - t)$ for $0 < t < b$, contradicting the $\liminf$ assumption. $\square$

Proposition 3.21 (A full aligned segment in two dimensions). *Let $f(x, y) = g(x, y) + cxy$, with $c \neq 0$ and g convex and continuous on $[L_x, U_x] \times [L_y, U_y]$. If the entire segment $\{a\} \times [L_y, U_y]$ is optimal, cutting at $x = a$ gives an exact certificate with at most two leaves. If a is an endpoint, the root is exact. The assertion uses the termwise model [equation \(30\)](#) with no other constraints.*

[Appendix A.6](#) proves this statement using convexity to identify the transverse slope along the segment. Its full-length hypothesis matters. Alignment alone is insufficient in higher dimensions, as the following example shows.

Example 3.22 (An aligned three-dimensional segment). On $D = F = [0, 1]^3$, fix $a \in (0, 1)$ and set

$$f = (x - a)^2 + (x - a)(y - z) + (y - z)^2.$$

Use the specified decomposition $g = (x - a)^2 + (y - z)^2 - a(y - z)$ and $\phi = xy - xz$, with graph $G = \{xy, xz\}$. The function g is convex. Writing $X = x - a$ and $Y = y - z$ gives $f = X^2 + XY + Y^2 \geq (X^2 + Y^2)/2$, so the optimum is zero and its set is the segment $\{(a, t, t) : 0 \leq t \leq 1\}$. Its tangent lies in the independent coordinates $\{y, z\}$ and its line constant τ is zero. Nevertheless

$$N_{\text{cover}} \asymp N_{\text{rect}} \asymp N_{\text{tree}} \asymp \epsilon^{-1/2}.$$

For the lower bound consider the strip $\{(x, t, t) : |x - a| \leq h, 0 \leq t \leq 1\}$, with parameter area $2h$. On it $m \leq h^2$. A valid test meets the strip in a parameter rectangle. At its center the two edge lower bounds give gap at least $w_x w_t / 2$, so its area is at most $2(h^2 + \epsilon)$. Summing and taking $h = \sqrt{\epsilon} \leq \min(a, 1 - a)$ gives $N_{\text{cover}} \geq 1/(2\sqrt{\epsilon})$.

For the upper bound, a dyadic cube of side s has $\Gamma_B \leq s^2/2$. An unpruned cube therefore contains a point with $|x - a| < s$ and $|y - z| < s$. At level j at most four x -intervals and at most $5 \cdot 2^j$ pairs of y, z intervals can do so. There are $O(2^j)$ unpruned cubes. All cubes are valid when $s_j^2/2 \leq \epsilon$. The geometric sum of active-parent counts up to this level gives $O(\epsilon^{-1/2})$ processed nodes, and coordinate cuts realize the dyadic construction. The fixed decomposition thus gives diverging certificates even for this convex quadratic objective.

4 Box-face integrals and analytic singularities

For an unconstrained smooth objective, the root-box faces give a complete description of certification complexity for a strictly positive separable quadratic gap. The description includes the full box. It also includes lower-dimensional faces, because a minimizer on the boundary can be sharp in its inward normal direction and flat along the boundary. A volume integral over the full box can then miss the part of the domain that requires the most subdivision.

Throughout this section, let

$$D = \prod_{i=1}^n [\ell_i, u_i], \quad \ell_i < u_i, \quad f_* = \min_D f, \quad m = f - f_*.$$

There are no constraints other than membership in D . The function f is continuously differentiable on a neighborhood of D , and its gradient is Lipschitz on D . Fix coefficients $\alpha_i > 0$ such that $f(x) + \sum_i \alpha_i x_i^2$ is convex on D . For a test box $B = \prod_i [a_i, b_i] \subseteq D$, use the exact separable α BB bound

$$L(B) = \min_{x \in B} f_B(x), \quad f_B(x) = f(x) - \sum_{i=1}^n \alpha_i (x_i - a_i)(b_i - x_i). \quad (40)$$

The assumed convexity makes f_B convex. Positive coefficients can always be chosen: if the gradient has Lipschitz constant M , taking $\alpha_i \geq M/2$ suffices, because the quadratic Taylor-remainder bound gives a supporting plane for $f + \sum_i \alpha_i x_i^2$ at every point of D . The coefficients and the representation (40) remain fixed as the tolerance changes. The separable quadratic underestimator is the established α BB construction (Adjiman et al., 1998).

A box is valid at tolerance $\varepsilon > 0$ when $L(B) \geq f_* - \varepsilon$. Here this has the exact pointwise form

$$m(x) + \varepsilon \geq \sum_i \alpha_i (x_i - a_i)(b_i - x_i) \quad (x \in B). \quad (41)$$

We use the separate certificate quantities N_{cover} , N_{rect} , and N_{tree} : the minimum number of valid boxes in a cover of D , in a rectangular partition of D , and among the leaves of an axis-parallel split tree, respectively. Thus

$$N_{\text{cover}}(\varepsilon) \leq N_{\text{rect}}(\varepsilon) \leq N_{\text{tree}}(\varepsilon). \quad (42)$$

The upper construction uses the fixed optimal incumbent f_* . Its cost therefore measures certification after this incumbent is available.

Let $\mathcal{F}_+(D)$ denote all closed faces of D of positive dimension, including D itself. A face F fixes some coordinates at root endpoints; write J_F for its free coordinates and $d_F = |J_F|$. Lebesgue measure in its affine hull is denoted by σ_F . Define

$$I_F(\varepsilon) = \int_F (m(x) + \varepsilon)^{-d_F/2} d\sigma_F(x), \quad \mathcal{I}_D(\varepsilon) = 1 + \sum_{F \in \mathcal{F}_+(D)} I_F(\varepsilon). \quad (43)$$

All comparison constants below may depend on this fixed instance, its dimension, and its coefficients, but not on ε .

Theorem 4.1 (Box-face characterization). *Under the preceding assumptions, let $T_{\text{dyad}}(\varepsilon)$ be the tree that halves every coordinate of each invalid box and tests its 2^n children. Then, as $\varepsilon \downarrow 0$,*

$$N_{\text{cover}}(\varepsilon) \asymp N_{\text{rect}}(\varepsilon) \asymp N_{\text{tree}}(\varepsilon) \asymp |T_{\text{dyad}}(\varepsilon)| \asymp \mathcal{I}_D(\varepsilon). \quad (44)$$

More explicitly, for every $\varepsilon > 0$,

$$N_{\text{cover}}(\varepsilon) \geq \max \left\{ 1, \max_{F \in \mathcal{F}_+(D)} \frac{d_F^{d_F/2}}{\pi^{d_F}} \left(\prod_{i \in J_F} \alpha_i \right)^{1/2} I_F(\varepsilon) \right\}. \quad (45)$$

There are finite constants C_0, C_1 , independent of ε , such that

$$|T_{\text{dyad}}(\varepsilon)| \leq C_0 + C_1 \sum_{F \in \mathcal{F}_+(D)} I_F(\varepsilon). \quad (46)$$

The same dyadic partition has a binary split-tree realization with at most a constant multiple of this number of nodes.

The proof is in [appendix B.3](#). Its lower bound is an arcsine integral on each root face. For the upper bound, an invalid box contains a point with $m + \varepsilon$ of order its squared side length. Nonnegativity of m on the closed root box controls the derivative toward a nearby face, even when there is no room for an outward step. Moving the point to that face gives a relative cube of comparable size inside a slightly larger sublevel set. Packing these relative cubes and summing their scales gives the face integrals. Optimal vertices require a separate argument: either every inward edge derivative is positive and the corner box is certified after a fixed refinement, or an edge integral accounts for its remaining levels. No separate $\log(1/\varepsilon)$ term is needed for vertices.

The theorem assumes neither quadratic growth nor analyticity. It also does not identify the finite values of the three certificate minima. The comparisons follow from a lower bound for arbitrary covers and an upper bound constructed by a tree. A total-node count and a leaf count differ by a fixed factor for these finite trees; additional relaxation evaluations or reduction rounds remain separate costs.

Remark 4.2 (Gap assumptions and owned certificates). The proof uses only two properties of the oracle: a bound implying $m + \varepsilon \geq \alpha q_B$ on every valid box, with $\alpha > 0$, and an upper-error bound $f_B \geq f - \alpha' q_B$, with $\alpha' > 0$ and $q_B = \sum_i (x_i - a_i)(b_i - x_i)$. Thus the characterization also holds under these two bounds, with modified constants. Strict positivity is essential to this scope. For example, choosing zero correction for a convex objective gives an exact root bound and can give a one-box certificate.

The per-face lower bound also applies to test boxes B_e with measurable owned sets $A_e \subseteq B_e$ covering D , when (41) is known only on A_e . The integral over each owned part is bounded by the same full-box arcsine integral. The event accounting from the certificate framework can therefore transfer this lower bound to allowed reductions. Such a transfer bounds the specified event or evaluation cost; it does not identify an owned family with a valid closed-box cover.

4.1 Analytic objectives and exact power–logarithm rates

Suppose now that f is real analytic on a neighborhood of D . On a face $F \in \mathcal{F}_+(D)$ with a nonzero restriction $h = m|_F$ and $\min_F h = 0$, define its real log canonical threshold and multiplicity (λ_F, θ_F) through the first positive pole and its order in

$$\zeta_F(z) = \int_F h(x)^{-z} d\sigma_F(x). \quad (47)$$

The integral initially converges for sufficiently small positive real z , and the definition uses its meromorphic continuation. The restriction must be considered on F as a domain in its own affine hull. A boundary zero is included, and no nonnegativity outside F is required.

The analytic input is classical. On these compact semianalytic domains, λ_F is a positive rational number, θ_F is a positive integer, and

$$Z_F(s) := \int_F e^{-sh(x)} d\sigma_F(x) \sim c_{Z,F} s^{-\lambda_F} (\log s)^{\theta_F-1}, \quad c_{Z,F} > 0. \quad (48)$$

This is the compact-domain asymptotic of Lin (2017, Section 2, Theorem 2.10). His domain treatment resolves the analytic function and the boundary inequalities together. A closed face has affine analytic boundary inequalities in its intrinsic coordinates and density one, so it satisfies those hypotheses. His absolute-value convention agrees with (47), because $h \geq 0$ on F . A restriction that is identically zero is handled separately below; it has no ordinary positive-pole RLCT pair of this form.

Lemma 4.3 (Evaluation of the regularized integral). *Let $h \geq 0$ be a nonzero analytic function on a positive-dimensional compact box F , with a zero on F , and suppose its Laplace integral satisfies (48) with parameters λ, θ, c_Z . For $a > 0$, put $I_a(\varepsilon) = \int_F (h + \varepsilon)^{-a} d\sigma_F$. Then*

$$I_a(\varepsilon) \sim \frac{c_Z \Gamma(a - \lambda)}{\Gamma(a)} \varepsilon^{\lambda-a} (\log(1/\varepsilon))^{\theta-1}, \quad \lambda < a, \quad (49)$$

$$I_a(\varepsilon) \sim \frac{c_Z}{\theta \Gamma(a)} (\log(1/\varepsilon))^\theta, \quad \lambda = a, \quad (50)$$

$$I_a(\varepsilon) \uparrow \int_F h^{-a} d\sigma_F < \infty, \quad \lambda > a. \quad (51)$$

Proof. The gamma integral and Tonelli's theorem give

$$I_a(\varepsilon) = \frac{1}{\Gamma(a)} \int_0^\infty s^{a-1} e^{-\varepsilon s} Z_F(s) ds. \quad (52)$$

Let $p = \theta - 1$. By the Laplace asymptotic and boundedness of Z_F near zero, for every $s > 0$, $Z_F(s) \leq C s^{-\lambda} (1 + \log(2 + s))^p$. If $a > \lambda$, substitute $s = u/\varepsilon$ in (52). For each $u > 0$, the integrand divided by $c_Z \varepsilon^{\lambda-a} (\log(1/\varepsilon))^p$ converges to $u^{a-\lambda-1} e^{-u}$. For $\varepsilon \leq e^{-1}$ it is bounded by a constant times $u^{a-\lambda-1} e^{-u} (2 + \log(2 + u))^p$. This function is integrable. Dominated convergence gives (49).

If $a = \lambda$, the part of the integral below a fixed $S > 1$ is bounded. Above S , use $Z_F(s) = c_Z s^{-a} (\log s)^p (1 + o(1))$. Writing $L = \log(1/\varepsilon)$, one has

$$\int_S^\infty e^{-\varepsilon s} \frac{(\log s)^p}{s} ds = \frac{L^{p+1}}{p+1} + O(L^p + 1).$$

Indeed, replacing $e^{-\varepsilon s}$ by one on $[S, 1/\varepsilon]$ changes the integral by at most $\varepsilon \int_S^{1/\varepsilon} (\log s)^p ds = O(L^p)$. On $[1/\varepsilon, \infty)$, the substitution $u = \varepsilon s$ bounds the integral by $O(L^p + 1)$. Bounding the $o(1)$ remainder above a sufficiently large fixed S by an arbitrary constant proves (50). Finally, if $a < \lambda$, the integrand in (52) with $\varepsilon = 0$ is integrable both at zero and infinity. Tonelli and monotone convergence give (51). $\square$

For each positive-dimensional face, define a rate without assigning an RLCT to the identically-zero case:

$$R_F(\varepsilon) = \begin{cases} \varepsilon^{-d_F/2}, & m|_F \equiv 0, \\ 1, & \min_F m > 0, \\ \varepsilon^{\lambda_F - d_F/2} (\log(1/\varepsilon))^{\theta_F-1}, & m|_F \not\equiv 0, \min_F m = 0, \lambda_F < d_F/2, \\ (\log(1/\varepsilon))^{\theta_F}, & m|_F \not\equiv 0, \min_F m = 0, \lambda_F = d_F/2, \\ 1, & m|_F \not\equiv 0, \min_F m = 0, \lambda_F > d_F/2. \end{cases} \quad (53)$$

Corollary 4.4 (Facewise analytic rate). *Under the assumptions of [theorem 4.1](#), with f real analytic near D , each certificate quantity and the dyadic node count satisfy*

$$N_{\text{cover}}(\varepsilon) \asymp N_{\text{rect}}(\varepsilon) \asymp N_{\text{tree}}(\varepsilon) \asymp |T_{\text{dyad}}(\varepsilon)| \asymp \max \left\{ 1, \max_{F \in \mathcal{F}_+(D)} R_F(\varepsilon) \right\}. \quad (54)$$

Proof. Apply [lemma 4.3](#) with $a = d_F/2$ whenever the restriction is nonzero and has a zero. If $m|_F \equiv 0$, its integral is exactly $\sigma_F(F)\varepsilon^{-d_F/2}$. If $\min_F m > 0$, its integral converges to a positive finite value. There are finitely many faces, so their sum is comparable to the largest term, with a baseline one. Now use [theorem 4.1](#). $\square$

The equality case $\lambda_F = d_F/2$ has one more logarithm than the Laplace asymptotic. The theorem retains this case even when $\theta_F > 1$; it needs no claim about whether such a face can dominate all of its subfaces. The constants in [lemma 4.3](#) are leading constants for the integrals. They do not assert that a discrete tree count has a limiting leading constant.

4.2 When the full-dimensional integral suffices

Proposition 4.5 (Full-box reduction). *Under the assumptions of [theorem 4.1](#), either of the following conditions implies*

$$N_{\text{cover}}(\varepsilon) \asymp N_{\text{rect}}(\varepsilon) \asymp N_{\text{tree}}(\varepsilon) \asymp |T_{\text{dyad}}(\varepsilon)| \asymp 1 + I_D(\varepsilon) : \quad (55)$$

- (i) *Every minimizer lies in the interior of D .*
- (ii) *m is nonnegative on a fixed open neighborhood of D , and its gradient is Lipschitz on a smaller closed neighborhood of D .*

Proof. In case (i), compactness gives $\min_{\partial D} m > 0$, so every proper face integral is bounded as $\varepsilon \downarrow 0$. Case (ii) is proved in [appendix B.4](#): nonnegativity beyond the box makes low-sublevel points on each face centers of full-dimensional inward cubes of side proportional to the square root of the sublevel height. Packing these cubes and integrating their sublevel volumes gives $I_F \leq C(1 + I_D)$ for every proper face. In either case, use [theorem 4.1](#). $\square$

Corollary 4.6 (Interior analytic minimizers). *Suppose f is analytic near D and every minimizer is interior. Let (λ, θ) be the RLCT pair on D . Then $\lambda \leq n/2$. If $\lambda < n/2$, all the quantities in (55) have rate $\varepsilon^{\lambda - n/2}(\log(1/\varepsilon))^{\theta-1}$. Equality $\lambda = n/2$ holds exactly when every minimizer has positive definite Hessian. In that case the minimizers are finite, $\theta = 1$, and the rate is $\log(1/\varepsilon)$. Moreover,*

$$I_D(\varepsilon) \sim \frac{(2\pi)^{n/2}}{\Gamma(n/2)} \left(\sum_{x \in \arg \min_D f} \det(\nabla^2 f(x))^{-1/2} \right) \log(1/\varepsilon). \quad (56)$$

Proof. At an interior minimizer x_* , the gradient vanishes and the Hessian is positive semidefinite. Taylor's theorem gives $m(x_* + h) \leq C\|h\|^2$, so the sublevel set of height t contains a ball of radius $c\sqrt{t}$. Hence $Z_D(s) \geq cs^{-n/2}$ and $\lambda \leq n/2$. If the Hessian has a unit null vector v , write $h = w + rv$ with $w \perp v$. Analyticity gives, for small h ,

$$m(x_* + w + rv) \leq C(\|w\|^2 + |r|^3).$$

For example, this follows by bounding the quadratic term on $v^\perp$ and the Taylor remainder $O(\|h\|^3)$. The corresponding anisotropic sublevel box has volume at least $ct^{(n-1)/2+1/3}$. Thus $Z_D(s) \geq cs^{-((n-1)/2+1/3)}$, forcing $\lambda \leq (n-1)/2 + 1/3 < n/2$.

If all Hessians are positive definite, each minimizer is isolated. Their closed set is compact, and an infinite set would have a nonisolated accumulation point, so there are finitely many. Outside fixed disjoint neighborhoods of these points, m has a positive lower bound. At each minimizer, rescale $h = s^{-1/2}z$ in its Laplace integral. The local quadratic lower bound gives an integrable Gaussian majorant, and Taylor expansion and dominated convergence give

$$Z_D(s) \sim (2\pi)^{n/2} s^{-n/2} \sum_{x \in \arg \min_D f} \det(\nabla^2 f(x))^{-1/2}.$$

Therefore $\lambda = n/2$ and $\theta = 1$. Apply [lemma 4.3](#) and [proposition 4.5](#) to obtain the stated rate and [\(56\)](#). $\square$

Under condition (ii) of [proposition 4.5](#), the same full-dimensional RLCT rate applies when m is nonzero. Every zero is then a critical point in an open neighborhood. A small ball centered at that zero meets D in a set containing a fixed fraction of its volume, so the preceding quadratic upper bound again gives $\lambda \leq n/2$. Boundary degeneracy need not satisfy the interior Hessian equivalence just proved.

4.3 Examples and the role of the boundary

Example 4.7 (An isolated degenerate minimum). On $D = [-1, 1]$, let $m(x) = x^4$ and fix a positive coefficient in [\(40\)](#). The minimum is isolated, but

$$I_D(\varepsilon) = \varepsilon^{-1/4} \int_{-\varepsilon^{-1/4}}^{\varepsilon^{-1/4}} (1 + z^4)^{-1/2} dz \asymp \varepsilon^{-1/4}.$$

The limiting integral over $\mathbb{R}$ is finite and positive. Thus all three certificate complexities have rate $\varepsilon^{-1/4}$, although the optimal set has dimension zero. Replacing x^4 by x^2 gives the critical logarithmic rate. A rule based only on the dimension of the optimal set therefore requires a growth hypothesis, such as quadratic growth; it does not apply to arbitrary isolated analytic minima.

Example 4.8 (A singular intersection gives a logarithmic factor). Let $m(x, y) = x^2 y^2$ on $[-1, 1]^2$. This function is nonnegative on $\mathbb{R}^2$, so [proposition 4.5](#) applies. Writing $t = \delta^2$, the sublevel area for $0 < \delta < 1$ is exactly

$$4 \left(\int_0^\delta 1 dx + \int_\delta^1 \frac{\delta}{x} dx \right) = 4\delta(1 + \log(1/\delta)).$$

It has rate $t^{1/2} \log(1/t)$. Directly, $\zeta_D(z) = 4/(1 - 2z)^2$ for $\operatorname{Re} z < 1/2$, so $(\lambda, \theta) = (1/2, 2)$. The regularized integral can also be evaluated directly:

$$I_D(\varepsilon) = \frac{4}{\sqrt{\varepsilon}} \int_0^1 \frac{\arctan(x/\sqrt{\varepsilon})}{x} dx \sim \pi \varepsilon^{-1/2} \log(1/\varepsilon).$$

To check the last asymptotic, split the integral at $x = \sqrt{\varepsilon}$. The lower part is bounded after rescaling. On the upper part, replacing the arctangent by $\pi/2$ changes it by at most $\sqrt{\varepsilon} \int_{\sqrt{\varepsilon}}^1 x^{-2} dx \leq 1$. Thus the certificate rate is $\varepsilon^{-1/2} \log(1/\varepsilon)$.

Example 4.9 (The full-box RLCT misses the dominant face). On the four-dimensional cube

$$D = [0, 0.9] \times [-0.4, 0.5]^3, \quad m(x, y, z, w) = x(1 - x) + y^4 + z^4 + w^4,$$

the unique minimizer is the origin, on the face $F = \{x = 0\} \cap D$. Taking $\alpha_i = 1$ in (40) makes every node underestimator convex, since the Hessian of m is $\text{diag}(-2, 12y^2, 12z^2, 12w^2)$.

The Laplace integral factors. For the first coordinate, substituting $x = r/s$ and using $x(1-x) \geq x/10$ gives

$$\int_0^{0.9} e^{-sx(1-x)} dx \sim s^{-1}.$$

For each quartic coordinate, substituting $y = s^{-1/4}v$ gives

$$\int_{-0.4}^{0.5} e^{-sy^4} dy \sim c_4 s^{-1/4}, \quad c_4 = \int_{\mathbb{R}} e^{-v^4} dv = 2\Gamma(5/4).$$

Consequently the full-box pair is $(\lambda_D, \theta_D) = (7/4, 1)$, and $I_D(\varepsilon) \asymp \varepsilon^{-1/4}$. On F , the pair is $(3/4, 1)$ and $d_F = 3$, so $I_F(\varepsilon) \asymp \varepsilon^{-3/4}$. Every proper face other than F has positive minimum: it fixes $x = 0.9$ or at least one quartic coordinate at a nonzero endpoint. The face characterization therefore gives

$$N_{\text{cover}}(\varepsilon) \asymp N_{\text{rect}}(\varepsilon) \asymp N_{\text{tree}}(\varepsilon) \asymp |T_{\text{dyad}}(\varepsilon)| \asymp \varepsilon^{-3/4}.$$

This also explains why nonnegativity merely on the root box is enough for the facewise theorem but not for the full-box reduction: $x(1-x)$ becomes negative immediately outside the face $x = 0$.

The analytic input describes how sublevel measure concentrates near a singular zero. The additional optimization argument identifies which intrinsic dimension must be paired with that concentration. For this oracle it is the maximum over root-box faces, and at boundary minima that maximum can be attained on a proper face. For face-exact relaxations such as termwise McCormick bounds, the gap has a different zero set; the geometric tools in the preceding section apply to that separate setting.

5 Changing the certificate representation

A spatial tree describes its final certificate by boxes in all the original variables. A decomposition certificate can reuse local boxes across many combinations of those variables. The distinction matters even when both certificates use exactly the same factor relaxations. We show this on a nonconvex path problem with a unique, nondegenerate interior minimum: every spatial certificate is exponentially large in the dimension, whereas a decomposition certificate has size $O(n \log(n/\epsilon))$. The comparison concerns certificate size. It does not charge the work of finding the minimizer or identify a local convex program with a spatial branch-and-bound node.

Decomposition and the reuse of independent subproblems are established ideas in AND/OR search and branch-and-bound (Marinescu and Dechter, 2009). Width also controls approximate formulations for polynomial optimization (Bienstock and Muñoz, 2018). Our purpose here is to isolate what changes when the local relaxation is fixed and the representation of its certificate changes.

5.1 Local boxes and separator minorants

In this section there are no constraints beyond a compact, nondegenerate box $D = \prod_{i=1}^n [L_i, U_i]$. Write

$$f(x) = \sum_{c \in \mathcal{C}} f_c(x_c), \quad f_* = \min_D f, \quad (57)$$

where $c \subseteq \{1, \dots, n\}$ is a factor's coordinate set and f_c is continuous on D_c . The factorization is fixed. For each factor and box B_c , the local oracle supplies a continuous convex function $f_{c,B_c} \leq f_c$ on B_c . The spatial oracle is

$$L(B) = \min_{x \in B} \sum_{c \in \mathcal{C}} f_{c,B_c}(x_c). \quad (58)$$

Let T be a rooted tree with root r and bags $V_t \subseteq \{1, \dots, n\}$. Every factor is assigned to one bag containing its coordinates. The bags cover all variables, and the bags containing any one variable form a connected subtree. Put

$$a_t = \sum_{c \text{ assigned to } t} f_c, \quad S_t = V_t \cap V_{p(t)} \quad (t \neq r), \quad S_r = \emptyset,$$

where $p(t)$ is the parent. The separator S_t contains the coordinates shared with the parent. We take $|V_t| \leq w + 1$ and $|S_t| \leq w$. Let $M = |T|$, let k be the maximum number of bags containing one variable, and let Δ be the maximum number of children of a bag. Width alone does not bound k .

For the subtree rooted at t , let $\phi_t(s)$ be its exact minimum with $x_{S_t} = s$. Connectedness of the bags containing each variable gives the recursion

$$\phi_t(s) = \min_{z \in D_{V_t}: z_{S_t} = s} \left\{ a_t(z) + \sum_{u \in \text{ch}(t)} \phi_u(z_{S_u}) \right\}, \quad \phi_r = f_*. \quad (59)$$

All these minima exist. The value functions are continuous, since each is the minimum of a continuous function over a fixed compact product box.

Definition 5.1 (Decomposition certificate). For each nonroot bag, choose a rectangular partition $\mathcal{P}_t$ of D_{S_t} and an affine function $\ell_{t,E}(s) = \lambda_{t,E}^\top s + \beta_{t,E}$ for each cell $E \in \mathcal{P}_t$. For each bag choose a rectangular partition $\mathcal{L}_t$ of D_{V_t} into local leaves. Boxes are closed and their interiors are disjoint; their shared boundaries are retained in every applicable test.

For every local leaf $B \in \mathcal{L}_t$ and child u , choose a continuous convex function $\psi_{u,B}$ on B_{S_u} such that

$$\psi_{u,B}(s) \leq \ell_{u,E}(s) \quad (s \in B_{S_u} \cap E, \ E \in \mathcal{P}_u). \quad (60)$$

Define

$$R_{t,B}(z) = \sum_{c \text{ assigned to } t} f_{c,B_c}(z_c) + \sum_{u \in \text{ch}(t)} \psi_{u,B}(z_{S_u}).$$

Require

$$R_{t,B}(z) \geq \ell_{t,E}(z_{S_t}) \quad (z \in B, \ z_{S_t} \in E) \quad (61)$$

for every nonempty leaf-cell intersection. At the root require $R_{r,B} \geq \ell_r$ on every leaf, for a scalar ℓ_r . The size is

$$N_{\text{dec}} = \sum_t |\mathcal{L}_t| + \sum_{t \neq r} |\mathcal{P}_t|.$$

The certificate proves tolerance ϵ if $\ell_r \geq f_* - \epsilon$. With a feasible incumbent U , the operational test is $\ell_r \geq U - \epsilon$.

Lemma 5.2 (Validity and reuse). *Every certificate satisfies $\ell_{t,E} \leq \phi_t$ on E and $\ell_r \leq f_*$. Moreover, for each $x \in D$ there are local leaves $B_t \ni x_{V_t}$ such that*

$$\ell_r \leq \sum_t \sum_{c \text{ assigned to } t} f_{c,(B_t)_c}(x_c). \quad (62)$$

Proof. Induct upward from the terminal bags. For $s \in E$, choose a minimizing z in (59) and a local leaf containing it. The child tests and the induction hypothesis give $\psi_{u,B}(z_{S_u}) \leq \phi_u(z_{S_u})$. Since factor relaxations are underestimators, the leaf test gives $\ell_{t,E}(s) \leq R_{t,B}(z) \leq \phi_t(s)$. The root is identical.

For (62), start at the root with a leaf containing x_{V_r} . Replace each child bound by the minorant of a cell containing x_{S_u} , using (60), and then use the leaf test in a child leaf containing x_{V_u} . Continue down the tree. Every cell minorant cancels with the corresponding child term. $\square$

There can be exponentially many compatible combinations of local leaves. The certificate stores their local members and the separator inequalities that justify their reuse, rather than listing every combination.

5.2 A small certificate under quadratic growth

The existence result uses one slope λ_t on every cell of separator S_t . Given the intercepts of its children, set

$$\psi_{u,B}(s) = \lambda_u^\top s + \min_{E \in \mathcal{P}_u: E \cap B_{S_u} \neq \emptyset} \beta_{u,E}, \quad (63)$$

$$\beta_{t,E} = \min_{B \in \mathcal{L}_t} \min_{z \in B: z_{S_t} \in E} \{R_{t,B}(z) - \lambda_t^\top z_{S_t}\}, \quad (64)$$

omitting empty intersections. At the root take $\ell_r = \min_{B \in \mathcal{L}_r} \min_B R_{r,B}$. These choices satisfy the certificate tests.

Theorem 5.3 (Certificate existence). *Assume $w \geq 1$, $k \geq 2$, and the following conditions:*

1. $f(x) - f_* \geq c_g \|x - x^*\|_2^2$ on D , for $c_g > 0$ and $x^* \in D$;
2. each a_t is C^1 on a neighborhood of D_{V_t} and has an H_a -Lipschitz gradient on that box, for a positive constant H_a ;
3. $0 \leq f_c(z) - f_{c,B_c}(z) \leq \alpha' q_{B_c}(z)$, where $q_{B_c}(z) = \sum_{i \in c} (z_i - l_i)(u_i - z_i)$. Let A bound the sum of the coordinate-set sizes of the relaxed factors assigned to any one bag.

Put

$$\begin{aligned} D_k &= \sum_{j=1}^{k-1} \Delta^j, & K_1 &= \max \left\{ 1, (k-1) \sum_{j=0}^{k-2} \Delta^j \right\}, \\ Q &= \frac{H_a^2 w k}{c_g} + \frac{H_a w}{2} + \alpha' A, & H_0 &= 768 K_1 Q + 3 \alpha' A. \end{aligned}$$

Choose $\theta = 2^{-\mu}$ for an integer $\mu \geq 1$ so that

$$4\theta \sqrt{(k-1)D_k} \leq \frac{1}{2}, \quad \theta^2 k \{48 K_1 (1 + \Delta) Q + \alpha' A\} \leq \frac{c_g}{2}. \quad (65)$$

For any $h > 0$ with $h^2 \leq \epsilon / (4 M H_0)$, there is a certificate of size

$$N_{\text{dec}} \leq 2M(4/\theta)^{w+1}(J+1), \quad J = \max\{0, \lceil \log_2(s_0/h) \rceil\}, \quad s_0 = \max_i (U_i - L_i). \quad (66)$$

It uses partitions centered at x^* and slopes

$$\lambda_{t,i} = \sum_{u \in \text{sub}(t): i \in V_u} \partial_i a_u(x_{V_u}^*) \quad (i \in S_t). \quad (67)$$

The proof is in Appendix C.1. The construction gives each local box B width at most $h + \theta \text{dist}_\infty(B, x^*)$. Separator copies can disagree, but bounded variable occurrence controls the accumulated disagreement. The slopes in (67) cancel its first-order cost. The remaining error is quadratic and can be paid by the growth margin. No smoothness of the exact value functions is required, and x^* may lie on the boundary.

For fixed $k, \Delta, c_g, H_a, \alpha' A$ and w , choosing $h^2 = \epsilon/(4MH_0)$ gives $O(M \log(M/\epsilon))$ local members as $\epsilon \downarrow 0$. Formula (66) displays the exponential dependence on width and the conditioning constants. It is an existence result with a known center and slopes. Locating this center and constructing the certificate without it are separate algorithmic questions.

5.3 An exponential separation with the same oracle

Theorem 5.4 (Fixed termwise path oracle). *For $n \geq 3$, let $D = [-1, 1]^n$ and*

$$f_n(x) = \sum_{i=1}^n (x_i^2 - \frac{1}{10}x_i^4) + \frac{4}{5} \sum_{i=1}^{n-1} x_i x_{i+1}. \quad (68)$$

Keep each unary factor exact. Relax each bilinear factor $\frac{4}{5}xy$ by the termwise alphaBB function

$$\frac{4}{5}xy - \frac{2}{5}\{(x - l_x)(u_x - x) + (y - l_y)(u_y - y)\}.$$

This is a convex underestimator. The objective is nonconvex, but has the unique, nondegenerate interior minimizer $x^ = 0$, with $f_* = 0$.*

Every spatial cover whose boxes pass (58) has size at least

$$\max \left\{ 0.068\sqrt{n}(2e/\pi)^{n/2} \log \frac{0.2}{n\epsilon}, \quad (5/3)^n \exp \left[-\frac{5}{9}(1 + \frac{5}{4}\epsilon) \right] \right\}. \quad (69)$$

The first expression is useful when $\epsilon < 0.2/n$. Consequently this bound applies to N_{cover} , and hence to N_{rect} and N_{tree} .

With path bags $V_t = \{t, t+1\}$, the same factor relaxations admit a decomposition certificate of size at most

$$U_n(\epsilon) = 3.36 \cdot 10^7 (n-1) \left\{ \frac{1}{2} \log_2 \frac{1.96 \cdot 10^6 (n-1)}{\epsilon} + 2 \right\} \quad (0 < \epsilon \leq 10^{-4}). \quad (70)$$

There is an absolute $c > 0$ such that, throughout this accuracy range,

$$\frac{N_{\text{cover}}(\epsilon)}{U_n(\epsilon)} \geq c \frac{(2e/\pi)^{n/2}}{\sqrt{n}}. \quad (71)$$

Appendix C.2 proves the assertions and both spatial lower bounds. Each valid spatial box must absorb the summed gaps of all the separately relaxed bilinear terms. A local path certificate absorbs those gaps in two-dimensional leaves and reuses the separator information. Even an optimal choice of spatial cuts cannot remove the lower bound. For a binary spatial tree with no additional reductions, N terminal boxes correspond to $2N - 1$ nodes. With tightening, the appropriate bound is on the owned certification events described in the certificate framework, including the certified removed regions; it is not a bound on leaves alone.

The same size distinction holds for termwise McCormick envelopes in both models. The second expression in (69) and the upper bound (70) still apply, as shown in the appendix. Thus at each fixed tolerance in the stated range the ratio is exponential in n . The alphaBB integral bound, and hence the uniform ratio (71), are not claimed for McCormick envelopes.

5.4 Size, queries, and checking work

The number of stored local members differs from the work needed to check them. For one nonroot leaf-cell pair, (61) is a convex minimization problem of dimension at most $w + 1$; a root leaf needs one such problem. For the common-slope construction, the child tests reduce to comparisons of intercepts. If $l_t = |\mathcal{L}_t|$ and $p_t = |\mathcal{P}_t|$, the number of local convex checks is at most

$$l_r + \sum_{t \neq r} l_t p_t \leq M(4/\theta)^{2w+1}(J+1)^2. \quad (72)$$

Only intersecting pairs need checking. If the leaves are aligned so that each separator projection meets at most a fixed number of cells, the number of checks is instead proportional to the number of leaves. The shell construction need not have that alignment: requiring it may refine the leaf partitions. Its unconditional checking bound therefore contains an additional logarithm.

The oracle in (58) supplies a factor relaxation on a specified box. One such query for each assigned factor and local leaf suffices: its returned function is reused on all cell intersections. If at most R factors are assigned to a bag, there are at most $R \sum_t l_t$ such queries. This is $O(M \log(M/\epsilon))$ for fixed parameters. If an oracle instead supplies only the result of a local minimization on each requested leaf-cell intersection, unaligned queries are counted by (72), giving the safe bound $O(M \log^2(M/\epsilon))$. Neither count specifies the arithmetic or bit cost of solving the convex programs.

The factorization and oracle restrictions are substantive. Aggregating terms, adding stronger cuts, or choosing different functional factors can change the gap and the certificate size. The theorem is not a lower bound for every representation of (68), nor a prediction of a general solver's runtime. It compares two representations of a certificate under one fixed local relaxation.

6 Objective-cutoff propagation and its representation

An objective cutoff can certify a box before its relaxation is solved. Its strength depends on the expression graph, and its cost depends on the propagation schedule. We first separate these two dependencies. Throughout this section, the graph contains the objective alone. Revising original constraints jointly with the objective, imposing new auxiliary-variable bounds, and branching on auxiliary variables define different schemes. Natural interval extensions and validated exclusion regions are established tools (Schichl et al., 2014). The results below concern objective-cutoff propagation and the certificates it can produce under a fixed graph.

Let $\mathcal{D}$ be a finite directed acyclic graph representing a continuous function f on a compact box D . Its nodes are variables, fixed constants, and continuous operations on the compact forward domains. Each operation has an elementary graph constraint E . Repeated references in such a constraint refer to the same coordinate. For $C \subseteq D$, let $Z_0(C)$ be the lifted box obtained by forward interval evaluation, and let $F_{\text{lo}}(C)$ be its root lower endpoint. Forward evaluation is assumed inclusion isotone and to use the exact image of each elementary operation over its input intervals. This permits dependency loss between operations.

For a lifted box Z , the exact revise $\rho_E(Z)$ replaces the coordinates in E by the interval hull of their projections from $E \cap Z$; it returns the empty box if this intersection is empty. Other coordinates remain unchanged. The closed cutoff revise intersects the root interval with $(-\infty, c]$. A box is *hull-consistent* if it is fixed by every elementary revise and by this cutoff revise. Hull consistency asks that interval endpoints have support within each elementary constraint. Those supports need not agree between constraints. An exact schedule is fair when every elementary revise and the cutoff revise occur infinitely often. For partial revises, fairness requires both the forward and backward revise of every operation, and the cutoff revise, infinitely often.

Theorem 6.1 (The objective-DAG bound). *There is a greatest hull-consistent box $Z^*(C, c) \subseteq Z_0(C)$. Every sequence of steps satisfying $\rho_E(Z) \subseteq \rho(Z) \subseteq Z$ preserves this box. A fair sequence of exact revises converges to $Z^*(C, c)$, and reaches an empty iterate if this limit is empty. The same conclusions hold for fair exact forward and backward partial revises.*

For $y \in C$, the minima

$$\pi_{\mathcal{D}}(C) = \min\{c : Z^*(C, c) \neq \emptyset\}, \quad (73)$$

$$\pi_{\mathcal{D}}(C, y) = \min\{c : y \in \Pi_x Z^*(C, c)\} \quad (74)$$

exist and satisfy

$$F_{\text{lo}}(C) \leq \pi_{\mathcal{D}}(C) \leq \pi_{\mathcal{D}}(C, y) \leq f(y). \quad (75)$$

Both bounds increase when C shrinks. The limiting propagation empties C exactly when $c < \pi_{\mathcal{D}}(C)$ and removes y exactly when $c < \pi_{\mathcal{D}}(C, y)$. Any allowed finite schedule removes only points satisfying this strict inequality.

The fixed-point viewpoint is standard in FBBT (Belotti et al., 2012). The proof for the present schedule and cutoff model is in [appendix D.1](#). The strict comparisons matter: a closed cutoff remains consistent at its threshold. Relaxation pruning, in contrast, can use $L(C) \geq f_* - \epsilon$. Computing the limiting box as a node oracle does not provide a finite stopping test for every nonempty limit.

Definition 6.2 (Flat objective sum). The root equation is $w = b + \sum_{j=1}^J a_j p_j$, with nonzero coefficients and *distinct* root-child node coordinates p_j . Each p_j is a base node $z_{k(j)}$ or a unary operation $p_j = \varphi_j(z_{k(j)})$ used only by the root. Write $t_j = a_j \varphi_j$, with $\varphi_j(z) = z$ for direct base terms. A base can feed several distinct unary nodes. Repeated references to the same root coordinate must first be collected into one coefficient. For base intervals $U_k = [s_k, t_k]$, set

$$\mu_j(U) = \min_{v \in U_{k(j)}} t_j(v), \quad h_j(U) = \max\{t_j(s_{k(j)}), t_j(t_{k(j)})\}, \quad (76)$$

$$\Phi(U) = b + \sum_j \mu_j(U) + \max_j \{h_j(U) - \mu_j(U)\}. \quad (77)$$

Theorem 6.3 (Exact flat-sum formula). *If $Z^*(C, c)$ is nonempty, its base intervals U satisfy $\Phi(U) \leq c$. If the bases are the original variables, then*

$$\pi_{\mathcal{D}}(C) = \min_{\emptyset \neq U \subseteq C} \Phi(U), \quad \pi_{\mathcal{D}}(C, y) = \min_{y \in U \subseteq C} \Phi(U), \quad (78)$$

where U ranges over closed subboxes, including degenerate ones.

Proof. In a consistent box, the signed term interval $T_j = a_j Z_{p_j}$ has lower endpoint at least μ_j , and upper endpoint at least h_j : each base endpoint must have support in its unary constraint. Supporting the upper endpoint of T_j in the root constraint gives $c \geq b + h_j + \sum_{i \neq j} \mu_i$. This proves the necessary inequality.

Conversely give the variables the intervals U , give each signed term the interval $[\mu_j, h_j]$, and give the root $[b + \sum_j \mu_j, \min\{c, b + \sum_j h_j\}]$. Unary constraints support their base endpoints and their term endpoints. Direct variable terms have their exact signed intervals. For the root, any term endpoint can be supported by putting all other terms at their minima; the resulting sum is at most $\Phi(U) \leq c$. The root endpoints are attainable in a sum of intervals. Distinct root coordinates make these root supports legitimate. This constructs a consistent box containing U . The necessary inequality applied to $\Pi_x Z^*$ proves both formulas. Continuity of term minima and endpoint values, on the compact set of subbox endpoints, proves attainment. $\square$

The endpoint maximum in (77) cannot be replaced by the full range maximum. For example, representing x^2 on $[-1, 1]$ by three distinct power nodes with coefficients $-3, 2, 2$ gives $\Phi([-a, a]) = -a^2$, and hence $\pi_{\mathcal{D}}([-1, 1]) = -1$. Indeed a same-side subinterval gives $\Phi = \min_U x^2 \geq 0$; a subinterval crossing zero gives $\Phi \geq -\max_U x^2 \geq -1$. The single-node square representation has bound zero. Moreover the formula does not apply to the root equation $w = x + x - x$: its repeated references give the exact equation $w = x$, not three independent root coordinates.

Corollary 6.4 (Sufficient exactness in exposed coordinates). *For one base coordinate, call an interval U one-sided if there are a term j_0 and an endpoint e such that $t_{j_0}(e) = h_{j_0}(U)$ and $t_j(e) = \mu_j(U)$ for every $j \neq j_0$. If every subinterval of its forward base range is one-sided, then*

$$\pi_{\mathcal{D}}(C) \geq \min_{z \in Z_0(C)_z} \left(b + \sum_j t_j(z) \right).$$

When the base is the original variable, this is equality with $\min_C f$.

More generally write $\hat{f}(z) = b + \sum_j t_j(z_{k(j)})$ and let $B_z = \prod_k Z_0(C)_{z_k}$. For each coordinate group define

$$e_k(U_k) = \min_{U_k} \sum_{j:k(j)=k} t_j - \sum_{j:k(j)=k} \mu_j \geq 0.$$

If on every subbox of B_z each coordinate group is one-sided and at most one e_k is positive, then $\pi_{\mathcal{D}}(C) \geq \min_{B_z} \hat{f}$. This bound is exact if $z(C) = B_z$.

Proof. In the first case $\Phi(U) \geq \hat{f}(e)$. For the second, put $G_k = \max_{j:k(j)=k} (h_j - \mu_j) - e_k$. One-sidedness gives $G_k \geq 0$, and direct substitution gives

$$\Phi(U) = \min_U \hat{f} + \max_k \left(G_k - \sum_{\ell \neq k} e_\ell \right).$$

Choose the sole coordinate with positive excess, or any coordinate if all excesses vanish. The last maximum is nonnegative. Apply the necessary part of [theorem 6.3](#) and (75). $\square$

Thus exposing a coordinate such as $z = x_1 - x_2$ can change certification. For example $h(z) = (z - 1)^2((z + 1)^2 + 1/2)$, written as $z^4 - 1.5z^2 - z + 1.5$, is one-sided on every positive interval: only z^4 increases. Adding a second exposed coordinate with one nonnegative square term preserves the sufficient criterion. These are sufficient conditions; they do not classify all exact graphs.

Theorem 6.5 (Local exactness and ideal node counts). *Assume $F = D$, f is L -Lipschitz in the sup norm, and $L(C) \geq \min_C f - \tau w(C)^2$, where $w(C)$ is the largest side length. Suppose a relatively open set R contains the minimizing set and, for some $w_0 > 0$, $\pi_{\mathcal{D}}(C) \geq f_*$ for all $C \subseteq R$ with $w(C) \leq w_0$. An ideal algorithm with incumbent f_* first decides whether the objective fixed point at cutoff $f_* - \epsilon$ is empty, then optionally contracts to a box containing its variable projection, then applies the relaxation test and widest-side bisection. Its node count is bounded independently of $\epsilon > 0$.*

Proof. If $R = D$, boxes of width at most w_0 are pruned by propagation. Otherwise $\delta = \min_{D \setminus R} (f - f_*) > 0$. Choose $0 < h \leq w_0$ with $Lh \leq \delta/2$ and $\tau h^2 \leq \delta/2$. A box of width at most h is either contained in R , or contains a point outside R . In the latter case $f - f_* \geq \delta/2$ throughout it, so any contracted box passes the relaxation test. Thus no such box is split. The integer potential $\sum_i \max\{0, \lceil \log_2(w_i(C)/h) \rceil\}$ never increases under contraction and falls by at least one at each split. If its root value is d , the tree has at most $2^{d+1} - 1$ nodes. $\square$

This theorem concerns the fixed-point oracle. Its decision may require many rounds. It is the exact-bound mechanism behind absence of clustering (Du and Kearfott, 1994; Wechsung et al., 2014), applied to a graph-dependent bound. No representation-free node lower bound can hold in this unconstrained model: the representation $f_* + |f - f_*|$ equals f on D and has forward lower bound f_* . It encodes the answer in its constant f_* . For a constrained optimum this identity need not hold at infeasible points.

Lemma 6.6 (A surviving-point witness). *Suppose the root is a sum of distinct child nodes, over any continuous DAG below them. Regard their signed expressions as $t_j(x)$. For every $y \in U \subseteq C$,*

$$\pi_{\mathcal{D}}(C, y) \leq b + \sum_j \min_U t_j + \max_j \left(\max_U t_j - \min_U t_j \right). \quad (79)$$

If each term is an expression tree with each variable used at most once, then its forward range is exact and

$$F_{\text{lo}}(C) \leq \pi_{\mathcal{D}}(C) \leq F_{\text{lo}}(C) + \max_j (\max_C t_j - \min_C t_j).$$

Proof. Give every nonroot node its exact range over U . Every value in each interval is attained by a point of U , and therefore has support within its elementary graph constraint, even if different values use different points. Give the root the interval from the proof of theorem 6.3, now using full term maxima. This gives a consistent witness with variable box U . For a single-use term, the children of each binary operation use disjoint variables, so their joint range is the product of their ranges. Induction proves exact forward evaluation, giving the second assertion. $\square$

Theorem 6.7 (Coordinatewise loss preserves geometric lower bounds). *Assume $F = D$ and the root is a sum of distinct term nodes with C^2 signed expressions t_j . Let $s_0 > 0$ be the largest root side. On a set $N \subseteq D$, assume, for every coordinate,*

$$d_i(y) := \sum_j |\partial_i t_j(y)| - 2 \max_j |\partial_i t_j(y)| \geq d_0 > 0.$$

On coordinate segments of radius $\rho > 0$ about these points, suppose $|\partial_{ii} t_j| \leq H_j$, and put $K_0 = \sum_j H_j/2 + \max_j H_j$, $\rho_ = \min\{\rho, d_0/(2K_0)\}$, $\vartheta = d_0 \rho_*/2$, interpreting $d_0/(2K_0) = +\infty$ if $K_0 = 0$. If propagation removes $y \in N \cap C$ at cutoff $f_* - \epsilon$, and $m(y) + \epsilon < \vartheta$, then*

$$\min\{y_i - l_i, u_i - y_i\} < \frac{2(m(y) + \epsilon)}{d_0}, \quad m(y) + \epsilon \geq \frac{d_0}{2ns_0} q_C(y). \quad (80)$$

Consider finite runs using coordinate splits, objective-only propagation phases with inherited lifted bounds, and a pointwise relaxation test that certifies $m(y) + \epsilon \geq \alpha q_C(y)$ on each discarded owned region. Let K count terminal boxes plus at most $2n$ removed-piece certificates per phase. Set $\alpha_{\text{eff}} = \min\{\alpha, d_0/(2ns_0)\}$. Then

$$K \geq 2^{-n} \mathcal{N}_{\infty} \left(E(\eta) \cap N, \sqrt{(\epsilon + \eta)/\alpha_{\text{eff}}} \right), \quad \epsilon + \eta < \vartheta, \quad (81)$$

$$K \geq \left(\frac{\alpha_{\text{eff}} n}{\pi^2} \right)^{n/2} \int_{N \cap \{m + \epsilon < \vartheta\}} (m(y) + \epsilon)^{-n/2} dy. \quad (82)$$

Here $E(\eta) = \{m \leq \eta\}$ and $\mathcal{N}_{\infty}(S, r)$ counts sup-norm balls of radius r covering S .

The full event and inheritance proof is in [appendix D.2](#). In particular, shared closed boundaries stay with the retained box; the discarded owner, rather than every point of its closed test box, satisfies the strict propagation comparison. A p -dimensional smooth piece of minimizers in N therefore forces $K = \Omega(\epsilon^{-p/2})$. An interior minimizer with a neighborhood contained in N and $m(y) \leq \Gamma\|y - x_*\|^2$ locally forces $K = \Omega(\log(1/\epsilon))$. These become node lower bounds when the number of phases per node is uniformly bounded. Arbitrarily many phases cannot be charged to a bounded number of nodes.

Coordinatewise no-dominance is stronger than the aggregate condition

$$\ell(y) = \sum_j \|\nabla t_j(y)\|_1 - 2 \max_j \|\nabla t_j(y)\|_1 > 0.$$

Aggregate loss confines removed points to thin face layers, rather than to vertex neighborhoods. Under uniform aggregate loss, an interior nondegenerate minimum in dimension $n \geq 2$ still forces a logarithmic event count; a minimizing arc whose tangent has every coordinate bounded away from zero still forces $\Omega(\epsilon^{-1/2})$ events. Precise hypotheses and complete proofs are in [appendix D.3](#). These weaker hypotheses do not establish the general $\epsilon^{-p/2}$ rate in dimension $p \geq 3$.

Finally, the node reduction in [theorem 6.5](#) can move work into propagation rounds. For the graph $s^2 - 2as + a^2$ on a fixed interval around $a > 0$, a forward-cutoff-backward schedule needs $\Omega(a\epsilon^{-1/2})$ rounds before emptying, although its limiting bound is exact. For $u = t^2$, $v = u^2$, $f = u - 2v$, a schedule with initial $0 < U_0 < 1/2$ on u empties in $O(\log \log(1/\epsilon))$ rounds. [Appendix D.4](#) proves both statements for the specified schedules. Neither is a bound for all graphs or all schedules. General criteria for bounded-round propagation remain open.

7 Selecting a tree from relaxation information

A small certificate does not imply that a branching rule will find a small tree. This section fixes the oracle and compares the tree selected by a rule with the smallest permitted certificate. The comparison counts bound evaluations. It does not count the work needed to compute a bound, discover an optimal incumbent, query a function germ, or compute an optimal cut.

Here $F = D = \prod_i [L_i, U_i]$, f is continuous, the incumbent is fixed at $f_* = \min_D f$, and $\epsilon > 0$. For a fixed $\alpha > 0$, the oracle is

$$f_C(y) = f(y) - \alpha q_C(y), \quad L(C) = \min_C f_C, \quad q_C(y) = \sum_i (y_i - l_i)(u_i - y_i). \quad (83)$$

Put $M_\epsilon(y) = f(y) - f_* + \epsilon$. A box is valid exactly when $M_\epsilon(y) \geq \alpha q_C(y)$ everywhere on C . At each invalid node a binary rule makes a strictly interior coordinate cut and processes both children. Thus $T = 2N - 1$ for a finished tree with N leaves. One exact bound evaluation is charged per node. Validity is hereditary, because $q_{C'} \leq q_C$ on $C' \subseteq C$. Uniformly small boxes are valid, since $M_\epsilon \geq \epsilon$.

The minimum rectangular-partition size is N_{rect} , and the minimum guillotine-tree leaf count is N_{tree} . In one dimension they agree, and $T_{\text{opt}} = 2N_{\text{rect}} - 1$. In higher dimensions a partition may require refinement to be realized by a tree. No convexity of f is assumed below. Even convexity of the underestimator in (83) is unnecessary for the one-dimensional theorem. If $f + \alpha\|y\|^2$ is convex, the oracle is a convex minimization oracle; otherwise exact minimization may itself be expensive.

Theorem 7.1 (One-dimensional minimizer branching). *On an interval, split each invalid node at any exact minimizer of f_C . For every valid partition with $N \geq 2$ intervals, the rule has at most $4N - 5$ splits and $8N - 9$ nodes. If $N = 1$, the root is valid and the count is one. Consequently*

$$T \leq 4T_{\text{opt}} - 5 \quad \text{when the root is invalid,} \quad T \leq 4T_{\text{opt}} \quad \text{on every instance.}$$

The rule terminates, and its bound permits arbitrary choices among tied minimizers, including choices using previous search history.

Proof. Use the shifted relaxation $\phi_C = M_\epsilon - \alpha q_C$. On an invalid interval its minimum is negative, while endpoint values are positive. Every selected minimizer is therefore strictly interior. Split points cannot repeat: an ancestor split is a descendant endpoint, and nonnested nodes have disjoint interiors.

Fix a certificate interval $J = [s, s']$ of length λ . An invalid node whose split lies inside J cannot be contained in J . Suppose nested split nodes $C_1 \supsetneq C_2$ both cross s , and their splits y_1, y_2 lie in (s, s') . The child containing s has upper endpoint y_1 , so $y_2 < y_1$. Write $d = s - l_{C_1} > 0$, $e = u_{C_1} - s > 0$, and $t_k = y_k - s$. Optimality at y_1 , invalidity at y_2 , and validity of J give

$$\begin{aligned} M_\epsilon(y_1) &\leq M_\epsilon(y_2) + \alpha[(t_1 + d)(e - t_1) - (t_2 + d)(e - t_2)], \\ M_\epsilon(y_2) &< \alpha(t_2 + d)(t_1 - t_2), \\ M_\epsilon(y_1) &\geq \alpha t_1(\lambda - t_1). \end{aligned}$$

For the second inequality, C_2 is contained in $[l_{C_1}, y_1]$. Cancellation yields $t_1(\lambda - t_1) < (t_1 - t_2)(e - t_1) < t_1(e - t_1)$. Thus $\lambda < e$: the larger node also crosses s' . Reflection gives the analogous right-endpoint statement.

Nodes crossing a given endpoint are nested. The crossing argument allows at most one split node crossing only the left endpoint of J , and at most one crossing only its right endpoint. At most one crosses both, since its split separates the two endpoints. Thus an interior certificate interval contains at most three split points. Each end interval contains at most one. The $N - 1$ interior certificate breakpoints can each be used once. The total is at most $(N - 1) + 2 + 3(N - 2) = 4N - 5$. The argument applies to every finite partial tree. An infinite run would have a finite partial tree exceeding this bound. Continuity gives a minimizer at every node, and a uniform valid partition supplies a finite comparison certificate. This proves termination and the node count. $\square$

Proposition 7.2 (Two-piece certificates and the known ratio range). *If $N_{\text{rect}} = 2$, minimizer branching uses at most five nodes, and this count is attained. Over all continuous one-dimensional instances, its worst-case competitive ratio lies in $[11/5, 4]$.*

Appendix E.1 proves the five-node refinement and gives a fully specified eleven-node instance with a three-piece optimum. The strict instancewise inequality in theorem 7.1 does not imply a supremum strictly below four. The exact worst-case ratio remains undetermined.

We next specify what a rule knows. A *local relaxation-information rule* sees its current box, α , ϵ , incumbent, exact bound, a selected relaxation minimizer, and the germ of f at that minimizer. A germ gives equality on some open neighborhood; it does not allow arbitrary queries over the box. The rule carries no objective-dependent information from other nodes. For the following root obstruction this last restriction is immaterial, but it matters in repeated ambiguities.

Proposition 7.3 (An information obstruction on a nonanalytic class). *There are two one-dimensional continuous objectives, with convex underestimators in (83), having identical root data, identical germs at the root minimizer, and identical functions near both root endpoints. Each has a three-node optimal tree, but their optimal first cuts differ. For every deterministic rule using these data, at least one of the two instances has ratio at least $5/3$. For every randomized root-cut rule, at least one has expected ratio at least $4/3$. The same statements hold in a class of C^∞ nonanalytic objectives.*

The construction and its smoothing proof are in [appendix E.2](#). It uses nonanalytic functions explicitly. For real-analytic objectives on a connected neighborhood, a germ determines the whole objective by uniqueness of analytic continuation; this information obstruction does not apply to that class. Together with [theorem 7.1](#), the deterministic minimax ratio in the stated nonanalytic information class belongs to $[5/3, 4]$.

Full function information can select an optimal certificate without information loss. In one dimension define $b(l)$ as the largest right endpoint for which $[l, b(l)]$ is valid. Validity is closed in the endpoint, so the largest endpoint exists. Starting at the left root endpoint, take this largest interval repeatedly. If s_j are endpoints of any optimal partition and x_j are greedy endpoints, induction gives $x_j \geq s_j$: either $x_{j-1} \geq s_j$, or $[x_{j-1}, s_j]$ is a valid subinterval of the j th comparison piece. Thus the greedy partition has the optimal size. Sufficiently short intervals are uniformly valid, so the construction terminates. In higher dimensions a full function oracle can choose the first cut of a minimum guillotine certificate, and repeat on its children. This statement counts the selected tree, not the cost of constructing the optimal certificate.

7.1 Safeguards and their unavoidable cost

A θ -safe cut of an interval of width w lies in $[l + \theta w, u - \theta w]$, where $0 < \theta < 1/2$. Every child then has width at most $(1 - \theta)w$. Along an arbitrary path its width halves within $\lceil \log 2 / [-\log(1 - \theta)] \rceil$ splits. This protects against relaxation points at or close to a bound. Exact minimizers at invalid nodes in the preceding one-dimensional model are already interior, but can be arbitrarily close to an endpoint.

For $f_a(y) = 2\alpha|y - a|$ on $[0, 1]$, the exact minimizer on an interval straddling a is a , and the interval is invalid exactly when $\alpha(a - l)(u - a) > \epsilon$. Every interval on one side of a is valid. If the root is invalid, cutting at a gives $T_{\text{opt}} = 3$.

Proposition 7.4 (Persistent clipping can lose logarithmically). *For fixed $0 \leq \lambda \leq 1$ and $0 \leq \theta < 1/2$, use the relative cut*

$$\sigma(p) = \text{clip}(\lambda p + (1 - \lambda)/2, \theta, 1 - \theta),$$

where p is the relaxation minimizer's relative position. Unless $(\lambda, \theta) = (1, 0)$, some fixed kink $a \in (0, 1/2)$ has $T = \Omega(\log(1/\epsilon))$ although $T_{\text{opt}} = 3$.

Proof. The continuous function $\sigma(p) - p/(1 - p)$ is positive at zero and negative at $1/2$, so it vanishes at a point $p \in (0, 1/2)$. Put $a = p$ and $\kappa = p/(1 - p) \in (0, 1)$. Symmetry of σ gives a kink-containing chain with relative positions alternating between p and $1 - p$, and width κ^k at depth k . All off-chain children are valid. The chain node is invalid when $\alpha p(1 - p)\kappa^{2k} > \epsilon$. Thus, for $\epsilon < \alpha p(1 - p)$, it uses at least

$$2 \left\lceil \frac{\log(\alpha p(1 - p)/\epsilon)}{2 \log(1/\kappa)} \right\rceil + 1$$

bound evaluations. The pure minimizer rule instead ends in three nodes. □

Proposition 7.5 (The price of safety and a recentring rule). *If $a < \theta$, put $J = \lceil \log(\theta/a) / \log(1/\theta) \rceil$. Every θ -safe rule, with any information and any randomization, has $T \geq 2J + 1$ on f_a whenever $\epsilon < \alpha a^2(1 - \theta)/\theta$. Consequently no fixed safety margin gives a uniformly bounded competitive ratio over kink locations as $\epsilon \downarrow 0$.*

For $0 < \theta \leq 1/3$, the following recentring rule nevertheless has a bounded count as $\epsilon \downarrow 0$ for each fixed kink. Cut at the minimizer p when it is safe; below the safe region cut at $l + \max\{\theta w, 2(p - l)\}$; above it use the reflected formula. If $d_0 = \min\{a, 1 - a\}$ and $J = \min\{j \geq 0 : d_0 \geq \theta^{j+1}\}$, it uses

$T \leq 2J + 3$. Its zero-tolerance chain hits the kink in exactly $J + 1$ splits, the minimum among safe chains that hit it.

Appendix E.4 proves these claims. The worst-case location in the first assertion may depend on ϵ ; the persistent-clamp counterexample instead uses one fixed location. Recentring gives an explicit tradeoff between width contraction and kink hitting. Its hitting optimality does not assert optimality among all finite-tolerance certificates, since a positive tolerance can prune before hitting the kink. All counts here are bound evaluations for the fixed oracle. They are not runtime comparisons between numerical minimization methods.

7.2 Coordinate choice and a positive restricted extension

For a separable objective write $f - f_* = \sum_i m_i(x_i)$, with $\min_{D_i} m_i = 0$. Let $N_i(b)$ be the least number of intervals partitioning D_i with $\min_J(m_i - \alpha q_J) \geq -b$. Slicing a rectangular certificate at minimizers of all other coordinates, and taking products of one-dimensional certificates, gives

$$\max_i N_i(\epsilon) \leq N_{\text{rect}}(\epsilon) \leq N_{\text{tree}}(\epsilon) \leq \inf_{\epsilon_i > 0, \sum_i \epsilon_i = \epsilon} \prod_i N_i(\epsilon_i). \quad (84)$$

The slice uses the whole tolerance; the product allocates it across axes. These inequalities alone do not settle coordinate selection.

Proposition 7.6 (Repeated coordinate ambiguity). *For every $n \geq 2$, there are n nonanalytic separable objectives on $[0, 1]^n$, at the common tolerance $\epsilon = 1/100$ and $\alpha = 1$, each with $T_{\text{opt}} = 3$. For every deterministic node-local relaxation-information rule, with coordinate-wise minimizer selection, at least one of these instances satisfies*

$$T \geq 2 \left\lceil \frac{2^{n+1} - n - 3}{n - 1} \right\rceil + 1.$$

For every distribution of such local rules, at least one instance has

$$\mathbb{E}T \geq \frac{2(2^{n+1} - n - 2)}{n} + 1.$$

The rules may also see the objective germ at every corner of the current box. They may not carry objective-dependent information from other nodes or use objective information outside these germs to resolve minimizer ties.

Appendix E.3 supplies the family and the counting proof. The result forces exponential dependence on dimension in any competitive constant for this information class. It does not show accuracy-dependent loss at a fixed dimension, or apply to analytic germs or rules that learn from siblings.

For example, on $f(x, z) = x^2$ in $[0, 1]^2$ with $\alpha = 1$, a rule that splits every coordinate with an interior relaxation minimizer has $\Omega(\epsilon^{-1/2} \log(1/\epsilon))$ processed boxes, whereas a guillotine certificate has $O(\epsilon^{-1/2})$ leaves. Appendix E.5 proves this separation. Selecting the correct point in each coordinate does not justify splitting all coordinates.

The ω rule instead selects a coordinate maximizing $q_{J_i}(y_i)$, where y_i is its selected exact relaxation minimizer, and cuts there. General constant competitiveness of this rule, even for arbitrary separable objectives in a fixed dimension at least two, remains open. The following extension isolates a class in which the phase accounting can be completed.

Theorem 7.7 (One arbitrary coordinate and sharp coordinates). *Scale to $\alpha = 1$. There is one arbitrary continuous nonnegative coordinate function with minimum zero on its original interval A , and r sharp coordinate functions with minima zero at interior kinks c_i . Fix a coordinate-wise minimizer selection depending only on its function and interval. A sharp coordinate means that on every interval straddling c_i , the selected minimizer is c_i with value $-q_J(c_i)$, and on either side the selected minimizer has zero gap and nonnegative relaxation value. Use the fixed optimal incumbent and exact separable oracle above. Explicitly, $f - f_* = m_0(x) + \sum_{i=1}^r m_i(z_i)$, and $N_A(b)$ is the least number of intervals partitioning A for which $\min_J(m_0 - q_J) \geq -b$.*

Set $C_0 = 4$ and, for $r \geq 1$, define

$$c_r = r[2r(r+1) + 1], \quad A_r = 4(1 + c_r), \quad C_r = (A_r + 1) + 2C_{r-1}(A_r + 2).$$

Then ω terminates with at most $C_r N_A(\epsilon)$ leaves and

$$T_\omega \leq 2C_r N_{\text{rect}}(\epsilon) - 1.$$

Coordinate-choice ties may be resolved arbitrarily. Every fixed coordinate-wise minimizer selection is permitted if every sharp-coordinate minimizer satisfies the stated properties; otherwise fix a selection satisfying them.

The complete induction and phase lemma are in [appendix E.6](#). The constants depend only on r and are conservative. For an explicit sharp coordinate on $[L_i, U_i]$, take $m_i(t) = \sigma_i |t - c_i| - (t - c_i)^2$, with $\sigma_i \geq 2(U_i - L_i)$ and $c_i \in (L_i, U_i)$. The left and right slopes of $m_i - q_{[l,u]}$ are $-\sigma_i - (l + u - 2c_i)$ and $\sigma_i - (l + u - 2c_i)$. They are strictly negative and positive respectively, so the kink or the nearest endpoint is the unique minimizer and the sharpness conditions hold. The arbitrary coordinate may be quadratic or nonconvex. The theorem does not cover two arbitrary coordinates or justify a global sum over phases for general separable objectives.

8 Admissible classes for integer certificates

For spatial certificates, the permitted pieces constrain how the domain can be covered. Integer certificates have a related distinction: a collection of convex pieces may cover the required integer points without covering all fractional points of the root relaxation. This section identifies the least number of pieces in that semantic model, then separates it from the cost of using variable or split disjunctions. The same distinction explains why an integer-space cut can shorten a particular tree without improving the lower bound forced by the projected relaxation.

8.1 The projected relaxation and the required integer points

Consider

$$\text{OPT} = \inf\{f(x, y) : (x, y) \in C, x \in \mathbb{Z}^n\}, \quad (85)$$

where y denotes the continuous variables. Let $F \subseteq \mathbb{Z}^n$ be the feasible integer projections, and let $f_F(x)$ be the infimum of the objective over the feasible continuous variables with integer part x . A fixed convex relaxation consists of a convex set $\widehat{K} \supseteq C$ and a jointly convex objective $\widehat{\phi}$ that underestimates f on C . Its projected value function is

$$\phi(x) = \inf\{\widehat{\phi}(x, y) : (x, y) \in \widehat{K}\}, \quad r(Q) = \inf_{x \in Q} \phi(x). \quad (86)$$

We assume that $\phi : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper. Infimal projection preserves convexity, although it need not preserve lower semicontinuity. The convention is $\inf \emptyset = +\infty$; in particular, $\phi = +\infty$

outside its effective domain K . The value $r(Q)$ is the node bound when branching restricts only the integer variables to Q .

Fix a set P with $F \subseteq P \subseteq \mathbb{Z}^n$. It records which integer points the certificate must preserve. Variable and split disjunctions on a bounded root box usually take P to be all integer points in that box; branching justified only on feasible points may take $P = F$. These choices can give different lower bounds. Fix also a threshold $\tau \in \mathbb{R} \cup \{+\infty\}$. An additive ε certificate uses $\tau = \text{OPT} - \varepsilon$, for finite OPT and $\varepsilon \geq 0$; infeasibility uses $\tau = +\infty$.

Definition 8.1 (*P-covering convex-piece certificate*). A finite rooted tree assigns a convex set $Q_v \subseteq \mathbb{R}^n$ to each node. The root contains P . Each internal node v satisfies

$$Q_v \cap P \subseteq \bigcup_{w \text{ child of } v} Q_w. \quad (87)$$

Its effective set is $\overline{Q}_v = \bigcap_{u \preceq v} Q_u$, where the intersection is over its root path. The tree is a τ certificate if $r(\overline{Q}_\ell) \geq \tau$ at every leaf ℓ .

The sets need not be closed. Replacing every node set by its effective set makes the tree nested and preserves (87): a required point in $\overline{Q}_v$ belongs to at least one child and to all its ancestors. The children must cover only $P \cap Q_v$. No coverage of all fractional points is required. This is the semantic certificate class used below. Variable branches $x_i \leq c$ or $x_i \geq c + 1$, integer splits $\pi^T x \leq c$ or $\pi^T x \geq c + 1$, and finite multiway integer hyperplane branches are subclasses when their integer disjunctions cover P .

A completed branch-and-bound run gives such a certificate if it preserves P and prunes only by relaxation infeasibility, a valid computed bound $\beta_v \leq r(\overline{Q}_v)$ satisfying $\beta_v \geq UB_v - \varepsilon$, or an attained relaxation optimum that is feasible for (85) and has the same original and relaxed objective. Indeed, every feasible incumbent satisfies $UB_v \geq \text{OPT}$. Bound pruning therefore gives $r(\overline{Q}_v) \geq \text{OPT} - \varepsilon$, and exact integral pruning gives $r(\overline{Q}_v) \geq \text{OPT}$. This argument permits weaker computed lower bounds and arbitrary node selection. Reductions that remove required integer points need the separate accounting below.

8.2 Class number and exact semantic attainment

Definition 8.2 (Admissibility and class number). A set $I \subseteq P$ is τ -admissible if

$$r(\text{conv } I) \geq \tau. \quad (88)$$

The class number $\kappa_\tau(P)$ is the least number of admissible sets covering P , or $+\infty$ if no finite cover exists. Set $\kappa_\tau(\emptyset) = 0$.

Admissibility is hereditary. A cover may therefore be turned into a partition by assigning every point to its first covering class. Pairwise conflicts provide useful lower bounds, but they can miss obstructions that involve more than two points. Define the midpoint graph on P by $a \sim b$ when $a \neq b$ and $\phi((a+b)/2) < \tau$. Define the segment graph by $a \sim b$ when $\inf_{[a,b]} \phi < \tau$. An admissible class is an independent set in both graphs. Thus

$$\omega(G_\tau^{\text{mid}}) \leq \omega(G_\tau^{\text{seg}}) \leq \chi(G_\tau^{\text{seg}}) \leq \kappa_\tau(P). \quad (89)$$

For every nonempty finite $S \subseteq P$ with positive denominator, there is also the counting bound

$$\kappa_\tau(P) \geq \frac{|S|}{\max\{|I \cap S| : I \text{ is admissible}\}}. \quad (90)$$

These follow by restricting an admissible partition to pairs and to S , respectively. If no admissible class contains a point of S , no admissible cover exists and $\kappa_\tau(P) = +\infty$. Midpoint conflicts and bounded occupancy of a leaf extend established branch-and-bound lower-bound mechanisms (Dey et al., 2023). Hiding sets give a related obstruction to small formulations (Kaibel and Weltge, 2015).

Theorem 8.3 (Class number and semantic trees). *Every P -covering convex-piece τ certificate has at least $\kappa_\tau(P)$ leaves. If $P \neq \emptyset$ and $k = \kappa_\tau(P) < \infty$, the semantic model of definition 8.1 attains this lower bound with k leaves. It also attains it with a full binary tree having $2k - 1$ nodes. If P is finite, all node sets in an attaining tree can be polytopes.*

Proof of the lower bound. Route each $p \in P$ from the root to a child containing it and continue until a leaf is reached. Let I_ℓ consist of the points routed to leaf ℓ . Then $I_\ell \subseteq \overline{Q}_\ell$, and convexity gives $\text{conv } I_\ell \subseteq \overline{Q}_\ell$. Consequently $r(\text{conv } I_\ell) \geq r(\overline{Q}_\ell) \geq \tau$. The nonempty I_ℓ form an admissible partition of P . $\square$

The attainment proof in appendix F.1 uses separation from the convex strict sublevel set $E_\tau = \{\phi < \tau\}$. Each class can be enlarged to a *hemispace*: a convex set whose complement is convex and which misses E_τ . Successively removing these hemispaces gives a binary chain. At its last internal node, the two remaining children cover the remaining points of P ; they need not cover its fractional points. The appendix proves the required hemispace separation for arbitrary disjoint convex sets, including nonclosed sets.

The root is part of the semantic certificate: it may be any convex set containing P . This convention matters when $k = 1$, since $r(\text{conv } P) \geq \tau$ can hold while a prescribed larger continuous root has bound below τ . For $k \geq 2$ the binary construction can begin at any prescribed convex root containing P . For $k = 1$, its one-node form requires the permitted integer-preserving root restriction. The lower bound holds without this extra permission.

Exact attainment is an existence statement about this specified certificate class. A solver may require disjunctions whose coverage of all node integer points has a short, explicit proof, and may restrict itself to variable or split branching. Finding a minimum admissible partition, separating its classes, and checking its integer coverage can be costly. Finite P makes the node sets polyhedral, but does not bound the number of vertices or the time required to construct and verify them. This separates certificate size from construction and checking costs, as in the continuous distinction between covers, rectangular partitions, and permitted trees.

8.3 Cuts and certified integer removals

Theorem 8.4 (Preserved classes and removal accounting). *Suppose all node bounds use the fixed projected value function ϕ .*

- (a) *Convex cuts in the integer variables that contain all required integer points at the node preserve the leaf lower bound $L \geq \kappa_\tau(P)$. In particular, cuts valid for all feasible integer points preserve $L \geq \kappa_\tau(F)$, even if the cuts give the node's integer hull. Convex feasibility reductions preserving those points have the same property.*
- (b) *Suppose a node's effective set undergoes a finite nested sequence $Q^{(0)} \supseteq \dots \supseteq Q^{(s)}$, and every removal has a convex test set D_i satisfying*

$$Q^{(i-1)} \cap P \subseteq Q^{(i)} \cup D_i, \quad r(Q^{(i-1)} \cap D_i) \geq \tau. \quad (91)$$

If the remaining tree is pruned at threshold τ , then

$$\kappa_\tau(P) \leq L + S, \quad S = \sum_v s_v. \quad (92)$$

The same bound holds for simultaneous certified pieces covering the removed integer points, each tested on the unreduced effective set.

- (c) *For box-bound changes, one removal is one change of one integer bound by any amount. One pass changing each bound at most once per node gives $N \geq \kappa_\tau(P)/(2n+1)$. One such pass only at the root gives $L \geq \kappa_\tau(P) - 2n$. For binary variables, iterated rounded bound changes give $N \geq \kappa_\tau(P)/(n+1)$.*

The proof in [appendix F.2](#) assigns the removed integer points to additional certified leaves. For general integer variables, iterated passes require the actual number S of bound changes; the proof does not give a factor depending only on dimension. Changes of the same bound at different times need not combine into a single admissible class.

Integer rounding gives a clean removal certificate. For example, suppose the current set is Q and incumbent-based tightening finds

$$\ell'_i = \lceil \inf\{x_i : x \in Q, \phi(x) \leq c\} \rceil, \quad c = UB - \varepsilon \geq \tau,$$

with a finite infimum. Every point of $Q \cap \{x_i \leq \ell'_i - 1\}$ has $\phi(x) > c$, so its infimum is at least c . Together with the retained set $Q \cap \{x_i \geq \ell'_i\}$, this closed slab covers the integer points of Q . Its endpoint is one integer unit below the retained bound. Thus the retained-boundary obstruction for continuous closed discarded slabs does not arise. Upper-bound rounding is analogous. Reduced-cost certificates and incumbent-based probing fit [\(91\)](#) when their tested pieces have the stated bound on the current node set.

The leaf bound can fail without the removal count. Take the root $[0, 1]$, $P = \{0, 1\}$, and $\phi(x) = (x - 0.4)^2$ on that box. The integer optimum is 0.16. At $\varepsilon = 0.01$, integers 0 and 1 conflict, so the class number is two. Tightening with the closed incumbent cutoff $\phi \leq UB = 0.16$ gives $x \leq 0.8$; integer rounding retains only $\{0\}$. This permits a one-node certificate, and the removed point $\{1\}$ is the second certificate piece. Strengthening the relaxation in continuous or epigraph variables changes ϕ and can change its class number. Symmetry or dominance reductions without a certificate of the form [\(91\)](#) are outside this accounting.

8.4 The cost of permitted disjunctions

Two elementary comparisons show why κ_τ cannot by itself predict the minimum tree size for a restricted branch class.

Proposition 8.5 (Linear descriptions and projected descriptions). *For a pure integer linear program with natural relaxation $K = \{x : Ax \leq b\}$, integral A, b with m rows, and objective $c^T x$, $\kappa_\tau(\mathbb{Z}^n) \leq m + 1$ for every finite $\tau \leq \text{OPT}$. If it is infeasible, $\kappa_{+\infty}(\mathbb{Z}^n) \leq m$. For a rational mixed-integer linear relaxation, if $\tau < \text{OPT}$ and the projected closed sublevel polyhedron $\{x : \exists y, (x, y) \in \widehat{K}, \widehat{\phi}(x, y) \leq \tau\}$ has a rational description with q rows, then $\kappa_\tau(\mathbb{Z}^n) \leq \max(q, 1)$.*

Proof. In the pure case, use the integer points in $\{c^T x \geq \text{OPT}\}$ and in $\{a_j^T x \geq b_j + 1\}$, $1 \leq j \leq m$. They cover $\mathbb{Z}^n$: an integer point with objective below OPT violates some row by at least one. Their hulls respectively have objective at least OPT or miss K . In the infeasible case, only the violated-row classes are needed. In the mixed case, scale the q projected rows to integral left-hand sides $g_j^T x \leq h_j$. The integer-free sublevel polyhedron is avoided by the covering halfspaces $g_j^T x \geq \lfloor h_j \rfloor + 1$. On each halfspace the projected value is at least τ , since every feasible continuous extension has value strictly above τ . If the sublevel polyhedron is empty, one class suffices. $\square$

Projection can hide exponentially many classes in a short formulation. The mixed-integer linear problem

$$\min \sum_{i=1}^n s_i, \quad s_i \geq x_i - \frac{1}{2}, \quad s_i \geq \frac{1}{2} - x_i, \quad x \in \mathbb{Z}^n$$

has $2n$ rows and projected function $\phi(x) = \|x - \frac{1}{2}\mathbf{1}\|_1$. Its optimum is $n/2$. For $0 \leq \varepsilon < 1/2$, all 2^n binary vectors pairwise conflict: vectors at Hamming distance d have midpoint value $(n-d)/2$. Conversely, the 2^n halfspaces $\sigma^T(x - \frac{1}{2}\mathbf{1}) \geq n/2$, $\sigma \in \{-1, 1\}^n$, cover $\mathbb{Z}^n$ and are admissible. Hence $\kappa_{\text{OPT}-\varepsilon}(\mathbb{Z}^n) = 2^n$.

Proposition 8.6 (A quadratic separation for variable branching). *Let n be odd, $h = (n+1)/2$, and minimize $\phi(x) = (\mathbf{1}^T x - n/2)^2$ over $\{0, 1\}^n$, with the natural box relaxation. For $0 \leq \varepsilon < 1/4$, the optimum is $1/4$ and the class number is two. The split $\mathbf{1}^T x \leq h-1$ or $\mathbf{1}^T x \geq h$ certifies the optimum with three nodes. Every variable-branching certificate has at least $\binom{n+1}{h}$ leaves. This count is attained when each branch fixes a free binary variable and every node whose bound reaches the threshold is immediately made a leaf.*

Proof. The two sum halfspaces cover the integer points and have bound $1/4$; the convex hull of all binary points has bound zero, so one class cannot suffice. A variable node fixing a ones and b zeros has bound zero precisely when $a, b \leq h-1$. Otherwise its bound is at least $1/4$. Contract redundant branches and make every already-certified node a leaf. At every remaining internal node a free variable is fixed, giving the two states $(a+1, b)$ and $(a, b+1)$. The resulting leaf count satisfies $T(a, b) = T(a+1, b) + T(a, b+1)$ for $a, b < h$, with boundary value one at $\max(a, b) = h$. Pascal's identity gives $T(a, b) = \binom{2h-a-b}{h-a}$, hence $T(0, 0) = \binom{n+1}{h}$. $\square$

This is the parity obstruction in a convex quadratic optimization form. It separates variable branching from both semantic branching and integer splits. It does not establish a comparable separation between general split trees and semantic trees for one convex quadratic.

Theorem 8.7 (Lattice-free interpretation). *Let $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and $\tau \leq \inf_{z \in \mathbb{Z}^n} \phi(z)$. If $E_\tau = \{\phi < \tau\}$ is empty, then $\kappa_\tau(\mathbb{Z}^n) = 1$. Otherwise it equals the least number of closed halfspaces covering $\mathbb{Z}^n$ and avoiding E_τ , and the least number of facets of a polyhedron M satisfying*

$$E_\tau \subseteq \text{int } M, \quad \text{int } M \cap \mathbb{Z}^n = \emptyset. \quad (93)$$

In particular, $\kappa_\tau(\mathbb{Z}^n) \leq 2^n$.

The proof is in [appendix F.3](#). The final bound uses the standard maximal lattice-free polyhedron theorem ([Basu et al., 2010](#)). This interpretation connects class number to relative relaxation complexity: the object to be enclosed is now a strict objective sublevel set. It retains the distinction between the number of separating pieces and the cost of realizing them through permitted integer disjunctions.

8.5 Haar-random closest-vector certificates

Let L be a unimodular lattice drawn from Haar probability on $\text{SL}_n(\mathbb{R})/\text{SL}_n(\mathbb{Z})$, $n \geq 2$. Conditional on L , let t be uniform on the quotient torus $\mathbb{R}^n/L$. For any lattice basis B , use the continuous relaxation

$$\phi(x) = \|Bx - t\|^2, \quad P = \mathbb{Z}^n, \quad \text{OPT} = \text{dist}(t, L)^2. \quad (94)$$

Let g_n be the radius of a Euclidean ball of volume one, so $g_n^2 \sim n/(2\pi e)$, and let $\lambda_1(L)$ be the shortest nonzero lattice-vector length. A unimodular change of basis maps integer points and

convex certificate pieces bijectively. All class-number bounds below are therefore independent of the chosen basis. This model is distinct from a matrix with independent Gaussian columns scaled to determinant one.

Lemma 8.8 (A deterministic occupancy bound). *Suppose $\tau > 0$ and $R^2 \leq \tau + \lambda_1(L)^2/2$. Every admissible class contains at most $n + 1$ lattice points Bz in the open ball $B^\circ(t, R)$. Consequently*

$$\kappa_\tau(\mathbb{Z}^n) \geq \frac{\#(L \cap B^\circ(t, R))}{n + 1}. \quad (95)$$

An admissible class lies on the far side of a supporting hyperplane to the ball of radius $\sqrt{\tau}$. Its intersection with $B^\circ(t, R)$ lies in a ball of radius at most $\lambda_1/\sqrt{2}$. The vectors from that ball's center to its lattice points have strictly negative pairwise inner products; there are at most $n + 1$ such vectors. The full proof is in [appendix F.5](#). This uses occupancy rather than only pairwise conflicts between all sampled points.

Theorem 8.9 (Random closest-vector class bounds). *In (94), put $\tau = \text{OPT} - \varepsilon$ and $q = \varepsilon/g_n^2$. Fix $\delta \in (0, 0.1)$ and suppose*

$$0 \leq q < (1 - \delta)^2, \quad R_-^2 = \frac{3}{2}(1 - \delta)^2 - q > 1.$$

Then, with probability at least $1 - \frac{3}{2}(1 - \delta)^n - 4R_-^{-n}$,

$$\kappa_\tau(\mathbb{Z}^n) \geq \frac{R_-^n}{2(n + 1)}. \quad (96)$$

For every $\gamma > 0$, an upper bound holds with probability at least $1 - (1 + \delta)^{-n} - e^{-\gamma n}$:

$$\kappa_\tau(\mathbb{Z}^n) \leq (e^{\gamma n} + Cn^2 \log(n + 1))(1 + (1 + \delta)^2 - q)^{n/2}, \quad (97)$$

where C is an absolute constant when $0 \leq q \leq 1/2$ and $0 < \delta < 0.1$. If $\tau \leq 0$, one class suffices.

For any deterministic tolerance sequence $q_n \rightarrow q_0 \in [0, 1/2)$, these bounds imply, with probability tending to one,

$$2^{(\frac{1}{2} \log_2(3/2 - q_0) - o(1))n} \leq \kappa_{\text{OPT} - \varepsilon_n}(\mathbb{Z}^n) \leq 2^{(\frac{1}{2} \log_2(2 - q_0) + o(1))n}. \quad (98)$$

The proof in [appendix F.4](#) combines torus unfolding with Siegel's mean-value theorem in the normalization stated by [Skenderi \(2021\)](#). Under the joint lattice-target law, the number of lattice points in a translated set of volume V has mean V and variance V . No Poisson independence assumption is needed. The upper bound covers nearby lattice points individually and distant points by halfspaces whose directions form a spherical cap cover.

At vanishing relative tolerance $q_n = o(1)$, (98) gives lower exponent $\frac{1}{2} \log_2(3/2) = 0.29248 \dots$ and upper exponent $1/2$. For fixed positive q_0 , both bounds depend on the tolerance. In particular, the upper exponent is strictly below $1/2$. Thus an exponent $1/2$ cannot hold uniformly for every fixed positive relative tolerance.

Proposition 8.10 (A midpoint-clique lower bound). *In (94), fix $\delta \in (0, 0.1)$, $\varepsilon \geq 0$, $q = \varepsilon/g_n^2$, and $\tau = \text{OPT} - \varepsilon$. Let $r_0^2 = (1 - \delta)^2 - q$, take $1 < \rho < 4/3$ with $\rho r_0^2 > 1$, and put $V = (\rho r_0^2)^{n/2}$. With probability at least*

$$1 - (1 - \delta)^n - 4/V - 2n \left(\frac{4(\rho - 1)}{\rho} \right)^{n/2},$$

the midpoint graph has a clique of size at least $V/4$. In particular, when $q_n = o(1)$ its clique number is at least $2^{(\frac{1}{2} \log_2(4/3) - o(1))n}$ with probability tending to one.

The proof in [appendix F.6](#) removes one endpoint of each nonconflicting pair from a ball of lattice points. Its exponential rate, $0.20752\dots$, is smaller than the occupancy rate in (96). These two lower bounds use different information about admissibility; neither establishes the exact exponent of the class number. Every $\mathbb{Z}^n$ -covering convex-piece tree inherits the class lower bound, including variable branching in any reduced basis and general integer splits. Certified incumbent reductions inherit (92); a claim about nodes requires its stated removal budget. Enumeration that omits out-of-radius values also requires the two certified integer tails, adding at most two leaves per internal node and giving a node lower bound of one third of the class bound.

8.6 Curvature conflicts and stronger block relaxations

The following finite construction places all conflicting points in the feasible integer set, and retains a unique optimum. It shows directly how changing a projected relaxation can eliminate an exponential class obstruction.

Use k mutually orthogonal two-dimensional feature blocks. Block j has unit features u_j, v_j , response y_j , and ridge $\lambda > 0$. There are $2k$ binary indicators z , with $\mathbf{1}^T z \leq k$. The perspective relaxation projects to

$$g(z) = \sum_{j=1}^k g_j(z_{u_j}, z_{v_j}), \quad g_j(a, b) = y_j^T \left(I_2 + \frac{au_j u_j^T + bv_j v_j^T}{\lambda} \right)^{-1} y_j, \quad (99)$$

on $0 \leq z \leq 1$, $\mathbf{1}^T z \leq k$, and is $+\infty$ elsewhere. For a binary support it equals the ordinary ridge objective on that support. Define $g_{ab,j} = g_j(a, b)$ for $a, b \in \{0, 1\}$, and

$$g_j^* = \min(g_{10,j}, g_{01,j}), \quad \Delta_j = |g_{10,j} - g_{01,j}|, \quad d_j = g_j^* - g_j(1/2, 1/2).$$

Theorem 8.11 (Curvature conflicts with a unique optimum). *Suppose*

$$\max_j (g_j^* - g_{11,j}) < \min_j (g_{00,j} - g_j^*), \quad \sum_j \Delta_j + \varepsilon < \min_j d_j. \quad (100)$$

Then $\text{OPT} = \sum_j g_j^$. If every $\Delta_j > 0$, the optimum is unique. All 2^k one-feature-per-block supports pairwise conflict at $\tau = \text{OPT} - \varepsilon$. Every F -covering convex-piece certificate using (99) has at least 2^k leaves, also with arbitrary convex cuts valid for the node's feasible indicators. With certified binary-bound reductions the node lower bound is $2^k/(2k+1)$.*

If each block relaxation is instead the closed convex hull of its four mixed-integer objective epigraphs, the root value equals OPT. For every k , rational data satisfying (100), with all $\Delta_j > 0$ and a fixed positive range of tolerances, can be chosen with encoding length polynomial in k .

The proof and the rational family are in [appendix F.7](#). The stronger root relaxation changes the projected value function in the epigraph variables. It therefore does not contradict preservation of classes under cuts involving only the indicator variables. The conclusion concerns these relaxation and operation classes, rather than a universal algorithmic lower bound.

8.7 A sufficient condition for one-path trees

The hard-class mechanism has a useful counterpart for easy certificates. Restrict the integer domain to a root box $\mathcal{B} = [\ell, u]$, with integral endpoints, and in this subsection set $P = \mathcal{B} \cap \mathbb{Z}^n$. Let (x°, y°) be a feasible point of objective U , and suppose $\phi(x^\circ) \geq U - \varepsilon$. Assume its incumbent value U is available whenever a node is processed. Write

$$\begin{aligned} r(\mathcal{B} \cap \{x_i \leq x_i^\circ - 1\}) &\geq U - \varepsilon & \text{if } \ell_i < x_i^\circ, \\ r(\mathcal{B} \cap \{x_i \geq x_i^\circ + 1\}) &\geq U - \varepsilon & \text{if } x_i^\circ < u_i. \end{aligned} \quad (\text{C1})$$

Lemma 8.12 (One-path certificate). *Under (C1), any proper variable-branching run that prunes nodes reaching the incumbent threshold before branching processes at most $2D + 1$ nodes, where $D \leq \sum_i (u_i - \ell_i)$. For binaries, $D \leq n$. If (x°, y°) is optimal, $\phi(x^\circ) = \text{OPT}$, and every nonempty wrong one-sided box has bound strictly greater than OPT , the same conclusion holds without an initial incumbent under best-bound selection using the exact node values r , provided processing the singleton $x = x^\circ$ obtains a feasible extension of value OPT .*

Proof. Every node not containing x° lies in a wrong one-sided root box. Its bound is at least $U - \varepsilon$, by monotonicity, so it is pruned. At each branched node exactly one child contains x° . Hence the branched nodes form a chain. Proper branching strictly shortens one integer range, giving its length bound and the total count $1 + 2D$. If the containing path reaches a singleton, its bound is at least $U - \varepsilon$, so it can be pruned. In the best-bound variant every node containing x° has bound at most $\phi(x^\circ) = \text{OPT}$, whereas every off-path node has strictly larger bound. Until the optimum is found, best-bound selection follows the containing path; at its singleton endpoint the stated processing hypothesis supplies OPT . All off-path nodes can then be pruned. $\square$

Condition (C1) is a sufficient certificate criterion, rather than a necessary condition for small trees. Probing each wrong one-sided box with this incumbent fixes every integer variable to x° ; the result uses one run node and up to $2n$ probe certificates. These one-sided classes and $\{x^\circ\}$ also give $\kappa_{\text{OPT} - \varepsilon}(P) \leq 2n + 1$: all their bounds are at least $U - \varepsilon \geq \text{OPT} - \varepsilon$.

Even within this regime, the class number can be much smaller than the variable tree. For $n \geq 2$, take $\phi(z) = \sum_i (z_i - a)^2$ on $[0, 1]^n$, $a = 1/(2n)$, and $\varepsilon = 1/(8n^2)$. The optimum is $1/(4n)$ at zero, and each wrong fixing has bound $(1 - a)^2 \geq \text{OPT}$. The two classes $\{0\}$ and $\{z \in \{0, 1\}^n : \mathbf{1}^T z \geq 1\}$ are admissible: the latter hull's minimum is attained at $\mathbf{1}/n$ and equals OPT . The root bound is zero, so the class number is exactly two. A node fixing only s zeros has bound $s/(4n^2)$ and is prunable only when $s = n$. The path to zero therefore has n branches and $n + 1$ leaves. This gives $2n + 1$ nodes despite a two-class semantic certificate.

9 Sparse regression: different thresholds for different certificates

Sparse regression gives a statistical example in which root exactness, short variable-branching trees, and large convex-piece certificates have different thresholds. The distinction depends on the relaxation and on the signal scaling. In particular, failure to certify the planted support does not establish failure to certify the global optimum.

9.1 Model and perspective relaxation

In this section, n is the sample size, p the number of features, and $1 \leq k < p$ the cardinality budget. Let $X \in \mathbb{R}^{n \times p}$ have independent standard Gaussian entries, with columns x_j , and let

$$y = X\beta^* + \sigma w, \quad w \sim N(0, I_n), \quad w \perp X. \quad (101)$$

The support $S^* = \text{supp } \beta^*$ is deterministic and has size k . The ridge parameter $\lambda > 0$ is deterministic. We use the unnormalized loss throughout:

$$F_\lambda(S) = \min_{b \in \mathbb{R}^{|S|}} \{\|y - X_S b\|^2 + \lambda \|b\|^2\}, \quad \text{OPT} = \min_{|S| \leq k} F_\lambda(S), \quad (102)$$

$$K = \{z \in [0, 1]^p : \mathbf{1}^T z \leq k\}, \quad g_\lambda(z) = y^\top (I_n + \lambda^{-1} X \text{diag}(z) X^\top)^{-1} y. \quad (103)$$

Multiplying both terms of the objective by $1/2$ has no effect on the results. Dividing the entire objective by n changes the ridge coefficient to λ/n .

The Boolean relaxation of [Pilanci et al. \(2015\)](#) has the equivalent perspective and dual representations below. The perspective connection is developed by [Xie and Deng \(2018\)](#).

$$g_\lambda(z) = \min_{\beta_j=0 \text{ if } z_j=0} \left\{ \|y - X\beta\|^2 + \lambda \sum_{z_j>0} \frac{\beta_j^2}{z_j} \right\}, \quad (104)$$

$$g_\lambda(z) = \max_{a \in \mathbb{R}^n} h(a, z), \quad h(a, z) = 2a^\top y - \|a\|^2 - \lambda^{-1} \sum_j z_j (x_j^\top a)^2. \quad (105)$$

These identities follow by completing the square in the corresponding ridge problems. The second identity makes g_λ convex in z . At $z = \mathbf{1}_S$, its value is $F_\lambda(S)$. Write

$$r(S_0, S_1) = \min\{g_\lambda(z) : z \in K, z_{S_0} = 0, z_{S_1} = 1\}, \quad R = r(\emptyset, \emptyset). \quad (106)$$

The two sets of fixed coordinates are disjoint. Infeasible nodes have value $+\infty$.

For a support S of size k , let

$$M_S = I_n + \lambda^{-1} X_S X_S^\top, \quad \hat{\beta}_S = (X_S^\top X_S + \lambda I)^{-1} X_S^\top y, \quad r_S = M_S^{-1} y = y - X_S \hat{\beta}_S. \quad (107)$$

The normal equations give $X_S^\top r_S = \lambda \hat{\beta}_S$. The following exact condition is the selected-support certificate in [Pilanci et al. \(2015, Corollary 2\)](#).

Proposition 9.1 (Exactness at a specified support). *For $|S| = k$, put $a_j = x_j^\top r_S$. Then*

$$R = F_\lambda(S) \iff \max_{j \notin S} |a_j| \leq \min_{i \in S} |a_i|. \quad (108)$$

Strict inequality implies that S is the unique optimal support and that every single wrong fixing has bound strictly above OPT.

Proof. The matrix in (103) is positive definite, and $\partial_j g_\lambda(\mathbf{1}_S) = -a_j^2/\lambda$. The first-order condition for minimizing this convex differentiable function over K says that $\mathbf{1}_S$ maximizes $\sum_j a_j^2 z_j$. This is precisely (108). For strictness, evaluate (105) at r_S . A removal of $i \in S$ must replace its squared correlation by a smaller one; forcing $j \notin S$ must displace a larger selected squared correlation. Both node bounds exceed $F_\lambda(S)$. Every other feasible support lies in one of these nodes, which proves uniqueness. $\square$

9.2 Single-variable certificates

At an optimal support S , condition C1 means that all single wrong fixings are certified:

$$r(\{i\}, \emptyset) \geq \text{OPT} - \varepsilon \quad (i \in S), \quad r(\emptyset, \{j\}) \geq \text{OPT} - \varepsilon \quad (j \notin S). \quad (109)$$

Strict C1 replaces the right sides by strict inequalities above OPT. The identity $g_\lambda(\mathbf{1}_T) = F_\lambda(T)$ supplies the surviving singleton bound required by the path lemma, [lemma 8.12](#). Thus C1 gives at most $2p + 1$ nodes for every variable-branching rule if an optimal incumbent is available when off-path nodes are examined. Strict C1 gives the same bound under best-bound search without an initial incumbent. Node relaxation solves, probing work, and their running times are separate costs. Dual sufficient tests for fixing one feature underpin the safe-screening rules of [Atamtürk and Gómez \(2020\)](#). Condition C1 here uses the exact single-fixing relaxation values.

The full condition is sufficient, but need not be necessary for a small certificate. If only the removal inequalities in (109) hold, branch on the k coordinates of S , always continuing through their value 1. Each sibling has a certified removal fixing, and the budget forces the final node to be $\mathbf{1}_S$. This is a certificate with $2k + 1$ nodes. Thus failure of the forced-in half of C1 alone is no lower bound on the minimum tree size.

A useful dual certificate modifies the residual to cap its unusually large null correlations. Fix S , set $a = X^\top r_S$, and choose a cap $q > 0$. Put

$$V = \{j \notin S : |a_j| > q\}, \quad H = X_V^\top M_S^{-1} X_V, \quad u = H^{-1}(a_V - q \operatorname{sign}(a_V)), \quad (110)$$

$$\Delta = M_S^{-1} X_V u, \quad \alpha = r_S - \Delta, \quad \Gamma = u^\top H u, \quad m = \min_{i \in S} |x_i^\top \alpha|, \quad M = \max_{j \notin S} |x_j^\top \alpha|. \quad (111)$$

The empty set V is allowed and gives $\Delta = 0, \Gamma = 0$.

Lemma 9.2 (Capped residual certificate). *If H is invertible, $M \leq m$, and $\Gamma < (m^2 - M^2)/\lambda$, then S is uniquely optimal and every single wrong fixing has bound at least*

$$F_\lambda(S) + (m^2 - M^2)/\lambda - \Gamma > F_\lambda(S). \quad (112)$$

Also $0 \leq F_\lambda(S) - R \leq \Gamma$.

Proof. Completing the square gives $h(\alpha, \mathbf{1}_S) = F_\lambda(S) - \Gamma$ and $\|\Delta\|^2 \leq \Gamma$; the cap satisfies $X_V^\top \alpha = q \operatorname{sign}(a_V)$. For any node, (105) gives the lower bound

$$2\alpha^\top y - \|\alpha\|^2 - \lambda^{-1} \left[\sum_{i \in S_1} c_i^2 + \operatorname{Top}_{k-|S_1|} \{c_i^2 : i \notin S_0 \cup S_1\} \right], \quad c = X^\top \alpha, \quad (113)$$

where Top_t sums the largest $\min(t, N)$ entries when N entries are available. Since all selected correlations are at least all null correlations, a wrong fixing lowers the bracket by at least $m^2 - M^2$ relative to $\sum_{i \in S} c_i^2$. This proves (112). The root bracket equals $\sum_{i \in S} c_i^2$, proving the root-gap bound. Every support other than S has a wrong fixing, so its value is larger. $\square$

9.3 Two sharp planted-support thresholds

For the easy regime, assume $\beta_i^* = \pm b$ on S^* , with arbitrary fixed signs, and fix $b, \sigma, C_0 > 0$. Define

$$b_\lambda = \frac{bn}{n + \lambda}, \quad \omega_\lambda^2 = n\sigma^2 + \frac{kb^2\lambda^2n}{(n + \lambda)^2}, \quad \tau_\lambda = \frac{\lambda b_\lambda}{\omega_\lambda}, \quad \Lambda_\lambda = \log(p\lambda/n). \quad (114)$$

The ratio τ_λ compares a selected correlation with the standard deviation of a null correlation. Indeed, conditional on (X_{S^*}, w) , the null correlations are independent $N(0, \|r_{S^*}\|^2)$ variables.

Assumption 9.3 (Easy-regime scaling). As $p \rightarrow \infty$,

$$\log^6 p \leq n \leq p, \quad 1 \leq k \leq C_0 n / \log p, \quad \sqrt{n} \leq \lambda \leq n / \log^2 p. \quad (115)$$

Theorem 9.4 (Root and single-variable certification). *Under assumption 9.3, fix $\delta \in (0, 1)$. The following statements hold with probability tending to one.*

- (a) *If $\tau_\lambda^2 \geq (2 + \delta) \log p$ eventually, then $R = \operatorname{OPT} = F_\lambda(S^*)$, S^* is uniquely optimal, and strict C1 holds. If $\tau_\lambda^2 \leq (2 - \delta) \log p$ eventually, then $R < F_\lambda(S^*)$.*
- (b) *If $\tau_\lambda^2 \geq (2 + \delta) \Lambda_\lambda$ eventually, then S^* is uniquely optimal and every single wrong fixing has bound at least $\operatorname{OPT} + \lambda b^2 / \log p$.*

(c) If $\tau_\lambda^2 \leq (2 - \delta)\Lambda_\lambda$ eventually and $k \geq 5000/\delta^2$, then all but at most $n/\log^2 p$ null features j satisfy $r(\emptyset, \{j\}) < F_\lambda(S^*) - \lambda b^2/2$.

The negative statements compare with $F_\lambda(S^*)$. They imply global root inexactness or C1 failure at S^* when S^* is optimal.

The proof in [appendix G.2](#) controls the true coefficients uniformly, then uses the capped residual for the positive C1 statement. The cap changes very few null correlations above the positive threshold. Below it, many null features jointly recover the price of consuming one unit of the cardinality budget. A single extreme null feature need not recover that price.

Corollary 9.5 (Sample sizes and a linear-tree window). *Within (115), the following bounds describe certification of S^* , with fixed positive slack in their constants. When the sample size is above $2k \log p$, root certification is achievable by a deterministic ridge; below that sample size it is impossible for each allowed deterministic ridge sequence. Single-variable certification is achievable when the sample size is above $2k \log(p/\sqrt{n})$; below this value the comparison in part (c) of [theorem 9.4](#) holds for each allowed deterministic ridge sequence, with a fixed sufficiently large k .*

More precisely, if $k = p^{\gamma+o(1)}$, $0 < \gamma < 1$, and $n = ak \log p$ with fixed $2 - \gamma < a < 2$, a suitable deterministic ridge gives a unique planted optimum, an inexact root, and at most $2p + 1$ nodes for every variable-branching rule under the incumbent or best-bound conventions above. For $a > 2$, a suitable ridge certifies the root. For $a < 2 - \gamma$, the single-variable planted comparison fails for each allowed deterministic ridge sequence; this does not assert a large tree.

The ridge choice and proof are given in [appendix G.3](#). The exact bound $\tau_\lambda^2 < n/k$ for $\sigma > 0$ explains the sample-size constants. In contrast, $\lambda = \sqrt{n}$ gives $\tau_\lambda \leq b/\sigma$, whereas both logarithmic thresholds diverge. At fixed per-entry signal-to-noise ratio, this ridge eventually fails both planted-support tests, subject to the hypotheses for their converses. This is an asymptotic statement. In particular, $\log^6 p \leq n \leq p$ excludes the moderate dimensions of the archived experiments discussed in [section 11](#); those observations do not estimate the constants in this theorem.

9.4 Local second-moment lifts

The same first-order thresholds survive specified stronger relaxations when the design has many more columns than rows. The result concerns unbounded, signed coefficients. Bounds on coefficients supply additional information and are outside this comparison. Second-moment convexification and rank-one strengthening have been studied by [Dong et al. \(2015\)](#) and [Atamtürk and Gómez \(2019\)](#); the comparison below specifies the local hulls and node conventions it covers.

Let $Q = X^\top X + \lambda I$ and define

$$\mathcal{F}(\beta, B) = \|y\|^2 - 2y^\top X\beta + \langle Q, B \rangle. \quad (116)$$

For $T \subseteq [p]$, let

$$\mathcal{H}_T = \text{cl conv}\{(\zeta, b, bb^\top) : \zeta \in \{0, 1\}^T, b_j = 0 \text{ whenever } \zeta_j = 0\}. \quad (117)$$

There is no cardinality constraint inside $\mathcal{H}_T$. For an integer $s \geq 1$, the projected root lift $\ell_s(z)$ minimizes (116) subject to

$$\begin{pmatrix} 1 & \beta^\top \\ \beta & B \end{pmatrix} \succeq 0, \quad (z_T, \beta_T, B_{TT}) \in \mathcal{H}_T \quad (|T| \leq s). \quad (118)$$

The node bound $L_s(S_0, S_1)$ minimizes $\ell_s(z)$ on the face in (106). In this definition, a zero fixing constrains z_j and β_j but leaves second moments available: $(0, 0, t) \in \mathcal{H}_{\{j\}}$ for every $t \geq 0$. This convention matters for the hard lower bound below.

The singleton lift L_1 is the relaxation with global positive semidefiniteness and $\beta_j^2 \leq z_j B_{jj}$. Every L_s dominates the perspective relaxation and is valid for the integer problem. Also L_s dominates selected block relaxations: if $Q = Q_0 + \sum_T Q_T$, where all matrices are positive semidefinite and each Q_T is supported on at most s coordinates, replacing $\beta_T^\top Q_T \beta_T$ by its exact closed indicator epigraph hull gives a bound no larger than L_s . The elementary hull proof is in [appendix G.4](#).

Theorem 9.6 (Thresholds for selected local lifts). *Under [assumption 9.3](#), let $s = s_p \geq 1$ satisfy $sn \log p = o(p)$. Consider any valid node relaxation $\mathcal{R}(S_0, S_1)$ satisfying*

$$r(S_0, S_1) \leq \mathcal{R}(S_0, S_1) \quad \text{at every node}, \quad \mathcal{R} \leq L_s \quad \text{at the root and all forced-in nodes } (\emptyset, \{j\}). \quad (119)$$

Then parts (a) and (b) of [theorem 9.4](#), including their positive margins, hold for $\mathcal{R}$. Its negative C1 statement holds with $k \geq 20000/\delta^2$ and at most $2n/\log^2 p$ exceptions. Consequently the planted-support sample-size thresholds and the linear-tree window of [corollary 9.5](#) also hold.

For the positive tree statement, monotonicity of $\mathcal{R}$ under fixings is unnecessary: an off-path node has $\mathcal{R}$ at least the monotone perspective bound of a single wrong fixing. Validity gives path-node bounds at most OPT. Column deletion at zero-fixed nodes is therefore compatible with this theorem if (119) holds: its upper comparisons occur only where no coordinate is fixed to zero.

The obstruction has a precise mechanism. A perspective point is the mean of a random feasible support. Its coefficient covariance has trace

$$\pi(z, \beta) = \sum_{z_j > 0} \beta_j^2 (1/z_j - 1). \quad (120)$$

An exact averaged fit would pay both the ridge variance $\lambda\pi$ and the design variance. The lift can cancel the latter using columns with $z_j = 0$ while paying an additional ridge cost $o(\lambda\pi)$. Local hulls do not detect the cancellation. The construction and both converse proofs are complete in [appendices G.4.1](#) and [G.5](#).

9.5 Large certificates in pure noise and at low total signal

For the hard regime, change the scaling. Let $k \rightarrow \infty$, $k/n \rightarrow 0$, $\lambda/n \rightarrow 0$, and

$$x_p = \frac{2}{n} \log \binom{p}{k} \rightarrow x \in (0, x_0), \quad e^{-x_0}(1 + 2x_0) = 1, \quad x_0 > 0. \quad (121)$$

The positive root is approximately 1.2564. The convex-piece certificates here branch on z : a node is a convex set in z -space, its children cover its feasible binary points, and its bound is the infimum of the fixed projected relaxation on that set intersected with K .

Theorem 9.7 (Conflict packing for perspective and pairwise hulls). *Under (121), consider either:*

- (a) *y independent of X , with $y \neq 0$ almost surely; or*
- (b) *the planted model (101), with $\|\beta^*\|^2/\sigma^2 \rightarrow \kappa$ and $\kappa < e^{-x}(1 + 2x) - 1$.*

There exist constants $c, \eta > 0$ such that, with probability tending to one, there is a family $\mathcal{C}$ of k -subsets with

$$|\mathcal{C}| \geq \exp(ck \log(p/k)) = \binom{p}{k}^{c+o(1)}. \quad (122)$$

For distinct $S, T \in \mathcal{C}$, both $g_\lambda((\mathbf{1}_S + \mathbf{1}_T)/2)$ and $\ell_2((\mathbf{1}_S + \mathbf{1}_T)/2)$ are strictly below $\text{OPT} - \eta Y$, where $Y = \|y\|^2$ in (a) and $Y = n\sigma^2$ in (b). Thus every convex-piece ε -certificate using either projected relaxation has at least $|\mathcal{C}|$ leaves whenever $0 \leq \varepsilon \leq \eta Y$.

For binary variable branching with certified incumbent reductions, the corresponding node count is at least $|\mathcal{C}|/(p+1)$ when each removed binary piece is charged to its node. The ℓ_2 assertion uses the retained-column node convention in (118); it does not cover deletion of zero-fixed helper columns.

Two binary points in one convex leaf bring their midpoint into the leaf. A midpoint below the required bound therefore prevents that leaf from containing both points. This is the conflict mechanism from section 8; it applies to variable, split, and other convex pieces under the stated fixed-oracle convention. The reduction count charges at most one removed piece per binary coordinate per node.

The proof in appendix G.6 selects highly correlated features, packs their k -subsets far apart, and bounds all pairwise union fits at once. The integer optimum receives a separate lower bound over all supports. For pairwise hulls, a uniform design event allows the completion of every union using its complement as helpers; this closes the additional uniformity step required by the stronger relaxation.

The hard theorem treats bounded *total* signal-to-noise ratio. With fixed b/σ and $k \rightarrow \infty$, the total ratio kb^2/σ^2 diverges, so it does not overlap the easy theorem's fixed per-coefficient scaling. For $k = p^{\gamma+o(1)}$ and $n = ak \log p$, (121) becomes $x = 2(1 - \gamma)/a$; the admissible pure-noise range is $a > 2(1 - \gamma)/x_0$. The proof gives a positive exponent, not a practical estimate of it or a general complexity lower bound for algorithms using other relaxations.

9.6 A richer moment matrix changes the variance price

The local-lift obstruction is not universal. To illustrate its scope, add moments Z and U of indicators and indicator-coefficient products. Consider the valid relaxation with objective (116) and

$$Y = \begin{pmatrix} 1 & z^\top & \beta^\top \\ z & Z & U \\ \beta & U^\top & B \end{pmatrix} \succeq 0, \quad \text{diag } Z = z, \quad \text{diag } U = \beta, \quad (123)$$

with $z \in K$, $Z\mathbf{1} \leq kz$, the usual binary-product inequalities $0 \leq Z_{jm} \leq \min(z_j, z_m)$, $Z_{jm} \geq z_j + z_m - 1$, and $U_{jm}^2 \leq Z_{jm}B_{mm}$ for $j \neq m$. The last inequalities follow from Cauchy-Schwarz for any genuine support/coefficient distribution.

Proposition 9.8 (Variance price in the richer lift). *At every feasible point of (123), put $A = \text{supp } z$ and $\nu_A = \lambda_{\min}(X_A^\top X_A)$, with $\nu_\emptyset = 0$. Then*

$$\mathcal{F}(\beta, B) \geq \|y - X\beta\|^2 + \lambda\|\beta\|^2 + (\lambda + \nu_A)\pi(z, \beta). \quad (124)$$

The Gram-matrix proof is given in appendix G.7. It uses both the full positive semidefinite moment matrix and the product constraints. Pairwise hulls with positive semidefiniteness only on $(1, \beta, B)$ do not supply this argument. The proposition does not give a sharp root threshold or a tree bound for this richer relaxation.

9.7 A fixed-dimensional noise-normalization check

Support recovery and relaxation exactness can differ even when the dimension is fixed and the sample size diverges. This gives a narrow check on the Gaussian exactness statement of Pilanci et al. (2015, Section 3.1, Theorem 2, p. 72).

Theorem 9.9 (Fixed dimension with per-entry noise). *Fix $1 \leq k < d$, $S \subseteq [d]$ with $|S| = k$, and $\beta^* \in \mathbb{R}^d$ with support S . Put $b_{\min} = \min_{i \in S} |\beta_i^*| > 0$ and fix $\sigma > 0$. For each n , use the Gaussian model (101) with $p = d$ and $\lambda = \sqrt{n}$. Then S is the unique optimal support with probability tending to one, but*

$$\lim_{n \rightarrow \infty} \mathbb{P}\{R = \text{OPT}\} = \left[1 - 2\bar{\Phi}(b_{\min}/\sigma)\right]^{d-k} < 1, \quad (125)$$

where $\bar{\Phi}(t) = \mathbb{P}\{N(0, 1) > t\}$.

The complete proof, including the global-optimum transfer, is in [appendix G.8](#). In the printed model, the noise variance per observation is σ^2 and the ridge is $\sqrt{n}$ for the unnormalized loss. The displayed sample-size requirement

$$n > c_0 \frac{\sigma^2 + \|\beta^*\|^2}{b_{\min}^2} \log d \quad (126)$$

holds eventually for every fixed $c_0 > 0$. Equation (125) therefore contradicts the claimed exactness probability $1 - 2e^{-c_1 n}$ for every fixed $c_1 > 0$ under those printed hypotheses. For example, $d = 2$, $k = 1$, $\beta^* = (1, 0)$, and $\sigma = 1$ give the limit $1 - 2\bar{\Phi}(1)$, approximately 0.683.

This conclusion concerns that Gaussian probability statement. The Boolean reformulation and its deterministic exactness condition remain the ones used above. A fixed finite coefficient-to-noise ratio, however large, cannot make the limit in (125) equal to one. Shrinking the per-entry noise defines a different statistical model and requires its own theorem.

10 Gaussian binary least squares

Binary least squares separates finding an optimal vertex from certifying its value with a particular relaxation. In the Gaussian model considered here, the planted vertex is the unique integer optimum above a logarithmic signal level. The box relaxation can nevertheless require a superpolynomial certificate. At a linear signal level, a sufficient condition for a linear variable-branching tree has a sharp threshold. A different, logarithmic threshold determines when the root box relaxation agrees with its nonnegative least-squares relaxation. These statements concern different events and different certificate classes.

10.1 Model and certificate conventions

Let $M = M_N \geq N$, let $H \in \mathbb{R}^{M \times N}$ have independent standard Gaussian entries, and let $w \sim N(0, I_M)$ be independent of H . For a deterministic $x^* \in \{-1, 1\}^N$ and $\rho > 0$, set

$$A = \sqrt{\rho/N} H, \quad y = Ax^* + w, \quad f(x) = \|y - Ax\|^2, \quad \text{OPT} = \min_{x \in \{-1, 1\}^N} f(x), \quad W = \|w\|^2. \quad (127)$$

Thus $W = f(x^*)$ and ρ is normalized per observation. Write $\beta_N = M/N$; asymptotic statements below assume $\beta_N \rightarrow \beta \in [1, \infty)$. All probabilities are uniform in the deterministic planted signs. Dividing the residual by a positive noise standard deviation σ replaces ρ by ρ/σ^2 .

The node oracle minimizes f over the node's convex set inside $[-1, 1]^N$. In particular, a variable-branching node minimizes f over the box obtained by fixing some coordinates to binary values. No identity $x_i^2 = 1$ is imposed on free coordinates. Diagonal quadratic reformulations, product lifts, and semidefinite relaxations change this oracle and are outside the certificate lower bound proved here.

For a target τ , define the class number

$$\kappa_\tau = \min \left\{ q : \{-1, 1\}^N = \bigcup_{j=1}^q I_j, \quad \min_{x \in \text{conv } I_j} f(x) \geq \tau \text{ for every } j \right\}. \quad (128)$$

Empty classes can be discarded. This is the minimum number of leaves of a certificate allowing arbitrary convex pieces that cover the binary vertices: a leaf containing I_j also contains its convex hull, and the convex hulls themselves give a certificate. Every variable or general split tree using this oracle has at least κ_τ leaves. Adding convex cuts valid for all the binary vertices assigned to a leaf preserves this argument, because those cuts contain their convex hull. Incumbent-based reductions also need certificates for the removed vertices; their cost is handled by the augmented accounting of the integer framework, rather than by treating the retained leaves alone as a cover.

Absorb the planted signs into the columns by writing $b_i = x_i^* a_i$, $B = A \text{diag}(x^*)$, and $u_i = 1 - x_i^* x_i$. Then

$$f(x) = F(u) := \|w + Bu\|^2, \quad u \in [0, 2]^N. \quad (129)$$

The planted vertex is $u = 0$, and flipping the coordinates in S gives $u = 2\mathbf{1}_S$. If a node fixes the wrong values on T and leaves the coordinates J free, its box value is

$$r(J, T) = \min_{u \in [0, 2]^J} \|v_T + B_J u\|^2, \quad v_T = w + 2B_T \mathbf{1}. \quad (130)$$

The residual v_T is independent of the free matrix B_J . Shared noise creates dependence between different nodes, but does not change this one-node independence.

10.2 Recovery and root exactness are different events

We first record the recovery range used when interpreting a bound relative to the planted value as a bound relative to OPT.

Proposition 10.1 (Planted recovery and a value comparison). *For every fixed $\delta > 0$, if $\beta_N \rho_N \geq (2 + \delta) \log N$, then x^* is the unique integer minimizer with probability tending to one. For every deterministic positive sequence ρ_N , with probability tending to one,*

$$0 \leq W - \text{OPT} \leq 4N \log(1 + e^{-0.14\beta_N \rho_N}) + 4 \log N. \quad (131)$$

The proof is a first-moment calculation over flipped sets, given in [Appendix H.2](#). At $\rho_N = \theta N$ with fixed $\theta > 0$, it in particular proves unique planted optimality without a box-decoder theorem. For square systems, the recovery comparison at a logarithmic signal level is consistent with the sufficient condition in [Hassibi et al. \(2014\)](#); we use the fixed margin $(2 + \delta) \log N$. Recovery is not asserted for every sequence of order $\log N$.

Even when x^* is the integer optimum, it usually does not minimize the root box relaxation. Let $R = \min_{x \in [-1, 1]^N} f(x)$.

Proposition 10.2 (Two root exactness events). *For every $M \geq N$ and $\rho > 0$,*

$$\mathbb{P}\{x^* \text{ minimizes } f \text{ on } [-1, 1]^N\} = 2^{-N}. \quad (132)$$

The root minimizer is unique almost surely, and $R = \text{OPT}$ if and only if that minimizer is a vertex. Moreover,

$$\begin{aligned} \mathbb{P}\{R = \text{OPT}\} &\leq \sum_{k=0}^N \binom{N}{k} 2^{k-N} (1 + 4\rho k/N)^{-M/2} \\ &\leq \left[\frac{1 + 2(1 + 4\rho)^{-\beta_N/2}}{2} \right]^N. \end{aligned} \quad (133)$$

The last bound decreases exponentially when $(1 + 4\rho)^{\beta_N/2} > 2$ with a fixed margin. The exact planted probability and the global vertex probability are distinct: at low signal levels another vertex can be the integer optimum. Rounding the box minimizer is a further event. In particular, a successful rounded decoder does not imply $R = \text{OPT}$. The proof in [Appendix H.3](#) uses only the box optimality conditions and a Gaussian exponential moment.

10.3 Finite Gaussian nonnegative least squares

The next results supply the one-node laws and the tail estimates needed for simultaneous conclusions. They also isolate the role of selecting a support from the data. The binomial face-dimension law and the chi-square projection mixture are classical Gaussian cone facts; see [Hug and Schneider \(2016\)](#) and the master Steiner formula of [McCoy and Tropp \(2014\)](#). We give a direct proof because its invariant-event identity also transfers coefficient laws. The Gaussian regression Student density itself is classical. The finite coefficient bound below is the ingredient used for the branch-and-bound applications.

Lemma 10.3 (An invariant-event identity). *Let $C \in \mathbb{R}^{M \times n}$, $n \leq M$, have full column rank almost surely, and let v be independent of C . Assume the joint column law of C is invariant under every independent column sign change. For each nonempty $S \subseteq [n]$, put*

$$d^S = -(C_S^T C_S)^{-1} C_S^T v, \quad L_S = \text{span}(C_S), \quad z^S = (I - P_{L_S})v.$$

For the empty support set $L_\emptyset = \{0\}$ and $z^\emptyset = v$. Assume that almost surely every coordinate of every d^S is nonzero, and $c_j^T z^S \neq 0$ for every $S \subseteq [n]$ and $j \notin S$. Let u° minimize $\|v + Cu\|^2$ over $u \geq 0$, and let $\mathcal{S} = \{j : u_j^\circ > 0\}$. If E_S is any measurable event invariant under every column sign change, then

$$\mathbb{P}\{\mathcal{S} = S, E_S\} = 2^{-n} \mathbb{P}(E_S). \quad (134)$$

In particular, $\mathcal{S}$ is uniform over all subsets of $[n]$.

Proof. For fixed C, v satisfying the hypotheses, the support is S precisely when $d_j^S > 0$ for $j \in S$ and $c_j^T z^S > 0$ for $j \notin S$. These are the necessary and sufficient first-order conditions for the strictly convex problem. A column sign change preserves L_S, z^S , changes the sign of the corresponding coordinate of d^S on S , and changes the sign of the corresponding residual inner product off S . Exactly one of the 2^n sign patterns satisfies all these conditions. Multiplying this pointwise sign count by the invariant indicator of E_S and taking expectations proves (134). Conditions indexed by an empty set are vacuous, so the argument includes the empty and full supports. $\square$

The explicit nonzero conditions matter: requiring only that v avoid column spans would not suffice. For independent Gaussian columns and independent isotropic Gaussian v of positive variance, the stated conditions hold automatically. Conditional on a full-rank matrix, each coefficient and each off-support residual inner product is a nonzero linear functional of v . There are finitely many such functionals.

Theorem 10.4 (Gaussian NNLS coefficient and projection laws). *Let $C = aH$, where $a > 0$ and $H \in \mathbb{R}^{M \times n}$ has independent standard Gaussian entries, $n \leq M$. Let $v \sim N(0, c^2 I_M)$ be independent of H , with $c > 0$, and define $u^\circ, \mathcal{S}$ as in [Lemma 10.3](#). Then $K = |\mathcal{S}| \sim \text{Bin}(n, 1/2)$. For a specified support S of size $s \geq 1$, conditional on $\mathcal{S} = S$, the positive coefficient vector has the law*

$$(u_j^\circ)_{j \in S} \stackrel{d}{=} \frac{c}{a} \frac{|Z_s|}{\sqrt{Q}}, \quad Z_s \sim N(0, I_s), \quad Q \sim \chi_{M-s+1}^2, \quad (135)$$

where Z_s, Q are independent and the absolute value is coordinatewise. Equivalently, its unconditioned signed regression vector d^S has density

$$p_s(d) = \left(\frac{a}{c}\right)^s \frac{\Gamma((M+1)/2)}{\pi^{s/2}\Gamma((M-s+1)/2)} \left(1 + \frac{a^2}{c^2}\|d\|^2\right)^{-(M+1)/2}. \quad (136)$$

Writing $\bar{\Phi}$ for the standard Gaussian upper-tail function, for $t \geq 0$ the maximum has the exact distribution

$$\mathbb{P}\{\max_j u_j^\circ \leq t\} = 2^{-n} \sum_{s=0}^n \binom{n}{s} \mathbb{E}_{Q \sim \chi_{M-s+1}^2} \left[1 - 2\bar{\Phi}\left(\frac{at\sqrt{Q}}{c}\right) \right]^s, \quad (137)$$

where the $s=0$ summand is one. In particular, the atom at zero is 2^{-n} .

If $X = \|Cu^\circ\|^2$ and $Z = \|v + Cu^\circ\|^2$, then conditional on $K = s$, X/c^2 and Z/c^2 are independent chi-square variables with s and $M-s$ degrees of freedom. Thus $Y = X/(X+Z)$ has the corresponding beta mixture, with point masses at the endpoints when a degree is zero. For $0 \leq t \leq 1$,

$$\mathbb{P}\{|Y - n/(2M)| > 2t\} \leq 6e^{-Mt^2/64}. \quad (138)$$

The auxiliary chi-square denominator in (135) is common to all coordinates on that support. It is not a set of independent one-coordinate denominators, nor a selected singular value. The representations concern one NNLS problem and assert no independence between different branch-and-bound nodes. The complete density and projection proofs are in [Appendix H.4](#).

Theorem 10.5 (A finite maximum-coefficient tail). *Under the hypotheses of [Theorem 10.4](#), for every $t > 0$ set $q = (1 + a^2t^2/c^2)^{-1/2}$. Then*

$$\mathbb{P}\{\max_j u_j^\circ > t\} \leq \frac{n}{2} q^{M-n+1} \left(\frac{1+q}{2}\right)^{n-1}. \quad (139)$$

Proof. On support S , the positive coordinates are d^S . The event $\max_{j \in S} |d_j^S| > t$ is invariant under column sign changes, so [Lemma 10.3](#) and a coordinate union bound give

$$\mathbb{P}\{\max_j u_j^\circ > t\} \leq 2^{-n} \sum_{S \neq \emptyset} \sum_{j \in S} \mathbb{P}\{|d_j^S| > t\}.$$

For a fixed support of size s and $j \in S$, let P project orthogonally to the other $s-1$ columns of H_S . The regression identity gives $d_j^S = -h_j^T P v / (aQ_j)$, where $Q_j = \|Ph_j\|^2 \sim \chi_{M-s+1}^2$. Conditional on H_S , this coefficient is $N(0, c^2/(a^2Q_j))$. The elementary bound $\mathbb{P}\{|G| > z\} \leq e^{-z^2/2}$ for a standard normal G and the chi-square Laplace transform therefore give

$$\mathbb{P}\{|d_j^S| > t\} \leq \mathbb{E} e^{-a^2t^2Q_j/(2c^2)} = q^{M-s+1}.$$

Consequently the required sum is at most

$$2^{-n} \sum_{s=1}^n \binom{n}{s} s q^{M-s+1} = \frac{n}{2} q^{M-n+1} \left(\frac{1+q}{2}\right)^{n-1}.$$

This argument includes $s = M$, where the regression denominator has one degree of freedom. $\square$

10.4 A sharp sufficient criterion for a linear variable tree

Let r_i be the box bound at the node fixing $x_i = -x_i^*$ and leaving all other coordinates free. Criterion C1 at x^* requires all these bounds to reach the target. Its usefulness is monotonicity: every node off the path to x^* is contained in one such single-wrong-fixing node. The path lemma, [Lemma 8.12](#), then gives at most $2N + 1$ nodes. For weak bounds $r_i \geq \text{OPT}$, the optimal incumbent must be available when off-path nodes are checked. For strict bounds $r_i > \text{OPT}$, best-bound search gives the same conclusion without an initial optimal incumbent. The remaining singleton has exact bound $f(x^*)$, as required by the path lemma.

Theorem 10.6 (Simultaneous C1 threshold for square and tall systems). *Assume $\rho_N = \theta N$ with fixed $\theta > 0$ and $\beta_N \rightarrow \beta \geq 1$. With probability tending to one, the orthant minimizer at every single-wrong-fixing node belongs to $[0, 2]^{N-1}$. Moreover,*

$$\max_{1 \leq i \leq N} \left| \frac{r_i - W}{N} - \left[2\theta(2\beta - 1) - \frac{1}{2} \right] \right| \xrightarrow{\mathbb{P}} 0. \quad (140)$$

Put $\theta_c = [4(2\beta - 1)]^{-1}$. For each fixed $\epsilon \in (0, 1)$, the planted vertex is uniquely optimal with probability tending to one, and the following conclusions hold:

- (a) If $\theta \geq (1 + \epsilon)\theta_c$, then all $r_i \geq \text{OPT} + \epsilon N/4$ with probability tending to one. Every variable-branching search satisfying the incumbent or best-bound qualification above has at most $2N + 1$ nodes.
- (b) If $\theta \leq (1 - \epsilon)\theta_c$, then all $r_i \leq \text{OPT} - \epsilon N/4$ with probability tending to one. Thus C1 fails at every single-wrong-fixing node.

Proof. Here the signed columns are independent $N(0, \theta I_M)$. At node i , $v_i = w + 2b_i \sim N(0, (1 + 4\theta)I_M)$ is independent of B_{-i} . Apply [Theorem 10.5](#) with $n = N - 1$, $a^2 = \theta$, $c^2 = 1 + 4\theta$, and $t = 2$. Writing

$$q_\theta = \sqrt{\frac{1 + 4\theta}{1 + 8\theta}}, \quad b_\theta = \frac{1 + q_\theta}{2} < 1,$$

a union bound over the N nodes bounds the probability of any upper-box violation by

$$\frac{N(N - 1)}{2} q_\theta^{M - N + 2} b_\theta^{N - 2}. \quad (141)$$

This tends to zero exponentially, including when $M = N$. No independence of node events is used.

Define the orthant value first: $\tilde{r}_i = \min_{u \geq 0} \|v_i + B_{-i}u\|^2$. By [Theorem 10.4](#), it has the marginal law

$$\tilde{r}_i / (1 + 4\theta) \mid K_i = s \sim \chi_{M-s}^2, \quad K_i \sim \text{Bin}(N - 1, 1/2).$$

Binomial and chi-square exponential bounds, followed by a union bound over i , give the following estimate. Specifically, take $s = \sqrt{2N \log N}$ in the binomial bounds and $x = 4 \log N$ in the conditional chi-square bounds; their total failure probability is $O(N^{-3})$.

$$\max_i \left| \tilde{r}_i - (1 + 4\theta) \left(M - \frac{N - 1}{2} \right) \right| = O_{\mathbb{P}}(\sqrt{N \log N}).$$

Also $W = M + O_{\mathbb{P}}(\sqrt{N \log N})$. On the event complementary to (141), strict convexity gives $r_i = \tilde{r}_i$ for every i . Hence, uniformly over i ,

$$r_i - W = N \left[2\theta(2\beta_N - 1) - \frac{1}{2} \right] + \frac{1 + 4\theta}{2} + O_{\mathbb{P}}(\sqrt{N \log N}), \quad (142)$$

which proves (140). The elementary Gaussian concentration estimates used here are proved in Appendix H.1.

Proposition 10.1 gives $\text{OPT} = W$ and unique planted optimality with probability tending to one. The limiting gap equals $\frac{1}{2}(\theta/\theta_c - 1)$, so either fixed margin leaves the claimed $\epsilon N/4$ gap for all sufficiently large N with probability tending to one. In the upper case, off-path bounds are strictly above OPT. Before the optimum is found, best-bound search always has a node on its path with bound at most OPT; it cannot expand an off-path node first. Once the optimum is found, those nodes are pruned. The incumbent version follows directly from Lemma 8.12. $\square$

This is a threshold for a sufficient criterion for a short tree. Failure of C1 does not imply a superlinear tree lower bound. The theorem gives no conclusion at $\theta = \theta_c$ or in a shrinking critical window. The simultaneous assertion follows from a finite coefficient tail; a per-decoder limiting theorem such as Hu and Lu (2020) would not by itself supply a rate that can be union-bounded over the N wrong-fixing nodes.

10.5 The logarithmic threshold for root box inactivity

Let $u_N^\circ(\rho)$ be the unique minimizer of

$$\left\| w_N + \sqrt{\rho/N} H_N u \right\|^2 \quad \text{over } u \geq 0,$$

where the planted signs have again been absorbed into H_N . Let $R_N^\circ(\rho)$ denote this orthant value, and $R_N(\rho)$ the value with $u \in [0, 2]^N$. We call equality of these values root box inactivity; at the equality endpoint some coordinate can equal its upper bound.

Theorem 10.7 (An exact random cutoff and its first-order scale). *For every realization of full-rank H_N and every $\rho > 0$, define*

$$\rho_{\text{box},N} = \frac{1}{4} \left(\max_j u_{N,j}^\circ(1) \right)^2. \quad (143)$$

Then, pathwise,

$$R_N(\rho) = R_N^\circ(\rho) \iff \rho \geq \rho_{\text{box},N}. \quad (144)$$

If $\beta_N \rightarrow \beta \geq 1$, then

$$\frac{\rho_{\text{box},N}}{\log N / (2\beta_N - 1)} \xrightarrow{\mathbb{P}} 1. \quad (145)$$

Consequently, for every fixed $\epsilon \in (0, 1)$ and every deterministic positive sequence ρ_N , root box inactivity holds with probability tending to one if eventually $\rho_N \geq (1 + \epsilon) \log N / (2\beta_N - 1)$. If eventually $\rho_N \leq (1 - \epsilon) \log N / (2\beta_N - 1)$, then $R_N(\rho_N) > R_N^\circ(\rho_N)$ with probability tending to one.

Proof. The substitution $z = \sqrt{\rho} u$ and uniqueness give $u_N^\circ(\rho) = \rho^{-1/2} u_N^\circ(1)$. The orthant minimizer lies in the box exactly when its maximum is at most 2. Since the box is a subset of the orthant and the minimizer is unique, this proves (144), including equality.

At $\rho = 1$, Theorem 10.4 gives the exact synthetic representation

$$\left(\max_j u_{N,j}^\circ(1) \right)^2 \stackrel{d}{=} \frac{N}{Q} \max_{j \leq K} |Z_j|^2, \quad K \sim \text{Bin}(N, 1/2), \quad Q \mid K \sim \chi_{M-K+1}^2,$$

with Q and the standard normal coordinates independent conditional on K . The value is zero at $K = 0$. Binomial and chi-square concentration give $K/N \rightarrow 1/2$ and $Q/N - (\beta_N - 1/2) \rightarrow 0$ in probability. Elementary normal tail bounds give $\max_{j \leq K} |Z_j|^2 / (2 \log N) \rightarrow 1$ in probability.

Appendix H.5 proves these limits explicitly, including the lower bound for the random number of coordinates. Thus

$$\frac{\rho_{\text{box},N}}{\log N/(2\beta_N - 1)} = \frac{\beta_N - 1/2}{Q/N} \frac{\max_{j \leq K} |Z_j|^2}{2 \log N} \xrightarrow{\mathbb{P}} 1.$$

The two fixed-margin conclusions follow from the pathwise cutoff. $\square$

A useful finite sufficient bound follows directly from Theorem 10.5:

$$\mathbb{P}\{\max_j u_{N,j}^\circ(\rho) > 2\} \leq \frac{N}{2} q^{M-N+1} \left(\frac{1+q}{2} \right)^{N-1}, \quad q = (1 + 4\rho/N)^{-1/2}. \quad (146)$$

For $\rho = O(\log N)$, the logarithm of this bound is $\log(N/2) - (2\beta_N - 1)\rho + o(1)$. The exact ratio β_N is needed for this $o(1)$ remainder; substituting its limit alone gives an $o(\log N)$ remainder under mere convergence. The finite bound and the exact mixture both apply to square systems. They describe equality with the orthant value. They do not establish planted recovery, successful rounding, a vertex root minimizer, or root equality with the integer optimum. The isolated value $\rho = 0$ is excluded: there the objective is constant in u and the strict-convexity cutoff argument does not apply.

10.6 Certificate complexity below a linear signal level

The lower bound is not obtained from C1 failure. It uses large families of flipped vertices and the value of their barycenter. On coordinates where the noise gives a descent direction, spreading a fixed number of flips among many coordinates retains the linear improvement while reducing the quadratic penalty. A certifiable class must therefore have substantial overlap among its flipped supports. An entropy estimate bounds the number of vertices that such a class can contain.

Theorem 10.8 (Matching safe certificate regimes). *Assume $\beta_N \rightarrow \beta \geq 1$, and let $0 \leq \varepsilon_N \leq \rho_N$. There are constants $c_\beta > 0$ and $\rho_0(\beta) > 0$ with the following properties.*

(a) *If $\rho_N \rightarrow \infty$ and $\rho_N = o(N)$, then with probability tending to one,*

$$\log \kappa_{\text{OPT} - \varepsilon_N} = \Theta_\beta \left(\frac{N}{\rho_N} \log \rho_N \right). \quad (147)$$

The minimum number of leaves of a variable-branching certificate has the same logarithmic order. A static variable order, fixed independently of the data, attains the upper order with an optimal incumbent.

(b) *For each fixed $\rho \geq \rho_0(\beta)$, with probability tending to one, both logarithms are $\Theta_{\beta,\rho}(N)$.*

In part (a), a static-order search given any incumbent value between OPT and W has the upper node bound for certifying that incumbent, and an optimal incumbent makes it a certificate for the stated target.

Proof outline. For the upper bound, take $K = \lceil (1 + o(1))N/[4(2\beta_N - 1)\rho_N] \rceil$, with the $o(1)$ chosen positive and large enough to dominate the concentration errors. Along a fixed variable order, simultaneously every node with K wrong fixings has bound at least W , with probability tending to one. The number of generated nodes is then at most

$$1 + 2 \sum_{i=1}^K \binom{N}{i} \leq 1 + 2(eN/K)^K. \quad (148)$$

In the present sublinear regime $K \rightarrow \infty$, so its logarithm is at most

$$(1 + o(1)) \frac{N}{4(2\beta - 1)\rho_N} \log \rho_N. \quad (149)$$

An optimal incumbent can only prune earlier. The full simultaneous proof is in [Appendix H.6](#); it does not require planted recovery.

For the lower bound, [Appendix H.8](#) proves

$$\log \kappa_{\text{OPT} - \varepsilon_N} \geq c_\beta \frac{N}{\rho_N} \log \rho_N - \log(8N) \quad (150)$$

in the divergent sublinear regime and for sufficiently large fixed ρ . It uses (131) when x^* has not been proved optimal. If $\rho_N \rightarrow \infty$ and $\rho_N = o(N)$, then $\log N = o((N/\rho_N) \log \rho_N)$: for $\rho_N \leq \sqrt{N}$ this is immediate, and for $\rho_N > \sqrt{N}$ the ratio is at most $2\rho_N/N$. Thus (150) gives the lower order in part (a). The lower bound applies to all convex-piece certificates, so also to variable trees. At fixed sufficiently large ρ , it gives $\Omega(N)$, while the full binary tree gives the upper $O(N)$ logarithm. $\square$

For example, $\rho_N = c \log N$ with any fixed $c > 0$ gives certificates of size

$$\exp(\Theta(N \log \log N / \log N));$$

planted recovery is justified by [Proposition 10.1](#) only when its constant condition is met. At $\rho_N = N^a$, $0 < a < 1$, the size is $\exp(\Theta(N^{1-a} \log N))$. A logarithmic root inactivity threshold therefore coexists with a superpolynomial box certificate.

The constants in the lower proof are conservative. In particular, the subtractive $\log(8N)$ in (150) prevents extending the matching statement to an arbitrary interval $\rho \leq c'N$ without checking constants. At a fixed linear signal level the C1 theorem is the positive result proved here; below its threshold, other short trees are not excluded. All these costs count certificates or nodes for the specified oracle, and exclude the time needed to solve node relaxations.

11 Computational evidence

The recorded computations examine three consequences of the theory: the dependence of spatial search on the tolerance and local geometry, the effect of branching points within a fixed solver, and the separation between root exactness and small variable-branching certificates in random designs. They provide finite-sample comparisons of specified implementations. The proofs in the preceding sections establish the asymptotic statements; the computations do not establish them by extrapolation.

11.1 Protocols and interpretation

The spatial studies use SCIP 10.0.2 through PySCIPOpt 6.2.1, with SoPlex 8.0.2 and single-threaded solves. The recorded environment is Python 3.13.11 on Linux WSL2; the branching benchmarks identify an Intel Xeon w5-2565X and Ipopt 3.14.19. The synthetic tolerance study does not record the Ipopt version. Several solves ran concurrently, so timing reflects the shared workload. We report node counts rather than time speedups. SCIP processed nodes, final certificate leaves, and nodes of the separate Python algorithms are different measurements. [Table 2](#) gives the principal SCIP cohorts and termination counts.

An **optimal** status denotes exhaustion of the solver's search under its numerical tests. A **gaplimit** status denotes its global gap stopping test. Neither status is an exact-arithmetic certificate

Table 2: Principal SCIP cohorts and termination counts. Each count is a run, not a distinct instance. **optimal** and **gaplimit** are solver statuses; limit counts are unfinished observations. The 120-second benchmark block contains only the core settings.

Cohort	Protocol	Records and statuses
Synthetic tolerance study	20 instances; seed 0; up to 13 tolerances; 120 s; 100,000 nodes	2,134 total 1,655 gap limit 420 optimal; 59 node limit
Branching benchmark, first campaign	57 instances; 5 settings; 3 seeds; 120 CPU s	855 total 797 optimal 57 time limit; 1 crash
Branching benchmark, second campaign	57 instances; 12 settings; 3 seeds; 60 CPU s	2,052 total 1,943 optimal 104 time limit; 5 crashes
Synthetic kink study	16 instances; 12 settings; 5 seeds; 4 tolerances; 60 s	3,840 total 2,024 gap limit; 1,816 optimal

of the mathematical optimum. In the tolerance study, **limits/absgap** is set to ϵ and the relative-gap limit to zero; ϵ is not a slack added to every node-pruning test. Feasibility tolerance is 10^{-9} , whereas the requested gaps range from 10^{-1} to 10^{-7} in half-decade steps. Recorded incumbents can lie as much as 5.5×10^{-9} below the known mathematical optimum because of accepted constraint violations. Some LP tolerance requests were also refused by SoPlex. These distinctions matter when comparing the solver with a theorem that assumes valid lower bounds and an exact incumbent.

The accompanying computational archive preserves the original generation and analysis code, run records, metadata, selected-instance lists, and numerical summaries. An inventory records source paths, byte counts, and SHA256 hashes; figure metadata identify every plotted observation and its original record. The package checks accounting and aggregation without invoking a solver. Such checks establish the integrity of the retained evidence, rather than the mathematical validity of floating-point bounds. External MINLP Lib model files and solver binaries are not included.

11.2 Tolerance, geometry, and solver representation

The tolerance study contains 20 synthetic instances with optima known by construction. It uses epigraph formulations $\min t$ subject to $f(x) \leq t$ and the instance constraints, with fixed expression forms. Each run has a 100,000-node limit and a 120-second limit; a tolerance sequence stops after its first limit. The setting named **model** supplies the known optimum as incumbent, disables heuristics, and uses best-first node selection without plunging. Ordinary propagation remains enabled. These three changes were made together, so their individual effects are not identified by a comparison with the default setting.

The main unconstrained families are built from

$$h_c(s) = (s - 1)^2((s + 1)^2 + c), \quad q_d(s) = (s + d)^4 + (s - d)^4 - 12d^2s^2 - 2d^4 = 2s^4.$$

The isolated **ison** objectives sum $h_{c_i}(x_i)$, with $(c_1, c_2, c_3, c_4) = (0.5, 0.7, 0.6, 0.8)$; **iso2c** adds $0.5(x_1 - x_2)^4$. The objectives for **linediag2** and **plane3** are $h_{0.5}(x_1 - x_2)$ and $h_{0.5}(x_1 + x_2 - x_3)$.

The axis-aligned `lineaxis2` adds the unsimplified zero $q_{0.375}(x_2) - 2x_2^4$ to $h_{0.5}(x_1)$. The `ring2` and `sphere3` objectives are $(\sum_i x_i^2 - 1)^2$ in two and three dimensions. The quartic objectives `qflat1`, `qflat2a`, `qflat2b`, and `qflat3` respectively use $q_{0.375}(x_1)$, $h_{0.5}(x_1) + q_{0.375}(x_2)$, $q_{0.375}(x_1) + q_{0.5}(x_2)$, and $h_{0.5}(x_1) + q_{0.375}(x_2) + q_{0.5}(x_3)$. These expressions are passed in the stated expanded or unsimplified forms, not replaced by their algebraically simpler identities. The exponential examples minimize $\log x_2 - x_1$ subject to $x_2 \geq e^{x_1}$, or the sum of two such independent copies. The two McCormick examples minimize $2|d| - d(x_2 - b)$ on $[0, 1]^2$, with $b = \sqrt{2} - 1$ and $d = x_1 - 1/3$ (aligned) or $d = x_1 - x_2 - 1/3$ (transversal). The archive supplies all coordinate bounds, which are asymmetric in the non-McCormick examples, and the generated solver formulations.

Figure 1(a) shows the predicted 1/2 and 1 exponent groups. For the line and sphere families with quadratic transverse growth, Section 3 predicts power $p/2$ for a p -dimensional optimal set. The constrained and McCormick examples additionally depend on their specific gap structure. Under `model`, the families with one-dimensional optimal sets have fitted slopes 0.50–0.52, compared with the predicted exponent 1/2; the families with two-dimensional optimal sets have slopes 1.01–1.05, compared with 1. The latter fits span only one or two decades; their available sequences cover two or three decades before the node budget limits further observations. For example, `sphere3` has 79, 747, 7,475, and 76,699 processed nodes at tolerances 10^{-1} through 10^{-4} . These observations support the geometric distinction in Section 3, within the tested formulations and search settings.

Degenerate isolated minima also show why dimension alone is insufficient. For local growth $\sum_i |t_i|^{q_i}$, the predicted power is $n/2 - \sum_i 1/q_i$. The quartic examples `qflat2b` and `qflat3` give fitted powers 0.49–0.51, compared with 1/2. For `qflat2a`, the predicted power is 1/4 and the `model` tail fit is 0.27. The one-dimensional quartic `qflat1` has tail and wide fits 0.31 and 0.33 under `model`, while controlled bisection gives 0.25 and 0.24. The records do not determine whether the discrepancy under LP-biased branching persists or reflects the finite tolerance window.

At isolated nondegenerate minima, `model` counts grow approximately additively with $\log(1/\epsilon)$: the archived fits give 9, 10, 40, and 92 nodes per decade for `iso2`, `iso2c`, `iso3`, and `iso4`. The default counts for `iso2` and `iso3` instead plateau at 95 and 341 nodes. Those runs exhaust their trees near the numerical feasibility scale. Their plateaus therefore do not establish tolerance-independent mathematical certificate sizes.

The archived slopes are least-squares fits of $\log_{10}$ nodes against $\log_{10}(1/\epsilon)$. A tail fit uses the last five available eligible records with $\epsilon \leq 10^{-3}$; a wide fit uses all eligible records with $\epsilon \leq 10^{-2}$. Both require at least three points spanning 0.99 decades. Node-limit, time-limit, and error records are excluded, whereas exhausted trees remain eligible. A short fit or an exhausted-tree plateau can therefore differ substantially from an asymptotic exponent. Limits also truncate the harder sequences, so the slopes describe the observed range rather than an uncensored distribution of solver work.

For seven instances satisfying the archived integral-bound hypotheses, recorded final leaves, including leaves still open at gap termination, were compared with the anisotropic integral lower bound. No recorded ratio is below one; the smallest is 1.59 on `qflat1`. Under `model`, the leaf-to-bound ratio ranges from 6.84 to 9.78 on `iso2`, 15.88 to 23.65 on `linediag2`, and 6.76 to 9.41 on `qflat2a`. The last range includes every eligible half-decade observation. This is a consistency check on recorded leaves and numerically evaluated bounds, not a certified inequality for the floating-point run.

Expression representation can change the effective relaxation. On `ring2`, keeping an outer square unexpanded gives one processed node at every tested tolerance, whereas `model` with the expanded expression uses 37,439 nodes at 10^{-6} . Shared auxiliary expressions likewise produce one-node runs on `condisk2`. Objective-cutoff propagation gives 15 nodes over 10^{-2} through 10^{-7} on `isofbtt2`, whose fourth-order secant gap lies outside the quadratic gap hypothesis. These examples concern

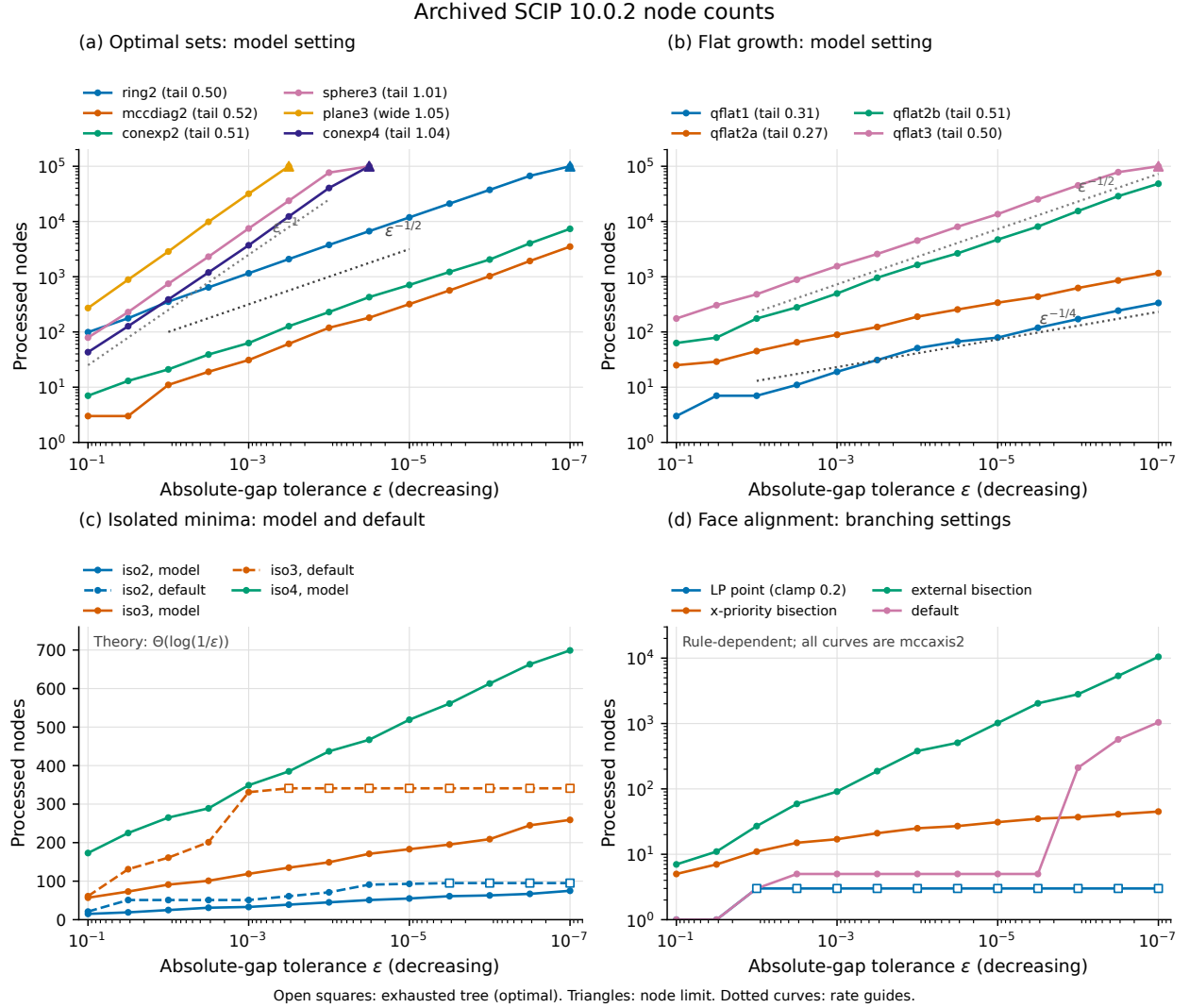


Figure 1: Recorded SCIP 10.0.2 processed-node counts: 230 observations in 19 separate instance/setting curves, covering 14 of the tolerance study’s 20 instances. Panels (a) and (b) use **model**: optimal incumbent, heuristics disabled, best-first selection, ordinary propagation enabled. Dotted guides are arbitrarily normalized theorem powers, not absolute predictions. Legend slopes are the archived finite-window fits described in the text. Panel (c) has a linear node axis to show additive growth and default-solver saturation; default **iso4** is omitted for legibility (3,431 nodes at 10^{-7}). Panel (d) compares branching settings on an aligned McCormick instance. External bisection and its x_1 -priority variant use best-first search with propagation, heuristics, and presolve disabled; LP-point and default settings use ordinary search. Tolerance axes are logarithmic and decrease to the right; other node axes are logarithmic. Open squares mark exhausted trees; triangles mark counts at the node limit. No observations are pooled or averaged. The requested absolute gap is a global stopping test, and these floating-point counts are not rigorous ϵ -certificates.

changes to the relaxation and its propagation, as in [Section 6](#); they do not contradict lower bounds for a fixed oracle satisfying that hypothesis.

Face alignment produces a different branching effect. On the aligned McCormick instance `mccaxis2` at $\epsilon = 10^{-7}$, LP-point branching uses 3 nodes, external bisection restricted to x_1 uses 45, and external bisection on both variables uses 10,469. The external bisection runs disable presolve and propagation and use best-first search, so the comparison identifies attainable regimes rather than isolating a single parameter. Its bounded, logarithmic, and power-like sequences illustrate the McCormick mechanism of [Section 3](#). The transversal instance `mccdiag2` has no bounded sequence among the tested settings. For these tests SCIP’s variable-lock reduction is disabled: with that reduction, `mccaxis2` becomes binary in one variable and is solved at the root, changing the permitted operations.

11.3 Branching points on a selected benchmark

The MINLPLib benchmark consists of 57 nonconvex instances selected by a default-solver screen, subject to size filters and a cap of three instances per family. Screened default solution times lie between 1 and 20 seconds; 24 instances have integer variables, and five perform no continuous branching. This is a selected, relatively short-running benchmark rather than a sample of all nonconvex problems. The two campaigns use the same instance names but different limits and settings. The first campaign has 855 core runs, $57 \times 5 \times 3$, at 120 CPU seconds. Its 171 optional, 228 tolerance-sweep, 486 screening, and 187 diagnostic records are separate from the core comparison; together all batches total 1,927 records. The second campaign has $57 \times 12 \times 3 = 2,052$ runs at 60 CPU seconds. Both use default numerical tolerances and zero requested absolute and relative gaps. We retain these campaigns as separate measurements.

Within each campaign, seed labels 0, 1, and 2 are matched across settings; the latter two change SCIP’s variable permutation and random-seed shift. The instance is the statistical unit. If n_{isk} is the observed count for instance i , setting s , and seed k , define

$$g_{is} = \exp \left\{ \frac{1}{|K_{is}|} \sum_{k \in K_{is}} \log(n_{isk} + 10) \right\}.$$

For a setting s and reference r , the reported ratio is

$$R_{s/r} = \exp \left\{ \frac{1}{|I|} \sum_{i \in I} \log \frac{g_{is}}{g_{ir}} \right\}. \quad (151)$$

Thus the shift is 10 nodes, and large instances do not dominate by their absolute counts. Intervals use 5,000 bootstrap resamples of paired instances, with bootstrap seed 1; they do not treat the three seed runs as independent benchmark instances. Reported p -values use a two-sided Wilcoxon signed-rank normal approximation on per-instance log ratios. They are unadjusted for the multiple comparisons.

Time-limited runs enter with their count at termination, which can understate completed-tree work. Crashes are handled as in the original analyses: the first campaign averages the available seed counts for that setting, whereas the second omits a paired instance if any required count is missing. [Table 3](#) reports the resulting paired cohort sizes and the full per-setting termination counts.

SCIP’s default point rule combines a relaxation point with midpoint pull 0.75, a relative-width trigger of 0.5, and clamp 0.2 away from the variable bounds. The first campaign compares this rule with zero midpoint pull (`1p`), zero pull and zero clamp (`1p_noclamp`), default pull with zero

Table 3: Separate MINLPLib branching comparisons. Ratios use shift 10 and (151); intervals are paired-instance bootstrap intervals. m is the number of instances contributing node ratios. Termination counts cover all 171 runs of the row’s setting, including any instance omitted from a paired ratio. O/T/C denote **optimal**, time limit, and crash.

Setting	Reference	m	Ratio	95% interval	O/T/C (of 171)
First campaign: 120 CPU s; SCIP selection					
default	—	—	1	—	170/1/0
lp	default	57	1.03	[0.83, 1.26]	166/5/0
lp_noclap	default	57	2.18	[1.43, 3.39]	138/32/1
mix_noclap	default	57	1.35	[1.03, 1.85]	154/17/0
mid	default	57	1.15	[1.03, 1.28]	169/2/0
Second campaign: 60 CPU s; $\mathbf{x}_\bullet$ settings use the plugin					
default	—	—	1	—	168/3/0
rclamp	default	56	0.96	[0.87, 1.06]	170/0/1
lp	default	57	1.01	[0.81, 1.21]	163/8/0
x_lp	—	—	—	—	164/7/0
x_recenter	x_lp	56	0.94	[0.86, 1.01]	162/8/1
x_recenter	default	56	0.92	[0.71, 1.14]	162/8/1
x_noclap	x_lp	56	2.00	[1.43, 2.96]	129/41/1

clamp (**mix_noclap**), and pure midpoint splitting (**mid**). The LP-point ratio is 1.03 with interval [0.83, 1.26], so this benchmark does not resolve an aggregate improvement. Pure midpoint splitting gives 1.15 with interval [1.03, 1.28] and $p = 0.009$. Seed variability is substantial: among the 56 instances with three completed default runs, the median ratio of largest to smallest shifted node count is 1.51.

Clamp removal gives ratio 2.18 with interval [1.43, 3.39] and 33 of 171 runs without an **optimal** termination, including one crash. The retained diagnostics identify a concrete mechanism: on 10 of the 16 instances with an LP/no-clamp failure, the branching variable’s LP value lies on a bound in 42–100% of the bounded continuous branchings. Without the clamp, the split lies only at the solver’s numerical offset from that bound and can create long chains of nearly unchanged boxes. Restricting the comparison to instances where all runs of both settings finish gives ratio 1.08 on 41 instances. That restriction removes much of the observed failure penalty and changes the question being measured.

The second campaign has two arms. Parameter-only settings use SCIP’s selection; settings prefixed $\mathbf{x}_\bullet$ use an external-branching plugin with a closely related selection rule. The plugin implements clipping, randomized clamping, recentring, and an incumbent-coordinate rule. Its selection and the point-formula inspection use SCIP 10.0.3 source, whereas the recorded executable is 10.0.2; the arms should therefore not be treated as an exact reproduction of one another. The randomized clamp ranges over [0.1, 0.3], and recentring uses safeguard parameter 0.2. Recentring gives 0.94 against the plugin LP-point rule and 0.92 against SCIP’s default, with intervals containing one. Randomized clamping also has an interval containing one. Removing the plugin clamp gives 2.00 with interval [1.43, 2.96] and 41 time limits. The two campaigns therefore agree on the boundary-chain failure mechanism, while providing no resolved general gain from the safe variants.

The second campaign also contains 3,840 synthetic kink runs: two families, eight kink positions, twelve settings, five seeds, and four tolerances 10^{-2} , 10^{-4} , 10^{-6} , 10^{-8} . These runs supply the optimum and disable heuristics, propagation, and weak-dual cutoff reductions, which would otherwise solve the one-dimensional kink at the root; feasibility tolerance is 10^{-9} . For the one-dimensional kink

at $a = 1/6$, LP-point clipping with clamp 0.2 uses 3, 7, 9, and 13 nodes over those tolerances, whereas the plugin recentring rule uses 5 throughout. At 10^{-8} on the McCormick kink with the same position, the ordinary LP/clipped rule averages 7,190 nodes and plugin recentring averages 21 over five seeds. These examples expose the trap mechanism in [Section 7](#); they do not imply the same improvement with ordinary propagation or on the selected benchmark. Randomized clamping also has a wide finite-sample spread: at a McCormick kink near $a = 0.0102$, its ordinary LP-point variant averages 766 nodes, with range 7–3,443, versus 26 with the fixed clamp. Independent exact rational model records, kept separate from SCIP, give 19,537 processed model nodes for widest-side clipping at $a = 1/6$, clamp $1/5$, and $\epsilon = 10^{-8}$.

11.4 Finite-size certificates in random designs

The random-design studies use their own Python node algorithms and Gaussian instance generators. The recorded stack is Python 3.13, NumPy 2.5, SciPy 1.18, and Clarabel 0.11; semidefinite comparisons additionally use CVXPY and, for larger sparse problems, SCS. Node decisions evaluate explicit dual bounds or Frank–Wolfe lower bounds rather than accept a solver’s reported objective alone. These formulas give a mathematical certificate when evaluated with controlled error; the archived evaluations are floating point. We therefore describe their recorded certificate decisions without treating them as interval-validated proofs.

For sparse regression, the main finite-size comparison has Gaussian design X , $k = 8$ planted coefficients of magnitude $b = 1$, per-entry noise standard deviation $\sigma = 0.5$, and

$$n = \text{round}(\alpha k \log p), \quad \lambda = 1.5\sigma\sqrt{2n \log p}/b.$$

Eight independent design/noise seeds are used in each cell; varying parameters on a shared seed does not produce independent new instances. The perspective lower bounds test the planted-support value, and a root certificate is the sufficient planted-point certificate discussed in [Section 9](#). C1 tests every single wrong fixing against that value. When it holds, its certificate also establishes planted optimality under the assumptions of that section. A failing wrong-fixing relaxation does not by itself show that a better integer support exists.

[Table 4\(a\)](#) shows a finite-size window with many C1 passes and no planted-point root certificate in the displayed cells. At nominal $\alpha = 1.75$, all eight instances pass C1 at every displayed p , while none passes the root criterion. This illustrates the distinction between a linear variable-branching certificate and an exact root relaxation. It does not measure the asymptotic constants of [Section 9](#): the computations have $p \leq 3,200$ and small k , outside the proved condition $(\log p)^6 \leq n \leq p$. In the revised records, 31 earlier undecided C1 cases were resolved, with 30 passes and one failure; the displayed decisions use those revised records.

The separate rule comparison contains 72 instances at $p = 100$ and 200. In three, every single deletion of a planted variable is prunable while full C1 fails because of forced-in null variables. On these three instances, largest- z branching uses 13, 13, and 17 nodes, and most-fractional branching uses 23, 17, and 23. The removal half of C1 and full C1 must therefore be distinguished, as must these two branching rules. The observations show small trees in both rules in these cases; they do not identify an asymptotic rule-dependent window.

The stronger sparse-relaxation comparison is a separate experiment: $k = 3$, $b = 1$, $\sigma = 0.5$, nested columns on a shared design per seed, and $\lambda = 1.5\sigma\sqrt{2n \log 200}/b$ fixed as p varies. With $n = 20$, the median fraction of the perspective root gap, measured relative to the numerically enumerated global optimum, closed by the optimal-perspective semidefinite relaxation is 51%, 25%, 12%, and 3% at $p = 40, 80, 160, 320$, respectively. Eight seeds are recorded per cell; the fractions use the seven, eight, eight, and eight runs, respectively, whose relative perspective root gap exceeds

Table 4: Recorded finite-sample C1 passes, with denominators. (a) Sparse regression: $k = 8$, $b = 1$, $\sigma = 0.5$, the scaled ridge rule in the text, and eight independent design/noise seeds per cell. The α values are nominal; the displayed n includes rounding. All eight planted-point root tests fail in each of these eight cells. (b) Binary least squares: square systems ($\beta = 1$), $\theta = \rho/N$, and the archived C1 decision tests. A dash means no recorded cell, not a failed test. Failed C1 tests do not prove large trees.

(a) Sparse regression				
p	n at $\alpha = 1.5$	C1	n at $\alpha = 1.75$	C1
100	55	8/8	64	8/8
200	64	8/8	74	8/8
400	72	6/8	84	8/8
1,600	89	7/8	103	8/8

(b) Binary least squares			
N	C1 at $\theta = 0.25$	C1 at $\theta = 0.40$	C1 at $\theta = 0.50$
100	0/8	—	5/8
200	0/8	—	4/8
400	0/6	—	6/6
800	—	2/4	4/4

10^{-4} . For $n = 40$ and $p = 3, 200$, the retained comparison uses the six of eight instances where C1 establishes planted optimality. Explicit feasible lifted points give a strictly better value than the planted point for the pairwise hull in four of those six. The relevant lift retains helper columns whose selectors are zero, as in the hard-side model in [Section 9](#). This selected finite-size example illustrates the helper-column mechanism; it neither estimates its asymptotic threshold nor establishes the exactness of a different stronger relaxation on the full instance.

For binary least squares, the generator is $y = \sqrt{\rho/N} Hx^* + w$, with $M = \beta N$ rows, standard Gaussian H and w independent, and $x^* \in \{-1, 1\}^N$. Node bounds use the box relaxation, and best-first search is supplied with x^* as incumbent. The C1 archive has 330 decisions, all labelled certified or refuted by the archived bound and feasible-point tests. [Table 4\(b\)](#) displays square-system cells without interpolating missing parameter values. At these sizes the transition cells lie near $\rho/N = 0.4$ – 0.5 , above the limiting C1 constant $1/4$ in [Section 10](#). Tall-system cells with $\beta = 2$ move from about 0.12 – 0.17 at $N \leq 400$ toward 0.10 – 0.12 at $N = 800$, compared with the limiting constant $1/12$. These are small-sample frequencies of a sufficient condition. C1 failure does not establish a lower bound above linear on every search tree.

The tiny-system root study uses 20,000 trials per cell for $N \in \{2, 4, 6, 8\}$, $\beta \in \{1, 2\}$, and $\rho \in \{1, 4, 16, 64\}$. The observed frequency that the planted x^* minimizes the box relaxation is close to 2^{-N} . The event that some binary vertex minimizes that relaxation is different and can have a larger frequency. Neither event should be identified with successful rounding of a fractional solution. In square systems with $\rho = 4 \log N$, the geometric-mean most-fractional node counts are 44, 97, 509, and 15,765 at $N = 32, 64, 128, 256$, from six, six, six, and four seeds, respectively. The static-order trees at $N = 256$ hit the 100,000-node budget in all four runs; those records are lower observations of work, not completed tree sizes. The rapid finite-size growth is consistent with a difficult box-certificate regime, but proves no growth law. In particular, the explicit hard-regime lower-bound constants are too small to give a useful numerical bound at the tested dimensions.

The benchmark intervals describe the selected instances and paired seed runs; the random-design frequencies describe their stated finite cells. Neither provides a universal branching recommendation

or a practical speedup guarantee.

12 Discussion

The results separate the size of a proof from the search and local work used to obtain it. They identify several reasons why a specified relaxation can require a large certificate, and several reasons why a search can be large when a small certificate exists. This distinction helps interpret both the mathematical examples and the archived computations.

12.1 Interpreting a certificate obstruction

For relaxations with a positive quadratic vertex gap, the covering profile in [section 3](#) measures how many near-optimal points must be placed near test-box vertices. With the hypotheses for the matching upper bound, it determines spatial certificate size up to a logarithmic factor. Quadratic growth around a regular positive-dimensional optimal set gives a polynomial rate, while an isolated degenerate minimum can give a polynomial rate for a different reason. The all-face characterization in [section 4](#) further shows that boundary geometry matters: a proper root face can govern the exponent even when the full-dimensional singularity has a smaller exponent.

These are obstructions for the stated relaxation. A choice of branching rule cannot produce a certificate below its minimum size. A stronger relaxation, a change of factorization, or a different certificate format can change that minimum. Tightening must be assessed with the operation costs: under the pointwise gap assumptions it preserves a lower bound on charged certificate events, while a node lower bound additionally requires bounded rounds and appropriate probe accounting.

McCormick examples make the placement of exactness particularly visible. The interaction graph identifies faces on which the separate bilinear gaps vanish, and transverse near-optimal strata can force many boxes. The aligned two-dimensional segment admits a two-leaf exact certificate, but the aligned three-dimensional example still has a polynomial count. Thus graph and tangent information supply useful sufficient tests and lower bounds; they do not yet classify every aligned or curved case. Convergence order alone omits this placement information.

For relaxed constraints, the obstruction comes from superoptimal infeasible points retained by an inner relaxation tube. The C^2 objective and compatible original exact and relaxed constraint data, LICQ, and the nonzero relaxed multiplier are substantive hypotheses of the KKT argument. Exact original-feasibility reduction lies outside its operation class because it can remove these witnesses directly. Whether specific original-constraint contractors remove the obstruction on curved active strata requires a separate analysis.

The decomposition comparison changes how local certificates are combined. A spatial leaf assigns one global box. An affine-separator certificate can combine local bag members without explicitly enumerating every global combination. The fixed path separation in [section 5](#) shows that this can remove an exponential dependence on dimension even with the same local factor relaxation and a unique nondegenerate minimizer. The existence proof uses a known center and separator slopes; locating them and checking intersecting local members have additional costs. It establishes a change in proof representation, with explicit structural assumptions, rather than a general algorithmic bound for every sparse problem.

For integer variables, the class number gives a corresponding convexity obstruction. Every permitted leaf must contain an admissible class of the required integer points. A large class number therefore persists under convex cuts preserving the required integer set for the fixed projected relaxation. Incumbent removals instead enter the augmented count. Conversely, the example with class number two and an exponential variable-branching tree demonstrates a restriction of the

disjunction class. The unrestricted semantic attainment theorem identifies the minimum proof size; it supplies no efficient method to construct the attaining convex pieces.

12.2 Finding a proof and paying for its tests

The one-dimensional minimizer theorem gives a case in which local relaxation information suffices to find a near-minimal tree: its exact-gap rule is 4-competitive for every continuous objective, with the specified incumbent convention. The information lower bounds have a different role. They use smooth nonanalytic functions that agree near the observed minimizer, and so apply to that stated information class. A real-analytic germ determines the function on a connected domain; the same indistinguishability construction cannot establish a lower bound there. In higher dimensions, all-coordinate splitting can lose a logarithmic factor even on a separable quadratic. Repeated coordinate ambiguity forces exponential dimension dependence for the specified node-local nonanalytic information class with coordinatewise minimizer selection and no learning from siblings. The sharp-coordinate theorem covers a restricted separable class.

Safeguards impose a geometric constraint even if a rule knows where it would prefer to cut. On kink families, an interior safety fraction can prevent the desired cut for a long chain. The recentring rule attains the minimum safe hitting time in this model; positive-tolerance pruning can stop the chain sooner. The exact one-dimensional gap model also guarantees that every relaxation minimizer at an invalid interval is strictly interior. Production relaxations can attain their minimum on variable bounds. That behavior, visible in the archived benchmark diagnostics, is a reason to retain safeguards that the idealized kink model does not capture.

Propagation creates a similar distinction between the ideal test and its implementation. The greatest hull-consistent fixed point defines the strength of objective-only cutoff propagation for a specified expression graph. Local exactness can yield a bounded ideal node count. The specified expanded-quadratic schedule nevertheless needs polynomially many rounds in $1/\epsilon$, while a particular lifted representation has a much smaller round count. The persistence results require their derivative-cancellation conditions and charge objective-only phases. Combining constraints, auxiliary-variable tightening, or other contractors changes the test and needs its own proof.

Probe tests and local decomposition checks are also part of finding or verifying a certificate. A bound formed by the minimum of probe-child bounds is justified by those child tests; a displayed one-node search does not absorb their cost. Likewise, an unaligned local leaf may meet many separator cells, each requiring a check. The accounting in this paper keeps such work visible without claiming that a relaxation evaluation or a convex program has uniform running time.

12.3 What the statistical transitions distinguish

The sparse-regression linear-tree window proves that root inexactness can coexist with a short certificate. Above the single-variable threshold, the planted support is uniquely optimal and the C1 path lemma applies; the root threshold occurs later. The converse compares relaxation bounds with the planted-support value. It becomes a statement about global inexactness only when that support is optimal. The low-total-signal certificate lower bound uses a separate asymptotic regime, and the selected local-lift conclusions retain their explicit sandwich and helper-column conventions.

Binary least squares separates three further events. The hard certificate law concerns minimum proof size in the divergent sublinear-SNR regime. The sharp linear-SNR transition concerns C1 at all single-wrong-fixing nodes, with fixed margins; it is a threshold for a sufficient linear-tree criterion. The logarithmic-SNR root transition concerns equality with the orthant relaxation, so it says when the upper box constraints are inactive. It implies neither an integral root optimum nor correct sign

rounding. Failure of C1 below its threshold likewise proves no tree lower bound on its own. The finite NNLS coefficient estimate is useful because it controls all node events simultaneously without assuming their independence.

The fixed-dimensional sparse-regression calculation also separates support recovery from relaxation exactness. Under fixed positive per-entry noise and ridge $\sqrt{n}$, the true support becomes optimal but the exactness probability approaches a constant below one. Changing the noise normalization changes this calculation. Such distinctions are necessary when transferring a published recovery or exactness statement into a branch-and-bound claim.

12.4 What the archived computations support

The archived computations in [section 11](#) illustrate the mechanisms at their recorded finite sizes and tolerances. Under controlled settings with a known optimal incumbent, best-first search, and heuristics switched off, synthetic node counts follow the predicted positive-dimensional exponents over the reported range. Those settings change several solver components together. Default settings can change the effective relaxation through propagation and reformulation, and numerical thresholds can mask small-tolerance behavior. The fitted slopes describe floating-point runs; they are not certified asymptotic exponents.

The benchmark comparisons provide no general gain from moving split points closer to relaxation minimizers. Removing the clamp roughly doubled the recorded node counts, while the recentering comparison had uncertainty compatible with similar performance. The midpoint comparison showed a modest node-count disadvantage in its separate recorded protocol. These comparisons support the relevance of bound-attaining relaxation solutions and safeguards, but do not rank branching rules across solver families.

Random-design transitions illustrate the difference between C1 and root certification. Their sizes do not satisfy the easy sparse-regression theorem’s asymptotic conditions, and their observed locations do not estimate its first-order constants. Some explicit hard-side constants are vacuous at all recorded sizes. The computations therefore complement the proofs with finite-size behavior and implementation effects. They establish no general practical speedup, and node counts alone do not compare methods whose node tests have different costs.

12.5 Open questions

The following questions remain within reach of the certificate viewpoint.

1. Can a natural relaxation-minimizer coordinate rule be competitive in every fixed dimension under an exact-gap oracle? The present sharp-coordinate theorem leaves two arbitrary separable coordinates and the general nonseparable case unresolved. Dimension dependence must be assessed within the stated information class.
2. Can the graph and face bounds be completed into a characterization for termwise McCormick certificates? Matching integrals leave logarithmic losses, and the curved example lacks a matching upper bound. Aligned higher-dimensional examples rule out a classification based only on whether the tangent obstruction vanishes.
3. Which expression graphs admit local exactness, and which retain geometric lower bounds under cutoff propagation? The sufficient classes here leave a gap. Round complexity for general fair schedules, combined contractors, and other representations also remains unresolved.
4. Which original-constraint reductions remove the relaxed-constraint obstruction on curved active strata, and at what evaluation or round cost? The current tube theorem intentionally excludes those operations.

5. For convex quadratic projected objectives, how large can the minimum split-branching tree be relative to the class number? Semantic attainment by arbitrary convex pieces does not resolve that restriction.
6. What finite-size bounds describe the statistical transition windows? Between the proved divergent sublinear-SNR certificate law and the fixed-margin C1 regime, the available statements do not give a complete size law. Failure of a sufficient test cannot fill that gap.
7. Can models that include relaxation minimizers on variable bounds explain the benchmark safeguard behavior and yield competitive guarantees with an explicit evaluation cost?

A Additional continuous geometry proofs

A.1 Uniform coordinate-sheet bounds

Lemma A.1 (Graph geometry and coordinate multiplicity). *Let $S = \{x + \phi(x) : x \in U\}$, where U is a convex subset of a d -dimensional linear space T and $\phi : U \rightarrow T^\perp$ is C^2 with $\|D^2\phi\| \leq H$. Then*

$$\text{dist}(z - y, T_y S) \leq \frac{H}{2} |z - y|^2 \quad (y, z \in S). \quad (152)$$

More generally, if a manifold has this inequality with $H = 1/\rho$, then each dominant coordinate projection in the definition preceding [lemma 3.2](#) is injective on subsets of diameter less than

$$r_\rho = \frac{2\rho}{1 + \binom{n}{d}^{1/2}}.$$

In particular its $M(S)$ is at most $\binom{n}{d}$ when its diameter is less than r_ρ . Uniformly bounded graph Hessians give a uniform choice of this diameter. A zero Hessian permits any diameter.

Proof. For $y = x + \phi(x)$ and $z = x' + \phi(x')$, the vector $(x' - x) + D\phi(x)(x' - x)$ lies in $T_y S$. Taylor's theorem on the segment in U bounds the remainder by $H|x' - x|^2/2 \leq H|z - y|^2/2$.

Suppose $y, z \in S^I$ have $y_I = z_I$. Decompose $z - y = t + \nu$ into its tangent and normal parts at y . The smallest singular value of $\pi_I|_{T_y S}$ is at least $J_I(y)$: all singular values are at most one and their product is J_I . Consequently

$$|t| \leq \binom{n}{d}^{1/2} |\pi_I t| = \binom{n}{d}^{1/2} |\pi_I \nu| \leq \binom{n}{d}^{1/2} |\nu|.$$

The two-point inequality gives $|z - y| \leq (1 + \binom{n}{d}^{1/2})|z - y|^2/(2\rho)$. Either $y = z$ or their distance is at least r_ρ . This proves the multiplicity bound. $\square$

A.2 Proof of the nonlinear constraint transfer

We prove [theorem 3.12](#) with the exact/relaxed constraint tuple specified there. Root bounds are exactly kept inequalities. Let E collect all exact equalities and active exact inequalities, including active root coordinates. LICQ supplies a vector e whose derivatives are zero along these constraints and every other active constraint except one selected relaxed constraint. Set its derivative to one for a relaxed inequality with positive multiplier, or to the sign of the multiplier for a relaxed equality with nonzero multiplier. Normalize to a unit vector e_0 . KKT stationarity gives

$$E'(z_*)e_0 = 0, \quad \nabla f(z_*) \cdot e_0 = -\sigma < 0.$$

Full row rank gives a fixed right inverse R of $E'(z_*)$. When E is empty no correction is needed. Otherwise apply the C^2 implicit function theorem to

$$H(y, t, \xi) = E(y + te_0 + R\xi) - E(y).$$

Its derivative in ξ at $(z_*, 0, 0)$ is the identity. On an open product neighborhood there is a C^2 solution $\xi(y, t)$, with $\xi(y, 0) = 0$ by uniqueness. Define

$$Z(y, t) = y + te_0 + R\xi(y, t).$$

Then $E(Z(y, t)) = E(y)$ and $Z(y, 0) = y$. Differentiating at $(z_*, 0)$ gives $\xi_t(z_*, 0) = 0$, and hence $Z_t(z_*, 0) = e_0$. Also $Z_y(y, 0) = I$ throughout a smaller neighborhood.

On a product neighborhood with compact closure, shrink the spatial and time ranges to obtain constants $\mu, c, t_0 > 0$, $C \geq 0$ such that for feasible y and $0 \leq t \leq t_0$,

$$|Z_t(y, t)| \leq 2, \quad f(Z(y, t)) \leq f(y) - \mu t + Ct^2, \quad (153)$$

$$|Z(y, t) - y| \leq 2t, \quad v(Z(y, t)) \leq ct. \quad (154)$$

Continuity of the strictly negative time derivative at $(z_*, 0)$ and a bound on the second derivative of $f \circ Z$ give the descent estimate. Bounded derivatives of the original relaxed functions and their nonpositive or zero values at y give the violation estimate. Active exact inequalities preserve their feasible value at y , exact equalities preserve zero, and active root bounds preserve their coordinate value. All inactive exact restrictions have uniform slack on the smaller neighborhood and retain it for small t . Thus $Z(y, t) \in D \cap P_{\text{ex}}$.

For the covering conclusion, put $a = \eta + \epsilon$ and $t = 3a/\mu$. If a is small enough that $t \leq t_0$ and $Ct^2 \leq a$, then for $y \in E(\eta)$,

$$f(Z(y, t)) \leq f_* + \eta - 3a + a = f_* - \epsilon - a < f_* - \epsilon. \quad (155)$$

The tube dichotomy therefore forces

$$\beta q_{C_e}(Z(y, t)) < v(Z(y, t)) \leq Aa, \quad A = 3c/\mu$$

in every owner containing the shifted point. It lies within $r = \sqrt{Aa/\beta}$ in sup norm of a test vertex. If also $a \leq c\mu/(12\beta)$, its displacement $6a/\mu$ is at most r . The original Y is consequently covered by at most $2^n K$ cubes of side $4r$. This proves [equation \(27\)](#), and one may take the effective coefficient $\alpha_{\text{eff}} = \mu\beta/(12c)$ in the side length $2\sqrt{a/\alpha_{\text{eff}}}$.

For the integral conclusion, LICQ gives a feasible active-stratum graph

$$y(x) = z_* + x + \phi(x), \quad x \in B_r \subset T, \quad \phi(0) = D\phi(0) = 0,$$

where T is its d -dimensional tangent space. The graph is feasible: all active inequalities are zero and inactive ones retain their slack. Put $m_T(x) = m(y(x))$. Because z_* minimizes on this graph, $m_T(0) = Dm_T(0) = 0$. Define

$$W_\epsilon(x) = Z\left(y(x), \frac{3(m_T(x) + \epsilon)}{\mu}\right), \quad \Psi_\epsilon(x) = \pi_T(W_\epsilon(x) - z_*).$$

These functions are jointly C^2 in an open neighborhood of $(0, 0)$; the time range can be kept inside the IFT domain. At $(0, 0)$ the derivative of the time parameter vanishes, so $D_x\Psi_0(0) = I$. Choose a fixed ball B_r and a fixed compact parameter range $0 \leq \epsilon \leq \epsilon_0$ such that

$$\|D_x\Psi_\epsilon - I\| \leq \frac{1}{2}, \quad |\Psi_\epsilon(0)| \leq r/8. \quad (156)$$

The mean-value formula along line segments gives $|\Psi_\epsilon(x) - \Psi_\epsilon(x')| \geq |x - x'|/2$. Thus Ψ_ϵ is injective and its determinant is at least 2^{-d} in absolute value.

Every image contains the common ball $V = B_{r/4} \subset T$. Indeed, for $u \in V$, the map $x \mapsto u - (\Psi_\epsilon(x) - x)$ is a contraction of the closed ball B_r into itself: its norm is at most $r/4 + r/8 + r/2 < r$. Its fixed point solves $\Psi_\epsilon(x) = u$. The inverse on V has derivative norm at most two. Differentiating the inverse identity gives

$$D^2\Psi_\epsilon^{-1} = -(D\Psi_\epsilon)^{-1}D^2\Psi_\epsilon[D\Psi_\epsilon^{-1}, D\Psi_\epsilon^{-1}],$$

so its second derivatives are bounded uniformly in ϵ . The shifted image over V is therefore a graph with uniformly bounded second derivatives:

$$u \mapsto z_* + u + \pi_{T^\perp}(W_\epsilon(\Psi_\epsilon^{-1}(u)) - z_*).$$

Restrict the original graph to a still smaller fixed patch S whose shifted projections lie in V . Its shifted images have uniformly small diameter, by the bounded derivative of W_ϵ and small ϵ_0 . [Lemma A.1](#) now gives $M(Z_\epsilon(S)) \leq \binom{n}{d}$ for the maps $Z_\epsilon(y) = Z(y, 3(m(y) + \epsilon)/\mu)$. This argument controls multiplicity uniformly even though the image need not be a translate of the original patch.

The area factor of W_ϵ is at least that of its tangential projection, hence at least 2^{-d} . The input graph's area factor is bounded above by $B_\phi = (1 + \sup \|D\phi\|^2)^{d/2}$. Consequently the Jacobian relative to stratum measure satisfies

$$J_{Z_\epsilon} \geq j_* := 2^{-d}/B_\phi > 0.$$

Finally shrink S and ϵ_0 so that the descent remainder is at most $m(y) + \epsilon$ throughout. The previous strict-descent calculation gives

$$f(Z_\epsilon(y)) < f_* - \epsilon, \quad v(Z_\epsilon(y)) \leq A(m(y) + \epsilon).$$

For each owner, the dichotomy implies $\beta q_{C_\epsilon}(Z_\epsilon(y)) < A(m(y) + \epsilon)$ on its preimage. Use the injective area formula and the lower Jacobian, followed by [lemma 3.2](#) on the shifted patch, to get

$$\int_{\{y \in S : Z_\epsilon(y) \in A_\epsilon\}} (m(y) + \epsilon)^{-d/2} d\mathcal{H}^d(y) \leq j_*^{-1} (A/\beta)^{d/2} C_{n,d} \binom{n}{d}.$$

Summing over a covering family proves the first inequality in [equation \(28\)](#). For a run the all-owner family is the one that covers these images; owners meeting F alone need not cover them.

Taylor's theorem gives $0 \leq m_T(x) \leq H|x|^2$ on a fixed parameter ball, with $H \geq 1$. Its input graph area factor is at least one, so

$$\int_S (m + \epsilon)^{-d/2} d\mathcal{H}^d \geq d\omega_d \int_0^{r'} t^{d-1} (Ht^2 + \epsilon)^{-d/2} dt,$$

where ω_d is the unit-ball volume. For $t \geq \sqrt{\epsilon/H}$ the integrand is at least $(2H)^{-d/2}/t$. The resulting logarithm proves the last assertion.

A.3 Why a single covering scale is insufficient

Example A.2 (Many near-optimal wells). Let $D = F = [0, 1]$, $t = y - 1/2$, and, for an integer $K \geq 2$, define

$$f_K(y) = K^{-2}(1 - \cos(2\pi Kt)) + K^{-2}(1 - e^{-K^2 t^2}).$$

Its optimum is zero at $t = 0$. Use $f_B = f_K - 32q_B$ and $R_B = B$. This is a convex underestimator because

$$f_K''(y) = 4\pi^2 \cos(2\pi Kt) + (2 - 4K^2 t^2)e^{-K^2 t^2} \geq -4\pi^2 - 4e^{-3/2} > -64.$$

Thus the lower and vertex-vanishing upper coefficients are both 32. The first and second derivatives are uniformly bounded on D , so the remaining Lipschitz and second-order constants can also be fixed across the family; feasibility imposes no error-bound difficulty.

For $|t| \leq 1/(2K)$, $1 - \cos(2\pi Kt) \geq 8K^2 t^2$, so $f_K \geq 8t^2$. For $|t| \geq 1/(2K)$ the second term is at least $(1 - e^{-1/4})K^{-2} > 0.22K^{-2}$. Hence for $\epsilon_K = 0.2K^{-2}$,

$$E(\epsilon_K) \subset \{|t| \leq \sqrt{\epsilon_K/8}\}.$$

For any fixed $c > 0$, its cover by side- $c\sqrt{\epsilon_K}$ intervals has size at most $2 + 1/(\sqrt{2}c)$, independently of K .

The points $t = k/K$, $|k| \leq K/2$, number at least K and all have $f_K \leq K^{-2}$. In [theorem 3.1](#) take $\eta = K^{-2}$. Its covering interval has side $2\sqrt{1.2/32}/K < 1/K$, so it covers at most one of these points. Consequently $N_{\text{cover}}(\epsilon_K) \geq K/2$. The ratio cannot be bounded by a uniform constant times $\log(1/\epsilon_K)$ times the single-scale covering count, since that expression is $O(\log K)$.

A.4 Proof of the regular KKT rates

Fix a global minimizer z_* . In its local feasible description, write A for the active inequalities, $\mu_j > 0$ for their multipliers, and λ_k for equality multipliers. Let $\mathcal{L} = f + \sum_{j \in A} \mu_j g_j + \sum_k \lambda_k h_k$, with all constraints zero at the minimizer. KKT gives $\nabla \mathcal{L}(z_*) = 0$. For feasible $y = z_* + \delta$, Taylor's theorem gives

$$m(y) = \frac{1}{2} \delta^\top H_{\mathcal{L}} \delta + o(|\delta|^2) + \sum_{j \in A} \mu_j (-g_j(y)). \quad (157)$$

If local quadratic growth failed, there would be feasible $y_k \rightarrow z_*$ with $m(y_k)/|\delta_k|^2 \rightarrow 0$, and, after a subsequence, $\delta_k/|\delta_k| \rightarrow u$ with $|u| = 1$. Boundedness of the Hessian and nonnegativity of the slacks in [equation \(157\)](#) imply $-g_j(y_k) = O(|\delta_k|^2)$ for each $j \in A$. Taylor expansion of these constraints and the equalities gives $u \in T$, the common nullspace of their gradients. If there are no active inequalities, the equalities alone give this conclusion. Positive definiteness of $H_{\mathcal{L}}$ on T then implies

$$\liminf_k \frac{m(y_k)}{|\delta_k|^2} \geq \frac{1}{2} u^\top H_{\mathcal{L}} u > 0,$$

a contradiction. If $T = \{0\}$, already $u \in T$ contradicts $|u| = 1$. Thus each minimizer has local quadratic growth. Disjoint neighborhoods of the finitely many minimizers, together with the positive minimum of m on their compact feasible complement, give global quadratic growth to the optimal set. The upper part of [proposition 3.7](#) gives the logarithmic upper bound.

For a minimizer whose active stratum has dimension $d \geq 1$, LICQ gives an injective C^2 coordinate graph $y(x)$ of that stratum. Inactive constraints have slack on a small patch, so the graph is feasible. All active constraints vanish there and hence $m(y(x)) = \mathcal{L}(y(x)) - f_* \leq C|x|^2$. Choose a d -coordinate projection invertible at the minimizer and shrink to a graph patch with bounded area factor. For every box, the arithmetic-geometric mean and arcsine calculation in [lemma 3.2](#) bounds $\int_{S \cap B} q_B^{-d/2}$ by a fixed constant using this projection alone. Also the radial calculation at the end of [appendix A.2](#) gives $\int_S (m + \epsilon)^{-d/2} \geq c \log(1/\epsilon)$. Validity therefore gives the objective-gap logarithmic lower bound. The tube lower bound is [theorem 3.12](#) for its compatible original tuple.

For a unique zero-dimensional active stratum, the active gradient matrix is square and nonsingular. Consequently

$$|\delta| \leq C \left(\sum_{j \in A} |\nabla g_j(z_*)\delta| + \sum_k |\nabla h_k(z_*)\delta| \right).$$

Taylor expansion and feasibility show

$$\sum_{j \in A} (-g_j(y)) \geq c|\delta| - C'|\delta|^2.$$

If A is nonempty, positive multipliers and [equation \(157\)](#) give local sharp growth $m(y) \geq c_s|y - z_*|$ after shrinking the neighborhood. If A is empty, the equalities give $|\delta| \leq C'|\delta|^2$, so z_* is locally isolated in F ; sharp growth holds vacuously there. Uniqueness and compactness extend sharp growth globally in both cases. This handles the empty-active-inequality case without an undefined minimum of multipliers.

To build an exact certificate under [equation \(29\)](#), choose a coordinate grid of side at most s through z_* , clipped at D . Take s small enough that $\alpha'ns^2/4 \leq v_0$. For $z \in R_C$, a nearest feasible y and sharp growth give

$$f_C(z) - f_* \geq c_s|z - z_*| - K'q_C(z), \quad K' = \alpha'(1 + L_f\kappa + c_s\kappa).$$

Every grid interval has $z_{*,i}$ outside it or at an endpoint, so $d_i^C(z) \leq |z_i - z_{*,i}|$. Hence

$$q_C(z) \leq s \sum_i d_i^C(z) \leq s\sqrt{n}|z - z_*|.$$

Choosing also $s \leq c_s/(K'\sqrt{n})$ gives $f_C(z) \geq f_*$ on every relaxation. If $\alpha' = 0$, the error bound gives feasible relaxed points and the root is exact directly. The finite clipped grid is a split-tree partition and proves the exact-certificate claim.

Finally, if a minimizer coordinate stays a relative distance at least $\delta \in (0, 1/2]$ from both endpoints of its dyadic interval, the objective lower gap gives $L(B) \leq f_* - \alpha\delta(1 - \delta)s_j^2$ for a box containing it. Such a box remains unpruned while this quantity exceeds ϵ , giving a logarithmic number of processed levels. For a coordinate position p/q in lowest terms with $q > 1$ odd, every dyadic fractional position is between $1/q$ and $1 - 1/q$. This establishes the stated example of the middle-position condition.

A.5 Convex centroids and fractional covers

Lemma A.3 (Centroid distance from supporting planes). *Let K be a bounded convex subset of a p -flat with positive relative volume. Its centroid belongs to K . For any linear functional of width W on K , its centroid is at distance at least $W/(p+1)$ in functional value from each supporting plane. The assertion holds when K is partly open.*

Proof. The closure has the same volume and first moments, since a convex body's boundary has zero relative volume. Its centroid is in its relative interior: a supporting hyperplane through a putative boundary centroid would put a nonnegative, nonzero linear functional above its minimum on a positive-volume subset, contradicting that centroid's value. The relative interiors of a convex set and its closure coincide and belong to the original convex set, so the centroid belongs to K .

Work with the closure and normalize the functional's minimum to zero. If $W = 0$ the conclusion is immediate. Choose an apex a at its maximum W . For $0 \leq t \leq W$, the homothetic image

$$(1 - t/W) \text{cl } K + (t/W)a \subset \text{cl } K \cap \{h \geq t\}$$

has volume $(1 - t/W)^p \text{vol}_p(K)$. Integrating these tail volumes gives the mean functional value at least $\int_0^W (1 - t/W)^p dt = W/(p+1)$. Apply the same argument to $W - h$ for the other supporting plane. $\square$

Lemma A.4 (Half-integral fractional vertex covers). *Every finite graph has an optimal fractional vertex cover with entries in $\{0, 1/2, 1\}$.*

Proof. Truncating an entry greater than one to one preserves every edge constraint and improves the objective. Thus an optimum exists in the compact polytope $0 \leq z_i \leq 1, z_i + z_j \geq 1$. Choose an optimal vertex of that polytope. On the vertices with $0 < z_i < 1$, form the graph of tight equations $z_i + z_j = 1$. A bipartite component admits a small perturbation adding t on one side and subtracting it on the other. All its tight equations remain satisfied; every other constraint involving these vertices is slack, so both signs of a sufficiently small perturbation stay feasible. This contradicts extremality. Every fractional component therefore has an odd cycle. Its tight equations force all entries around the cycle, and hence throughout the component, to be $1/2$. All remaining entries are zero or one. $\square$

A.6 Proof of the two-dimensional aligned certificate

We prove [proposition 3.21](#). On the optimal segment, $q(y) := g(a, y) = f_* - cay$ is affine. For a feasible transverse sign $\sigma \in \{-1, 1\}$ define

$$h_\sigma(y) = \lim_{t \downarrow 0} \frac{g(a + \sigma t, y) - q(y)}{t}, \quad L_y < y < U_y.$$

Convexity in x gives existence of the derivative. The minimum condition for m along the feasible transverse direction gives a finite lower bound; a fixed positive secant gives a finite upper bound. Thus these derivatives are finite on the open interval.

For two interior values y_0, y_1 , choose $0 < \lambda < 1$ and interior y_2 with $y_1 = \lambda y_0 + (1 - \lambda)y_2$. For small $t > 0$ convexity gives

$$g(a + \sigma t, y_1) \leq \lambda g(a + \sigma t/\lambda, y_0) + (1 - \lambda)q(y_2).$$

Subtract the affine identity for $q(y_1)$, divide by t , and let $t \downarrow 0$. The right-hand difference quotient is exactly the one defining $h_\sigma(y_0)$, so $h_\sigma(y_1) \leq h_\sigma(y_0)$. Reversing the roles gives equality. Each applicable transverse derivative of g is therefore constant along the open segment. Write the nonnegative transverse derivatives of m as

$$r_+(y) = h_+ + cy, \quad r_-(y) = h_- - cy.$$

For $c > 0$, their nonnegativity throughout the interval forces

$$r_+(y) \geq c(y - L_y), \quad r_-(y) \geq c(U_y - y).$$

Convexity of $x \mapsto m(x, y)$ with $m(a, y) = 0$ then gives

$$m(x, y) \geq c(x - a)(y - L_y) \quad (x \geq a), \quad m(x, y) \geq c(a - x)(U_y - y) \quad (x \leq a).$$

On the respective child boxes these expressions are among the endpoint products whose minimum is the positive-coefficient gap in [equation \(32\)](#). Hence $\Gamma_B \leq m$ on both children. For $c < 0$, the same nonnegativity argument gives

$$r_+(y) \geq |c|(U_y - y), \quad r_-(y) \geq |c|(y - L_y),$$

which supplies the negative-coefficient gap products instead. Continuity extends the inequalities to the segment endpoints and the closed box boundaries. At an endpoint a only the feasible sign is needed and its box is the root. The pointwise pruning criterion [equation \(31\)](#) proves exactness in every case.

B Proofs for the box-face characterization

B.1 A per-face lower bound

Let F be a root face with free-coordinate set J and dimension $d \geq 1$. If a test box $B = \prod_i [a_i, b_i] \subseteq D$ meets F , each coordinate fixed on F is also an endpoint of the corresponding interval of B . Its gap factor therefore vanishes on $F \cap B$. For the free coordinates, [\(41\)](#) and the arithmetic–geometric mean inequality give, almost everywhere on the relative interior of a nondegenerate $F \cap B$,

$$(m + \varepsilon)^{-d/2} \leq d^{-d/2} \left(\prod_{i \in J} \alpha_i \right)^{-1/2} \prod_{i \in J} ((x_i - a_i)(b_i - x_i))^{-1/2}.$$

Each one-dimensional factor has integral

$$\int_{a_i}^{b_i} \frac{dt}{\sqrt{(t - a_i)(b_i - t)}} = \pi.$$

It follows that

$$\int_{F \cap B} (m + \varepsilon)^{-d/2} d\sigma_F \leq \pi^d d^{-d/2} \left(\prod_{i \in J} \alpha_i \right)^{-1/2}. \quad (158)$$

If a free-coordinate interval has zero width, the intersection has zero d -measure and contributes zero instead. Summing [\(158\)](#) over any valid cover proves [\(45\)](#). The same argument integrates over $F \cap A_e$ for an owned set $A_e \subseteq B_e$ and then bounds it by the product integral over $F \cap B_e$; full validity on the unowned boundary is unnecessary.

B.2 Projection and relative sublevel cubes

We first work on a cube $D = \prod_i [L_i, L_i + s_0]$. Choose a positive Lipschitz constant M for ∇m , increasing it if necessary. Convexity of D gives the two-sided Taylor estimate

$$\left| m(z) - m(y) - \nabla m(y)^\top (z - y) \right| \leq \frac{M}{2} \|z - y\|_2^2 \quad (y, z \in D). \quad (159)$$

Fix $0 < s \leq s_0/2$ and $y \in D$. Move each coordinate whose distance from a root endpoint is strictly less than s to that endpoint. If both endpoints qualify at $s = s_0/2$, choose either; under the strict inequality they cannot both qualify. Let F be the resulting face, let $d = \dim F$, and let y' be the moved point. Every free coordinate of y' has distance at least s from both endpoints. In particular, the relative cube

$$Q_F(y', s) = \{z \in \text{aff } F : \|z - y'\|_\infty \leq s\}$$

is contained in F .

For a moved coordinate i , let η_i be the sign pointing from y_i to its chosen endpoint. There is room for the step $y - \eta_i s e_i$ in the opposite direction. Nonnegativity and [\(159\)](#) imply

$$0 \leq m(y - \eta_i s e_i) \leq m(y) - s \eta_i \partial_i m(y) + \frac{M}{2} s^2, \quad \eta_i \partial_i m(y) \leq \frac{m(y)}{s} + \frac{M}{2} s.$$

Coordinates already at their chosen endpoint have zero displacement and require no derivative test. Summing the first-order terms for the other moved coordinates and using (159) gives

$$m(y') \leq (1 + n - d)m(y) + (n - d)Ms^2. \quad (160)$$

For each free coordinate, both steps $y' \pm se_i$ are feasible. The same argument gives $|\partial_i m(y')| \leq m(y')/s + Ms/2$. For every $z \in Q_F(y', s)$,

$$\begin{aligned} m(z) &\leq (1 + d)m(y') + dMs^2 \\ &\leq (1 + d)(1 + n - d)m(y) + ((1 + d)(n - d) + d)Ms^2 \\ &\leq K_n(m(y) + Ms^2), \quad K_n = \frac{(n + 2)^2}{4}. \end{aligned} \quad (161)$$

Both coefficients in the preceding line are at most K_n . These inequalities use nonnegativity only on the closed cube.

B.3 Dyadic packing and the vertex estimate

Let $s_j = s_0 2^{-j}$ and $\bar{\alpha} = \max_i \alpha_i$. Put

$$\Lambda_0 = \bar{\alpha}n/4, \quad \Lambda_1 = K_n(\Lambda_0 + M), \quad \Lambda_2 = \Lambda_0 + \Lambda_1.$$

An invalid level- j cube B has a point $y \in B$ with

$$m(y) + \varepsilon < \sum_i \alpha_i (y_i - a_i)(b_i - y_i) \leq \Lambda_0 s_j^2. \quad (162)$$

Thus no box remains invalid once $\Lambda_0 s_j^2 \leq \varepsilon$, and the dyadic tree is finite. For $j \geq 1$, apply (161) to the witness with $s = s_j$. It gives a face F and a point y' at sup-distance less than s_j from y , with

$$Q_F(y', s_j) \subseteq F \cap \{m \leq \Lambda_1 s_j^2\}.$$

For each face, choose a maximal subset of these projected witnesses whose mutual sup-distances exceed s_j . This is a finite selection from the invalid boxes at that level. On a face of dimension $d \geq 1$, the relative cubes of radius $s_j/2$ about the selected points have disjoint interiors, measure s_j^d , and lie in the displayed sublevel set. Hence the number selected is at most

$$s_j^{-d} \sigma_F \{m \leq \Lambda_1 s_j^2\}.$$

By maximality, every projected witness is within s_j of a selected point. Its original witness is within $2s_j$, and its entire grid box is within $3s_j$. A cube of radius $3s_j$ meets at most 8^n level- j grid boxes. Consequently the number of invalid boxes charged to positive-dimensional faces is at most

$$8^n \sum_{F \in \mathcal{F}_+(D)} s_j^{-d_F} \sigma_F \{m \leq \Lambda_1 s_j^2\}. \quad (163)$$

Shared grid boundaries only decrease the useful packing volume and are covered by this deliberately loose grid-count constant.

Whenever a level contributes, (162) implies $\varepsilon < \Lambda_0 s_j^2$, so the sublevel in (163) lies in $\{m + \varepsilon \leq \Lambda_2 s_j^2\}$. For $t > 0$ and $d \geq 1$, the geometric series gives

$$\sum_{j \geq 1} s_j^{-d} \mathbf{1}\{t \leq \Lambda_2 s_j^2\} \leq \frac{1}{1 - 2^{-d}} \left(\frac{\Lambda_2}{t} \right)^{d/2} \leq 2\Lambda_2^{d/2} t^{-d/2}.$$

Tonelli's theorem therefore bounds the sum over levels in (163) by

$$2 \cdot 8^n \sum_{F \in \mathcal{F}_+(D)} \Lambda_2^{d_F/2} I_F(\varepsilon). \quad (164)$$

If the projected face is a vertex v , the witness is strictly within s_j of v in every coordinate. It is therefore in the unique corner grid box at that vertex. Let $V_v(\varepsilon)$ count the levels $j \geq 1$ at which that corner box is invalid. The projection estimate also applies when all coordinates of a corner-box witness are moved to v with distances at most s_j : the opposite length- s_j steps stay in D because $2s_j \leq s_0$. It gives $m(v) \leq (1+n)m(y) + nMs_j^2 \leq \Lambda_1 s_j^2$ on every such level. For $m(v) > 0$, this allows only finitely many levels, independently of ε .

It remains to control V_v when $m(v) = 0$. Use inward edge coordinates $t_i \geq 0$ at v and write $g_i \geq 0$ for the inward derivatives. Put $g = \min_i g_i$ and choose an edge E attaining this minimum. On a corner box of side s ,

$$m(v+t) \geq g\|t\|_1 - \frac{M}{2}\|t\|_2^2 \geq \left(g - \frac{M}{2}s\right)\|t\|_1, \quad \sum_i \alpha_i t_i (s - t_i) \leq \bar{\alpha}s\|t\|_1.$$

The box is valid whenever $s \leq g/(\bar{\alpha} + M/2)$. Independently, it is valid whenever $\Lambda_0 s^2 \leq \varepsilon$. Define

$$\rho = \max\{g/M, \sqrt{\varepsilon}\}, \quad c_* = \min\{M/(\bar{\alpha} + M/2), \Lambda_0^{-1/2}\} > 0.$$

An invalid corner box must have $s > \max\{g/(\bar{\alpha} + M/2), \sqrt{\varepsilon/\Lambda_0}\} \geq c_*\rho$. Counting dyadic scales, with $\log_+ t = \max\{0, \log t\}$, gives

$$V_v(\varepsilon) \leq 1 + \frac{\log_+(1/c_*)}{\log 2} + \frac{\log_+(s_0/\rho)}{\log 2}. \quad (165)$$

Along E , Taylor's estimate gives $m(v+te) \leq gt + Mt^2/2$. For $\rho \leq t \leq s_0$ this is at most $3Mt^2/2$ and $\varepsilon \leq t^2$. It follows, including the case $\rho > s_0$ where the right side is zero, that

$$I_E(\varepsilon) \geq (3M/2 + 1)^{-1/2} \log_+(s_0/\rho). \quad (166)$$

Equations (165) and (166) absorb the vertex count into an edge integral and a fixed constant. There are finitely many vertices, and each edge can be selected by at most its two endpoints.

Every invalid dyadic node has exactly 2^n children. If H is the number of invalid nodes, the total number tested is $1 + 2^n H$. The root contributes at most one invalid node. Combining this identity, (164), the fixed contribution of nonoptimal vertices, and the optimal-vertex estimate proves (46) on a cube. Replacing each simultaneous coordinate refinement by n successive binary coordinate splits gives the same terminal partition. The intermediate binary nodes add at most a fixed multiple of the dyadic work. This proves the tree claim.

For a general box, set $w_i = u_i - \ell_i$ and use $x_i = \ell_i + w_i z_i$ to map $[0, 1]^n$ onto D . The transformed coefficients are $\alpha_i w_i^2 > 0$, and the transformed gradient is Lipschitz. On each face,

$$I_F(\varepsilon) = \left(\prod_{i \in J_F} w_i \right) \int_{\tilde{F}} (\tilde{m} + \varepsilon)^{-d_F/2} d\sigma_{\tilde{F}}.$$

These are fixed positive factors. The transformation preserves covers, rectangular partitions, and coordinate split trees, and maps the unit dyadic tree to coordinate-wise halving on D . The upper bound therefore holds on D with changed constants. The per-face lower bound in appendix B.1 was already proved in the original coordinates. A finite sum and its maximum are comparable. Together with (42), these bounds prove all the comparisons in theorem 4.1.

B.4 Dominance by the full-box integral

We prove case (ii) of [proposition 4.5](#). By the preceding diagonal change of coordinates it suffices to use $D = [0, 1]^n$. Choose $r > 0$ such that $m \geq 0$ and ∇m has positive Lipschitz constant M on $D + \overline{B}_2(0, r)$. For $z \in D$ with $m(z) \leq t \leq Mr^2/2$, a descent step of length r in the negative gradient direction first gives

$$\|\nabla m(z)\| \leq t/r + Mr/2 \leq Mr.$$

If the gradient is zero this conclusion is immediate. Thus the step $-\nabla m(z)/M$ is admissible, and the descent estimate and nonnegativity give the sharper inequality

$$\|\nabla m(z)\|^2 \leq 2Mm(z) \leq 2Mt. \quad (167)$$

Put $t_0 = \min\{Mr^2/2, 1/4\}$ and $K = 1 + \sqrt{2Mn} + Mn/2$. For a proper face F of dimension $d \geq 1$, write

$$V_F(t) = \sigma_F\{z \in F : m(z) \leq t\}, \quad V_D(t) = \text{vol}\{z \in D : m(z) \leq t\}.$$

For $0 < t \leq t_0$, put $s = \sqrt{t}$ and select a maximal set of points in $F \cap \{m \leq t\}$ separated by more than $2s$ in sup norm. Compactness gives a finite such set, say of size P . Its relative cubes of radius $2s$ cover the sublevel set, so $V_F(t) \leq P(4s)^d$.

For each selected point z , choose an axis-parallel cube of side s contained in D and having z as a vertex. Such a cube exists because $s \leq 1/2$: in each free coordinate choose a direction with at least $1/2$ of room, and in each fixed coordinate point inward. Every point x of this cube has distance at most $\sqrt{n}s$ from z . Equation (167) and Taylor's estimate give $m(x) \leq Kt$. The selected cubes have disjoint interiors, because their vertex sup-distances exceed $2s$. Consequently

$$V_D(Kt) \geq Ps^n \geq 4^{-d}t^{(n-d)/2}V_F(t) \quad (0 < t \leq t_0). \quad (168)$$

Let $a = d/2$, $b = n/2$, and assume $0 < \varepsilon < t_0$. The elementary layer-cake identity is

$$I_F(\varepsilon) = a \int_0^\infty (t + \varepsilon)^{-a-1} V_F(t) dt.$$

The part $t \geq t_0$ is bounded by $\sigma_F(F)t_0^{-a}$. On $[0, \varepsilon]$, monotonicity bounds the contribution by $\varepsilon^{-a}V_F(\varepsilon)$. By (168) this is at most $4^d\varepsilon^{-b}V_D(K\varepsilon)$, while

$$I_D(\varepsilon) \geq ((K+1)\varepsilon)^{-b}V_D(K\varepsilon).$$

On $[\varepsilon, t_0]$, the volume transfer and $t \geq (t + \varepsilon)/2$ bound the integrand by a constant times $(t + \varepsilon)^{-b-1}V_D(Kt)$. Since $Kt + \varepsilon \leq K(t + \varepsilon)$, this is at most a constant times $(Kt + \varepsilon)^{-b-1}V_D(Kt)$. Substituting $u = Kt$ and using the layer-cake identity for I_D bounds its integral by $CI_D(\varepsilon)$. Thus $I_F(\varepsilon) \leq C_0 + C_1I_D(\varepsilon)$ for every proper positive-dimensional face. This proves the claimed dominance and completes [proposition 4.5](#).

C Proofs for decomposition certificates

C.1 Configurations and shell partitions

Lemma C.1 (Unfolding the local inequalities). *Use (63)–(64). A configuration chooses a local leaf B_t and a point $z^t \in B_t$ for each bag. For each nonroot bag it also chooses a cell E_t such that $z_{S_t}^t \in E_t$ and $E_t \cap (B_{p(t)})_{S_t} \neq \emptyset$. Then*

$$\ell_r = \min_{\text{configurations}} \left\{ \sum_t \sum_{c \text{ assigned to } t} f_{c, (B_t)_c}(z_c^t) + \sum_{t \neq r} \lambda_t^\top (z_{S_t}^{p(t)} - z_{S_t}^t) \right\}. \quad (169)$$

Proof. For a fixed root leaf and point, each child term selects a cell meeting the root leaf projection and pays its intercept. The intercept is the minimum over the child's leaf and point restricted to that cell. Substitute its definition. The child's positive slope term, evaluated at the parent point, combines with the negative slope term in its intercept to give the corresponding summand in (169). Repeat for every child. At each step, child choices have independent coordinate copies; no equality of copies is imposed. A sum of independent minima is the minimum over the product of their choices. There are finitely many boxes and all point domains are compact. Continuity supplies every minimum and completes the recursive substitution. $\square$

For a box B , write $\omega(B)$ for its largest side length. For a point p , let $Q(p, a) = p + [-a, a]^d$. For an empty coordinate set, the coordinate box is the singleton in $\mathbb{R}^0$, its partition has one cell, and $\omega(B)$ and all empty-coordinate norms are zero. Interiors are always taken relative to the corresponding coordinate space. These conventions cover empty separators.

Lemma C.2 (Shell partition). *Let X be a nondegenerate box in $\mathbb{R}^d$ with side lengths at most $s_0 > 0$, $p \in X$, $h > 0$, and $\theta = 2^{-\mu}$ for an integer $\mu \geq 0$. There is a rectangular partition of X with at most $(J + 1)(4/\theta)^d$ members, where $J = \max\{0, \lceil \log_2(s_0/h) \rceil\}$, such that every member satisfies*

$$\omega(B) \leq \max\{h, \theta \operatorname{dist}_\infty(B, p)\} \leq h + \theta \operatorname{dist}_\infty(B, p). \quad (170)$$

Proof. Partition $Q(p, h)$ into 2^d cubes of side h with vertex p . For $j = 1, \dots, J$, put a grid of side $g_j = \theta 2^{j-1}h$ on $Q(p, 2^j h)$, anchored at its lower corner. There are $4/\theta$ grid cells per coordinate. The boundary of $Q(p, 2^{j-1}h)$ is on grid lines, because $1/\theta$ is an integer. Retain the grid cells outside the interior of that smaller cube, omitting the cells contained in it. Each retained cell has $\operatorname{dist}_\infty(B, p) \geq 2^{j-1}h$ and side g_j , so it satisfies (170).

These levels cover $Q(p, 2^J h)$ with disjoint interiors. This cube contains X , because $p \in X$ and $2^J h \geq s_0$. Intersect the cells with X and discard degenerate intersections. Clipping decreases widths and increases distance from p . The nondegenerate intersections still cover X : they cover its interior, and their finite closed union therefore contains its closure. Each level contributes at most $(4/\theta)^d$ boxes. The central level contributes $2^d \leq (4/\theta)^d$. $\square$

Proof of Theorem 5.3. Use Lemma C.2 in every bag and separator, centered at the corresponding coordinates of x^* . Use the slopes (67) and the maximal intercepts (64). We bound every configuration in (169).

For each coordinate i , let $\operatorname{top}(i)$ be the bag nearest the root that contains it. Define a consistent point $x \in D$ by $x_i = z_i^{\operatorname{top}(i)}$. Connectedness of the bags containing i ensures that its other copies are connected to this copy through at most $k - 1$ edges containing i in their separators. Put

$$\begin{aligned} X_* &= \|x - x^*\|_2, \quad \rho_t = \|x_{V_t} - x_{V_t}^*\|_\infty, \quad r_t = \|x_{V_t} - x_{V_t}^*\|_2, \\ d_t &= \|z_{S_t}^t - z_{S_t}^{p(t)}\|_\infty \quad (t \neq r), \quad d_r = 0, \quad \delta_t = \|z^t - x_{V_t}\|_\infty. \end{aligned}$$

Empty-separator drifts are zero. Since every variable occurs in at most k bags,

$$\sum_t \rho_t^2 \leq \sum_t r_t^2 \leq k X_*^2. \quad (171)$$

Let $\operatorname{anc}_j(t)$ denote the j th ancestor, with missing ancestor drifts set to zero. Telescoping each coordinate's copies gives

$$\delta_t \leq \sum_{j=0}^{k-2} d_{\operatorname{anc}_j(t)}. \quad (172)$$

The nonseparator coordinates have their topmost bag at t , so only separator coordinates can differ between z^t and x_{V_t} .

The width estimate in (170), evaluated at the bag copy, gives

$$\omega(B_t), \omega(E_t) \leq 2h + \theta(\rho_t + \delta_t),$$

where the harmless $2h$ also covers a zero-dimensional separator. Choose a point in $E_t \cap (B_{p(t)})_{S_t}$. Comparing each copy with that point yields

$$d_t \leq 4h + \theta(\rho_t + \rho_{p(t)} + \delta_t + \delta_{p(t)}).$$

Use (172) for t and its parent, then absorb the term θd_t using $\theta \leq 1/2$. With $b_t = 8h + 2\theta(\rho_t + \rho_{p(t)})$, this gives

$$d_t \leq b_t + 4\theta \sum_{j=1}^{k-1} d_{\text{anc}_j(t)}. \quad (173)$$

Write (173) as $d \leq b + \Gamma d$ entrywise, with vectors indexed by nonroot bags. The nonnegative matrix Γ has nonzero entries only in ancestor columns, so it is nilpotent. Its row sums are at most $4\theta(k-1)$ and its column sums at most $4\theta D_k$: a node has at most D_k descendants in the next $k-1$ generations. For any matrix, $\|\Gamma\|_2^2 \leq \|\Gamma\|_1 \|\Gamma\|_\infty$; thus (65) gives $\|\Gamma\|_2 \leq 1/2$. Iterating the entrywise inequality and using nilpotence gives $\|d\|_2 \leq 2\|b\|_2$. By $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$ and the child-degree bound,

$$\sum_t b_t^2 \leq 12 \left\{ 16Mh^2 + \theta^2(1+\Delta) \sum_t \rho_t^2 \right\}.$$

Consequently

$$\sum_t d_t^2 \leq 48\{16Mh^2 + \theta^2(1+\Delta)kX_*^2\}, \quad \sum_t \delta_t^2 \leq K_1 \sum_t d_t^2. \quad (174)$$

For the second assertion, square (172) by Cauchy–Schwarz. An edge drift occurs for its own bag and for at most $\sum_{j=1}^{k-2} \Delta^j$ descendants, giving precisely the bound K_1 .

The chosen slopes have an exact telescoping identity:

$$\sum_t \nabla a_t(x_{V_t}^*)^\top (z^t - x_{V_t}) = \sum_{t \neq r} \lambda_t^\top (z_{S_t}^t - z_{S_t}^{p(t)}). \quad (175)$$

To verify it, fix i and telescope $z_i^t - x_i$ along its path to $\text{top}(i)$. The coefficient of an edge difference $z_i^u - z_i^{p(u)}$ is the sum of $\partial_i a_t(x^*)$ over bags below u containing i . This is $\lambda_{u,i}$. Sum over i . Thus the exact-factor version of a configuration value equals

$$f(x) + \sum_t \{a_t(z^t) - a_t(x_{V_t}) - \nabla a_t(x_{V_t}^*)^\top (z^t - x_{V_t})\}.$$

Only at most w separator coordinates differ in each bracket. Lipschitz gradients, Taylor’s formula on the segment between its two points, and $\|z^t - x_{V_t}\|_2 \leq \sqrt{w} \delta_t$ bound the bracket below by $-H_a \sqrt{w} r_t \delta_t - (H_a w/2) \delta_t^2$. The factor-relaxation loss in bag t is at most $\alpha' A \omega(B_t)^2/4$. Hence the relaxed configuration value is at least $f(x) - E$, where

$$E \leq \sum_t \left\{ H_a \sqrt{w} r_t \delta_t + \frac{H_a w}{2} \delta_t^2 + \frac{\alpha' A}{4} \omega(B_t)^2 \right\}.$$

Young’s inequality gives

$$H_a \sqrt{w} r_t \delta_t \leq \frac{c_g}{4k} r_t^2 + \frac{H_a^2 w k}{c_g} \delta_t^2.$$

Also $\omega(B_t)^2 \leq 3(4h^2 + \theta^2 \rho_t^2 + \theta^2 \delta_t^2)$. Using (171) and $\theta \leq 1/2$ therefore gives

$$E \leq \frac{c_g}{4} X_*^2 + Q \sum_t \delta_t^2 + 3\alpha' AMh^2 + \frac{3}{4} \alpha' A\theta^2 k X_*^2.$$

Now apply (174) and (65):

$$E \leq \frac{3c_g}{4} X_*^2 + MH_0 h^2.$$

Quadratic growth shows that every configuration is at least $f_* + (c_g/4)X_*^2 - MH_0 h^2 \geq f_* - \epsilon/4$. Lemma C.1 proves the required root bound. Finally, each bag partition has at most $(J+1)(4/\theta)^{w+1}$ leaves and each separator partition no more than that many cells. Summing gives (66). $\square$

C.2 The path separation

Proof of Theorem 5.4. The unary function $\varphi(t) = t^2 - t^4/10$ has $\varphi''(t) = 2 - 6t^2/5 \in [4/5, 2]$ on $[-1, 1]$. Using $x_i^4 \leq x_i^2$ and $2|x_i x_{i+1}| \leq x_i^2 + x_{i+1}^2$ gives

$$f_n(x) \geq \frac{1}{10} \|x\|_2^2. \quad (176)$$

Thus zero is the unique global minimizer. Its Hessian is the positive definite tridiagonal matrix $H_n = \text{tridiag}(4/5, 2, 4/5)$, whose eigenvalues are $2 + (8/5) \cos(j\pi/(n+1))$, $1 \leq j \leq n$. At the all-ones point the Hessian instead has diagonal $4/5$ and the same off-diagonals. Its smallest eigenvalue is $4/5 - (8/5) \cos(\pi/(n+1)) < 0$ for $n \geq 3$. Continuity of the Hessian gives a negative direction at nearby interior points, proving nonconvexity. The bilinear alphaBB underestimator has Hessian $\begin{pmatrix} 4/5 & 4/5 \\ 4/5 & 4/5 \end{pmatrix}$, so it is convex.

We first prove the accuracy-dependent spatial bound. Put $\alpha_1 = \alpha_n = 2/5$ and $\alpha_i = 4/5$ for $2 \leq i \leq n-1$. A valid box B satisfies, at every $x \in B$,

$$f_n(x) + \epsilon \geq \sum_i \alpha_i (x_i - l_i)(u_i - x_i). \quad (177)$$

For a nondegenerate box, the arithmetic-geometric mean inequality implies

$$\int_B (f_n + \epsilon)^{-n/2} dx \leq n^{-n/2} \prod_i \alpha_i^{-1/2} \prod_i \int_{l_i}^{u_i} \frac{dx_i}{\sqrt{(x_i - l_i)(u_i - x_i)}} = n^{-n/2} \pi^n \prod_i \alpha_i^{-1/2}.$$

Degenerate boxes have zero n -dimensional measure. Summing over any finite valid cover gives

$$N \geq (n/\pi^2)^{n/2} \left(\prod_i \alpha_i \right)^{1/2} \int_D (f_n + \epsilon)^{-n/2} dx. \quad (178)$$

Since $f_n(x) \leq \frac{1}{2} x^\top H_n x$ and $\lambda_{\min}(H_n) \geq 2/5$, the ellipsoid $\{x : \frac{1}{2} x^\top H_n x \leq 1/5\}$ lies in the unit Euclidean ball and hence in D . The substitution $x = \sqrt{2} H_n^{-1/2} y$ and polar coordinates give from (178)

$$N \geq 2(2n/\pi)^{n/2} \left(\frac{\prod_i \alpha_i}{\det H_n} \right)^{1/2} \frac{J_n(\sqrt{1/(5\epsilon)})}{\Gamma(n/2)}, \quad J_n(T) = \int_0^T t^{n-1} (1+t^2)^{-n/2} dt. \quad (179)$$

The determinant recurrence $\det H_j = 2 \det H_{j-1} - (16/25) \det H_{j-2}$, with initial values 1, 2, yields

$$\det H_n = \frac{4}{3} (8/5)^n - \frac{1}{3} (2/5)^n \leq \frac{4}{3} (8/5)^n, \quad \prod_i \alpha_i = (4/5)^n / 4.$$

For $T \geq \sqrt{n}$,

$$J_n(T) \geq e^{-1/2} \int_{\sqrt{n}}^T \frac{dt}{t} = \frac{1}{2} e^{-1/2} \log(T^2/n),$$

because $(1 + t^{-2})^{-n/2} \geq e^{-n/(2t^2)} \geq e^{-1/2}$. The standard upper Stirling bound

$$\Gamma(n/2) \leq \sqrt{4\pi/n} (n/(2e))^{n/2} e^{1/(6n)}$$

then bounds the coefficient of $\sqrt{n}(2e/\pi)^{n/2} \log(1/(5n\epsilon))$ below by $\sqrt{3}e^{-1/2-1/(6n)}/(8\sqrt{\pi}) > 0.068$ for $n \geq 3$. This proves the first expression in (69) when its logarithm is nonnegative; otherwise that expression is below zero.

For the second bound, define $d_i^B(x) = \min\{x_i - l_i, u_i - x_i\}$. The alphaBB gap of each edge is at least

$$\frac{2}{5} \{(x_i - l_i)(u_i - x_i) + (x_{i+1} - l_{i+1})(u_{i+1} - x_{i+1})\} \geq \frac{4}{5} d_i^B(x) d_{i+1}^B(x).$$

Hence every valid box satisfies $\frac{4}{5} \sum_{i < n} d_i^B(x) d_{i+1}^B(x) \leq f_n(x) + \epsilon$. Intersect such a box with D , writing the intersection as $\prod_i [z_i - h_i, z_i + h_i]$, with $0 \leq h_i \leq 1$. If it has positive volume, put $s_i = 1 - h_i \in [0, 1]$. Its center obeys $|z_i| \leq s_i$ and $d_i^B(z) \geq h_i$. The upper quadratic bound on f_n therefore implies

$$\frac{4}{5} \sum_{i < n} (1 - s_i)(1 - s_{i+1}) \leq \sum_i s_i^2 + \frac{4}{5} \sum_{i < n} s_i s_{i+1} + \epsilon.$$

Cancel the cross terms and use that each index has degree at most two:

$$n - 1 \leq \sum_i \left(\frac{5}{4} s_i^2 + 2s_i \right) + \frac{5}{4} \epsilon. \quad (180)$$

For $0 \leq s < 1$,

$$\log(1 - s) \leq \log(3/5) - \frac{5}{9} \left(\frac{5}{4} s^2 + 2s - 1 \right). \quad (181)$$

Indeed, the derivative of $\log(1 - s) + (5/9)(5s^2/4 + 2s)$ is $(2 - 5s)/(18(1 - s))$, so its maximum is at $s = 2/5$; there $5s^2/4 + 2s = 1$. Summing (181) and using (180) gives

$$\frac{\text{vol}(B \cap D)}{\text{vol}(D)} = \prod_i (1 - s_i) \leq (3/5)^n \exp\left[\frac{5}{9} \left(1 + \frac{5}{4} \epsilon\right)\right].$$

Zero-volume intersections obey the same bound. Volume subadditivity over the cover proves the second expression in (69).

Both lower bounds also hold for rectangular owned certification events (A_e, B_e) covering D , with $A_e \subseteq B_e$ and the corresponding gap test valid at owned points. For the integral proof, integrate over A_e and bound its arcsine integral by the integral over B_e . For the volume proof, use the center of the closure of $A_e \cap D$. If this owner has positive volume, its center is in its interior and hence in the owner; distances to the faces of B_e are at least its half-widths. Open or closed endpoint choices do not change its volume. Thus the same per-event estimates apply. This is the owned event count, which can include removed-region certificates and virtual probe events.

For the decomposition upper bound, assign the unary factor at coordinate t and the edge $(t, t+1)$ to bag t ; assign the final unary factor also to the final bag. Then $M = n - 1$, $w = 1$, $k = 2$, $\Delta = 1$, $D_k = K_1 = 1$, $c_g = 1/10$, $H_a = 14/5$, and $\alpha'A = 4/5$. The Hessian norm of each bag objective is at most $2 + 4/5 = 14/5$. Thus $Q = 159$ and $H_0 = 122114.4$. The choice $\theta = 2^{-10}$ satisfies (65). Choose $h^2 = \epsilon/(488457.6(n - 1))$. All slopes vanish because every factor gradient vanishes at zero. Theorem 5.3 gives

$$N_{\text{dec}} \leq 2(n - 1)4096^2 \left\{ \frac{1}{2} \log_2 \frac{1953830.4(n - 1)}{\epsilon} + 2 \right\},$$

which is bounded by (70).

For the uniform ratio, let $a = \sqrt{2e/\pi}$ and split into two cases. If $\epsilon \leq 0.04/n^2$, then $\log(0.2/(n\epsilon)) \geq \frac{1}{2} \log(1/\epsilon)$ and $n \leq 0.2\epsilon^{-1/2}$. Since $\epsilon \leq 10^{-4}$, (70) is at most $C_1 n \log(1/\epsilon)$ for an absolute constant C_1 . The first spatial bound consequently gives $N/U_n \geq c_1 a^n / \sqrt{n}$. If $0.04/n^2 < \epsilon \leq 10^{-4}$, then $1/\epsilon < 25n^2$, so $U_n \leq C_2 n \log n$. The second spatial bound gives $N/U_n \geq c_2 (5/3)^n / (n \log n)$. Since $(5/3)/a > 1$, the positive sequence $((5/3)/a)^n / (\sqrt{n} \log n)$ tends to infinity. Its infimum over integers $n \geq 3$ is therefore positive. Decreasing the constant proves (71) in both cases.

Finally, for $b > 0$ the McCormick gap on a rectangle equals

$$b \min\{(x - l_x)(y - l_y), (u_x - x)(u_y - y)\}.$$

It is at least $b d_x d_y$, which is all the center-volume proof needs. It is at most $\frac{b}{2} \{(x - l_x)(u_x - x) + (y - l_y)(u_y - y)\}$: use $\min\{a_1 b_1, a_2 b_2\} \leq \sqrt{a_1 a_2 b_1 b_2} \leq (a_1 a_2 + b_1 b_2)/2$ with $a_1 = x - l_x$, $a_2 = u_x - x$, $b_1 = y - l_y$, $b_2 = u_y - y$. Thus it satisfies the upper error hypothesis with $\alpha' = 2/5$, and the decomposition upper bound is unchanged. The strictly positive coordinate gap in (177) is not available, so the accuracy-dependent integral bound is not transferred. $\square$

D Proofs for objective-cutoff propagation

D.1 Greatest fixed boxes and threshold attainment

Proof of theorem 6.1. An elementary revise is monotone and deflationary. Take the box hull H of the union of all consistent boxes inside $Z_0(C)$, including the empty box. Each such box A satisfies $A = \rho_E(A) \subseteq \rho_E(H)$. Thus $H \subseteq \rho_E(H)$, and deflation gives equality. Its root interval is below c . This proves existence and maximality. If a step starts from a box containing H , monotonicity gives $H = \rho_E(H) \subseteq \rho_E(Z) \subseteq \rho(Z)$, so every step preserves it.

Exact revises are continuous along decreasing compact boxes. To see this, let $Y_k \downarrow Y$. The closed sets $E \cap Y_k$ are nested and compact. If their intersection is empty, one is already empty. Otherwise maxima and minima of every coordinate converge to the extrema on $E \cap Y$: take extremizers, extract a convergent subsequence, and use closedness. Consequently $\rho_E(Y) = \bigcap_k \rho_E(Y_k)$. For a fair run Z_k , let $Z_\infty = \bigcap_k Z_k$. Along the infinitely many applications of E , this identity implies $Z_\infty \subseteq \rho_E(Z_\infty)$. Every constraint and the cutoff therefore fix the limit. Preservation and maximality give $Z_\infty = H$. An empty intersection of nested nonempty compact boxes is impossible, so an empty limit is reached at a finite iterate.

A forward partial revise updates the parent projection of $E \cap Z$; a backward partial revise updates its child projections. Each is monotone, deflationary, continuous from above, and preserves consistent boxes. A box fixed by both has all the projections required by the exact revise. The same fair-limit argument therefore applies to these partial steps.

Shrinking C or decreasing c shrinks $Z^*(C, c)$, by maximality. For a decreasing sequence $c_k \downarrow c$ with nonempty fixed boxes, their intersection is nonempty and fixed at cutoff c . If all their variable projections contain y , the intersection also contains y in its projection. Thus the two nonempty-cutoff sets are closed upper sets. The exact lifted evaluation at y is a singleton consistent box at cutoff $f(y)$. No box is consistent below $F_{lo}(C)$. This proves attainment and (75). The closed upper sets are precisely $[\pi_{\mathcal{D}}(C), \infty)$ and $[\pi_{\mathcal{D}}(C, y), \infty)$; preservation and convergence prove the strict emptiness and removal statements. $\square$

We will also use a consequence of topological forward evaluation. Every consistent box Z with constants fixed satisfies

$$Z \subseteq Z_0(\Pi_x Z). \quad (182)$$

Indeed variables and constants satisfy this inclusion first. Supporting each operation-node endpoint shows that its interval is contained in its forward operation image. Induction through the graph proves the claim.

D.2 First-order loss and owned event accounting

For a centered subbox $U = y + \prod_i [-\delta_i, \delta_i]$ define $a_j = \sum_i |\partial_i t_j(y)| \delta_i$. If every second partial of t_j is bounded in absolute value by H_j , Taylor's theorem gives

$$\min_U t_j \leq t_j(y) - a_j + \frac{H_j}{2} \|\delta\|_1^2, \quad \max_U t_j - \min_U t_j \leq 2a_j + H_j \|\delta\|_1^2.$$

The first inequality uses the corner opposing the derivative signs; the second bounds the two extreme values. The witness lemma yields

$$\pi_{\mathcal{D}}(C, y) \leq f(y) - \left(\sum_j a_j - 2 \max_j a_j \right) + \left(\frac{1}{2} \sum_j H_j + \max_j H_j \right) \|\delta\|_1^2. \quad (183)$$

For a coordinate segment only the corresponding second partial is needed.

Point localization in theorem 6.7. Put $g = m(y) + \epsilon$ and choose $\delta = \min\{y_i - l_i, u_i - y_i, \rho_*\}$. By strict removal and (183),

$$f_* - \epsilon < \pi_{\mathcal{D}}(C, y) \leq f(y) - d_0 \delta + K_0 \delta^2 \leq f(y) - d_0 \delta / 2.$$

Therefore $\delta < 2g/d_0 < \rho_*$, which gives the nearest-face bound. Each coordinate product is at most $w(C)$ times its nearest-face distance. Summing gives $q_C(y) < 2nw(C)g/d_0$, hence (80). A degenerate box has $q_C = 0$ in its degenerate directions and causes no difficulty. $\square$

For completeness we establish the propagation event statement with inherited bounds. A phase starts with variable box B_0 , ends with a retained box B_f , and consists of consecutive objective-only runs. Each restart uses the forward box of its current variable box, intersected with final lifted boxes inherited from earlier runs at this node or its ancestors. Its *inheritance chain* includes all those earlier phases recursively. Let c_Φ be the smallest cutoff in this chain. All cutoffs are at least $f_* - \epsilon$, even if incumbent values change.

Every run in phase Φ preserves $H_\Phi = Z^*(B_0, c_\Phi)$. Prove this in time order. An earlier phase Ψ feeding this phase has $B_0 \subseteq B_{0,\Psi}$ and $c_\Phi \leq c_\Psi$, so $H_\Phi \subseteq H_\Psi$ by monotonicity. Its inherited final box contains H_Φ . Earlier runs in the same phase also contain H_Φ . Their variable projections contain $\Pi_x H_\Phi$, so restarted forward boxes contain H_Φ by (182). Finally H_Φ is consistent for each larger cutoff used in this phase. Preservation proves the induction. Thus $\Pi_x H_\Phi \subseteq B_f$.

If $B_f = \prod_i [a_i, b_i]$ is nonempty, give each removed point to its first coordinate outside $[a_i, b_i]$. The resulting owners have preceding coordinates in their retained intervals, the first offending coordinate in $[l_i, a_i)$ or $(b_i, u_i]$, and subsequent coordinates in their original intervals. There are at most $2n$ convex rectangular owners. Their closures are the test boxes. Carry the retained boundaries forward. For an owner with test box $S \subseteq B_0$, monotonicity gives $Z^*(S, c_\Phi) \subseteq H_\Phi$. Every owned $y \notin B_f$ therefore satisfies $\pi_{\mathcal{D}}(S, y) > c_\Phi \geq f_* - \epsilon$. If a phase empties its box, preservation implies $H_\Phi = \emptyset$; its whole terminal owner satisfies this strict test. Splits similarly assign their shared boundary to one child. Following each point through the finite run partitions the domain into terminal and discarded owners, each satisfying either the relaxation point test or the propagation test. Extra

variable-space reductions may be included if their discarded owners have the stated relaxation point certificate; each such reduction adds at most $2n$ pieces to the accounting.

Every point of $N \cap \{g < \vartheta\}$ consequently satisfies $g \geq \alpha_{\text{eff}} q_C$ in its own test box. If $g \leq \epsilon + \eta$, the nearest endpoint in each coordinate lies within $r = \sqrt{(\epsilon + \eta)/\alpha_{\text{eff}}}$, since a coordinate product is at least the square of that nearest distance. The 2^n vertex balls of each test box cover its certified points. This proves (81).

For a nondegenerate test box, arithmetic-geometric mean gives

$$\int_C q_C(y)^{-n/2} dy \leq n^{-n/2} \prod_{i=1}^n \int_{l_i}^{u_i} ((y_i - l_i)(u_i - y_i))^{-1/2} dy_i = n^{-n/2} \pi^n.$$

Thus its certified owner contributes at most $(\pi^2/(\alpha_{\text{eff}} n))^{n/2}$ to the integral. Degenerate boxes have zero n -dimensional measure. Sum over owners to prove (82). A compact smooth p -dimensional minimizing patch has covering number $\Omega(r^{-p})$, by a local nonsingular coordinate chart. For the logarithmic consequence, integrate over a small ball: on radii between $\sqrt{\epsilon}$ and a fixed radius the radial integrand is bounded below by a constant times r^{-1} . If there are at most P_0 phases per node and no other reductions, $K \leq (1 + 2nP_0)T$. Without this hypothesis the conclusion remains an event lower bound.

D.3 Aggregate loss at faces

Proposition D.1 (Two consequences of aggregate loss). *Assume the objective-only event model above and the relaxation point test with $\alpha > 0$. Suppose on a set N , $\ell(y) \geq \ell_0 > 0$, and on centered cubes of radius ρ_1 the remainder in (183) is at most $K_1 \delta^2$ when all half-widths equal δ . Put $\rho_* \leq \min\{\rho_1, \ell_0/(2K_1)\}$ and $\vartheta = \ell_0 \rho_*/2$. Here $\rho_1 > 0$, $\rho_* > 0$, and $\ell_0/(2K_1) = +\infty$ if $K_1 = 0$.*

If $S \subseteq N \cap E(\eta)$ is a compact C^1 embedded arc whose unit tangent satisfies $|\tau_i| \geq \tau_0 > 0$ for every coordinate, and $\epsilon + \eta < \vartheta$, then

$$K \geq \frac{\tau_0 \mathcal{H}^1(S)}{2^{n+1} \sqrt{(\epsilon + \eta)/\alpha} + 8n(\epsilon + \eta)/\ell_0}. \quad (184)$$

Alternatively let $n \geq 2$, $\rho_0 > 0$, $\gamma > 0$, and $N = Q_\infty(x_, \rho_0)$, with $Q_\infty(x_*, 2\rho_0) \subseteq D$. Assume on the larger cube $m \geq \gamma \|y - x_*\|_\infty^2$ and $|\partial_i m| \leq \ell_0/8$. Take also $\rho_* \leq \rho_0$. There is a finite constant C , independent of ϵ , such that*

$$K \geq C^{-1} \int_{N \cap \{m + \epsilon < \vartheta\}} (m + \epsilon)^{-n/2} dy. \quad (185)$$

If $m \leq \Gamma \|y - x_\|_\infty^2$ locally as well, this is $\Omega(\log(1/\epsilon))$.*

Proof. For a propagation-certified point choose $\delta = \min\{\min_i(y_i - l_i, u_i - y_i), \rho_*\}$. The cube version of (183) gives $\delta < 2g/\ell_0 < \rho_*$, where $g = m + \epsilon$. Thus one face is within $2g/\ell_0$. Along the arc each tangent coordinate has constant sign, so each coordinate is strictly monotone. A coordinate slab of half-width r meets an arc of length at most $2r/\tau_0$. Relaxation-certified points are in 2^n vertex balls of radius $r_R = \sqrt{(\epsilon + \eta)/\alpha}$; propagation-certified points are in $2n$ face slabs of half-width $r_F = 2(\epsilon + \eta)/\ell_0$. Their total arc length per event is at most $(2^{n+1}r_R + 4nr_F)/\tau_0$. Sum to obtain (184).

For the integral, examine a face layer $y = (l_1 + s, z)$ with $0 \leq s < 2g(y)/\ell_0 < \rho_0$, and put $y' = (l_1, z)$. The segment from y' to y lies in the larger cube. If $M(z) = m(y') + \epsilon$, the gradient hypothesis gives $|M - g(y)| < g(y)/4$. Therefore

$$s < \frac{8M(z)}{3\ell_0}, \quad g(y)^{-n/2} \leq (5/4)^{n/2} M(z)^{-n/2}.$$

Integrating first in s bounds this layer by

$$\frac{8}{3\ell_0}(5/4)^{n/2} \int_{\|z-z_*\|_\infty \leq 2\rho_0} M(z)^{1-n/2} dz.$$

For $n = 2$ the last integrand is one. For $n > 2$, use $M \geq \gamma\|z - z_*\|_\infty^2$. In either case the integral is at most $\gamma^{1-n/2}(n-1)2^n\rho_0$: the sup-norm radial integral is $\int_0^{2\rho_0} t^{2-n}(n-1)2^{n-1}t^{n-2} dt$. There are $2n$ layers. Relaxation-certified points contribute at most $(\pi^2/(\alpha n))^{n/2}$ per event by the preceding arcsine estimate. Adding these finite constants and summing the owned regions proves (185). The local upper quadratic bound makes its integral logarithmic by the same radial estimate used above. $\square$

D.4 Two schedule-dependent round estimates

Proposition D.2 (A slow expanded quadratic). *Let $a > 0$, $0 < \epsilon \leq a^2/16$, and use the distinct term nodes s^2 and s in $f = s^2 - 2as + a^2$. A round consists of a forward pass, cutoff $-\epsilon$, and a backward sum pass followed by the power revise. Start from $[a - y_0, a + x_0]$, with $0 < x_0, y_0 \leq a/4$ fixed. If $z_0 = \min\{x_0, y_0\}$, the box remains nonempty for at least*

$$\frac{a}{4\sqrt{\epsilon}} \left(\arctan \frac{z_0}{\sqrt{\epsilon}} - \arctan \frac{4\sqrt{\epsilon}}{a} \right) - 1$$

rounds. As $\epsilon \downarrow 0$ this is $(\pi/8 - o(1))a\epsilon^{-1/2}$. Its fixed point is nevertheless empty.

Proof. While the current endpoints straddle a , write them as $a - y, a + x$. Backward propagation from the sum uses the old forward term ranges and gives

$$y' = y - \frac{y^2 + \epsilon}{2a}, \quad x' = \sqrt{a^2 + 2ax - \epsilon} - a \geq x - \frac{x^2 + \epsilon}{a}.$$

The last inequality follows by squaring; its difference reduces to $(x^2 + \epsilon)(1 + 2x/a - (x^2 + \epsilon)/a^2) \geq 0$. For $4\epsilon/a \leq z \leq a/4$, the decrement $\Delta = (z^2 + \epsilon)/a$ is at most $z/2$. Both updated distances are therefore positive whenever both old distances are at least $4\epsilon/a$, so that the round is nonempty. In this range the potential $\theta(z) = \arctan(z/\sqrt{\epsilon})$ falls by at most

$$\Delta \frac{\sqrt{\epsilon}}{\epsilon + (z - \Delta)^2} \leq \frac{4\sqrt{\epsilon}}{a}.$$

Apply this separately to x and y until their minimum first falls below $4\epsilon/a$. The displayed number of rounds follows, with one round allowed for crossing the threshold. The graph is one-sided on this positive interval, so its limiting bound is zero and its fixed point at cutoff $-\epsilon$ is empty. $\square$

Proposition D.3 (Fast contraction at an exposed endpoint). *For $u = t^2$, $v = u^2$, $f = u - 2v$, use forward-cutoff-backward rounds with cutoff $-\epsilon$. If the initial upper bound on u satisfies $0 < U_0 < 1/2$, the box empties in $O(\log \log(1/\epsilon))$ rounds as $\epsilon \downarrow 0$.*

Proof. Each forward pass gives $v \leq U_k^2$. The backward sum pass gives $u \leq 2U_k^2 - \epsilon$, hence $2U_{k+1} \leq (2U_k)^2$ while nonempty. Induction yields $2U_k \leq (2U_0)^{2^k}$. The next pass is empty once $(2U_0)^{2^{k+1}} < 2\epsilon$, because $u \geq 0$. For $0 < \epsilon < 1/2$, taking $k = 1 + \max\{0, \lceil \log_2(\log(1/(2\epsilon))/\log(1/(2U_0))) \rceil\}$ is more than sufficient and has the stated order. $\square$

E Proofs for branching comparisons

E.1 Small certificates and the eleven-node example

The five-node assertion in [proposition 7.2](#). Let s be the breakpoint of a two-piece valid partition of $[L, U]$. If the root split is s , the tree has three nodes. By reflection suppose it is $y_1 > s$. Its right child is valid. If its left child is invalid, let y_2 be its selected minimizer. The crossing lemma in the proof of [theorem 7.1](#) excludes $y_2 > s$: it would force the root to cross its own right endpoint. If $y_2 = s$, both new children are valid. Otherwise $y_2 < s$, and $[L, y_2]$ is valid.

Suppose $[y_2, y_1]$ is invalid, witnessed by a point z with $M_\epsilon(z) < \alpha(z - y_2)(y_1 - z)$. Optimality of y_2 in $[L, y_1]$ gives

$$M_\epsilon(y_2) < \alpha[(z - y_2)(y_1 - z) + (y_2 - L)(y_1 - y_2) - (z - L)(y_1 - z)] = \alpha(y_2 - L)(z - y_2).$$

Validity of $[L, s]$ at y_2 therefore implies $z > s$. Optimality of y_1 in the root gives

$$M_\epsilon(y_1) < \alpha[(z - y_2)(y_1 - z) + (y_1 - L)(U - y_1) - (z - L)(U - z)] = \alpha(y_1 - z)[(U - y_1) - (y_2 - L)].$$

Validity of $[s, U]$ gives $M_\epsilon(y_1) \geq \alpha(y_1 - s)(U - y_1) > 0$. If the bracket on the right is nonpositive there is an immediate contradiction; otherwise dropping $y_2 - L > 0$ implies $z < s$. This contradiction proves the middle interval valid. There are at most two splits and five nodes. The information pair below attains five. $\square$

Here is an explicit example for the lower endpoint $11/5$. Take $\alpha = 1$, $\epsilon = 1/100$, and root $[0, 1]$. Interpolate H linearly between the knots below, giving it values $H(x) = x^2 + M(x)$ at those knots, and set $f(x) = H(x) - x^2 - \epsilon$:

x	0	1/16	3/16	7/16	1/2	9/16	13/16	15/16	1
$M(x)$	41/1600	41/1600	823/20000	1/100	1/100	1/100	823/20000	41/1600	41/1600

On each affine piece of H , f is concave. Its minimum is attained at a knot, so $f_* = 0$. The slopes of H are nondecreasing, and the shifted node function is $H(x) - (l + u)x + lu$, which is affine on each piece. Direct exact substitution gives the following invalid nodes and unique minimizers:

$[l, u]$	y	$\min(H - (l + u)x + lu)$
$[0, 1]$	1/2	-6/25
$[0, 1/2]$	3/16	-2791/160000
$[3/16, 1/2]$	7/16	-9/1600
$[1/2, 1]$	13/16	-2791/160000
$[1/2, 13/16]$	9/16	-9/1600

The six resulting leaves are valid: the shifted function is affine between knots, and its values at all contained knots and the leaf endpoints are nonnegative. Thus every minimizer selection uses eleven nodes. A valid three-piece certificate has breakpoints $2/5$ and $3/5$. To verify its validity, again evaluate $H - (l + u)x + lu$ at its endpoints and all contained knots. The side minima are $209/160000$, and the middle minimum is zero at $1/2$. No two-piece certificate is possible. Put $a = 12209/30000$. Testing its left interval at $3/16$ forces $s \leq a$ when $s \geq 3/16$; when $s < 3/16$ this conclusion holds automatically. Testing its right interval at $13/16$ forces $s \geq 1 - a$ when $s \leq 13/16$; otherwise this conclusion holds automatically. Since $1 - a > a$, the two requirements contradict each other. Hence $N_{\text{rect}} = 3$ and $T/T_{\text{opt}} = 11/5$.

E.2 Two indistinguishable root instances

Proof of proposition 7.3. Set $\alpha = 1$, $\epsilon = 1/10000$, and $C_{ab}(y) = (a + b)y - ab$. Define

$$\begin{aligned} g_-(y) &= 143/300 + (4/5)(y - 3/5), & g_+(y) &= 143/300 + (6/5)(y - 3/5), \\ e_0(y) &= \epsilon + y/6, & e_1(y) &= (5/3)y - 2/3 + \epsilon, \\ H_A &= \max\{C_{0,1/3}, C_{1/3,1}, g_-, g_+, e_0, e_1\}, \\ H_B &= \max\{C_{0,3/5}, C_{3/5,1}, g_-, g_+, e_0, e_1\}, \\ f_i &= H_i - y^2 - \epsilon \quad (i \in \{A, B\}). \end{aligned}$$

Both H_i are convex. On $[0, 1/6]$, $e_0 - y^2 \geq \epsilon$; on $[2/3, 1]$, $e_1 - y^2 \geq \epsilon$. On the two intervening intervals $[1/6, 3/5]$ and $[3/5, 2/3]$, use $g_- - y^2$ and $g_+ - y^2$ respectively. These are concave quadratics whose endpoint values exceed ϵ . Thus $f_i \geq 0$. At zero and one, $H_i = e_0$ and e_1 respectively, so both objectives have minimum zero.

The root shifted function is $H_i - y$. The common lower envelope $\max(g_-, g_+) - y$ has its unique minimum $-37/300$ at $3/5$, and both H_i equal this envelope there. Hence both root minimizers are uniquely $3/5$, and both actual relaxation bounds are $-37/300 - \epsilon$. Strict comparisons of the displayed affine lines show that both functions equal e_0 near zero, e_1 near one, and $\max(g_-, g_+)$ near $3/5$. Their root germs and endpoint neighborhoods therefore agree. All node underestimators are convex, since $f_{[l,u]} = H_i - (l + u)y + lu - \epsilon$.

For each instance, domination of its two chord lines gives a valid partition at its named breakpoint $c_A = 1/3$ or $c_B = 3/5$. Those breakpoints are unique. At $y = 1/200$, $H_i = C_{0,c_i}$ with strict domination over its other lines; at $y = 19/20$, $H_i = C_{c_i,1}$ with strict domination. If a two-piece partition has breakpoint s , validity at the first test point forces $s \leq c_i$ unless $s < 1/200$, which already satisfies this inequality. The second test forces $s \geq c_i$ unless $s > 19/20$, which already satisfies it. Thus $s = c_i$. The root is invalid. Root data force the same deterministic cut on both instances, and at least one cut is wrong. That instance needs at least five nodes, against an optimum of three. For a randomized cut, the probabilities of cutting exactly c_A and c_B sum to at most one. Some wrong-cut event has probability at least $1/2$, giving expected count at least $3 + 2(1/2) = 4$.

To obtain smooth nonanalytic functions, extend each maximum of lines to the real line and convolve with the same symmetric nonnegative C^∞ kernel of integral one, positive inside its support $[-\zeta, \zeta]$. Choose $\zeta > 0$ small enough that the endpoint neighborhoods and the two tight-chord test neighborhoods stay affine, and that the common neighborhood of $3/5$ contains the convolution window. Symmetry gives zero first moment, so convolution preserves these affine pieces. Jensen's inequality for convex H_i makes its convolution no smaller than H_i . Nonnegativity, endpoint minima, chord domination, and the distinct unique certificate cuts all persist. Near $3/5$ the common function is a line of slope one plus $(1/5)|y - 3/5|$. Its symmetric convolution minus y has derivative strictly increasing through zero at $3/5$, hence a unique minimum there. Convexity excludes any other minimum outside that neighborhood. The functions still agree on the needed open neighborhoods, and all root data still agree. The preceding information argument applies. Each smoothed H_i is affine on an endpoint neighborhood and nonaffine near the smoothed V at $3/5$. If it were analytic, continuation would force it to be affine throughout. Thus each smoothed H_i , and therefore each smoothed $f_i = H_i - y^2 - \epsilon$, is nonanalytic. $\square$

E.3 Repeated ambiguity across coordinates

Proof of proposition 7.6. Fix $\epsilon = 1/100$, $c = 1/(5n)$, and put

$$\delta = \frac{\epsilon}{4(n-1)}, \quad \varphi = (n-1)c - \epsilon/2, \quad \varphi' = c - \frac{\epsilon}{2(n-1)} = \frac{\varphi}{n-1}.$$

On $[0, 1]$ define

$$\begin{aligned}
h_-(t) &= 1/2 - c + (1 - \delta)(t - 1/2), \\
h_+(t) &= 1/2 - c + (1 + \delta)(t - 1/2), \\
H_R(t) &= \max\{h_-, h_+, t/2 + \varphi, 3t/2 - 1/2 + \varphi\}, \\
H_N(t) &= \max\{h_-, h_+, t/2 - \varphi', 3t/2 - 1/2 - \varphi'\}, \\
R(t) &= H_R(t) - t^2, \quad N(t) = H_N(t) - t^2, \\
f_i(x) &= R(x_i) + \sum_{j \neq i} N(x_j), \quad i = 1, \dots, n.
\end{aligned}$$

The coordinate underestimators are maxima of affine functions after subtracting q_J , and hence convex.

Both root coordinate relaxations have unique minimizer $1/2$ and value $-c$. Indeed at $1/2$ the common lines dominate the floor lines: the R margin is $1/4 + \epsilon/2 - nc = 11/200 > 0$, and the N margin is $1/4 - \epsilon/(2(n-1)) > 0$. Around this point $H - t$ has left slope $-\delta$ and right slope δ . Convexity makes its minimum unique. Thus $R = N$ near $1/2$. Near zero and one the respective floor lines dominate. At zero their strict margins above h_- are $nc - \epsilon/2 - \delta/2 > 0$ for R and $3\epsilon/(8(n-1)) > 0$ for N . The symmetric comparison holds at one. Consequently $R - N = \varphi + \varphi'$ near both endpoints.

The floor lines give $R \geq \varphi$ and $N \geq -\varphi'$ on both halves: subtracting t^2 leaves the nonnegative products $t(1/2 - t)$ and $(t - 1/2)(1 - t)$. Equality holds at the endpoints. Thus $\min R = \varphi$, $\min N = -\varphi'$, and $\min f_i = 0$. Furthermore the R relaxation value on either half is exactly φ , while on any interval containing zero or one the N relaxation value is at most $-\varphi'$ by evaluation at that endpoint. Cutting coordinate i at $1/2$ therefore makes both children valid: their shifted relaxation value is $\varphi - (n-1)c + \epsilon = \epsilon/2$. The root is invalid, since its shifted value is $-nc + \epsilon < 0$. Hence every f_i has a three-node optimum.

Call a box a corner box if each coordinate interval contains a root endpoint. Let S be its set of coordinates already cut; the others are free and still have interval $[0, 1]$. For every possible rigid label $i \notin S$, each used coordinate is an N coordinate, while each free coordinate has the same unique root minimizer and value whether it is R or N . Coordinate-wise selection therefore gives the same full minimizer and bound for all possible labels. At the minimizer the free coordinates lie at $1/2$, where $R = N$, so the objective germs also agree. Near each corner, the difference between two labelled objectives is $(R - N)(x_i) - (R - N)(x_{i'}) = 0$, because both differences are the same constant at root endpoints. The corner germs agree as well. Every such corner box is invalid: if $d = |S| \leq n - 1$, its shifted value is at most

$$-(n-d)c - d\varphi' + \epsilon = -nc + \epsilon + \frac{d\epsilon}{2(n-1)} \leq -1/5 + 3\epsilon/2 < 0.$$

Fix a deterministic local rule. Until the rigid label has been cut, the common corner-box data force the same decision for every remaining label. These decisions define a reference tree of corner boxes. Cutting a free coordinate creates two corner children and removes that label from the free set. Re-cutting a used coordinate has one corner child, with the same free set, and one noncorner child. If such a re-cut chain never ends while a free label remains, that instance has an infinite tree and satisfies the conclusion. Otherwise contract the re-cut chains. The contracted reference tree has at least 2^d branching nodes after d free coordinates have been cut.

Let I_i count reference corner nodes at which label i is still free. Every such node is reached and is internal in the actual tree for f_i , by induction and the invalidity just proved. Thus $T_i \geq 2I_i + 1$.

Each branching node at level d contributes to $n - d$ labels, so

$$\sum_i I_i \geq \sum_{d=0}^{n-1} 2^d (n - d) = 2^{n+1} - n - 2.$$

If the root cuts coordinate j , then $I_j = 1$. Averaging over the other $n - 1$ labels gives the deterministic bound. For a distribution of local rules, average this last sum inequality over the random rule and then over the n labels. At least one label has expected count at least $2(2^{n+1} - n - 2)/n + 1$. The reference construction uses node-locality: objective-dependent information from another subtree could identify the rigid label and invalidate the repeated indistinguishability argument. $\square$

E.4 Safety lower bounds and recentring

For the kink objective on $[0, 1]$, the derivative of the underestimator on the left of a is $-2\alpha + \alpha(2y - l - u) < 0$, and on its right is $2\alpha + \alpha(2y - l - u) > 0$. These signs also show that a one-sided interval has its minimizer at the endpoint nearest a and a nonnegative value. Thus the invalidity test and off-chain validity used in [propositions 7.4](#) and [7.5](#) are exact.

Proof of [proposition 7.5](#). For $a < \theta$, every one of the first J cuts on its containing chain lies to its right. Indeed after k cuts the left endpoint is zero and the upper endpoint is at least θ^k . For $k < J$, $\theta^{k+1} > a$, so every admissible cut exceeds a . Also $(a - l)(u - a) > a^2(1 - \theta)/\theta$ at those nodes. The tolerance hypothesis makes each node invalid, proving $T \geq 2J + 1$. A chain that hits a needs one further split. This proof holds samplewise for randomized rules. Choosing $a = \sqrt{2\theta\epsilon/(\alpha(1 - \theta))} < \theta$ for sufficiently small ϵ makes the node count logarithmic in $1/\epsilon$, while $T_{\text{opt}} = 3$.

The recentring rule is safe: its offset from the nearby endpoint is between θw and $2\theta w$, and $2\theta \leq 1 - \theta$. Let $d = \min\{(a - l)/w, (u - a)/w\}$. If $d < \theta/2$, an ordinary clamp leaves a containing child of width θw and new distance $d/\theta < 1/2$. If $\theta/2 \leq d < \theta$, the offset $2dw$ makes the kink the midpoint of its containing child. If $d \geq \theta$, the next cut hits the kink. Put $m = \min\{j \geq 0 : d_0 \geq \theta^{j+1}/2\}$. After m ordinary clamps the distance is at least $\theta/2$. If it is at least θ , then $m = J$ and one more cut hits the kink. Otherwise $J = m + 1$: $d_0 < \theta^{m+1}$, but $d_0 \geq \theta^{m+1}/2 \geq \theta^{m+2}$. One recentring cut and one exact kink cut finish. In either case the zero-tolerance chain takes exactly $J + 1$ splits. The preceding safety argument, reflected if necessary, makes this the minimum hitting count. For any positive tolerance the chain can stop earlier, giving $T \leq 2J + 3$; for every fixed kink equality holds at all sufficiently small positive tolerances, because its finitely many pre-hit gap products are positive. $\square$

E.5 Why splitting all coordinates can overrefine

Proposition E.1 (A two-dimensional multisection separation). *For $f(x, z) = x^2$ on $[0, 1]^2$ with $\alpha = 1$, multisection at all interior exact coordinate minimizers has $\Omega(\epsilon^{-1/2} \log(1/\epsilon))$ internal nodes, whereas $N_{\text{tree}} = O(\epsilon^{-1/2})$. Count every processed box, whether a split has two or four children.*

Proof. The coordinate relaxation values are

$$F_x([l, u]) = \begin{cases} lu - (l + u)^2/8, & u > 3l, \\ l^2, & u \leq 3l, \end{cases} \quad F_z([c, d]) = -(d - c)^2/4.$$

The selected x minimizer is $(l + u)/4$ in the first case and l otherwise; the z minimizer is always the midpoint.

For an upper certificate put $h = 2\sqrt{\epsilon}$ and use x strips $[0, h]$, $[h, 2h]$, $[2h, 4h]$, and so on, truncating at one. The first strip has shifted x value $\epsilon/2$, so z intervals of width at most $\sqrt{2\epsilon}$ suffice. Each later strip $[l, u]$ has $u \leq 2l$, hence $F_x = l^2$, and z intervals of width at most $2l$ suffice. Their counts sum to at most $\lceil 1/\sqrt{2\epsilon} \rceil + 1/(2\sqrt{\epsilon}) + O(\log(1/\epsilon)) = O(\epsilon^{-1/2})$. These cuts form a guillotine certificate.

In the multisection tree every depth- D box has a z interval of width 2^{-D} . For each x history all 2^D such intervals occur together, since validity depends only on the z width. The left x spine is $[0, 4^{-k}]$. Its right sibling $J_k = [l, 4l]$, $l = 4^{-(k+1)}$, appears at depth $k+1$. Inside J_k , let $r_0 = l$ and $r_{j+1} = (r_j + 4l)/4$. The right descendants $[r_j, 4l]$ split at r_{j+1} , with $r_j \uparrow 4l/3$. Their left pieces $L_j = [r_{j-1}, r_j]$ appear at depth $k+1+j$ and are never split in x again. Their shifted x values are strictly below $\epsilon + 16^{-k}/9$. At depth $D \leq 2k$, the z value is at most $-16^{-k}/4$. All these boxes are therefore invalid when $\epsilon \leq (5/36)16^{-k}$; the strict x estimate handles equality. Hereditary validity ensures that their ancestors are also invalid, so the indicated boxes really occur. With $K = \lfloor \log_{16}(5/(36\epsilon)) \rfloor$, count only depth $2k$ and $j = 1, \dots, k-1$. These disjoint histories give at least

$$\sum_{k=2}^K (k-1)4^k \geq (K-1)4^K = \Omega(\epsilon^{-1/2} \log(1/\epsilon))$$

internal nodes as $\epsilon \downarrow 0$. A tree with at least two children per internal node has at least $2I + 1$ total nodes. The binary certificate above has $2N - 1$ nodes, proving the unbounded ratio. $\square$

E.6 A phase bound and the sharp-coordinate induction

Fix a continuous $m \geq 0$ on an interval A , with a fixed exact minimizer selection for every subinterval. Define $F(J) = \min_J(m - q_J)$ and $\omega(J) = q_J(y_J)$. The one-dimensional split-count argument applies unchanged to a continuous $m \geq 0$ at any budget $b > 0$: it needs only a positive shifted objective, and does not require a zero of m on the current interval. Let $\tau > 0$, let $\kappa \geq 1$ be an integer, take $\beta \in \mathbb{R}$, and put $b = \beta + \kappa\tau > 0$. Define a pruned minimizer tree S that splits J precisely when $F(J) < -\beta$ and $\omega(J) \geq \tau$.

Lemma E.2 (Phase count). *Let $N = N_A(b)$, and put $c_\kappa = \kappa[2\kappa(\kappa+1)+1]$. The number of internal nodes of S is at most $4N - 5 + c_\kappa(4N - 4)$ when $N \geq 2$, and at most c_κ when $N = 1$.*

Proof. First take an interval $\ell = [p_0, q_0]$ with $F(\ell) \geq -b$. For a split node $J = [p, q] \subseteq \ell$ with selected point y , write $L = y - p$, $R = q - y$, $g_L = p - p_0$, $g_R = q_0 - q$. Validity of ℓ at budget b and the split conditions imply

$$Lg_R + Rg_L + g_Lg_R < \kappa\tau \leq \kappa LR. \quad (186)$$

Thus $g_R < \kappa R$, $g_L < \kappa L$, and $|J| > |\ell|/(\kappa+1)$. On a nested chain of split nodes contained in ℓ , each left-child step removes a right piece of length $R \geq \tau/|\ell|$. After n_L such steps $g_R \geq n_L\tau/|\ell|$. At a further left-child step, (186) gives $n_LL < \kappa|\ell|$, and its split child has $L > |\ell|/(\kappa+1)$. Hence there are at most $\kappa(\kappa+1)$ left steps, and the same number of right steps. A split-node chain contains at most $2\kappa(\kappa+1)+1$ nodes. For any finite portion of the split-node forest inside ℓ , terminal split boxes are disjoint in their interiors and all longer than $|\ell|/(\kappa+1)$, so there are at most κ of them. The forest is covered by their ancestor chains. There are at most c_κ split nodes inside ℓ . This applies to every finite partial tree, so also excludes an infinite forest.

Let T_b be the ordinary minimizer tree stopped when $F \geq -b$. Any split node of S with $F < -b$ belongs to T_b , because all ancestors also have $F < -b$ and the selections agree. There are at most $4N - 5$ such nodes by the one-dimensional theorem. Every remaining node lies inside the first ancestor with $F \geq -b$, a leaf of T_b . That tree has at most $4N - 4$ leaves when $N \geq 2$; each contains

at most c_κ additional split nodes. When $N = 1$, the root itself is budget- b valid and the forest bound applies directly. $\square$

Proof of theorem 7.7. After a sharp coordinate is cut at its kink, each half has relaxation value exactly zero: it contains a zero-valued kink, and the one-sided hypothesis gives a nonnegative minimum. Its selected gap is zero. It can never be selected on an invalid node while any positive gap remains; some positive gap must remain because all $m_i \geq 0$.

We prove the leaf bound by induction on the number r of unsplit sharp coordinates, allowing already-split sharp coordinates as above. For $r = 0$, the one-dimensional theorem gives at most $4N_A(\epsilon)$ leaves, including the case $N_A = 1$. For $r \geq 1$, freeze the unsplit sharp intervals until the first sharp split on each branch. Their gaps $\omega_j > 0$ are constants and their values are $-\omega_j$. Put

$$\tau = \max_j \omega_j, \quad \beta = \epsilon - \sum_j \omega_j, \quad b = \beta + r\tau \geq \epsilon.$$

Every arbitrary-coordinate split in this initial phase satisfies $F(A') < -\beta$ and $\omega(A') \geq \tau$, so its phase is a subtree of the phase tree in lemma E.2 with $\kappa = r$. Since $N_A(b) \leq N_A(\epsilon) = N$, that lemma bounds its internal nodes by $A_r N$, also when $N_A(b) = 1$. Its frontier partitions A into $H \leq (A_r + 1)N$ intervals A_j .

Refine an optimal budget- ϵ partition of A by these $H - 1$ frontier breakpoints. Hereditary validity means every refined piece is still valid. Therefore

$$\sum_j N_{A_j}(\epsilon) \leq N + H - 1 \leq (A_r + 2)N. \quad (187)$$

A frontier node is either valid or splits one sharp coordinate at its kink. In the latter case its two children have $r - 1$ unsplit sharp coordinates and the same arbitrary interval A_j . Apply induction to them. Overcounting valid frontier nodes by H gives a final leaf bound

$$H + 2C_{r-1} \sum_j N_{A_j}(\epsilon) \leq [(A_r + 1) + 2C_{r-1}(A_r + 2)]N = C_r N.$$

The phase lemma and induction prove termination as well.

Finally slice any original rectangular certificate at every sharp kink. On a box meeting that slice, the other coordinate values are zero and their gap terms are nonnegative, so its arbitrary-coordinate interval is valid at budget ϵ . These intervals cover A . An interval cover by hereditary-valid intervals gives a partition with no more pieces: repeatedly select, among covering intervals containing the current endpoint, one with the farthest right endpoint, and take the corresponding subinterval. Finite covers guarantee progress, and no covering interval can be selected twice. Thus $N_A(\epsilon) \leq N_{\text{rect}}(\epsilon)$, proving the claimed node comparison. Every phase and selection in this proof uses the fixed coordinate-wise oracle; sharpness for a different tied minimizer cannot be inferred from sharpness for the selected one. $\square$

F Proofs for admissible integer classes

F.1 Hemispace separation and semantic attainment

We first prove the separation fact needed when a projected convex value function has a nonopen strict sublevel set. A hemispace in an affine space V is a convex subset of V whose complement in V is also convex. It may assign different parts of a supporting hyperplane to its two sides.

Lemma F.1 (Hemisphere separation). *If A, B are disjoint convex subsets of a finite-dimensional affine space V , there is a hemisphere $G \subseteq V$ with $A \subseteq G$ and $G \cap B = \emptyset$.*

Proof. We prove the assertion by induction on $\dim V$. If $A = \emptyset$, take $G = \emptyset$; if $B = \emptyset$, take $G = V$. These cases also settle dimension zero. Suppose that both are nonempty and $\dim V > 0$.

Identify the direction space of V with a Euclidean space. There is a nonzero linear functional g on it and a finite α such that $g(a) \geq \alpha \geq g(b)$ for all $a \in A, b \in B$. Here is the finite-dimensional separation argument, including its boundary case. The convex set $C = A - B$ does not contain zero. If $0 \notin \text{cl } C$, the nearest point p of the nonempty closed convex set $\text{cl } C$ to zero satisfies $\langle p, c \rangle \geq \|p\|^2 > 0$ for every $c \in C$. If $0 \in \text{cl } C$ and $\text{cl } C$ lies in a proper linear subspace, any nonzero functional orthogonal to that subspace gives $g(c) = 0$. In the remaining case $\text{cl } C$ is full-dimensional, and zero is a boundary point of $\text{cl } C$. To justify the latter assertion, an interior point of the closure of a full-dimensional convex set lies in that set: choose a small simplex around the point in the interior of the closure, approximate its vertices by points of the convex set, and keep the point in the interior of the approximating simplex. Convexity then includes the point. Since $0 \notin C$, it cannot be interior to $\text{cl } C$. Choose $q_j \notin \text{cl } C$ tending to zero, and let p_j be its nearest point in $\text{cl } C$. Since $0 \in \text{cl } C$, $p_j \rightarrow 0$. The unit vectors $(p_j - q_j)/\|p_j - q_j\|$ have a convergent subsequence, with unit limit v . The projection inequalities $\langle p_j - q_j, c - p_j \rangle \geq 0$ yield $\langle v, c \rangle \geq 0$ for all $c \in C$. Thus in every case there is a nonzero g with $g(a) \geq g(b)$ for all a, b . Fixing one point of each set shows that $\sup_B g$ and $\inf_A g$ are finite; choose α between them. On the affine space, g is interpreted as a nonconstant affine functional after choosing an origin.

Let $H = \{g = \alpha\}$. It has dimension one less than V . By induction, there is a hemisphere G' of H containing $A \cap H$ and missing $B \cap H$. Put

$$G = \{g > \alpha\} \cup G'.$$

For a strict convex combination of two points of G , either one point has $g > \alpha$ and the combination does too, or both lie in the convex set G' . Hence G is convex. Its complement is $\{g < \alpha\} \cup (H \setminus G')$, which is convex by the same argument. The points of A either have $g > \alpha$ or belong to $A \cap H \subseteq G'$. The points of B have $g \leq \alpha$, and those on H miss G' . This proves the induction. $\square$

Attainment in theorem 8.3. Take an admissible partition $I_1, \dots, I_k$. The strict sublevel set $E_\tau = \{x : \phi(x) < \tau\}$ is convex; for $\tau = +\infty$ it is the effective domain of the proper projected function. Admissibility means $\text{conv } I_j \cap E_\tau = \emptyset$. A depth-one tree whose children are these convex hulls covers P and has k certified leaves.

For a binary tree, use lemma F.1 to find hemispaces $G_j \supseteq \text{conv } I_j$ with $G_j \cap E_\tau = \emptyset$. When $k = 1$, G_1 itself is a valid root and a certified leaf. It contains $P = I_1$ and gives one node.

When $k \geq 2$, begin with any convex root N_0 containing P . For $j = 1, \dots, k-2$, split N_{j-1} into

$$N_{j-1} \cap G_j \quad \text{and} \quad N_j = N_{j-1} \cap (\mathbb{R}^n \setminus G_j).$$

The first child is a leaf and the second continues the chain. Both are convex, nested in their parent, and together cover that parent. At the last internal node N_{k-2} , use the two leaves $N_{k-2} \cap G_{k-1}$ and $N_{k-2} \cap G_k$. To check their required coverage, take $p \in P \cap N_{k-2}$. It is outside $G_1, \dots, G_{k-2}$, and hence outside $I_1, \dots, I_{k-2}$. Because the I_j partition P , $p \in I_{k-1} \cup I_k \subseteq G_{k-1} \cup G_k$. This verifies (87), including its restriction to integer points at the final split. Every leaf lies in a G_j , so it misses E_τ and has bound at least τ . The full binary chain has k leaves and $2k-1$ nodes. If a constructed leaf were empty of required points, routing to the others would contradict minimality of k .

For finite P , replace each node set S by $S' = \text{conv}(S \cap P)$. This is a polytope or the empty set. Convexity gives $S' \subseteq S$, while $S' \cap P = S \cap P$: one inclusion follows from $S' \subseteq S$, the other because

every point of $S \cap P$ is among the hull's generators. Thus every node's required points and every coverage relation are unchanged. Effective sets only shrink, so leaf bounds cannot decrease. This gives the claimed polyhedral certificate. $\square$

If E_τ is nonempty and open, its disjoint convex class hulls can be separated by closed halfspaces. Indeed, weak separation gives $g^T x \geq \alpha$ on the class hull and $g^T x \leq \alpha$ on E_τ ; openness and $g \neq 0$ make the latter inequality strict. Thus a closed halfspace containing the hull misses E_τ . If ϕ is differentiable and its minimum on a class hull is attained at $\hat{x}$, first-order optimality gives the explicit halfspace

$$\{x : \nabla \phi(\hat{x})^T (x - \hat{x}) \geq 0\}.$$

Convexity puts its value at least $\phi(\hat{x}) \geq \tau$ throughout that halfspace. If the gradient is zero, the entire space has value at least τ , so the class number is one. These special forms do not replace the hemispace proof when E_τ is nonopen.

F.2 Operation accounting

Proof of theorem 8.4. Normalize the run's node sets to their effective sets. For part (a), route required integer points to leaves as in the class-number lower-bound proof. A routed point is required at every ancestor. It belongs to each convex cut added on its path, by cut validity at the node where it was added. Thus the convex hull of the points assigned to a leaf is contained in the leaf's relaxation domain after all such cuts. The pruning bound makes that hull admissible for the original ϕ . The same argument applies to a convex reduction preserving the required integer points. If the cuts preserve only feasible integer projections, use $P = F$; it is not legitimate to retain a larger required set without its preservation hypothesis.

For part (b), replace each reduction sequence at a node by a chain. At stage i , let $Q^{(i-1)}$ have the two children $Q^{(i-1)} \cap D_i$ and $Q^{(i)}$. The first is a certified leaf, and the second continues. Both are convex and nested, and (91) is precisely their integer-coverage condition. Attach the run's original children to the last retained set, or make it a leaf if the run pruned it. The expanded tree has at most $L + S$ leaves and is a P -covering τ certificate. Applying theorem 8.3 proves (92). Simultaneous certificates instead give extra leaf children at the unreduced node, with the same count.

In one pass, at most $2n$ bounds change per node, so $\kappa_\tau(P) \leq L + 2nN \leq (2n + 1)N$. A pass at the root alone gives $\kappa_\tau(P) \leq L + 2n$.

For binaries, consider integer-rounded changes which remove a remaining binary value. Before a node empties, each free variable can be fixed only once, giving at most n removals. A further change of an already fixed variable empties the node; such a run node is a leaf and has at most $n + 1$ removals. Its final retained set has no required integer points, so the expansion can drop that retained leaf: at the last step all required points belong to the certified removed child. If e run leaves are emptied, the expanded leaf count is at most $L + S - e \leq L + nN$. Hence $\kappa_\tau(P) \leq L + nN \leq (n + 1)N$.

For general integer bounds, each genuine rounded change shortens the sum of the integer widths by at least one while the node stays nonempty. There can be up to $\sum_i (u_i - \ell_i)$ such changes and one final emptying change at that node. This depends on its widths rather than only its dimension, and supplies no dimension-only estimate for iterated passes. $\square$

If an enumeration node creates children only for a finite integer range $a \leq \pi^T x \leq b$, and its omitted values have relaxation bound at least τ , append the two certified tails $\pi^T x \leq a - 1$ and $\pi^T x \geq b + 1$. This converts the run into a P -covering certificate. There are at most two added leaves per internal node, so its expanded leaf count is at most $L + 2N_{\text{int}} \leq 3N$. Consequently the

original run has at least $\kappa_\tau(P)/3$ nodes. The validity of the tail bounds is an assumption about the enumeration's radius tests, not an automatic consequence of skipping a value.

F.3 The lattice-free facet interpretation

Proof of [theorem 8.7](#). A finite convex function on $\mathbb{R}^n$ is continuous. Thus the nonempty strict sublevel set E_τ is open and convex. Each admissible class hull is disjoint from it. The open-set separation argument above supplies a closed halfspace containing that hull and missing E_τ . An admissible partition therefore gives a halfspace cover of the integer lattice. Conversely, the integer points of any closed halfspace missing E_τ are admissible, because their hull stays in the halfspace. This identifies the first two minima.

Write a halfspace cover as $H_j = \{x : g_j^T x \geq \alpha_j\}$, with $g_j \neq 0$. Its open complementary polyhedron

$$O = \{x : g_j^T x < \alpha_j \text{ for all } j\}$$

contains E_τ and has no integer point. Because it is nonempty and open, its closure is $M = \{x : g_j^T x \leq \alpha_j \text{ for all } j\}$ and $\text{int } M = O$. To see closure equality, fix $x_0 \in O$; the segment from any point of the closed system to x_0 satisfies every row strictly except possibly at its first endpoint. Each genuine facet of M comes from one of these rows, so there are no more facets than covering halfspaces. Conversely, the closed complementary halfspaces of the facets of any full-dimensional polyhedron satisfying [\(93\)](#) cover $\mathbb{Z}^n$ and miss E_τ . The minima therefore agree.

For the bound 2^n , use the maximal lattice-free polyhedron theorem ([Basu et al., 2010](#), Theorem 2): every nonempty open convex set avoiding $\mathbb{Z}^n$ is contained in a full-dimensional maximal lattice-free convex set; every such maximal set is a polyhedron with at most 2^n facets. This is the standard external input in this argument. Since E_τ is open, its inclusion in that polyhedron places it in the interior. Its facet complements give the desired admissible cover. If E_τ is empty, the whole lattice is one admissible class. $\square$

One constrained variant also follows from the same standard input. Suppose ϕ is lower semi-continuous, $\phi(z) \geq \text{OPT}$ for every $z \in \mathbb{Z}^n$, $\tau < \text{OPT}$, and the closed sublevel set $C_\tau = \{\phi \leq \tau\}$ is bounded. It is compact, convex, and integer-free. If nonempty, its distance η from $\mathbb{Z}^n$ is positive. The open convex set $C_\tau + B^\circ(0, \eta/2)$ is lattice-free. Facet complements of a maximal lattice-free polyhedron enclosing this set cover the lattice and avoid C_τ , so $\kappa_\tau(\mathbb{Z}^n) \leq 2^n$ here as well. Lower semicontinuity and the strict threshold are needed in this compact-neighborhood argument.

There is also an elementary parity bound on midpoint cliques whenever $\phi(z) \geq \tau$ at every integer point: two points congruent modulo two have an integer midpoint and cannot conflict. Every midpoint clique then has at most 2^n vertices. This does not bound the segment graph or the class number without additional hypotheses.

F.4 Random-lattice moments and scale bounds

The probabilistic proofs use the following standard external theorem, with the normalization of [\(94\)](#): for a nonnegative integrable test function h on $\mathbb{R}^n$ and Haar-random unimodular L , $n \geq 2$,

$$\mathbb{E}_L \sum_{a \in L \setminus \{0\}} h(a) = \int_{\mathbb{R}^n} h(a) da. \quad (188)$$

This is Siegel's mean-value theorem; see [Skenderi \(2021, Theorem 3.2 and Remark 3.3\)](#) for this normalization and test-function class. All moment statements used here follow from [\(188\)](#) and the elementary torus calculation below.

Let $A \subset \mathbb{R}^n$ be bounded and measurable, with volume V , and write $N_A = \#\{v \in L : v - t \in A\}$. A fundamental domain D for L has volume one. Its translates $v - D$, $v \in L$, tile $\mathbb{R}^n$ up to null sets. Tonelli's theorem gives, for any nonnegative measurable h ,

$$\mathbb{E}_t \sum_{v \in L} h(v - t) = \sum_{v \in L} \int_{v-D} h(x) dx = \int h(x) dx. \quad (189)$$

For two such functions h_1, h_2 , write the second lattice point as $v + a$ and apply (189) to obtain

$$\mathbb{E}_t \left(\sum_v h_1(v - t) \right) \left(\sum_w h_2(w - t) \right) = \sum_{a \in L} \int h_1(x) h_2(x + a) dx. \quad (190)$$

With both functions equal to $\mathbf{1}_A$, the $a = 0$ term is V . Applying (188) to the other terms gives

$$\mathbb{E}_{L,t} N_A = V, \quad \mathbb{E}_{L,t} N_A^2 = V + V^2, \quad \text{Var}_{L,t}(N_A) = V. \quad (191)$$

The integral of the intersection-volume function equals V^2 , again by Tonelli. Similarly, the expected number of ordered distinct pairs in A at distance less than s is

$$\int_{\|a\| < s} \text{vol}(A \cap (A - a)) da. \quad (192)$$

Since a ball of radius rg_n has volume r^n , Markov's inequality and (191) give

$$\mathbb{P}\{\text{dist}(t, L) < (1 - \delta)g_n\} \leq (1 - \delta)^n, \quad (193)$$

$$\mathbb{P}\{\text{dist}(t, L) > (1 + \delta)g_n\} \leq (1 + \delta)^{-n}. \quad (194)$$

For the second inequality, a ball of volume $V = (1 + \delta)^n$ has no lattice point on the event in question; Chebyshev bounds that void event by V/V^2 . Nonzero lattice vectors occur in opposite pairs, so (188) and Markov also give

$$\mathbb{P}\{\lambda_1(L) < (1 - \delta)g_n\} \leq \frac{1}{2}(1 - \delta)^n. \quad (195)$$

No independence between these three events is used.

F.5 Cap occupancy and class bounds

Proof of lemma 8.8. Consider the finite subset of lattice points from an admissible class in $B^\circ(t, R)$, translated by $-t$. If it is empty there is nothing to prove. Its compact convex hull has a point p of minimum norm. Since it lies in the full class hull, admissibility gives $\|p\| \geq \sqrt{\tau} > 0$. For every point u in the finite subset, differentiation along the segment from p to u gives $\langle p, u - p \rangle \geq 0$. Put $e = p/\|p\|$. Then $\langle e, u \rangle \geq \|p\| \geq \sqrt{\tau}$, and hence

$$\|u - \sqrt{\tau}e\|^2 = \|u\|^2 - 2\sqrt{\tau}\langle e, u \rangle + \tau < R^2 - \tau \leq \lambda_1(L)^2/2.$$

Distinct such points are separated by at least $\lambda_1(L)$. Their vectors v, w from the center $\sqrt{\tau}e$ therefore obey

$$\langle v, w \rangle = \frac{1}{2}(\|v\|^2 + \|w\|^2 - \|v - w\|^2) < 0.$$

If one vector is zero, it is the only point, since the enclosing ball has radius less than $\lambda_1(L)$. Otherwise there are at most $n + 1$ vectors with pairwise strictly negative inner products. To prove this, suppose $n + 2$ existed. Their affine dependence has coefficients summing to zero, with both positive and negative coefficients. Splitting these coefficients gives disjoint nonempty index sets S, T and positive coefficients with $w_0 = \sum_{i \in S} \alpha_i v_i = \sum_{j \in T} \beta_j v_j$. Taking the inner product of the two expressions yields $\|w_0\|^2 = \sum_{i \in S, j \in T} \alpha_i \beta_j \langle v_i, v_j \rangle < 0$, a contradiction. Finally apply (90) to all the lattice points in the outer ball. $\square$

Lower bound in [theorem 8.9](#). Take the outer ball of radius $g_n R_-$, of volume $V = R_-^n$. On the events

$$\text{OPT} \geq (1 - \delta)^2 g_n^2, \quad \lambda_1(L) \geq (1 - \delta) g_n, \quad N_{B^\circ(0, g_n R_-)} \geq V/2,$$

we have $\tau > 0$ and

$$g_n^2 R_-^2 - \tau \leq g_n^2 [\tfrac{3}{2}(1 - \delta)^2 - q - (1 - \delta)^2 + q] = \tfrac{1}{2}(1 - \delta)^2 g_n^2 \leq \lambda_1(L)^2/2.$$

The cap lemma gives $\kappa_\tau \geq V/[2(n+1)]$. The first two failure probabilities total at most $\frac{3}{2}(1 - \delta)^n$ by [\(193\)](#) and [\(195\)](#). Chebyshev and [\(191\)](#) bound the third by $4/V$. Their union bound proves [\(96\)](#). $\square$

Lemma F.2 (An elementary spherical cap cover). *For θ in a compact subinterval of $(0, \pi/2)$, the unit sphere S^{n-1} has a cover by at most $Cn^2 \log(n+1)(\sin \theta)^{-n}$ spherical caps of angular radius θ , with C uniform over that interval.*

Proof. Take a maximal chordal $1/n$ -separated subset Y of the sphere. The Euclidean balls of radius $1/(2n)$ about its points are disjoint and lie in the ball of radius $1 + 1/(2n)$. Comparing their volumes gives $|Y| \leq (1 + 2n)^n$. Maximality makes it a chordal $1/n$ -net. For unit vectors, an angular separation α has chord length $2 \sin(\alpha/2)$, and $\arcsin x \leq \pi x/2$ on $[0, 1]$; the angular mesh is therefore at most $\pi/(2n)$.

Let $\varphi = \theta - \pi/(2n)$. For sufficiently large n it exceeds $1/n$ and is less than $\pi/2$. A uniform random cap center covers a fixed point within angle φ with probability

$$\sigma_n(\varphi) = \frac{\int_0^\varphi \sin^{n-2} u \, du}{\int_0^\pi \sin^{n-2} u \, du} \geq \frac{\sin^{n-2}(\varphi - 1/n)}{\pi n}.$$

For M independent centers, a fixed net point is missed with probability at most $e^{-M\sigma_n(\varphi)}$. Choose $M > n \log(1 + 2n)/\sigma_n(\varphi)$. A union bound shows that some choice of centers covers every net point within angle φ . The angular mesh then covers the full sphere within θ . Uniformly over the specified angle interval,

$$\sin^{-n} \left(\theta - \frac{\pi/2 + 1}{n} \right) \leq C_0 \sin^{-n} \theta$$

by the mean-value bound for $\log \sin \theta$ on a slightly larger compact interval. Combining the estimates gives the stated M bound. Increase C to cover the finitely many smaller dimensions, using an ordinary finite angular net there. $\square$

Upper bound in [theorem 8.9](#). If $\tau \leq 0$, the nonnegative quadratic objective makes the entire lattice one admissible class. Suppose $\tau > 0$. In Euclidean coordinates centered at the target, each halfspace

$$H_e = \{u : \langle e, u \rangle \geq \sqrt{\tau}\}, \quad \|e\| = 1,$$

is admissible. Every lattice point $u = v - t$ has norm at least $\sqrt{\text{OPT}} \geq \sqrt{\tau}$, so the halfspace with $e = u/\|u\|$ covers it.

Put $a_+ = (1 + \delta)^2 - q > 0$ and $R_+^2 = 1 + a_+$. On $\text{OPT} \leq (1 + \delta)^2 g_n^2$, $\tau \leq a_+ g_n^2$. Cover each point with $\|u\| < g_n R_+$ by its own halfspace. The number of these points has mean R_+^n , so it is at most $e^{\gamma n R_+^n}$ except on an event of probability $e^{-\gamma n}$.

Choose θ by $\sin \theta = 1/R_+$. A far point $\|u\| \geq g_n R_+$ whose direction is within θ of e satisfies

$$\langle e, u \rangle \geq g_n R_+ \cos \theta = g_n \sqrt{a_+} \geq \sqrt{\tau}.$$

A cap cover supplied by [lemma F.2](#) therefore covers all far lattice points by at most $Cn^2 \log(n+1) R_+^n$ admissible halfspaces. When $0 \leq q \leq 1/2$ and $0 < \delta < 0.1$, these angles lie in a fixed compact

subinterval of $(0, \pi/2)$, so C is absolute. Combining the near and far covers gives (97). Equation (194) supplies the remaining failure probability.

For (98), take $\delta_n \downarrow 0$ and $\gamma_n \downarrow 0$ slowly enough that $n\delta_n, n\gamma_n \rightarrow \infty$, for example $n^{-1/3}$, and use $q_n \rightarrow q_0 < 1/2$. The lower outer-ball volume is exponential, and its base tends to $\sqrt{3/2 - q_0} > 1$. The upper base tends to $\sqrt{2 - q_0}$, while its prefactor has logarithm $o(n)$. The explicit failure bounds tend to zero, proving the assertion. $\square$

F.6 Midpoint clique estimate

Proof of proposition 8.10. Divide all Euclidean lengths by g_n . A ball of normalized radius r then has volume r^n under the rescaled, unit-ball-normalized measure. Write $R^2 = \rho r_0^2$ and $s^2 = 4(\rho - 1)r_0^2$. On $\text{OPT}/g_n^2 \geq (1 - \delta)^2$, a pair of translated lattice points u, w is conflicting if $\|u + w\| < 2r_0$. If $\|u\|, \|w\| < R$ and they are not conflicting, the parallelogram identity yields

$$4r_0^2 \leq \|u + w\|^2 = 2\|u\|^2 + 2\|w\|^2 - \|u - w\|^2 < 4R^2 - \|u - w\|^2.$$

Thus every nonconflicting pair in this ball has distance less than s . Let N be its point count and let M count all unordered distinct pairs in the ball at distance less than s . Removing one endpoint from each such pair leaves a midpoint clique of size at least $N - M$ on the stated radius event.

Equations (192) and symmetry give $\mathbb{E}M = J/2$, where, in the normalized coordinates,

$$J = \int_{\|a\| < s} \text{vol}(B_R \cap (B_R - a)) da = \int_{B_R} \text{vol}(B_R \cap B(x, s)) dx.$$

The normalized measure here means the original physical volume after rescaling; its unit ball has measure one. For $z \in B_R \cap B(x, s)$ and $0 \leq \eta \leq 1$, averaging the squared-distance inequalities gives

$$\|z - \eta x\|^2 + \eta(1 - \eta)\|x\|^2 < (1 - \eta)R^2 + \eta s^2.$$

Consequently the intersection volume is at most $(A - b\|x\|^2)^{n/2}$, where

$$\eta = \frac{2 - \rho}{\rho}, \quad A = (1 - \eta)R^2 + \eta s^2, \quad b = \eta(1 - \eta), \quad a^2 = A - bR^2 = \frac{4(\rho - 1)}{\rho}r_0^2.$$

The function $r^{n-1}(A - br^2)^{n/2}$ is increasing for $0 \leq r \leq R$. Indeed, its logarithmic derivative is nonnegative exactly when $(n-1)A \geq (2n-1)br^2$; it suffices to check $r = R$, giving $bR^2 \leq (1 - 1/n)a^2$. Substitution reduces this to $\rho \geq 2/n$, which holds for $n \geq 2$ and $\rho > 1$. Radial integration therefore gives

$$J \leq n \int_0^R r^{n-1}(A - br^2)^{n/2} dr \leq nR^n a^n = nV a^n.$$

Chebyshev gives $\mathbb{P}\{N < V/2\} \leq 4/V$, and Markov gives

$$\mathbb{P}\{M \geq V/4\} \leq \frac{4\mathbb{E}M}{V} \leq 2na^n \leq 2n \left(\frac{4(\rho - 1)}{\rho} \right)^{n/2},$$

using $r_0 \leq 1$. Together with (193), these prove the displayed probability bound and the clique size $V/4$.

For $q_n = o(1)$, choose $\delta_n = \max\{q_n, n^{-1/3}\}$ and $\rho_n = 4/3 - n^{-1/3}$ for sufficiently large n . Their limiting outer volume base is $\sqrt{4/3}$; all three failure bounds tend to zero. Polynomial prefactors and the factor $1/4$ give only $o(n)$ in the logarithm, proving the limiting exponent. $\square$

F.7 The two-feature curvature construction

For a block with feature matrix $X = [u \ v]$, the projected perspective value is

$$g(z) = \inf_{\beta} \left\{ \|y - X\beta\|^2 + \lambda \sum_{i=1}^2 \frac{\beta_i^2}{z_i} \right\} = y^T (I + X \operatorname{diag}(z) X^T / \lambda)^{-1} y, \quad (196)$$

where $\beta_i^2/0$ is zero for $\beta_i = 0$ and $+\infty$ otherwise. To prove the identity, restrict to the positive coordinates of z and minimize the resulting strictly convex quadratic. Its value is $y^T y - y^T X_S (X_S^T X_S + \lambda D^{-1})^{-1} X_S^T y$, where $D = \operatorname{diag}(z_S)$. Multiplying the matrices verifies that this equals the right side of (196); zero coordinates contribute nothing. A useful direct proof of convexity is the dual formula

$$g(z) = \sup_w \left\{ 2y^T w - \|w\|^2 - \lambda^{-1} \sum_i z_i (x_i^T w)^2 \right\}.$$

The maximum is attained at $w = (I + X \operatorname{diag}(z) X^T / \lambda)^{-1} y$. It is a supremum of affine functions of z , so g is convex. At binary z , the original minimization is exact ridge regression on its selected columns. Orthogonal blocks make these formulas add as in (99).

Proof of theorem 8.11. Choose π strictly between the two sides of the first inequality in (100). Ridge regression cannot worsen when a feature is added, so $g_{11,j} \leq g_j^*$. Hence $\pi > 0$. For an integer block selection $p \in \{0, 1\}^2$,

$$g_j(p) \geq g_j^* - \pi(\mathbf{1}^T p - 1). \quad (197)$$

This holds by the choice of π for the zero- and two-feature selections, strictly in both cases, and by the definition of g_j^* for one-feature selections. Summing gives, for every support with at most k features,

$$g(z) \geq \sum_j g_j^* - \pi(\mathbf{1}^T z - k) \geq \sum_j g_j^*.$$

Choosing the better feature in each block attains equality. Equality requires exactly one feature per block. If every $\Delta_j > 0$, each block's better feature is unique, proving uniqueness of the optimum.

For two different one-per-block supports, let D be the nonempty set of blocks on which they differ. Their midpoint selects both features with weight one half on D and agrees with one binary feature elsewhere. It belongs to the root relaxation and has value at most

$$\sum_{j \in D} (g_j^* - d_j) + \sum_{j \notin D} (g_j^* + \Delta_j) \leq \text{OPT} - \min_j d_j + \sum_j \Delta_j < \text{OPT} - \varepsilon.$$

Thus all 2^k supports form a midpoint clique of feasible indicators. The class-number lower bound and the binary removal accounting give the claimed leaves and nodes.

For the stronger relaxation, every point of a block's original mixed-integer objective epigraph satisfies the affine inequality $t_j \geq g_j^* - \pi(\mathbf{1}^T z_j - 1)$, by (197). It remains valid on the closed convex hull of that epigraph. Summing it and using the root budget gives $\sum_j t_j \geq \sum_j g_j^* = \text{OPT}$. The optimal binary support is feasible for the hull relaxation, so its root value is exactly OPT.

It remains to construct a uniform rational family. Take $\lambda = 1$, $u = (1, 0)$, $v = (3/5, 4/5)$, and baseline response $y = u + v$. Direct inversion in (196) gives

$$g_{00} = \frac{16}{5}, \quad g_{10} = g_{01} = \frac{48}{25}, \quad g_{11} = \frac{16}{13}, \quad g(1/2, 1/2) = \frac{16}{9}.$$

The midpoint deficit is $d_0 = 32/225 > 0$, and the difference between the first and second feature improvements is $192/325 > 0$. Choose one sufficiently small positive rational η , independent of k , and let

$$s_j = 1 + \eta j/k, \quad e_j = \eta j/k^3, \quad y_j = s_j(u + v) + e_j u, \quad 1 \leq j \leq k. \quad (198)$$

All these responses lie in a common neighborhood of $u + v$. Continuity of the four binary values and the midpoint value gives, after choosing η small enough, the global first inequality in (100) and $d_j \geq d_0/2$ for every j, k . This is a global inequality across blocks: shrink the neighborhood until its smallest zero-to-one improvement exceeds its largest one-to-two improvement, using the positive baseline margin.

Sherman–Morrison gives $g_{10,j} = \|y_j\|^2 - (u^T y_j)^2/2$ and $g_{01,j} = \|y_j\|^2 - (v^T y_j)^2/2$. Their difference is strictly positive in favor of u , because $u^T y_j - v^T y_j = (1 - 3/5)e_j > 0$ and both inner products are positive. On the same compact neighborhood, their sum is bounded, so $0 < \Delta_j \leq C_0 e_j$ for a constant C_0 independent of k, j . Consequently

$$\sum_j \Delta_j \leq C_0 \eta \frac{k+1}{2k^2} \leq C_0 \eta.$$

Choose η still smaller so that $C_0 \eta < d_0/4$. Every tolerance $0 \leq \varepsilon < d_0/4$ now satisfies the second inequality in (100), and every Δ_j is positive. The feature coordinates and responses are rational with denominators polynomial in k ; their bit lengths and the total instance encoding length are polynomial in k . This proves the final assertion. $\square$

G Proofs for sparse regression

G.1 Gaussian estimates and planted residuals

We use the following standard concentration inequalities. For $Z \sim N(0, 1)$, $t > 0$, and the standard normal density ϕ ,

$$\mathbb{P}\{|Z| > t\} \leq e^{-t^2/2}, \quad \frac{t}{1+t^2} \phi(t) \leq \bar{\Phi}(t) \leq \frac{\phi(t)}{t}. \quad (199)$$

Laurent and Massart (2000, Lemma 1 and (4.3)–(4.4), p. 1325) give, for $Q_d \sim \chi_d^2$ and $t > 0$,

$$\mathbb{P}\{Q_d > d + 2\sqrt{dt} + 2t\} \leq e^{-t}, \quad \mathbb{P}\{Q_d < d - 2\sqrt{dt}\} \leq e^{-t}. \quad (200)$$

For a d by m standard Gaussian matrix A , $d \geq m$, each of

$$s_{\max}(A) \leq \sqrt{d} + \sqrt{m} + t, \quad s_{\min}(A) \geq \sqrt{d} - \sqrt{m} - t \quad (201)$$

fails with probability at most $e^{-t^2/2}$. These bounds follow from Davidson and Szarek (2001, Theorem II.13, p. 353) with the correction in Davidson and Szarek (2003, item 5, p. 1819). Their matrix has entry variance $1/d$; multiplying it by $\sqrt{d}$ gives the standard Gaussian normalization used here. We square the latter lower bound only when its right side is nonnegative. Finally, if N is binomial with mean μ , then $\mathbb{P}\{N > 2\mu + 3t\} \leq e^{-t}$ and $\mathbb{P}\{N < \mu/2\} \leq e^{-\mu/8}$.

One additional estimate will be useful. For independent standard Gaussian variables Z_j and $q > 0$, put $T_j = (|Z_j| - q)_+^2$. Completing the square gives

$$m(q) = \mathbb{E}e^{T_j/4} - 1 = 2\sqrt{2}e^{q^2/2}\bar{\Phi}(\sqrt{2}q) - 2\bar{\Phi}(q) \leq 2\phi(q)/q, \quad (202)$$

$$\mathbb{P}\left\{\sum_{j=1}^N T_j > 4[Nm(q) + t]\right\} \leq e^{-t}. \quad (203)$$

Indeed, on $z > q$, $-z^2/2 + (z - q)^2/4 = -(z + q)^2/4 + q^2/2$. Integrating over the two tails gives the first identity and (199) gives its bound. Markov's inequality, $\mathbb{E}e^{\sum_j T_j/4} = (1 + m(q))^N \leq e^{Nm(q)}$, gives the second line.

Lemma G.1 (Uniform planted fits). *Under assumption 9.3, with $S = S^*$,*

$$\max_{i \in S} \left| \frac{\hat{\beta}_i}{b_\lambda} - 1 \right| = o_{\mathbb{P}}(1), \quad \frac{\|r_S\|^2}{\omega_\lambda^2} = 1 + o_{\mathbb{P}}(1). \quad (204)$$

These errors can be bounded by deterministic quantities $O(\sqrt{k/n} + \sqrt{\log p/n} + \log^{-2} p)$ on events whose probabilities tend to one.

Proof. Put $G = X_S^\top X_S$, $\theta = \sqrt{k/n} + 2\sqrt{\log p/n}$, and $v = M_S^{-1} X_S \beta_S^*$. By (201), with probability tending to one, the eigenvalues of G are $n(1 + O(\theta))$. The push-through identity gives

$$v = \lambda X_S (G + \lambda I)^{-1} \beta_S^*, \quad \|v\|^2 = \lambda^2 (\beta_S^*)^\top (G + \lambda I)^{-1} G (G + \lambda I)^{-1} \beta_S^*.$$

Hence $\|v\|^2 = (1 + O(\theta)) k b^2 \lambda^2 n / (n + \lambda)^2$. On the orthogonal complement of range X_S , M_S^{-1} is the identity; on the column space its eigenvalues are between zero and one. Conditional chi-square concentration gives

$$n - k - O(\sqrt{n \log p}) \leq \|M_S^{-1} w\|^2 \leq n + O(\sqrt{n \log p} + \log p).$$

Conditional on X_S , $v^\top M_S^{-1} w$ is centered normal with standard deviation at most $\|v\|$. Its magnitude is at most $\|v\| \sqrt{2 \log p}$ with probability tending to one. Since $r_S = v + \sigma M_S^{-1} w$, and $2\sigma \|v\| \sqrt{2 \log p} \leq \sqrt{2 \log p/n} (n\sigma^2 + \|v\|^2)$, this proves the residual estimate.

For the coefficient estimate, remove coordinate i and write $A_i = M_{S \setminus \{i\}}^{-1}$ and $y_{(i)} = y - \beta_i^* x_i$. Eliminating the remaining coefficients gives

$$\hat{\beta}_i = \frac{\beta_i^* q_i + \xi_i}{\lambda + q_i}, \quad q_i = x_i^\top A_i x_i, \quad \xi_i = x_i^\top A_i y_{(i)}. \quad (205)$$

The matrix A_i lies between the orthogonal complement projector and the identity. By (200) and a union bound, simultaneously in $i \in S$, $q_i = n + O(k + \sqrt{n \log p} + \log p)$. Conditional on $(X_{S \setminus \{i\}}, w)$, ξ_i is normal with variance $\|A_i y_{(i)}\|^2$. The preceding signal calculation, applied to a subset, and eigenvalue interlacing bound

$$\|A_i y_{(i)}\| \leq C b \sqrt{k} \lambda / \sqrt{n} + \sigma \|w\|.$$

Gaussian tails and a union bound then show that the noise term in (205), relative to b_λ , is at most

$$C \left[\frac{\sigma}{b} \sqrt{\frac{\log p}{n}} + \frac{\lambda}{n} \sqrt{\frac{k \log p}{n}} \right].$$

The change from $b q_i / (\lambda + q_i)$ to b_λ is bounded relatively by $C \theta \lambda / n$. Under (115), all these errors tend to zero and obey the stated bound. The union bounds have failure $O((k + 1)/p)$, which tends to zero because $k/p = O(1/\log p)$. $\square$

G.2 Proof of the perspective thresholds

Proof of [theorem 9.4\(a\)](#). Condition on (X_S, w) and put $a = X^\top r_S$, $m_0 = \lambda \min_{i \in S} |\hat{\beta}_i|$. The null variables $Z_j = a_j / \|r_S\|$, $j \notin S$, are independent standard normal variables. By [lemma G.1](#), $m_0 / \|r_S\| = (1 + o_{\mathbb{P}}(1))\tau_\lambda$. For the positive assertion, a union bound gives

$$\mathbb{P}\{\max_{j \notin S} |Z_j| > \sqrt{2 \log p + 2 \log \log p}\} \leq 1 / \log p.$$

The selected threshold is larger with probability tending to one; [proposition 9.1](#) proves all conclusions. For the negative assertion, take $t = \sqrt{(2 - \delta/2) \log p}$. On the high-probability fit event, $t > m_0 / \|r_S\|$. Moreover,

$$\mathbb{P}\{\max_{j \notin S} |Z_j| \leq t \mid X_S, w\} \leq \exp[-2(p - k)\bar{\Phi}(t)] \rightarrow 0,$$

since the exponent's magnitude is at least $cp^{\delta/4}/\sqrt{\log p}$. The exactness condition fails. $\square$

Proof of [theorem 9.4\(b\)](#). Use the cap $q = (1 - \zeta)m_0$, $\zeta = 1/\log p$, in [\(110\)](#), and write $t = q/\|r_S\|$. The fit event and the hypothesis give $t^2 \geq (2 + \delta/2)\Lambda_\lambda$ eventually. Because $\Lambda_\lambda \geq \frac{1}{2} \log p$, define

$$\Pi = 2p\phi(t)/t \leq C \frac{n}{\lambda t} (p\lambda/n)^{-\delta/4} \leq C \frac{n}{\lambda} p^{-\delta/8} \leq C\sqrt{n} p^{-\delta/8}.$$

Conditional on (X_S, w) , [\(203\)](#) and the binomial bound imply

$$\sum_{j \notin S} (|Z_j| - t)_+^2 \leq 4(\Pi + \log p), \quad |V| \leq 2\Pi + 3 \log p = o(\sqrt{n}). \quad (206)$$

Also $\max_{j \notin S} |Z_j| \leq 2\sqrt{\log p}$ with probability tending to one.

Decompose each null column as $x_j = (a_j / \|r_S\|^2) r_S + x_j^\perp$. Conditional on $(X_S, w, (a_j)_{j \notin S})$, the perpendicular components are independent centered Gaussian vectors on $r_S^\perp$. Thus choosing V by its correlations does not bias these components. Let P project onto $(\text{range } X_S + \text{span } r_S)^\perp$. This space has dimension $n - k - 1$ almost surely. On it, M_S^{-1} is the identity; consequently $H = X_V^\top M_S^{-1} X_V \succeq X_V^\top P X_V$. Conditional singular-value concentration and [\(206\)](#) give

$$\lambda_{\min}(H) \geq n(1 - o(1)), \quad \|X_V^\perp\|_{\text{op}} \leq \sqrt{n}(1 + o(1)).$$

The parallel component has squared operator norm at most $4|V| \log p = o(n)$, so $\|X_V\|_{\text{op}} \leq \sqrt{n}(1 + o(1))$. The construction is therefore defined and

$$\Gamma \leq \frac{4\|r_S\|^2(\Pi + \log p)}{n(1 - o(1))}, \quad \gamma_1 := \frac{\sqrt{\Gamma}}{\|r_S\|} = o(1/\log p). \quad (207)$$

For $j \in V$, the new absolute correlation equals q . For $j \notin S \cup V$, condition also on the selected perpendicular columns, which determine Δ . The remaining perpendicular columns are still independent Gaussian vectors. A union bound gives $|(x_j^\perp)^\top \Delta| \leq 2\sqrt{\Gamma \log p}$ simultaneously. It follows that

$$M \leq q(1 + \gamma_1) + 2\sqrt{\Gamma \log p} \leq m_0(1 - \zeta + 3\gamma_1),$$

using $m_0 / \|r_S\| \geq t$ and $t^2 \geq \log p$.

For the selected correlations,

$$X_S^\top \Delta = \lambda(G + \lambda I)^{-1} X_S^\top X_V u.$$

The spectral estimates above and $\|u\| \leq 2\|r_S\|\sqrt{\Pi + \log p}/[n(1 - o(1))]$ imply

$$\frac{\max_{i \in S} |x_i^\top \Delta|}{m_0} \leq C \left[\frac{\sigma}{b} \sqrt{\frac{\Pi + \log p}{n}} + \sqrt{\frac{k}{n}} \frac{\lambda}{n} \sqrt{\Pi + \log p} \right] = o(1/\log p).$$

For the last equality, the $\log p$ terms use $n \geq \log^6 p$, $k \log p/n \leq C_0$, and $\lambda/n \leq \log^{-2} p$. For the Π terms use $\Pi \leq C(n/\lambda)p^{-\delta/8}$ and $\lambda \geq \sqrt{n}$. Therefore

$$m^2 - M^2 \geq (2 - o(1))\zeta m_0^2.$$

By (207),

$$\frac{\Gamma}{(m^2 - M^2)/\lambda} \leq (1 + o(1)) \frac{4\lambda(\Pi + \log p)}{n\zeta t^2} \rightarrow 0.$$

Indeed ζt^2 is bounded away from zero, $\lambda\Pi/n = O(p^{-\delta/8})$, and $\lambda \log p/n \leq 1/\log p$. lemma 9.2 now gives every wrong-fixing bound at least $F_\lambda(S) + (2 - o(1))\lambda b^2/\log p$. This proves the stated margin and uniqueness. $\square$

We first establish a deterministic feasible point for the negative single-variable statement.

Lemma G.2 (A forced-in primal point). *Fix S of size k and $j \notin S$. Put $b_+ = \max_{i \in S} |\hat{\beta}_i|$ and $a = X^\top r_S$. Choose $V' \subseteq [p] \setminus (S \cup \{j\})$, $q \geq \lambda b_+$, and define*

$$e_l = (|a_l| - q)_+, \quad Q_e = \sum_{l \in V'} e_l^2, \quad A = \|X_{V'}\|_{\text{op}}^2, \quad v_l = s \text{sign}(a_l) e_l / A, \quad t_l = \lambda |v_l| / q, \quad T = \sum_l t_l.$$

If $0 < s \leq 1$, $t_l \leq 1$, and $T < k - 1$, then

$$r(\emptyset, \{j\}) \leq F_\lambda(S) + \lambda b_+^2 \left[1 + \frac{(1 + T)^2}{k - 1 - T} \right] - (2s - s^2)Q_e/A. \quad (208)$$

Proof. Take $z_i = 1 - (1 + T)/k$ on S , $z_j = 1$, $z_l = t_l$ on V' , and zero otherwise. Keep $\hat{\beta}$ on S , take coefficients v on V' , and zero at j . The budget is exactly k . Substitution in (104) bounds the objective by

$$F_\lambda(S) + \lambda \|\hat{\beta}\|^2 \frac{1 + T}{k - 1 - T} - 2 \sum_l |v_l| |a_l| + \|X_{V'} v\|^2 + q \sum_l |v_l|.$$

Use $\|\hat{\beta}\|^2 \leq k b_+^2$ and $k(1 + T)/(k - 1 - T) = 1 + T + (1 + T)^2/(k - 1 - T)$. The term $\lambda b_+^2 T$ is at most $q \sum_l |v_l|$. The last terms are consequently at most $-2 \sum_l |v_l| e_l + A \|v\|^2 = -(2s - s^2)Q_e/A$. $\square$

Proof of theorem 9.4(c). Let $d_0 = \delta/16$, $q = \lambda b_+(1 + 1/\log p)$, $t' = q/\|r_S\|$, and $t_0 = (1 + d_0)t'$. On the fit event, $t_0^2 \leq (2 - \delta/2)\Lambda_\lambda$. The number of nulls with $|a_l|/\|r_S\| \geq t_0$ has conditional binomial mean μ satisfying

$$\mu \geq c \frac{n}{\lambda} (p\lambda/n)^{\delta/4} / \sqrt{\log p}.$$

Here one may use $1 + t_0$ in the denominator of the elementary Gaussian lower-tail bound; this also handles bounded t_0 . In particular $\mu/(n/\lambda) \rightarrow \infty$. With probability tending to one, at least $\mu/2$ such nulls exist. Let $N' = \min(\lfloor \mu/2 \rfloor, \lfloor n/\log^2 p \rfloor)$ and let V' contain the N' largest null absolute correlations. This set is common to all forced-in indices $j \notin S \cup V'$. On V' , $e_l \geq d_0 q$.

Conditioning on all correlations leaves its perpendicular columns independent Gaussian. Since $N' \leq n/\log^2 p$ and $\max_l |Z_l| \leq 2\sqrt{\log p}$, the same decomposition as above gives $A = \|X_{V'}\|_{\text{op}}^2 \leq$

$n(1 + o(1))$. Both choices of N' satisfy $N' \geq 3(1 + o(1))n/(d_0^2\lambda)$ eventually: the first because $\mu/(n/\lambda) \rightarrow \infty$, and the second because $\sqrt{n}/\log^2 p \rightarrow \infty$. Thus $Q_e/A \geq 3\lambda b_+^2$ for all sufficiently large p . Set $s = 3\lambda b_+^2 A/Q_e \leq 1$. Then $(2s - s^2)Q_e/A \geq 3\lambda b_+^2$ and

$$T = \frac{3\lambda^2 b_+^2}{qQ_e} \sum_l e_l \leq 3/d_0.$$

Also $\max_l t_l = o(1)$. To see this, $Q_e \geq N' d_0^2 q^2$, $\max e_l \leq 2\|r_S\|\sqrt{\log p}$, and $N' \gg \sqrt{\log p}$. The ratio $q/\|r_S\|$ is bounded below: writing the square of (114) as a harmonic mean,

$$\tau_\lambda^{-2} = \frac{\sigma^2(n + \lambda)^2}{\lambda^2 b^2 n} + \frac{k}{n},$$

shows a fixed positive lower bound for τ_λ under (115). The fit estimate transfers it to $q/\|r_S\|$.

Since $k \geq 5000/\delta^2$, $(1+T)^2/(k-1-T) \leq 1/2$; indeed $2(1+48/\delta)^2 + 1 + 48/\delta \leq 4851/\delta^2$. lemma G.2 gives the node bound at most $F_\lambda(S) - \frac{3}{2}\lambda b_+^2$. The coefficient estimate implies $b_+ = b(1 + o(1))$, proving the claimed strict inequality with margin $\lambda b^2/2$. There are at most $N' \leq n/\log^2 p$ exceptional indices. $\square$

G.3 Ridge optimization

Proof of corollary 9.5. The preceding harmonic-mean identity shows exactly that $\tau_\lambda^2 < n/k$. Also, uniformly for the allowed ridges, writing $\lambda = (\sigma/b)\sqrt{Tn}$ gives

$$\tau_\lambda^2 = (1 + O(\log^{-2} p)) \frac{Tn}{n + kT}. \quad (209)$$

Let $\Lambda = \log p$ for the root test, and $\Lambda = \log(p/\sqrt{n})$ for the C1 test. For any fixed $e \in (0, 1)$, if $n \geq (2 + e)k\Lambda$, set

$$A = (2 + e/2)(1 + e/8)\Lambda, \quad T = \frac{A}{1 - Ak/n}, \quad \lambda = (\sigma/b)\sqrt{Tn}.$$

Because $(2 + e/2)(1 + e/8) < 2 + e$, the denominator is bounded away from zero and $T = \Theta_e(\log p)$. The inequalities $\log^6 p \leq n \leq p$ imply that this ridge is between $\sqrt{n}$ and $n/\log^2 p$ eventually. In the C1 case, $\Lambda_\lambda = \Lambda + \frac{1}{2} \log(T\sigma^2/b^2) = (1 + o(1))\Lambda$. Formula (209) then gives either positive conclusion of theorem 9.4 with a smaller fixed slack than e , for example $e/4$ eventually. It need not meet the threshold hypothesis with the original slack e .

If $n \leq (2 - e)k\Lambda$, the exact inequality $\tau_\lambda^2 < n/k$ gives the negative root comparison. For C1, also use $\Lambda_\lambda \geq \log(p/\sqrt{n})$. Apply the respective negative theorem with slack $e/2$ and its corresponding lower bound on k . For $k = p^{\gamma+o(1)}$ and $n = ak \log p$, $\log(p/\sqrt{n}) = (1 - \gamma/2 + o(1)) \log p$. In $2 - \gamma < a < 2$, the positive C1 theorem proves S^* uniquely optimal, while the negative root theorem gives $R < F_\lambda(S^*)$. The root is therefore globally inexact, and the path lemma gives the tree bound. The other regimes follow in the same way. $\square$

G.4 Hull calculus and deterministic completion

Zero extension maps generators of $\mathcal{H}_T$ to generators of a larger hull. Moreover, for $N \succeq 0$,

$$(z, b, B) \in \mathcal{H}_T \implies (z, b, B + N) \in \mathcal{H}_T. \quad (210)$$

For a rank-one $N = vv^\top$, mix (z, b, B) with the generator $(\mathbf{1}_T, v/\sqrt{t}, vv^\top/t)$ with weight t and let $t \downarrow 0$. The first moments converge to (z, b) and the second moment to $B + vv^\top$. Closedness and a decomposition of N into rank-one terms prove (210). The singleton hull is $0 \leq z \leq 1$, $B \geq 0$, $b^2 \leq zB$. Consequently the global positive semidefinite constraint and the singleton hulls give, for $E = B - \beta\beta^\top$, $E \succeq 0$ and $E_{jj} \geq \beta_j^2(1/z_j - 1)$. This proves $\ell_s(z) \geq g_\lambda(z)$; integer generators prove validity. Convexity follows by projecting a convex feasible epigraph.

For a block T and a positive semidefinite block matrix Q_T , every generator satisfies $b^\top Q_T b = \langle Q_T, bb^\top \rangle$. Convex combinations and limits imply that the exact closed indicator epigraph hull has value at most $\langle Q_T, B_{TT} \rangle$ on $\mathcal{H}_T$. The residual matrix $Q_0 \succeq 0$ has $\langle Q_0, B \rangle \geq \beta^\top Q_0 \beta$. Summing proves the selected-block dominance stated in the text.

Lemma G.3 (A support distribution). *For $z \in K$, with any prescribed zero and one fixings, there is a random feasible binary support with indicator ζ and mean z . For $\beta_j = 0$ where $z_j = 0$, define $\xi_j = \beta_j \zeta_j / z_j$ when $z_j > 0$ and zero otherwise. Then $\mathbb{E}\xi = \beta$ and $D = \text{Cov } \xi$ satisfies $\text{tr } D = \pi(z, \beta)$. On each coordinate subset, $(z, \beta, \beta\beta^\top + D)$ restricts to a point of its local hull.*

Proof. Every vertex of K is binary. Indeed, two fractional coordinates can be perturbed in opposite directions, while a lone fractional coordinate can be perturbed if the budget is slack. If the budget is tight and only one coordinate is fractional, its value would equal the difference of two integers, a contradiction. The same argument applies to a face with fixings. Express z as a convex combination of its binary vertices. The coefficient expectation and variance follow coordinatewise: $\text{Var } \xi_j = \beta_j^2(1/z_j - 1)$. The second-moment points are actual convex combinations of generators. $\square$

G.4.1 Completion using zero-indicator helper columns

Let $F \supseteq \text{supp } z$, and embed the covariance D from lemma G.3 in the F block. Let H be a disjoint helper set and assume $W = X_H X_H^\top \succ 0$. Define

$$P_H = I - X_H^\top W^{-1} X_H, \quad q_m = x_m^\top W^{-1} x_m, \quad \theta_F = \|X_F^\top W^{-1} X_F\|_{\text{op}}, \quad Y_F = X_F D^{1/2}.$$

The matrix P_H is the orthogonal projector onto $\ker X_H$. For a slack $a > 0$, set

$$G_F = \sqrt{1+a} D^{1/2}, \quad C = -X_H^\top W^{-1} X_F G_F,$$

and form G by placing G_F on F , C on H , and zeros elsewhere. Then $XG = 0$.

Lemma G.4 (Local completion). *Let $J \succeq 0$ have range in $\ker X_H$ and be embedded in the helper block. Put $B = \beta\beta^\top + GG^\top + J$. Its lifted objective is*

$$\mathcal{F}(\beta, B) = \|y - X\beta\|^2 + \lambda\|\beta\|^2 + \lambda[(1+a)\pi + \|C\|_F^2 + \text{tr } J], \quad \|C\|_F^2 \leq (1+a)\theta_F\pi. \quad (211)$$

Two choices ensure local hull feasibility:

- (a) *For hulls of order s , suppose $\eta_H = \max_{M \subseteq H, |M| \leq s-1} \|X_M^\top W^{-1} X_M\|_{\text{op}} < 1$. Set $J = tP_H$, where*

$$t = \frac{1+a}{a(1-\eta_H)} \max_{|M| \leq s-1} \|C_M\|_{\text{op}}^2.$$

Then B is feasible for all hulls of order at most s , and $\text{tr } J = t(|H| - n)$. For $s = 1$ take $J = 0$.

- (b) *For pairwise hulls, suppose explicitly that $q_{\max} = \max_{m \in H} q_m < 1$. Set*

$$J = P_H \text{diag} \left(\frac{\|C_m\|^2}{a(1-q_m)^2} : m \in H \right) P_H.$$

Then B is feasible for pairwise hulls and

$$\text{tr } J \leq \frac{\|C\|_F^2}{a(1 - q_{\max})}.$$

Thus the additional cost above the perspective objective is at most $\lambda\pi[a + (1+a)\theta_F(1 + 1/[a(1 - q_{\max})])]$.

Proof. Positive semidefiniteness is immediate from $GG^\top + J \succeq 0$. Since $XG = 0$ and $X_H J = 0$, its design-variance cost vanishes. The trace is $(1 + a) \text{tr } D + \|C\|_F^2 + \text{tr } J$, giving (211). Also $\|C\|_F^2 = (1 + a) \text{tr}(Y_F^\top W^{-1} Y_F) \leq (1 + a)\theta_F \pi$.

For a tested subset T , put $T' = T \cap F$ and $M = T \cap H$. The random-support moment on T' belongs to $\mathcal{H}_{T'}$. By zero extension and (210), it is enough that the additional block

$$\begin{pmatrix} C_M C_M^\top + J_{MM} & C_M G_F^\top \\ G_F C_M^\top & aD \end{pmatrix}$$

is positive semidefinite, before restriction to T' . If either block is absent, this is immediate. For (a), $J_{MM} \succeq t(1 - \eta_H)I$. If $C \neq 0$ and $t > 0$, its Schur complement is bounded by $\|C_M\|^2(1 + a)D/[t(1 - \eta_H)] \preceq aD$. If the maximum in t is zero, every relevant C_M vanishes; the blocks are then diagonal and positive semidefinite. This handles the zero case without invoking a positive definite helper covariance.

In (b), a mixed tested pair has $M = \{m\}$ and $J_{mm} \geq \|C_m\|^2/a$, because the diagonal contribution of P_H is $1 - q_m$. The Schur complement is at most $(1 + a)\|C_m\|^2 D/(\|C_m\|^2 + J_{mm}) \preceq aD$. If $C_m = 0$, the cross block vanishes. Finally, $P_H^2 = P_H$ gives $\text{tr } J = \sum_m \|C_m\|^2/[a(1 - q_m)]$. $\square$

G.5 Proof of the local-lift easy thresholds

Split the null features, in a deterministic order, into two halves H_1, H_2 . For a helper half H , all constructions below are measurable with respect to $\mathcal{A}_H = \sigma(X_{[p] \setminus H}, w)$. The helper matrix is independent of this information. Let $m = |H|$, and define

$$w_- = (\sqrt{m} - \sqrt{n} - 2\sqrt{\log p})^2, \quad v_+ = n + 2\sqrt{3n \log p} + 6 \log p, \quad a_p = 1 + 2\sqrt{3 \log p} + 6 \log p.$$

Under $sn \log p = o(p)$, the right side before squaring in w_- is positive eventually, and $w_- = m(1 - o(1))$. With probability tending to one, $\lambda_{\min}(W) \geq w_-$ and $\max_{j \in H} \|x_j\|^2 \leq v_+$. For any fixed finite number of $\mathcal{A}_H$ -measurable matrices Y , simultaneously for every $j \in H$,

$$\|x_j^\top W_{-j}^{-1} Y\|^2 \leq a_p \|W_{-j}^{-1} Y\|_F^2, \quad W_{-j} = W - x_j x_j^\top. \quad (212)$$

To prove this, condition on $(X_{H \setminus \{j\}}, \mathcal{A}_H)$. The remaining x_j is standard Gaussian, so the weighted chi-square bound, obtained by applying its exponential moment, gives $x_j^\top A x_j \leq \text{tr } A + 2\sqrt{3 \log p} \|A\|_F + 6 \log p \|A\|_{\text{op}}$ for $A = W_{-j}^{-1} Y Y^\top W_{-j}^{-1}$. The last expression is at most $a_p \text{tr } A$. A union bound proves (212).

Sherman–Morrison, $W_{-j} \succeq (w_- - v_+)I$, and (212) imply

$$\max_{|M| \leq s-1} \|C_M\|^2 \leq \frac{(1 + a)(s - 1)a_p \|Y_F\|_F^2}{(w_- - v_+)^2}, \quad \eta_H \leq (s - 1)v_+/w_-.$$

If $\|X_F\|_{\text{op}}^2 \leq 16n$, then $\|Y_F\|_F^2 \leq 16n\pi$. Choose the completion slack $a = \sqrt{sn \log p/p}$. lemma G.4(a) gives

$$\ell_s(z) \leq P_e(z, \beta) := \|y - X\beta\|^2 + \lambda \|\beta\|^2 + \lambda(1 + e)\pi(z, \beta), \quad e = O(\sqrt{sn \log p/p}) = o(1). \quad (213)$$

The constant is uniform for the finite number of matrices used below. It does not require independence of W and its own individual columns: (212) used leave-one-out conditioning.

Proof of theorem 9.6. The positive assertions follow from perspective domination and validity. For the negative root assertion, use H_1 as helpers and H_2 as candidates. On the fit event a candidate $l \in H_2$ satisfies $|a_l| \geq (1+c)m_0$ with probability tending to one, where $c = \delta/12$ and $m_0 = \lambda \min_S |\hat{\beta}_i|$. This follows from the Gaussian maximum lower bound at $\sqrt{(2-\delta/2)\log p}$, whose ratio to the selected threshold is eventually greater than $1+c$. Choose $i \in S$ attaining m_0 , set $\bar{e} = c/4$, and take

$$t = \min(c/8, c\lambda/(8v_+)), \quad v = ta_l/[\lambda(1+\bar{e})], \quad z = \mathbf{1}_S - te_i + te_l, \quad \beta = \hat{\beta} + ve_l.$$

Use the two-support mixture on S and $S \setminus \{i\} \cup \{l\}$. All these objects are $\mathcal{A}_{H_1}$ -measurable. The global column-norm bound and the fixed X_S spectral bound give $\|X_{S \cup \{l\}}\|^2 \leq 16n$. Eventually $e \leq \bar{e}$ in (213). Direct substitution gives

$$P_{\bar{e}}(z, \beta) \leq F_\lambda(S) - \frac{t}{\lambda} \left[\frac{a_l^2}{1+\bar{e}} - \frac{(1+\bar{e})m_0^2}{1-t} - \frac{ta_l^2\|x_l\|^2}{\lambda} \right].$$

Dividing the bracket by a_l^2 bounds it below by

$$(1 - c/4) - \frac{1 + c/4}{(1+c)^2(1-c/8)} - c/8 \geq c/8 > 0.$$

Here $(1+c/4)/(1-c/8) \leq 1+c/2$ and $(1+c/2)/(1+c)^2 \leq 1/(1+c) \leq 1-c/2$. Thus $\mathcal{R}(\emptyset, \emptyset) < F_\lambda(S)$.

For the forced-in converse, repeat the construction of lemma G.2 separately in each candidate half, using the other half as helpers. Let $d_0 = \delta/16$, $q = \lambda b_+(1 + 1/\log p)$ and select the N' largest absolute correlations in the candidate half, with $N' = \min(\lfloor \mu/2 \rfloor, \lfloor n/\log^2 p \rfloor)$ as before. Half the number of candidates changes no limiting estimate. For any nonexceptional j in this half, put $\bar{e} = d_0/2$ and $e'_l = (|a_l| - (1+\bar{e})q)_+$ on V' . Then $e'_l \geq (d_0/2)q$. Keep the definitions of v_l, t_l, T in the primal lemma, using e'_l in place of e_l . Substitution in $P_{\bar{e}}$ gives

$$P_{\bar{e}}(z, \beta) \leq F_\lambda(S) + (1+\bar{e})\lambda b_+^2 \left[1 + \frac{(1+T)^2}{k-1-T} \right] - (2s_0 - s_0^2)Q'/A, \quad (214)$$

where $Q' = \sum_l (e'_l)^2$ and $v_l = s_0 \text{sign}(a_l)e'_l/A$. For clarity, the ridge increment on S is $(1+\bar{e})\lambda\|\hat{\beta}\|^2(1+T)/(k-1-T)$, and on V' it is at most $(1+\bar{e})q\sum_l |v_l|$. The same rearrangement as in lemma G.2 then gives (214).

Use a support mixture that includes j always and chooses a support of size at most $k-1$ on $S \cup V'$. Its coefficient on j is zero, so its covariance has a zero j row and column. Set $F = S \cup V' \cup \{j\}$, padding D with that zero row. The matrix $Y_F = X_F D^{1/2}$ is the same for all nonexceptional j in this half, and is independent of the helper half. The norm of X_F is bounded uniformly by $16n$ in squared operator norm: the bounds for X_S and $X_{V'}$, together with the global column-norm bound for x_j , suffice. Thus (213) applies simultaneously and $\mathcal{R}(\emptyset, \{j\}) \leq P_{\bar{e}}(z, \beta)$.

As in the perspective proof, eventually $N' \geq 12(1+o(1))n/(d_0^2\lambda)$. Take $s_0 = 3\lambda b_+^2 A/Q' \leq 1$. Then $T \leq 6/d_0 = 96/\delta$, and $\max_l t_l = o(1)$. The bound $k \geq 20000/\delta^2$ ensures $(1+T)^2/(k-1-T) \leq 1/2$. Formula (214) is consequently at most $F_\lambda(S) + \frac{3}{2}(1+\bar{e})\lambda b_+^2 - 3\lambda b_+^2 < F_\lambda(S) - \lambda b_+^2$. This proves the desired margin, with at most $2N'$ exceptions over the two halves. The sample-size statements follow from appendix G.3, again with a smaller fixed threshold slack. $\square$

G.6 Conflict packing and the retained-helper hard theorem

Lemma G.5 (Integer optimum lower bound). *Let $q = k/n < 1$, and let $\rho \in (q, 1)$ satisfy*

$$\frac{n}{2} [q \log(q/\rho) + (1-q) \log((1-q)/(1-\rho))] = \log \binom{p}{k} + \log(1/\alpha).$$

If y is independent of X , then with probability at least $1 - \alpha$, $\text{OPT} \geq (1 - \rho)\|y\|^2$. In the planted model the same probability bound holds for $\text{OPT} \geq (1 - \rho)\sigma^2\|P_{S^}^\perp w\|^2$.*

Proof. For a fixed k -subset T , the span of its columns is a uniform k -dimensional subspace. Conditional on an independent nonzero vector v , the squared projection fraction has the law $A/(A+C)$, where $A \sim \chi_k^2$ and $C \sim \chi_{n-k}^2$ are independent. For $\rho > q$, exponential Markov applied to $(1-\rho)A - \rho C$ gives

$$\mathbb{P}\{A/(A+C) \geq \rho\} \leq (1 - 2t(1-\rho))^{-k/2} (1 + 2t\rho)^{-(n-k)/2}.$$

Choosing $2t(1-\rho) = 1 - q/\rho$ yields the exponential of minus the left side of the displayed equation. Union-bound over all k -subsets. Ridge values dominate projection residuals, and padding any smaller support to size k cannot increase its value. This proves the independent case.

For the planted case, for each T let $v_T = X_{S^* \setminus T} \beta_{S^* \setminus T}^* + \sigma w$. It is independent of X_T , and $y - v_T$ lies in its span. Therefore the same fixed-support probability bound applies. Moreover $\|v_T\| \geq \sigma\|P_{S^*}^\perp w\|$. The union bound again gives the result. Independence of the different residuals is unnecessary. $\square$

Proof of theorem 9.7. Put $\alpha = 1/k$. Since $k/n \rightarrow 0$ and $x_p \rightarrow x$, the solution ρ in lemma G.5 satisfies $1 - \rho = e^{-x} + o(1)$. This follows from $q \log(q/\rho) + (1-q) \log((1-q)/(1-\rho)) = -\log(1-\rho) + o(1)$, uniformly on compact subsets of $(0, 1)$. In case (a),

$$\text{OPT} \geq (e^{-x} - o(1))\|y\|^2 \tag{215}$$

with probability tending to one.

Choose constants $\mu, \vartheta \in (0, 1)$ and $0 < c < (1 - \mu)\vartheta$ such that, with $B_0 = 1 + 2\sqrt{cx} + 2cx$,

$$\frac{(1 + \mu)(1 - \vartheta)x}{B_0} > e^x - 1. \tag{216}$$

These exist because $2x > e^x - 1$: let μ tend to one and ϑ, c tend to zero, then use a strict margin. Write $M = \lceil k(p/k)^\vartheta \rceil$ and $m = \lceil \mu k \rceil$. Conditional on y , the correlations $c_j = x_j^\top (y/\|y\|)$ are independent standard normal variables. The number whose absolute values exceed $t_M = \Phi^{-1}(M/p)$ has mean $2M$, and is at least M with probability at least $1 - e^{-M/4}$. Let W_0 be the M largest absolute correlations. The binomial coefficient estimates $(p/k)^k \leq \binom{p}{k} \leq (ep/k)^k$ imply

$$\log(p/k) = \frac{x + o(1)}{2} \frac{n}{k} \rightarrow \infty, \quad t_M^2 = 2(1 - \vartheta) \log(p/k)(1 + o(1)).$$

Every $U \subseteq W_0$ with $|U| \geq k + m$ therefore has

$$s_U := \sum_{j \in U} c_j^2 \geq (1 + \mu)(1 - \vartheta)xn(1 + o(1)). \tag{217}$$

Greedily pack the k -subsets of W_0 in a fixed lexicographic order so that $|S \setminus T| \geq m$ for distinct members. Each chosen support deletes at most $\sum_{d < m} \binom{k}{d} \binom{M-k}{d}$ candidates, so the family has size at least the total number of candidates divided by this sum. Using $\binom{k}{d} \leq 2^k$ and $\binom{M-k}{d} \leq (eM/d)^d$,

$$\log |\mathcal{C}_0| \geq (1 - \mu)\vartheta k \log(p/k) - O(k) - O(\log(p/k)).$$

Since $k \rightarrow \infty$, select the first $\lceil \exp(ck \log(p/k)) \rceil$ members to obtain $\mathcal{C}$. This code is measurable in the correlations before the perpendicular components of the columns are revealed.

Conditional on y and all c_j , $X^\perp = (I - \widehat{y}\widehat{y}^\top)X$ has independent Gaussian columns on $\widehat{y}^\perp$, where $\widehat{y} = y/\|y\|$. For a fixed union U , $\|X_U^\perp c_U\|^2/s_U$ has law χ_{n-1}^2 . Union-bound over the at most $|\mathcal{C}|^2$ pairs, with $t = \log(|\mathcal{C}|^2/\alpha) = cn(1 + o(1))$ in (200). With probability tending to one, all pairs satisfy

$$\|X_U^\perp c_U\|^2/s_U \leq nB_0(1 + o(1)). \quad (218)$$

No independence between different unions is used.

For $z = (\mathbf{1}_S + \mathbf{1}_T)/2$, perspective fitting on $U = S \cup T$ has ridge coefficient λ on the intersection and 2λ on the symmetric difference. It is bounded above by the fit with coefficient 2λ on all of U . Taking $\beta = tc_U$ and minimizing this scalar t gives

$$g_\lambda(z) \leq \|y\|^2 \left[1 - \frac{s_U}{s_U + \|X_U^\perp c_U\|^2/s_U + 2\lambda} \right]. \quad (219)$$

The identity follows by writing $X_U = \widehat{y}c_U^\top + X_U^\perp$ and expanding the squared loss. Since $|U| \geq k + m$ and $\lambda = o(n)$, (217)–(219) give, uniformly over pairs,

$$g_\lambda(z) \leq \|y\|^2 \left[\frac{B_0}{B_0 + (1 + \mu)(1 - \vartheta)x} + o(1) \right].$$

The coefficient in brackets is strictly below e^{-x} by (216). Together with (215), this leaves a fixed positive margin. Choose η strictly smaller than that margin.

For case (b), perform the correlation and code construction on $[p] \setminus S^*$, conditionally on (X_{S^*}, w) . There are $p - k = p(1 - o(1))$ candidates, so all estimates are unchanged. The optimum bound is now $\text{OPT} \geq (e^{-x} - o(1))n\sigma^2$, while $\|y\|^2 = (1 + \kappa + o_\mathbb{P}(1))n\sigma^2$. Choose the parameters instead to satisfy

$$(1 + \kappa) \frac{B_0}{B_0 + (1 + \mu)(1 - \vartheta)x} < e^{-x}.$$

They exist because $\kappa < e^{-x}(1 + 2x) - 1$. This proves the perspective conflicts in both cases.

It remains to justify the same conflicts for ℓ_2 uniformly. The hard hypotheses imply $\log p = o(n)$ and $p/n \rightarrow \infty$. Indeed $\log(p/k) \asymp n/k$, $\log k = o(n)$, and $p/n = (p/k)/(n/k) \rightarrow \infty$. There is a simultaneous event of probability tending to one on which

$$\lambda_{\min}(XX^\top) = p(1 - o(1)), \quad \max_j \|x_j\|^2 = n(1 + o(1)), \quad \max_{|U| \leq 2k} \|X_U\|_{\text{op}}^2 = O(n). \quad (220)$$

The first two follow from (201) applied to $X^\top$ and from (200), with union bounds. For the third, pad to $h = 2k$ columns and apply the largest singular-value bound with deviation $\sqrt{4h \log(ep/h)}$. Union-bound over at most $\exp(h \log(ep/h))$ subsets. The failure is at most $\exp(-h \log(ep/h))$, and the squared bound $(\sqrt{n} + \sqrt{h} + \sqrt{4h \log(ep/h)})^2$ is $O(n)$.

For every such U , use $H = [p] \setminus U$ as helpers. On (220),

$$\lambda_{\min}(X_H X_H^\top) \geq \lambda_{\min}(XX^\top) - \|X_U\|^2 = p(1 - o(1)).$$

Consequently $\theta_U = O(n/p)$ and $q_{\max} = O(n/p) < 1$, uniformly in U . Apply lemma G.4(b) with $a = \sqrt{n/p}$. For the two-support mixture with mean $z = (\mathbf{1}_S + \mathbf{1}_T)/2$ and any coefficient supported on U , $\pi(z, \beta) \leq \|\beta\|^2$. The completion therefore gives

$$\ell_2(z) \leq \|y - X_U \beta\|^2 + (2 + e_p)\lambda \|\beta\|^2, \quad e_p = O(\sqrt{n/p}) = o(1),$$

simultaneously for every code pair. All support-selection dependence is handled by the deterministic uniform event; the helpers need not be independent of the selected union. Taking again $\beta = tc_U$ replaces 2λ in (219) by $(2 + e_p)\lambda = o(n)$. The same strict margin proves the pairwise-hull conflicts. Shrinking η once gives the common assertion.

Convexity of a leaf containing two code points forces their midpoint into that leaf, contradicting its certificate bound. Charging certified removals as additional convex pieces gives at least $|C|$ terminal or removed pieces. For binaries each node can fix each coordinate at most once, hence creates at most p such removed pieces; leaves plus removals are at most $(p + 1)$ times the node count. The lifted construction retains its helper second moments at zero fixings. Deleting those columns invalidates this feasible point and is not a consequence of the argument. $\square$

G.7 Variance price for the richer moment lift

Proof of proposition 9.8. Represent the positive semidefinite matrix Y as the Gram matrix of $v_0, v_{z_j}, v_{\beta_j}$, with $\|v_0\| = 1$. Set $w_j = v_{z_j} - z_j v_0$ and $u_j = v_{\beta_j} - \beta_j v_0$. Then $B - \beta\beta^\top = \text{Gram}(u)$. Let $F = \{j : 0 < z_j < 1\}$ and let V be the span of its w_j . For $m \notin A = \text{supp } z$, positive semidefiniteness and $U_{mm} = \beta_m$ give $\beta_m^2 \leq z_m B_{mm} = 0$. The product cones and $Z_{jm} \leq z_m = 0$ give $U_{jm} = 0$ for $j \neq m$. Therefore $\langle u_m, w_j \rangle = U_{jm} - z_j \beta_m = 0$: the zero-indicator coefficient vectors are orthogonal to V .

For $j \in F$, $\langle u_j, w_j \rangle = \beta_j(1 - z_j)$ and $\|w_j\|^2 = z_j(1 - z_j)$. Put $g_j = P_V u_j$ for every coordinate. Orthogonal decomposition gives $\text{Gram}(u) \succeq \text{Gram}(g)$, and $g_m = 0$ off A . Thus

$$\langle Q, B - \beta\beta^\top \rangle \geq \langle X_A^\top X_A + \lambda I, \text{Gram}_A(g) \rangle \geq (\nu_A + \lambda) \sum_{j \in F} \|g_j\|^2.$$

Cauchy–Schwarz yields $\|g_j\|^2 \geq \beta_j^2(1 - z_j)/z_j$. Summing and adding the deterministic coefficient-fit objective proves (124). Coordinates with $z_j = 1$ contribute zero to π , so none are missing. $\square$

G.8 The fixed-dimensional per-entry-noise limit

Proof of theorem 9.9. Here d, k, β^*, σ are fixed and $\lambda = \sqrt{n}$. Finite-dimensional laws of large numbers give $X^\top X/n \rightarrow I_d$, $X^\top w/n \rightarrow 0$, and $\|w\|^2/n \rightarrow 1$ in probability. For every T with $|T| \leq k$,

$$\frac{F_\lambda(T)}{n} = \frac{\|y\|^2}{n} - \left(\frac{X_T^\top y}{n} \right)^\top \left(\frac{X_T^\top X_T}{n} + \frac{\lambda}{n} I \right)^{-1} \left(\frac{X_T^\top y}{n} \right) \rightarrow \sigma^2 + \|\beta_{S \setminus T}^*\|^2.$$

For the empty support the subtraction is zero. There are finitely many supports; convergence holds simultaneously. Every $T \neq S$ in this family misses at least one selected coordinate and has limiting gap at least $b_{\min}^2$ from S . Let U_n be the event that S is uniquely optimal. Then $\mathbb{P}(U_n) \rightarrow 1$. Each restricted ridge fit is unique, and on U_n its coefficients on S are nonzero, since a zero coefficient would give the same value on a proper subset. Thus the global cardinality-constrained coefficient vector is unique as well.

Write $C_n = X_S^\top X_S$, $A_n = C_n + \sqrt{n}I$, and $r = M_S^{-1}y$. Conditional on X_S ,

$$\hat{\beta}_S \sim N(A_n^{-1}C_n\beta_S^*, \sigma^2 A_n^{-1}C_n A_n^{-1}).$$

The mean tends to β_S^* and n times the covariance tends to $\sigma^2 I_k$. In particular $\hat{\beta}_S \rightarrow \beta_S^*$. The exact normal-equation identity gives

$$m_n := \min_{i \in S} |x_i^\top r| = \sqrt{n} \min_{i \in S} |\hat{\beta}_i|, \quad m_n/\sqrt{n} \rightarrow b_{\min}.$$

For the residual, its signal part is $q_n = \sqrt{n}X_S A_n^{-1}\beta_S^*$ and

$$\|q_n\|^2 = n(\beta_S^*)^\top A_n^{-1}C_n A_n^{-1}\beta_S^* \rightarrow \|\beta_S^*\|^2.$$

The matrix $M_S^{-1} - I$ acts only on the fixed k -dimensional column space and has norm at most one. Conditional on X_S , $\|P_S w\|^2 \sim \chi_k^2$. Consequently $\|r - \sigma w\| = O_{\mathbb{P}}(1)$, and

$$\frac{\|r\|^2}{n} \rightarrow \sigma^2, \quad \frac{m_n}{\|r\|} \rightarrow b_{\min}/\sigma$$

in probability.

Compute the planted-support exactness probability before conditioning on its optimality. Conditional on $\mathcal{A}_n = \sigma(X_S, w)$, the null columns are independent standard Gaussian vectors, whereas r and m_n are fixed. The residual is nonzero almost surely, since M_S is invertible and the noise has a density. Therefore the exact condition (108) gives

$$\mathbb{P}\{R = F_\lambda(S) \mid \mathcal{A}_n\} = \left[1 - 2\bar{\Phi}(m_n/\|r\|)\right]^{d-k}.$$

Equalities between a null absolute correlation and m_n have conditional probability zero. Thus the weak inequality used in the exact convex first-order condition and the strict selected inequality in the published certificate give the same probability. The displayed random variable is bounded and converges in probability to the right side of (125), so its expectations converge to that constant. This calculation uses conditional independence; the unscaled null correlations share a random variance and need not be unconditionally independent.

Finally, always $R \leq \text{OPT} \leq F_\lambda(S)$. The event $R = F_\lambda(S)$ implies $R = \text{OPT}$, and on U_n the two events coincide. Hence

$$0 \leq \mathbb{P}\{R = \text{OPT}\} - \mathbb{P}\{R = F_\lambda(S)\} \leq \mathbb{P}(U_n^c) \rightarrow 0.$$

This proves the global exactness limit without conditioning the Gaussian correlations on an event involving the null columns. $\square$

H Proofs for Gaussian binary least squares

H.1 Elementary concentration estimates

We use the following estimates at finite dimensions. If G is standard normal, $Q_d \sim \chi_d^2$, and $K_n \sim \text{Bin}(n, 1/2)$, then for $z, x, s \geq 0$,

$$\mathbb{P}\{|G| > z\} \leq e^{-z^2/2}, \tag{221}$$

$$\mathbb{P}\{Q_d > d + 2\sqrt{dx} + 2x\} \leq e^{-x}, \quad \mathbb{P}\{Q_d < d - 2\sqrt{dx}\} \leq e^{-x}, \tag{222}$$

$$\mathbb{P}\{|K_n - n/2| > s\} \leq 2e^{-2s^2/n}. \tag{223}$$

The convention $Q_0 = 0$ makes the second line valid for zero degrees. For completeness, the first inequality follows from $2e^{z^2/2}\overline{\Phi}(z) \leq 1$: the expression equals one at zero and is decreasing, since $z\overline{\Phi}(z) \leq (2\pi)^{-1/2}e^{-z^2/2}$ by integrating $te^{-t^2/2}$ over $t \geq z$. The Gaussian integral gives $\mathbb{E}e^{tQ_d} = (1-2t)^{-d/2}$ for $t < 1/2$. For $0 < t < 1/2$ the upper centered moment satisfies

$$\log \mathbb{E}e^{t(Q_d-d)} \leq \frac{dt^2}{1-2t}.$$

For every $t \geq 0$ the lower centered moment satisfies

$$\log \mathbb{E}e^{-t(Q_d-d)} \leq dt^2.$$

The inequalities follow by expanding $-\log(1-2t)$ and by $\log(1+2t) \geq 2t-2t^2$, respectively. Exponential Markov bounds, with $t = \sqrt{x/d}/(1+2\sqrt{x/d})$ for the upper tail and $t = \sqrt{x/d}$ for the lower tail, give (222) for $d > 0$; the case $d = 0$ is immediate. Finally, $\mathbb{E}e^{t(K_n-n/2)} = \cosh(t/2)^n \leq e^{nt^2/8}$; optimizing Markov's inequality gives (223).

Lemma H.1 (Concentration of the projection mixture). *Let $X \sim \chi_s^2$ and $Z \sim \chi_{M-s}^2$ be independent and $Y = X/(X+Z)$, where $0 \leq s \leq M$ and $M \geq 1$. For $0 \leq t \leq 1$,*

$$\mathbb{P}\{|Y - s/M| > t\} \leq 4e^{-Mt^2/64}.$$

If instead $s \sim \text{Bin}(n, 1/2)$, $n \leq M$, and X, Z have these laws conditional on s , then

$$\mathbb{P}\{|Y - n/(2M)| > 2t\} \leq 6e^{-Mt^2/64}.$$

Proof. The statement is trivial at $t = 0$. Put $x = Mt^2/64$. Except on an event of probability at most $2e^{-x}$, the chi-square estimates give

$$X \leq s + \delta_1, \quad Z \geq M - s - \delta_2, \quad \delta_1 = 2\sqrt{sx} + 2x, \quad \delta_2 = 2\sqrt{(M-s)x}.$$

Here $\delta_1 \leq M(t/4 + t^2/32)$ and $\delta_2 \leq Mt/4$. Monotonicity of the ratio, or the bound $Y \leq 1$ when the displayed lower bound for Z is negative, gives

$$Y - \frac{s}{M} \leq \frac{(M-s)\delta_1 + s\delta_2}{M(M + \delta_1 - \delta_2)} \leq \frac{t/4 + t^2/32}{1 - t/4} \leq \frac{3t}{8} < t.$$

Interchanging X, Z proves the lower tail with the same failure bound. For the mixture, (223) bounds $\mathbb{P}\{|s - n/2| > Mt\}$ by $2e^{-2M^2t^2/n} \leq 2e^{-2Mt^2}$. On the complement, $|s/M - n/(2M)| \leq t$. Combining the two estimates proves the result. $\square$

Lemma H.2 (An elementary Gaussian operator bound). *Suppose $M_N/N \rightarrow \beta < \infty$, and define*

$$C_\beta = 2 \left(\sqrt{\beta + 1} + \sqrt{2(\log 5 + 1)} \right). \quad (224)$$

For an $M_N \times N$ independent standard Gaussian matrix H_N , $\mathbb{P}\{\|H_N\|_{\text{op}} > C_\beta \sqrt{N}\} \leq e^{-N}$ for all sufficiently large N .

Proof. A maximal $1/2$ -separated subset $\mathcal{V}$ of the unit sphere in $\mathbb{R}^N$ is a $1/2$ -net of size at most 5^N . Indeed, its disjoint radius- $1/4$ balls lie in the radius- $5/4$ ball; comparing volumes gives the bound. For any matrix, $\|H\|_{\text{op}} \leq 2 \max_{z \in \mathcal{V}} \|Hz\|$ by approximating a maximizing unit vector by a net point. Each $\|Hz\|^2$ has law χ_M^2 . Apply (222) with $x = (\log 5 + 1)N$ and note that $M + 2\sqrt{Mx} + 2x \leq (\sqrt{M} + \sqrt{2x})^2$. The union bound leaves probability at most e^{-N} . Eventually $M \leq (\beta + 1)N$, giving the stated constant. $\square$

H.2 Recovery and the value gap

We prove [Proposition 10.1](#). For a deterministic $c \in \mathbb{R}^N$, $Bc \sim N(0, a_c^2 I_M)$ independently of w , where $a_c^2 = \rho \|c\|^2 / N$. Direct integration over one coordinate gives

$$\mathbb{E} e^{-\lambda(F(c)-W)} = \left[1 + (2\lambda - 4\lambda^2)a_c^2\right]^{-M/2} \quad (0 < \lambda < 1/2).$$

For example, conditional on $Bc = z$, integration over w gives $\exp((2\lambda^2 - \lambda)\|z\|^2)$, and the remaining Gaussian integral gives the formula. At $\lambda = 1/4$, exponential Markov gives

$$\mathbb{P}\{F(c) \leq W - s\} \leq e^{-s/4} \left(1 + \frac{\rho \|c\|^2}{4N}\right)^{-M/2} \quad (s \geq 0). \quad (225)$$

At a vertex $c = 2\mathbf{1}_S$. Hence

$$\mathbb{P}\{\exists S \neq \emptyset : F(2\mathbf{1}_S) \leq W - s\} \leq e^{-s/4} \Sigma_N, \quad \Sigma_N = \sum_{k=1}^N \binom{N}{k} (1 + \rho k/N)^{-M/2}. \quad (226)$$

First consider recovery. Since every term of Σ_N decreases in ρ , it suffices to set $\rho = (2 + \delta) \log N / \beta_N$, reducing δ if necessary so $0 < \delta \leq 1$. Write $x_k = \rho k/N$ and set $x_0 = \delta/4$. If $x_k \leq x_0$, then $\log(1 + x_k) \geq x_k(1 - x_0/2)$, and

$$\binom{N}{k} (1 + x_k)^{-M/2} \leq \frac{[N e^{-(\beta_N \rho/2)(1-\delta/8)}]^k}{k!} \leq \frac{N^{-\delta k/4}}{k!}.$$

The sum of these terms tends to zero. For $x_k > x_0$, put $t = k/N$. Using $\binom{N}{k} \leq (eN/k)^k$, the logarithm of the term divided by N is at most

$$t \log(e/t) - \frac{\beta_N}{2} \log(1 + \rho t).$$

Choose $t_1 > 0$ so small that $t_1 \log(e/t_1) \leq \frac{1}{4} \log(1 + x_0)$. For $t \leq t_1$ the display is at most $-\frac{1}{4} \log(1 + x_0)$, since $\beta_N \geq 1$. For $t \geq t_1$ it is at most $1 - \frac{1}{2} \log(1 + \rho t_1)$, which tends to $-\infty$. The large- x_k sum is therefore $O(Ne^{-cN})$ for some $c > 0$. With $s = 0$, [\(226\)](#) shows that no other vertex has value at most W with probability tending to one. This proves uniqueness.

For the uniform value comparison, split at $x_k = 7$ instead. On $[0, 7]$, concavity gives $\log(1 + x) \geq (\log 8)x/7 > 0.28x$. Thus the small- x_k sum is at most

$$\sum_{k=1}^N \binom{N}{k} e^{-0.14\beta_N \rho k} = (1 + e^{-0.14\beta_N \rho})^N - 1.$$

For $x_k > 7$, the same entropy bound gives exponent at most $N(1 - \frac{1}{2} \log 8) = -c_0 N$, where $c_0 > 0$. Set $s = 4N \log(1 + e^{-0.14\beta_N \rho}) + 4 \log N$ in [\(226\)](#). The probability is at most $N^{-1} + Ne^{-c_0 N}$, proving [\(131\)](#). The binomial expression, rather than its looser exponential upper bound, is needed for the displayed logarithmic gap.

H.3 The two root exactness probabilities

We prove [Proposition 10.2](#). Put $\zeta_i = b_i^T w$. The first-order conditions for $u = 0$ to minimize the convex function F on $[0, 2]^N$ are $\zeta_i \geq 0$ for all i . Conditional on $w \neq 0$, these inner products are independent centered nondegenerate Gaussian variables. Thus their signs are independent fair coins, proving (132). The matrix has full column rank almost surely, making the objective strictly convex. A root value equal to the integer optimum therefore forces the unique root minimizer to be an optimal vertex, and a vertex root minimizer conversely forces value equality.

Fix a flipped vertex indexed by S , $|S| = k$, and write $z = B_S \mathbf{1}$ and $v = w + 2z$. Its box optimality conditions are $b_i^T v \geq 0$ off S and $-b_i^T v \geq 0$ on S . Given (w, B_S) , the $N - k$ off-support conditions hold with probability $2^{-(N-k)}$. Summing the conditions on S gives the necessary condition $-z^T w - 2\|z\|^2 \geq 0$. Here $z \sim N(0, a^2 I_M)$ independently of w , with $a^2 = \rho k/N$. For $\lambda > 0$ in the range where the moment exists,

$$\mathbb{E} e^{\lambda(-z^T w - 2\|z\|^2)} = \left(1 + 4\lambda a^2 - \lambda^2 a^2\right)^{-M/2}.$$

At $\lambda = 2$, Markov's inequality bounds the necessary event by $(1 + 4\rho k/N)^{-M/2}$. Its indicator is measurable with respect to (w, B_S) , so multiplication by the conditional off-support probability is valid. Summing over vertices gives the first inequality in (133); for $k = 0$ use the already proved probability 2^{-N} . Finally, $\log(1 + 4\rho k/N) \geq (k/N) \log(1 + 4\rho)$ by concavity. The binomial theorem then gives the second inequality.

H.4 Coefficient density and projection mixture

We prove [Theorem 10.4](#). Full column rank and the nonzero conditions of [Lemma 10.3](#) hold almost surely: rank-deficient Gaussian matrices have Lebesgue measure zero, and, conditional on a full-rank matrix, each required functional of v is nondegenerate Gaussian. The sign-orbit identity gives a uniform support and hence $K \sim \text{Bin}(n, 1/2)$.

Fix a nonempty support S of size s . Conditional on H_S ,

$$d^S \sim N\left(0, \frac{c^2}{a^2} (H_S^T H_S)^{-1}\right).$$

Multiplying its conditional density by the Gaussian matrix density shows that its unconditional density at d is proportional to

$$\left(\frac{a}{c}\right)^s \int_{\mathbb{R}^{M \times s}} \det(H^T H)^{1/2} \exp\left[-\frac{1}{2} \text{tr}\left(H^T H \left(I_s + \frac{a^2}{c^2} dd^T\right)\right)\right] dH.$$

Put $T = I_s + (a^2/c^2) dd^T$ and change variables $H = GT^{-1/2}$. The transformation of the M rows has Jacobian $\det(T)^{-M/2}$; the Gram determinant contributes $\det(T)^{-1/2}$. The exponential becomes $\exp(-\text{tr}(G^T G)/2)$. Thus all dependence on d is

$$\det(T)^{-(M+1)/2} = \left(1 + \frac{a^2}{c^2} \|d\|^2\right)^{-(M+1)/2}.$$

The matrix integral is finite, since Hadamard's inequality bounds $\det(G^T G)^{1/2}$ by the product of the column norms, whose Gaussian moments are finite. This argument also covers $M = s$.

To compute the normalizing constant, take independent $Z_s \sim N(0, I_s)$ and $Q \sim \chi_\nu^2$, with $\nu = M - s + 1$. For $D = (c/a)Z_s/\sqrt{Q}$, condition on $Q = q$ and change variables in the Gaussian density. Integration over q gives

$$\begin{aligned} p_D(d) &= \frac{(a/c)^s}{2^{(s+\nu)/2} \pi^{s/2} \Gamma(\nu/2)} \int_0^\infty q^{(s+\nu)/2-1} e^{-q(1+(a^2/c^2)\|d\|^2)/2} dq \\ &= \left(\frac{a}{c}\right)^s \frac{\Gamma((s+\nu)/2)}{\pi^{s/2} \Gamma(\nu/2)} \left(1 + \frac{a^2}{c^2} \|d\|^2\right)^{-(s+\nu)/2}. \end{aligned}$$

Since $s + \nu = M + 1$, this is (136). It is a normalized density with the shape just derived for d^S , so the two laws agree.

For any Borel set $D \subseteq [0, \infty)^s$, the event $|d^S| \in D$ is invariant under every column sign change. Lemma 10.3, divided by $\mathbb{P}\{S = S\} = 2^{-n}$, therefore gives

$$\mathcal{L}((u_j^\circ)_{j \in S} \mid S = S) = \mathcal{L}(|d^S|).$$

This proves (135). Conditional on Q , the probability that its s absolute normal coordinates are all below $at\sqrt{Q}/c$ is $[1 - 2\Phi(at\sqrt{Q}/c)]^s$. Mixing over the binomial support size gives (137). At $t = 0$ every positive-size summand vanishes, leaving the atom 2^{-n} ; the explicitly defined zero-size summand avoids an ambiguous 0^0 .

For the projection statement, fix S and set $X_S = \|P_{L_S} v\|^2$ and $Z_S = \|(I - P_{L_S})v\|^2$. These quantities are invariant under all column sign changes. Conditional on C , their scaled laws are independent χ_s^2 and χ_{M-s}^2 ; these laws depend only on the dimensions. The invariant-event identity transfers their joint distribution to the event $S = S$. On that event the fitted vector is $Cu^\circ = -P_{L_S} v$ and the residual is $(I - P_{L_S})v$. Thus the same joint law holds for X, Z conditional on the selected support. Supports of the same size have identical laws, proving the claimed mixture and (138) by Lemma H.1.

This proof transfers sign-invariant quantities from a fixed support. It does not treat the signed regression vector as independent of the event selecting its support.

H.5 The matching root inactivity scale

Here we complete the asymptotic argument in Theorem 10.7. Use its synthetic representation with $K \sim \text{Bin}(N, 1/2)$, $Q \mid K \sim \chi_{M-K+1}^2$, and conditionally independent normal coordinates Z_j . (223) gives $K/N \rightarrow 1/2$ in probability and $\mathbb{P}\{N/4 \leq K \leq 3N/4\} \rightarrow 1$. On this event the chi-square degree $d = M - K + 1$ is between $N/4$ and $(\beta + 1)N + 1$ for all sufficiently large N . For each fixed $\eta > 0$, choosing a sufficiently small constant $c_\eta > 0$ and $x = c_\eta N$ in (222) bounds $\mathbb{P}\{|Q - d| > \eta N \mid K\}$ by $2e^{-c_\eta N}$ uniformly on that event. Therefore

$$\frac{Q}{N} - \left(\beta_N - \frac{1}{2}\right) = \frac{Q - (M - K + 1)}{N} - \left(\frac{K}{N} - \frac{1}{2}\right) + \frac{1}{N} \xrightarrow{\mathbb{P}} 0.$$

For $\delta \in (0, 1)$, the Gaussian tail bound and $K \leq N$ give

$$\mathbb{P}\{\max_{j \leq K} |Z_j|^2 > 2(1 + \delta) \log N\} \leq N^{-\delta}.$$

For the reverse inequality, integrate the normal density over $[z, z + 1/z]$ for $z \geq 1$. The density throughout this interval is at least a fixed positive constant times $e^{-z^2/2}$, so $\mathbb{P}\{|G| > z\} \geq cz^{-1}e^{-z^2/2}$. At $z = \sqrt{2(1 - \delta) \log N}$, this yields

$$\mathbb{P}\{|G| > z\} \geq c_\delta \frac{N^{-(1-\delta)}}{\sqrt{\log N}}.$$

Conditional on $K \geq N/4$, independence of the normal coordinates then bounds the probability of no exceedance by

$$\exp \left[-\frac{c_\delta N^\delta}{4\sqrt{\log N}} \right].$$

It follows that $\max_{j \leq K} |Z_j|^2 / (2 \log N) \rightarrow 1$ in probability. Since $\beta_N - 1/2 \geq 1/2$, the denominator Q/N stays bounded away from zero in probability, and the ratio in (145) follows.

For completeness, when $\rho = O(\log N)$ the logarithm of (146) has the asserted precision. Taylor expansion gives

$$\log q = -2\rho/N + O(\rho^2/N^2), \quad \log \frac{1+q}{2} = -\rho/N + O(\rho^2/N^2).$$

Multiplying by $M - N + 1$ and $N - 1$, respectively, gives

$$\log \left[\frac{N}{2} q^{M-N+1} \left(\frac{1+q}{2} \right)^{N-1} \right] = \log(N/2) - (2\beta_N - 1)\rho + O(\rho/N + \rho^2/N).$$

Boundedness of β_N and $\rho = O(\log N)$ make the remainder $o(1)$.

H.6 The static variable-order upper bound

We prove the upper bound in Theorem 10.8 directly from the orthant residual mixture. Assume $\rho_N \rightarrow \infty$ and $\rho_N = o(N)$, and abbreviate $\rho = \rho_N$. A permutation fixed independently of the data can be taken as the identity by Gaussian column exchangeability. Define

$$a_N = \frac{\log(e\rho)}{\rho} + \frac{\log N}{N} + \frac{\rho}{N}, \quad \eta_N = a_N^{1/8}, \quad K = \left\lceil \frac{(1 + \eta_N)N}{4(2\beta_N - 1)\rho} \right\rceil.$$

Then $\eta_N \rightarrow 0$, $K \rightarrow \infty$, $K/N \rightarrow 0$, and

$$\log \binom{N}{K} \leq K \log(eN/K) = O(N \log(e\rho)/\rho) = o(N). \quad (227)$$

Put $L_N = \log(2(N+1)\binom{N}{K}) + 2 \log N$, so $L_N/M = O(a_N)$. At a depth- d node with wrong set $T \subseteq [d]$ of size K , the free set is $\{d+1, \dots, N\}$. Its residual is independent of its free matrix and has law $N(0, c_K^2 I_M)$, where $c_K^2 = 1 + 4\rho K/N$. This variance stays bounded because $\rho K/N = (1 + \eta_N)/[4(2\beta_N - 1)] + O(\rho/N)$. Writing $\tilde{r}_{d,T}$ for the orthant value, Theorem 10.4 gives its marginal representation

$$\tilde{r}_{d,T}/c_K^2 \mid J_{d,T} = s \sim \chi_{M-s}^2, \quad J_{d,T} \sim \text{Bin}(N-d, 1/2).$$

This also includes no free coordinates by taking $J_{d,T} = 0$.

For a fixed node, the binomial upper tail gives $J_{d,T} \leq (N-d)/2 + \sqrt{NL_N}$ except on probability at most e^{-2L_N} . Conditional on this index, the chi-square lower tail gives $\tilde{r}_{d,T}/c_K^2 \geq M - J_{d,T} - 2\sqrt{ML_N}$ except on probability at most e^{-L_N} ; replacing its degree by M in the square root weakens the lower bound. There are at most $(N+1)\binom{N}{K}$ such fixed-order nodes. A union bound therefore proves that, with probability tending to one, simultaneously

$$\tilde{r}_{d,T} \geq c_K^2 \left(M - \frac{N-d}{2} \right) - O(\sqrt{NL_N}) \geq c_K^2 (M - N/2) - O(\sqrt{NL_N}).$$

The box value is at least the orthant value. Moreover,

$$c_K^2(M - N/2) - M = 2\rho K(2\beta_N - 1) - \frac{N}{2} \geq \frac{\eta_N N}{2}.$$

The error is $O(N\sqrt{a_N})$, while $W = M + O_{\mathbb{P}}(\sqrt{N \log N})$. Since $\sqrt{a_N} = o(\eta_N)$, every K -wrong node has box bound at least W with probability tending to one. It is pruned for any incumbent of value at most W . Dependence between these node values and W does not matter: the events are intersected by a union bound.

No node with more than K wrong fixings is generated. Every node that branches has at most $K - 1$ wrong fixings and depth less than N . Counting such potential parents, including ones that would already have been pruned, gives

$$\#\text{nodes} \leq 1 + 2 \sum_{d=0}^{N-1} \sum_{j=0}^{K-1} \binom{d}{j} = 1 + 2 \sum_{i=1}^K \binom{N}{i}.$$

For $K \leq N/2$, put $t = K/N$. Since $t^{-i} \leq t^{-K}$ for $i \leq K$,

$$\sum_{i=0}^K \binom{N}{i} \leq t^{-K}(1+t)^N \leq (eN/K)^K.$$

This proves (148). As $K \rightarrow \infty$, rounding has relative error $o(1)$ and

$$K \log(eN/K) = (1 + o(1)) \frac{N}{4(2\beta - 1)\rho} \log \rho.$$

The logarithm of the node count has the same upper bound because the last expression diverges. With an optimal incumbent, the leaves certify OPT , and hence $\text{OPT} - \varepsilon_N$. This establishes the needed upper certificate order without assuming $\text{OPT} = W$.

H.7 An entropy bound for overlapping supports

Lemma H.3 (Overlap forces an entropy loss). *Let $\mathcal{F}$ be a nonempty family of k -subsets of an n -set, where $1 \leq k < n$, and let p_i be the fraction of its members containing i . Suppose $0 < \lambda \leq 1$, $\sum_i p_i^2 \geq \lambda k$, and $\lambda/2 \geq e^2 k/n$. Then*

$$\log \frac{\binom{n}{k}}{|\mathcal{F}|} \geq \frac{\lambda k}{2} \log \frac{n}{ek} - \log(8\sqrt{k}). \quad (228)$$

Proof. Let S be uniform on $\mathcal{F}$, represented by its indicator vector. Entropy subadditivity gives $\log |\mathcal{F}| \leq \sum_i h(p_i)$, where $h(p) = -p \log p - (1-p) \log(1-p)$, with $0 \log 0 = 0$. Indeed, the entropy chain rule writes the joint entropy as a sum of conditional entropies, and conditioning cannot increase entropy. For $q = k/n$, $\sum_i p_i = k$ implies

$$\sum_i h(p_i) = nh(q) - \sum_i d(p_i \| q),$$

where $d(p \| q) = p \log(p/q) + (1-p) \log((1-p)/(1-q))$. Using $\log x \geq 1 - 1/x$ on the second term gives

$$d(p \| q) \geq \phi(p) := p \log(p/q) + q - p \geq 0.$$

The last inequality follows since ϕ is convex with minimum zero at $p = q$. Put $L = \log(1/(eq)) \geq 1$. We claim $\phi(p) \geq Lp(p - \lambda/2)$ for $0 \leq p \leq 1$. For $p \leq \lambda/2$ this follows from nonnegativity. For $p \geq \lambda/2$ it suffices that $\log(p/q) - 1 - L(p - \lambda/2) \geq 0$. This function is concave; at $p = \lambda/2$ it equals $\log(\lambda/(2q)) - 1 \geq 1$, and at $p = 1$ it equals $L\lambda/2 \geq 0$. The claim follows throughout the interval. Summing gives

$$\sum_i d(p_i \| q) \geq L \left(\sum_i p_i^2 - \frac{\lambda}{2} \sum_i p_i \right) \geq \frac{L\lambda k}{2}.$$

Thus $\log |\mathcal{F}| \leq nh(q) - L\lambda k/2$.

We also need a lower estimate for $\binom{n}{k}$. For $V \sim \text{Bin}(n, q)$, $k = nq$ is a mode, as follows by taking ratios of consecutive probabilities. Since $\text{Var}(V) = nq(1 - q) \leq k$, Chebyshev's inequality places probability at least $1/2$ on $|V - k| < \sqrt{2k}$. This interval contains at most $4\sqrt{k}$ integers, each with probability at most $\mathbb{P}\{V = k\}$. Therefore

$$\binom{n}{k} q^k (1 - q)^{n-k} = \mathbb{P}\{V = k\} \geq \frac{1}{8\sqrt{k}}.$$

Taking logarithms gives $\log \binom{n}{k} \geq nh(q) - \log(8\sqrt{k})$. Combining the estimates proves (228). $\square$

H.8 The convex-piece lower bound

We prove (150) in precisely the two regimes used in Theorem 10.8: divergent $\rho_N = o(N)$, or fixed sufficiently large ρ . Constants here are intentionally conservative, allowing an elementary operator estimate. Set

$$p = \Phi(-1) - \Phi(-2) > 0, \quad d_\beta = \frac{\beta p}{64e^2 C_\beta^2}, \quad c_\beta = \frac{\beta p}{256e^2 C_\beta^4}, \quad (229)$$

where C_β is defined in (224). Take $\rho_0 \geq e$ large enough that

$$4 \log(1 + e^{-0.07\beta\rho_0}) < d_\beta/2. \quad (230)$$

The fixed-SNR conclusion uses this choice and not a numerical threshold from a sharper spectral estimate.

For $w \neq 0$, decompose each signed standard Gaussian column as

$$\tilde{h}_i = g_i \hat{w} + h_i^\perp, \quad \hat{w} = w/\|w\|, \quad g_i = \tilde{h}_i^T \hat{w}.$$

Conditional on w , the g_i are independent standard normals, and their orthogonal components are independent Gaussian vectors in $\hat{w}^\perp$, independent of the g_i . Let

$$T = \{i : -2 \leq g_i \leq -1\}, \quad n' = |T|, \quad a = \sqrt{\rho/N} \|w\|.$$

With probability tending to one, $n' \geq pN/2$, and on T , $a \leq -\zeta_i \leq 2a$, where $\zeta_i = b_i^T w = ag_i$. Binomial concentration proves the first assertion; it is valid after conditioning on w because the conditional law does not depend on w . By the chi-square estimates, also with probability tending to one,

$$\frac{\beta N}{2} \leq W \leq 2\beta N, \quad \frac{1}{2}\sqrt{\beta\rho} \leq a \leq 2\sqrt{\beta\rho}. \quad (231)$$

The coarse factors cover $\beta_N \rightarrow \beta$. Let $H^\perp$ be the matrix of all orthogonal components. The deterministic inequality $\|H^\perp\|_{\text{op}} \leq \|\tilde{H}\|_{\text{op}}$ and [Lemma H.2](#) give, on another event of probability tending to one,

$$\|H^\perp\|_{\text{op}} \leq C_\beta \sqrt{N}. \quad (232)$$

This uses the full matrix, so no conditioning on the selected set T is needed. We intersect these events by a union bound.

Put

$$\lambda = \frac{\sqrt{\beta}}{4C_\beta^2\sqrt{\rho}}, \quad k = \left\lfloor \frac{\lambda n'}{2e^2} \right\rfloor, \quad P' = \{x^S : S \subseteq T, |S| = k\}. \quad (233)$$

In either stated regime, $k \rightarrow \infty$, so eventually $1 \leq k < n'$. Also $\lambda \leq 1$ and $k/n' \leq \lambda/(2e^2)$. The lower bound on n' and [\(231\)](#) imply

$$\begin{aligned} ak &\geq \frac{1}{2}\sqrt{\beta\rho} \left(\frac{\lambda p N}{4e^2} - 1 \right) \\ &= \frac{\beta p N}{32e^2 C_\beta^2} - \frac{1}{2}\sqrt{\beta\rho} \geq d_\beta N \end{aligned}$$

for all sufficiently large N . The final inequality uses $\sqrt{\rho} = o(N)$, which holds in both regimes.

Let $\Delta = W - \text{OPT}$. In the divergent regime, [Proposition 10.1](#) gives $\Delta = o(N)$ with probability tending to one; also $\varepsilon_N \leq \rho_N = o(N)$. At a fixed $\rho \geq \rho_0$, eventually $\beta_N \geq \beta/2$, and [\(131\)](#) and [\(230\)](#) give $\Delta < d_\beta N/2 + 4 \log N$ with probability tending to one, while $\varepsilon_N \leq \rho$ is bounded. Consequently, in either regime,

$$\Delta + \varepsilon_N \leq ak \quad (234)$$

holds with probability tending to one. All the following statements are deterministic on the intersection of these events.

Consider any class I admissible for the target $\text{OPT} - \varepsilon_N$, and suppose $I \cap P'$ is nonempty. Let $\mathcal{F}_I = \{S : x^S \in I \cap P'\}$ and let p_i be the fraction of these sets containing i ; set $p_i = 0$ outside T . Then $\sum_i p_i = k$, and the barycenter of these vertices has error coordinates $c_i = 2p_i$. It belongs to $\text{conv } I$. Define $X = -\zeta^T c$. Since all selected coordinates lie in T , $2ak \leq X \leq 4ak$. The orthogonal decomposition gives the exact identity

$$F(c) - W = -2X + \frac{X^2}{W} + \frac{\rho}{N} \|H^\perp c\|^2. \quad (235)$$

Indeed, the component of Bc parallel to w is $(\zeta^T c / \|w\|) \hat{w}$. The choices above yield

$$\frac{X}{W} \leq \frac{4ak}{W} \leq \frac{2}{e^2 C_\beta^2} < \frac{1}{2}.$$

Thus $-2X + X^2/W \leq -3X/2 \leq -3ak$. Also [\(232\)](#) bounds the last term in [\(235\)](#) by $4\rho C_\beta^2 \sum_i p_i^2$. Admissibility forces $F(c) \geq \text{OPT} - \varepsilon_N = W - \Delta - \varepsilon_N$, and therefore

$$\sum_i p_i^2 \geq \frac{3ak - \Delta - \varepsilon_N}{4\rho C_\beta^2} \geq \frac{ak}{2\rho C_\beta^2} \geq \lambda k. \quad (236)$$

The last step uses $a \geq \frac{1}{2}\sqrt{\beta\rho}$. Every admissible class hence meets P' in a family satisfying [Lemma H.3](#).

If m is the largest possible size of such an intersection, then covering P' requires at least $\binom{n'}{k}/m$ classes. Applying (228) gives

$$\log \kappa_{\text{OPT} - \varepsilon_N} \geq \frac{\lambda k}{2} \log \frac{n'}{ek} - \log(8\sqrt{k}). \quad (237)$$

Now

$$\log \frac{n'}{ek} \geq \log \frac{2e}{\lambda} \geq \frac{1}{2} \log \rho, \quad \lambda k \geq \frac{\lambda^2 p N}{4e^2} - \lambda = \frac{\beta p N}{64e^2 C_\beta^4 \rho} - \lambda.$$

Substitution gives

$$\log \kappa_{\text{OPT} - \varepsilon_N} \geq c_\beta \frac{N}{\rho} \log \rho - \frac{\lambda}{4} \log \rho - \log(8\sqrt{k}).$$

The quantity $\lambda \log \rho$ is bounded for $\rho \geq \rho_0$. Since $k \leq N$, the two negative terms are at most $\log(8N)$ for all sufficiently large N . This proves (150).

The proof covers arbitrary convex pieces, because it uses only a class's vertices and their barycenter. It does not use a prescribed variable order, a branching rule, or an assumption that the planted vertex is optimal. The value comparison (234) is what connects the planted descent directions to the target at the actual integer optimum.

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